

Fundamental algebra and analysis II

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Chapter 1

Language of category theory

1.1 Universe

1.1.1 Definition. We call *universe* any set U whose elements are sets, which satisfies the following conditions:

- (a) $\emptyset \in U$;
- (b) If $A \in U$ and $B \in A$, then $B \in U$ (in particular, B is also a set);
- (c) If $A \in U$, then $\mathcal{P}(A) \in U$;
- (d) If $I \in U$ and $(A_i)_{i \in I} \in U^I$, then $\bigcup_{i \in I} A_i \in U$.

1.1.2 Axioms. Every set belongs to some universe.

1.1.3 Proposition. *Let U be a universe.*

- (1) *If $A \in U$ and $B \subseteq A$, then $B \in U$.*
- (2) *For any $A \in U$, one has $\{A\} \in U$.*
- (3) *If A and B are elements of U , then $\{A, B\} \in U$.*
- (4) *If A and B are elements of U , then the ordered pair (A, B) , that is defined as*

$$\{\{A\}, \{A, B\}\},$$

belongs to U .

- (5) *If X and Y are elements of U , then the Cartesian product $X \times Y$, that is defined as*

$$\{(x, y) \mid x \in X, y \in Y\},$$

belongs to U .

- (6) If X and Y are elements of U , then the set of all correspondences from X to Y belongs to U .
- (7) If X and Y are elements of U , then $Y^X \in U$.
- (8) Let I be a non-empty set in U and let $(X_i)_{i \in I} \in U^I$, then $\prod_{i \in I} X_i$ belongs to U .

Proof. (1) By the condition (c) in Definition 1.1.1, one has $\mathcal{P}(A) \in U$. Since B is an element of $\mathcal{P}(A)$, one has $B \in U$.

(2) By the condition (c) in Definition 1.1.1, one has $\mathcal{P}(A) \in U$. Since $\{A\}$ is a subset of $\mathcal{P}(A)$, by (1) we deduce that $\{A\} \in U$.

(3) By the condition (c) in Definition 1.1.1, one has

$$\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\} \in U.$$

By (2), one has $\{A\} \in U$ and $\{B\} \in U$. Hence, by the condition (d) in Definition 1.1.1, we deduce

$$\{A, B\} = \{A\} \cup \{B\} \in U.$$

(4) By (2), one has $\{A\} \in U$. By (3), one has $\{A, B\} \in U$. Still by (3), we deduce that

$$(A, B) := \{\{A\}, \{A, B\}\} \in U.$$

(5) For any $x \in X$ and $y \in Y$, by the condition (b) of Definition 1.1.1, one has $x \in U$ and $y \in U$. By (4) one has $(x, y) \in U$. Hence, by (2) one has $\{(x, y)\} \in U$. For any $x \in X$, by (4) one has

$$\{x\} \times Y = \bigcup_{y \in Y} \{(x, y)\} \in U.$$

Therefore, by the condition (d) in Definition 1.1.1, we deduce

$$X \times Y = \bigcup_{x \in X} \{x\} \times Y \in U.$$

(6) Note that a correspondence from X to Y is of the form $f = ((X, Y), \Gamma)$, where Γ is an element of $\mathcal{P}(X \times Y)$. By (5), $X \times Y \in U$. By (1), we deduce that $\Gamma \in U$. Hence, by (4) we obtain that $((X, Y), \Gamma) \in U$ and then by (2) we get

$$\{((X, Y), \Gamma)\} \in U.$$

By the condition (c) in Definition 1.1.1 and the fact $X \times Y \in U$, we deduce that $\mathcal{P}(X \times Y) \in U$. Therefore, by the condition (d) in Definition 1.1.1, the set of all correspondences from X to Y , which can be written as

$$\bigcup_{\Gamma \in \mathcal{P}(X \times Y)} \{((X, Y), \Gamma)\},$$

belongs to U .

(7) follows from (6) and (1) since Y^X is a subset of the set of all correspondences from X to Y .

(8) Let $X = \bigcup_{i \in I} X_i$. Note that $\prod_{i \in I} X_i$ is a subset of X^I . By (7), X^I belongs to U . Hence, by (1), $\prod_{i \in I} X_i \in U$. $\square$

1.2 Category

1.2.1 Definition. Let U be a universe. We call U -category C the data that consist of:

- (i) a set $\text{Obj}(C)$ (the elements of which are called *objects*);
- (ii) for any $(X, Y) \in \text{Obj}(C) \times \text{Obj}(C)$, a set $C(X, Y) \in U$ (the elements of which are called *morphisms* from X to Y);
- (iii) for any triple $(X, Y, Z) \in \text{Obj}(C)^3$, a composition mapping

$$C(X, Y) \times C(Y, Z) \longrightarrow C(X, Z), \quad (f, g) \longmapsto g \circ f,$$

satisfying the following axioms:

- (a) for any $X \in \text{Obj}(C)$, there exists a morphism $\text{Id}_X \in C(X, X)$ (called the *identity morphism* of X) such that, for any object Y of C , one has

$$\begin{aligned} \forall f \in C(X, Y), \quad f \circ \text{Id}_X &= f, \\ \forall g \in C(Y, X), \quad \text{Id}_X \circ g &= g, \end{aligned}$$

that, is, $f \circ \text{Id}_X = f$ and $\text{Id}_X \circ g = g$ whenever defined.

- (b) for any $(X, Y, Z, W) \in \text{Obj}(C)^4$, $f \in C(X, Y)$, $g \in C(Y, Z)$, $h \in C(Z, W)$, one has

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

If $\text{Obj}(C)$ belongs to U , then we say that C is U -small.

If $\text{Obj}(C)$ is finite and if $C(X, Y)$ is finite for any $(X, Y) \in \text{Obj}(C)^2$, then we say that C is a *finite category*.

1.2.2 Example. Let U be a universe. Then the sets in U and the mappings between sets form a category $U\text{-Set}$. The set of objects of the category $U\text{-Set}$ is U . For any $(A, B) \in U \times U$, the set

$$U\text{-Set}(A, B)$$

is just the set of all mappings from A to B .

1.2.3 Remark. In practice we often omit the mention of the universe U in which we consider the category theory. With this abbreviation, a U -category is simply called a *category*. A U -small U -category is simply called a *small category*. Similarly, the category of sets in U and the mappings between sets is denoted as **Set**.

1.2.4 Notation. (1) We denote by **Top** the category of topological spaces and continuous mappings between topological spaces.

(2) We denote by **Gp** the category of groups and homomorphisms of groups. We denote by **Mon** the category of monoids and homomorphisms of monoids. We denote by **Ab** the category of abelian groups and homomorphism of groups.

(3) We denote by **Ring** the category of unitary rings and homomorphism of unitary rings, and by **CRing** the category of commutative, unitary rings and homomorphisms of unitary rings.

(4) Let K be a unitary ring. We denote by ${}_K\mathbf{Mod}$ the category of left K -modules. If M and N are left K -modules, we usually denote the set of left K -module homomorphisms from M to N as $\text{Hom}_K(M, N)$.

(5) Let K be a commutative unitary ring. We denote by $\mathbf{Alg}_K$ the category of K -algebras and homomorphism of K -algebras. Similarly we denote by $\mathbf{CAlg}_K$ the category of commutative K -algebras and homomorphism of K -algebras.

1.2.5 Remark. Let K be a commutative unitary ring. Recall that a K -algebra is by definition a K -module A together with a bilinear mapping

$$A \times A \longrightarrow A$$

such that A equipped with this bilinear mapping forms a unitary ring.

1.2.6 Notation. Let C be a category, X and Y be objects of C . We often use the expression

$$f : X \longrightarrow Y$$

or

$$X \xrightarrow{f} Y$$

to denote a morphism f in $C(X, Y)$.

1.2.7 Proposition. *Let C be a category and X be an object of C . The identity morphism of X is unique.*

Proof. Let Id_X and Id'_X be both identity morphisms of X . By definition one has

$$\text{Id}_X = \text{Id}_X \circ \text{Id}'_X = \text{Id}'_X$$

□

1.2.8 Definition. Let C be a category and $(X, Y) \in \text{Obj}(C)^2$. Let $f \in C(X, Y)$. We say that f is *left invertible* if there exists $g : Y \rightarrow X$ such that

$$g \circ f = \text{Id}_X,$$

In this case we say that g is a *left inverse* of f . We say that f is *right invertible* if there exists $h : Y \rightarrow X$ such that

$$f \circ h = \text{Id}_Y,$$

In this case we say that h is a *right inverse* of f .

1.2.9 Proposition. Let C be a category and $(X, Y) \in \text{Obj}(C)^2$. Let $f \in C(X, Y)$. Suppose that f is left invertible and right invertible. Then f has only one left inverse and one right inverse, and the left inverse and right inverse of f are equal to each other.

Proof. Let g be a left inverse and h be a right inverse of f . One has

$$g = g \circ \text{Id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{Id}_X \circ h = h$$

□

1.2.10 Definition. Let C be a category and $(X, Y) \in \text{Obj}(C)^2$. Let $f \in C(X, Y)$. If f is left invertible and right invertible, then there exists a unique morphism $f^{-1} \in C(Y, X)$ such that

$$f \circ f^{-1} = \text{Id}_Y, \quad f^{-1} \circ f = \text{Id}_X.$$

In this case we say that f is an *isomorphism* and that f^{-1} is the *inverse* of f .

If there exists an isomorphism in $C(X, Y)$, then we say that X and Y are *isomorphic*.

1.2.11 Example. We consider a category C having only one object $*$, namely $\text{Obj}(C)$ is the one point set $\{*\}$. In this case, $C(*, *)$ forms a monoid under the composition law. The identity morphism of $*$ is the neutral element of this monoid, and isomorphisms are just invertible element of the monoid.

1.2.12 Proposition. *Let C be a category. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are isomorphisms, then $g \circ f$ is an isomorphism, and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof. One has

$$\begin{aligned} (f^{-1} \circ g^{-1}) \circ (g \circ f) &= f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{Id}_Y \circ f = f^{-1} \circ f = \text{Id}_X, \\ (g \circ f) \circ (f^{-1} \circ g^{-1}) &= g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{Id}_Y \circ g^{-1} = g \circ g^{-1} = \text{Id}_Z. \end{aligned}$$

□

1.2.13 Definition. Let C be a category. We define the *opposite category* C^{op} of C as follows: the category C^{op} has the same objects as C , and, for any $(X, Y) \in \text{Obj}(C) \times \text{Obj}(C)$,

$$C^{\text{op}}(X, Y) := C(Y, X).$$

The composition mapping

$$C^{\text{op}}(X, Y) \times C^{\text{op}}(Y, Z) \longrightarrow C^{\text{op}}(X, Z)$$

is given by the composition mapping

$$C(Y, X) \times C(Z, Y) \longrightarrow C(Z, X), \quad (f, g) \longmapsto f \circ g$$

in the structure of C .

Note that, if we use an arrow $X \xrightarrow{f} Y$ to denote a morphism in C , it suffices to reverse the arrow to denote the same morphism viewed as a morphism in C^{op} .

1.2.14 Definition. A diagram of morphisms in a category is a directed graph whose vertices are objects and arrows are morphisms. A diagram is *commutative* if for any two vertices X, Y , the morphisms obtained by composing morphisms along different paths coincide. For example, in a category C , the diagram

$$\begin{array}{ccc} & Y & \\ f \nearrow & & \searrow g \\ X & \xrightarrow{h} & Z \end{array}$$

is commutative if and only $h = g \circ f$. The diagram

$$\begin{array}{ccc} X & \xrightarrow{a} & Y \\ b \downarrow & & \downarrow c \\ Z & \xrightarrow{d} & W \end{array}$$

is commutative if and only if

$$d \circ b = c \circ a.$$

1.3 Functors

1.3.1 Definition. Let C and D be categories. We call *functor* from C to D any data that consist of:

- (i) a mapping on objects $F : \text{Obj}(C) \rightarrow \text{Obj}(D)$;
- (ii) for any pair $(X, Y) \in C \times C$, a mapping of sets

$$F : C(X, Y) \rightarrow D(FX, FY),$$

such that:

- (a) $F(\text{Id}_X) = \text{Id}_{F(X)}$;
- (b) for any morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

$$\text{in } C, F(g \circ f) = F(g) \circ F(f).$$

Usually, we use the expression $F : C \rightarrow D$ to denote a functor F from C to D .

1.3.2 Example. Consider

$$P : \mathbf{CRing} \rightarrow \mathbf{Set}$$

such that for all $A \in \text{Obj}(\mathbf{CRing})$

$$P(A) = \{(x, y, z) \in A^3 \mid x^2 + y^2 = z^2\}$$

and for all homomorphism $f : A \rightarrow B$ of commutative rings,

$$P(f) : P(A) \longrightarrow P(B), \quad (x, y, z) \longmapsto (f(x), f(y), f(z))$$

1.3.3 Example. Let C be a category and X be an object of C . We denote by h_X the functor from C to $\mathbf{Set}$ that sends any object Y of C to the set $C(X, Y)$, and send any morphism $f : Y \rightarrow Z$ to the mapping

$$h_X(f) : C(X, Y) \longrightarrow C(X, Z), \quad g \longmapsto f \circ g.$$

1.3.4 Example. Let C be a category, and $\text{Id}_C : C \rightarrow C$ be a mapping that send $X \in \text{Obj}(C)$ to X and $f : X \rightarrow Y$ to f . This is a functor, called the *identity functor* of C .

1.3.5 Example. Let C, D, E be categories, $F : C \rightarrow D$ be functors. We define a functor

$$G \circ F : C \longrightarrow E$$

such that for all $X \in \text{Obj}(C)$,

$$(G \circ F)(X) = G(F(X))$$

and for all morphism $f : X \rightarrow Y$ in C ,

$$(G \circ F)(f) = G(F(f)).$$

1.3.6 Definition. Let C and D be categories, F and G be functors from C to D . We call *natural transformation* (morphism of functors) from F to G any family

$$(\varphi_X : F(X) \longrightarrow G(X))_{X \in \text{Obj}(C)}$$

such that, for any morphism $f : X \rightarrow Y$ in C , the diagram

$$\begin{array}{ccc} F(X) & \xrightarrow{\varphi_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\varphi_Y} & G(Y) \end{array}$$

is commutative, namely

$$G(f) \circ \varphi_X = \varphi_Y \circ F(f).$$

We denote by $\text{Nat}(F, G)$ the set of all natural transformations from F to G . We often use the expression $\varphi : F \rightarrow G$ to signify that φ is a natural transformation from F to G .

If F, G and H are functors from C to D , $\varphi = (\varphi_X)_{X \in \text{Obj}(C)}$ is a natural transformation from F to G , and $\psi = (\psi_X)_{X \in \text{Obj}(C)}$ is a natural transformation from G to H , then

$$\psi \circ \varphi := (\psi_X \circ \varphi_X)_{X \in \text{Obj}(C)}$$

is a natural transformation from F to H . We call it the *composite* of ψ and φ .

1.3.7 Example. Let C and D be categories, F be a functor from C to D . Then $(\text{Id}_{F(X)})_{X \in \text{Obj}(C)}$ is a natural transformation from F to itself. It is denoted as Id_F and is called the *identity natural transformation* of F .

1.3.8 Remark. Let C and D be categories. Let $\mathbf{Fun}(C, D)$ be the data composed of the set (which does not necessarily belong to the universe with respect to which the categories C and D are considered) of all functors from C to D and natural transformations between them. Then $\mathbf{Fun}(C, D)$ forms a category (with respect to some suitable universe). It is called the category of functors from C to D .

1.3.9 Definition. Let C and D be two categories, F and G be functors from C to D . If every φ_X , $X \in \text{Obj}(C)$ is an isomorphism, we say that φ is a *natural isomorphism*. In this case the family

$$(\varphi_X^{-1} : G(X) \rightarrow F(X))_{X \in \text{Obj}(C)}$$

is a natural transformation from G to F . It is denoted as φ^{-1} and is called the inverse of φ . Note that

$$\varphi^{-1} \circ \varphi = \text{Id}_F, \quad \varphi \circ \varphi^{-1} = \text{Id}_G.$$

If there exists a natural isomorphism from F to G , we say that the functors F and G are *isomorphic*.

1.3.10 Remark. An isomorphism of functors between the categories C and D is an isomorphism in the category $\mathbf{Fun}(C, D)$.

1.3.11 Proposition. Let U be a universe, C be a small U -category and D be a U -category. Then for any functors F and G from C to D , the set of natural transformations from F to G belongs to U .

Proof. The set of natural transformations from F and G is a subset of

$$\prod_{X \in \text{Obj}(C)} D(F(X), G(X)).$$

Since C is assumed to be a small category, one has $\text{Obj}(C) \in U$. Moreover, since D is a U -category, for any $X \in \text{Obj}(C)$, one has

$$D(F(X), G(X)) \in U.$$

Hence

$$\prod_{X \in \text{Obj}(C)} D(F(X), G(X)) \in U.$$

Therefore, by (1) of Proposition 1.1.3, the set of natural transformations from F to G belongs to U . □

1.3.12 Corollary. Let U be a universe. Let C be a U -small category and D be a U -category. Then $\mathbf{Fun}(C, D)$ is a U -category.

1.3.13 Remark. In the particular case where D is the category of sets, the functor category $\mathbf{Fun}(C, \mathbf{Set})$ is also denoted as $\widehat{C}$.

1.4 Yoneda Lemma

1.4.1 Theorem (Yoneda Lemma). *Let C be a category, F be a functor from C to $\mathbf{Set}$, and $X \in C$. Then the mapping*

$$\beta : \text{Nat}(h_X, F) \longrightarrow F(X), \quad \psi \longmapsto \psi_X(\text{Id}_X)$$

is a bijection.

Proof. We construct a mapping

$$\alpha : F(X) \longrightarrow \text{Nat}(h_X, F)$$

as follows. Given $x \in F(X)$, we define $\alpha(x)$ as the family

$$\alpha(x)_Y : C(X, Y) \rightarrow F(Y), \quad f \longmapsto F(f)(x),$$

where $Y \in \text{Obj}(C)$.

We check that α is a natural transformation, namely, for any morphism $u : A \rightarrow B$ in C , one has

$$F(u) \circ \alpha(x)_A = \alpha(x)_B \circ h_X(u).$$

Indeed, for any $f \in C(X, A)$,

$$\begin{aligned} (F(u) \circ \alpha(x)_A)(f) &= F(u)(F(f)(x)) = F(u \circ f)(x) \\ &= \alpha(x)_B(u \circ f) = (\alpha(x)_B \circ h_X(u))(f). \end{aligned}$$

Now we show these two mappings α and β are inverse to each other. Let $\psi \in \text{Nat}(h_X, F)$ and $x = \psi_X(\text{Id}_X)$. Since the diagram

$$\begin{array}{ccc} C(X, X) & \xrightarrow{\psi_X} & F(X) \\ h_X(f) \downarrow & & \downarrow F(f) \\ C(X, Y) & \xrightarrow{\psi_Y} & F(Y) \end{array}$$

is commutative, for any $f : X \rightarrow Y$, one has

$$\alpha(x)_Y(f) = F(f)(x) = F(f)(\psi_X(\text{Id}_X)) = \psi_Y(h_X(f)(\text{Id}_X)) = \psi_Y(f).$$

Hence $\alpha(x) = \psi$. Conversely,

$$\alpha(x)_X(\text{Id}_X) = F(\text{Id}_X)(x) = \text{Id}_{F(X)}(x) = x.$$

□

1.4.2 Notation. Let C be a category, X and Y be objects of C and $f : X \rightarrow Y$ be a morphism. We denote by $v(f)$ the natural transform from h_Y to h_X corresponding to f by the bijection

$$\text{Nat}(h_Y, h_X) \longrightarrow h_X(Y) = C(X, Y)$$

defined by Yoneda lemma.

Note that, for any $W \in \text{obj } C$,

$$v(f)_W : C(Y, W) \longrightarrow C(X, W)$$

sends $u \in C(Y, W)$ to $u \circ f$ (see the construction of the mapping α in the proof of Theorem 1.4.1).

1.4.3 Proposition. *Let C be a category.*

(1) *Let X be an object of C . One has*

$$\text{Id}_{h_X} = v(\text{Id}_X).$$

(2) *Let X, Y and Z be objects of C , $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be morphisms. Then*

$$v(g \circ f) = v(f) \circ v(g).$$

(3) *Let X and Y be object of C , and $f : X \rightarrow Y$ be a morphism. Then f is an isomorphism if and only if $v(f)$ is an isomorphism. Moreover, in the case where f is an isomorphism, one has*

$$v(f^{-1}) = v(f)^{-1}.$$

Proof. (1) For any $W \in \text{obj } C$ and any $u \in C(X, W)$, one has

$$v(\text{Id}_X)_W(u) = u \circ \text{Id}_X = u.$$

Hence $v(\text{Id}_X) = \text{Id}_{h_X}$.

(2) For any $W \in \text{obj } C$ and any $u \in C(Z, W)$, one has

$$v(g \circ f)_W(u) = u \circ (g \circ f) = (u \circ g) \circ f = v(f)_W(u \circ g) = v(f)_W(v(g)_W(u)).$$

(3) Suppose that f is an isomorphism. Then, by (2) and (1),

$$\begin{aligned} v(f^{-1}) \circ v(f) &= v(f \circ f^{-1}) = v(\text{Id}_Y) = \text{Id}_{h_Y}, \\ v(f) \circ v(f^{-1}) &= v(f^{-1} \circ f) = v(\text{Id}_X) = \text{Id}_{h_X}, \end{aligned}$$

Therefore, $v(f)$ is an isomorphism, and $v(f^{-1}) = v(f)^{-1}$. Conversely, suppose that $v(f)$ is an isomorphism and φ is its inverse. Let $g : Y \rightarrow X$ be the morphism such that $v(g) = \varphi$. Then

$$v(g) \circ v(f) = v(f \circ g) = \text{Id}_{h_Y}, \quad v(f) \circ v(g) = v(g \circ f) = \text{Id}_{h_X},$$

which leads to

$$g \circ f = \text{Id}_X, \quad f \circ g = \text{Id}_Y.$$

Hence f is an isomorphism. $\square$

1.4.4 Remark. In the case where C is a small category, the mapping $X \rightarrow h_X$ together with the bijections

$$C(Y, X) \longrightarrow \text{Nat}(h_X, h_Y)$$

defines a functor from C^{op} to $\widehat{C}$. This functor is called the *Yoneda functor*.

1.5 Representable functors

1.5.1 Definition. Let C be a category, F be a functor from C to **Set**. If there exists an object X of C such that F and h_X are isomorphic, then we say that the functor F is *representable* and that X is an object that *represents* the functor F . An isomorphism from h_X to F is called a *representation* of the functor F by the object X .

Let F be a representable functor from C to **Set** and X be an object of C that represents F . Let $\psi : h_X \rightarrow F$ be an isomorphism. Then the element $\psi_X(\text{Id}_X) \in F(X)$ is called the *universal object* of the representation ψ .

1.5.2 Proposition. Let C be a category, F be a functor from C to **Set**. Assume that F is representable and that $\varphi : h_X \rightarrow F$ is an isomorphism of functors that represents F , where X is an object of C . For any object Y of C and any $y \in F(Y)$, there exists a unique morphism $f : X \rightarrow Y$ such that $F(f)(x) = y$. Actually the morphism f is given by $\varphi_Y^{-1}(y)$.

Proof. We first prove the uniqueness of f in assuming the existence. Since φ is a natural transformation from h_X to F , the diagram

$$\begin{array}{ccc} h_X(X) & \xrightarrow{\varphi_X} & F(X) \\ h_X(f) \downarrow & & \downarrow F(f) \\ h_X(Y) & \xrightarrow{\varphi_Y} & F(Y), \end{array}$$

which can also be written as

$$\begin{array}{ccc} C(X, X) & \xrightarrow{\varphi_X} & F(X) \\ f \circ - \downarrow & & \downarrow F(f) \\ C(X, Y) & \xrightarrow{\varphi_Y} & F(Y), \end{array}$$

is commutative. Therefore, one has

$$y = F(f)(x) = F(f)(\varphi_X(\text{Id}_X)) = \varphi_Y(f \circ \text{Id}_X) = \varphi_Y(f),$$

which leads to

$$f = \varphi_Y^{-1}(y).$$

The same computation shows that $f := \varphi_Y^{-1}(y)$ satisfies the equality $F(f)(x) = y$. Thus the statement is proved. $\square$

1.5.3 Corollary. *Let C be a category and F be a functor from C to **Set**. Then the object representing F is unique up to a unique isomorphism. Namely, if X and Y are objects of C representing F , and if $\varphi : h_X \rightarrow F$ and $\psi : h_Y \rightarrow F$ are corresponding representations, then there exists a unique isomorphism $f : X \rightarrow Y$ such that $F(f)$ sends the universal object of φ to that of ψ .*

Proof. Let x and y be respectively the universal objects of φ and ψ . By Proposition 1.5.2, $f := \varphi_Y^{-1}(y)$ is the unique morphism in $C(X, Y)$ such that $F(f)(x) = y$. It remains to check that f is an isomorphism.

Note that, by Proposition 1.5.2, $g := \psi_X^{-1}(x)$ is the unique morphism in $C(Y, X)$ such that $F(g)(y) = x$. Therefore, one has

$$F(g \circ f)(x) = (F(g) \circ F(f))(x) = F(g)(F(f)(x)) = F(g)(y) = x.$$

By Proposition 1.5.2, we obtain that $g \circ f = \text{Id}_X$. Similarly, one has

$$F(f \circ g)(y) = F(f)(F(g)(y)) = F(f)(x) = y.$$

Hence, by Proposition 1.5.2, we obtain that $f \circ g = \text{Id}_Y$. Therefore, f is an isomorphism. $\square$

1.5.4 Example. Consider the forgetful functor F from **Mon** to **Set** that sends a monoid M to its underlying set, and sends a homomorphism of monoids $f : M \rightarrow N$ to itself (viewed as a mapping between sets). This functor is representable by the additive monoid $\mathbb{N}$ of natural numbers. In fact, for any element x of a monoid M , there exists a unique homomorphism of monoids $\mathbb{N} \rightarrow M$ that sends 1 to x . Therefore, the mapping

$$\varphi_M : h_{\mathbb{N}}(M) = \mathbf{Mon}(\mathbb{N}, M) \longrightarrow F(M) = M, \quad f \longmapsto f(1)$$

is a bijection, and the family $\varphi = (\varphi_M)_{M \in \text{Obj}(\mathbf{Mon})}$ forms a representation of the forgetful functor M .

1.5.5 Example. Let K be a commutative unitary ring and I be a set. The functor F from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends any K -module W to the set W^I is representable by the direct sum $K^{\oplus I}$. Recall that $K^{\oplus I}$ is defined as the set of $(a_i)_{i \in I} \in K^I$ such that $\{i \in I \mid a_i \neq 0\}$ is a finite set. For any K -module W , the mapping

$$W^I \longrightarrow \mathbf{Mod}_K(K^{\oplus I}, W)$$

sending $(x_i)_{i \in I}$ to

$$((a_i)_{i \in I} \in K^{\oplus I}) \longmapsto \sum_{i \in I} a_i x_i$$

is a bijection. These mappings define an isomorphism from the functor F to $h_{K^{\oplus I}}$.

1.5.6 Example. Let K be a unitary ring and V be a right K -module. Let V' be a right sub- K -module. Let F be the functor from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends a right K -module W to the set of $f \in \mathbf{Mod}_K(V, W)$ such that

$$\forall x \in V', \quad f(x) = 0_W.$$

Then the functor is representable by V/V' . In fact, for any $f \in F(W)$, there exists a unique homomorphism of right K -modules $\tilde{f}: V/V' \rightarrow W$ that

$$\forall x \in V, \quad \tilde{f}([x]) = f(x).$$

Exercises

1. Consider the identity functor

$$F: \mathbf{Set} \longrightarrow \mathbf{Set}$$

that sends a set X to itself (i.e., $F(X) = X$), and sends a mapping $f: X \rightarrow Y$ to itself. Prove that this functor is representable by $\{\emptyset\}$.

2. Consider the functor

$$F: \mathbf{Set} \longrightarrow \mathbf{Set}$$

that sends a set X to $\{\emptyset\}$, and sends a mapping $f: X \rightarrow Y$ to the identity mapping of $\{\emptyset\}$. Prove that this functor is representable by the empty set.

3. Consider the functor

$$F: \mathbf{Set}^{\text{op}} \longrightarrow \mathbf{Set}$$

that sends a set X to $\{\emptyset\}$, and sends any morphism of $\mathbf{Set}^{\text{op}}$ to the identity mapping of $\{\emptyset\}$. Prove that this functor is representable by $\{\emptyset\}$.

4. Let n be a positive integer. Consider the functor

$$F: \mathbf{Set} \longrightarrow \mathbf{Set}$$

that sends a set X to the n -fold Cartesian product X^n , and sends a mapping $f: X \rightarrow Y$ to the morphism $F(f): X^n \rightarrow Y^n$ defined by

$$F(f)(x_1, \dots, x_n) = (f(x_1), \dots, f(x_n)).$$

Prove that this functor is representable by $\{1, 2, \dots, n\}$.

5. Let X be a set. Consider the functor

$$F: \mathbf{Set}^{\text{op}} \longrightarrow \mathbf{Set}$$

that sends any set Y to the set $C(Y, X)$ of all correspondences from Y to X , and sends any mapping of sets $f: Y \rightarrow Z$ (viewed as a morphism from Z to Y in $\mathbf{Set}^{\text{op}}$) to

$$C(Z, X) \longrightarrow C(Y, X), \quad g \longmapsto g \circ f.$$

Prove that the functor F is representable by the power set of X .

6. Define the functor of power set

$$F: \mathbf{Set}^{\text{op}} \longrightarrow \mathbf{Set}$$

that sends a set X to its power set $\mathcal{P}(X)$, and sends any mapping of sets $f: X \rightarrow Y$ (viewed as a morphism from Y to X in $\mathbf{Set}^{\text{op}}$) to the mapping

$$F(f): \mathcal{P}(Y) \longrightarrow \mathcal{P}(X), \quad S \longmapsto f^{-1}(S).$$

Prove that this functor is representable by the 2-element set $\{0, 1\}$.

7. Let A be a set, and R be a binary relation on A . Let

$$F: \mathbf{Set} \longrightarrow \mathbf{Set}$$

be the functor that sends a set X to the set of mappings $f: A \rightarrow X$ such that

$$\forall (a, b) \in A \times A, \quad a R b \implies f(a) = f(b).$$

Prove that this functor is representable by the quotient set $A/\sim$, where $\sim$ is the equivalence relation generated by R .

8. Let I be a set and $(A_i)_{i \in I}$ be a family of sets. Let $\coprod_{i \in I} A_i$ be the disjoint union of $(A_i)_{i \in I}$, which is defined as the set of $(i, x) \in I \times \bigcup_{i \in I} A_i$ such that $x \in A_i$. Consider the functor

$$F: \mathbf{Set} \longrightarrow \mathbf{Set}$$

that sends any set X to

$$\prod_{i \in I} \mathbf{Set}(A_i, X)$$

and any mapping $f: X \rightarrow Y$ to

$$\prod_{i \in I} \mathbf{Set}(A_i, X) \longrightarrow \prod_{i \in I} \mathbf{Set}(A_i, Y), \quad (g_i)_{i \in I} \longmapsto (f \circ g_i)_{i \in I}.$$

Prove that the functor F is representable by $\coprod_{i \in I} A_i$.

9. Let S be a subset of a set A . Consider the functor

$$F: \mathbf{Set}^{\text{op}} \longrightarrow \mathbf{Set}$$

that sends a set X to the set of morphisms $f \in \mathbf{Set}(X, A)$ such that $\text{Im}(f) \subseteq S$. Prove that this functor is representable by the subset S of A .

10. Let n be a positive integer. Consider the functor

$$F: \mathbf{Mon} \longrightarrow \mathbf{Set}$$

that sends a monoid M to the n -fold Cartesian product M^n , and sends a morphism of monoids $f: M \rightarrow N$ to the morphism $F(f): M^n \rightarrow N^n$ defined by

$$F(f)(m_1, \dots, m_n) = (f(m_1), \dots, f(m_n)).$$

Prove that this functor is representable by the free monoid on n generators (i.e., the monoid of finite strings over an n -letter alphabet under concatenation).

11. Let $\mathbb{N}_*$ be the multiplicative monoid of positive integers and P be the set of all prime numbers. Consider the functor

$$F: \mathbf{CMon} \longrightarrow \mathbf{Set}$$

from the category of commutative monoids to the category of sets, that sends a commutative monoid M to M^P and sends a morphism of monoids $f: M \rightarrow N$ to

$$M^P \longrightarrow N^P, \quad (x_p)_{p \in P} \longmapsto (f(x_p))_{p \in P}.$$

Prove that the functor F is representable by $\mathbb{N}_*$.

12. Consider the functor

$$F: \mathbf{Mon} \longrightarrow \mathbf{Set}$$

that sends a monoid M to the set

$$F(M) := \{m \in M \mid m^2 = m\}$$

of its idempotent elements, and sends a morphism of monoids $f: M \rightarrow N$ to the restriction of f to the set of idempotent elements of M . Prove that this functor is representable by the 2-element monoid $\{e, a\}$ with neutral element e , that satisfies $a^2 = a$.

13. Consider the functor

$$F: \mathbf{Mon} \longrightarrow \mathbf{Set}$$

that sends a monoid M to the set $M^\times$ of its invertible elements, and sends a morphism of monoids $f: M \rightarrow N$ to the restriction of f to $M^\times$. Prove that this functor is representable by the additive monoid of integers $(\mathbb{Z}, +, 0)$.

14. Consider the forgetful functor

$$F: \mathbf{Gp} \longrightarrow \mathbf{Set}$$

that sends a group G to its underlying set (i.e., $F(G) = G$), and sends a morphism of groups $f: G \rightarrow H$ to itself as a morphism of sets. Prove that this functor is representable by the additive group of integers $(\mathbb{Z}, +, 0)$.

15. Let I be a set. Consider the functor

$$F: \mathbf{Ab} \longrightarrow \mathbf{Set}$$

that sends an abelian group A to A^I and sends a morphism of groups $f: A \rightarrow B$ to

$$A^I \longrightarrow B^I, \quad (a_i)_{i \in I} \longmapsto (f(a_i))_{i \in I}.$$

Prove that the functor F is representable by the free abelian group $\mathbb{Z}^{\oplus I}$.

16. Let G be a group, and S be a set of elements of G . Consider the functor

$$F: \mathbf{Gp} \longrightarrow \mathbf{Set}$$

that sends a group H to the set of group homomorphisms $f: G \rightarrow H$ such that $f(x) = e_H$ for each $x \in S$. Prove that this functor is representable by the quotient group G/N , where N is the minimal normal subgroup of G containing S .

17. Let n be a positive integer. Consider the functor

$$T_n: \mathbf{Ab} \longrightarrow \mathbf{Set}$$

that sends an abelian group A to the set of all n -torsion elements

$$T_n(A) = \{x \in A \mid n \cdot x = 0_A\}$$

of A , and sends a morphism of abelian groups $f: A \rightarrow B$ to the restriction of f to the n -torsion elements of A . Prove that this functor is representable by the abelian group $\mathbb{Z}/n\mathbb{Z}$.

18. Consider the functor

$$T: \mathbf{Ab} \longrightarrow \mathbf{Set}$$

that sends an abelian group A to the set of all torsion elements

$$T(A) = \{x \in A \mid n \cdot x = 0_A \text{ for some } n \geq 1\}$$

of A , and sends a morphism of abelian groups $f: A \rightarrow B$ to the restriction of f to the torsion elements of A .

Suppose that T is representable by an abelian group X , and that $\varphi: h_X \rightarrow T$ is an isomorphism of functors that represents T .

$$\begin{array}{ccccc} X & h_X(X) = \mathbf{Ab}(X, X) & \xrightarrow{\varphi_X} & T(X) & \\ f \downarrow & h_X(f) \downarrow & & \downarrow T(f) & \\ A & h_X(A) = \mathbf{Ab}(X, A) & \xrightarrow{\varphi_A} & T(A) & \end{array}$$

Then for any abelian group A and any element $y \in T(A)$, there is a unique homomorphism $f = \varphi_A^{-1}(y) \in \mathbf{Ab}(X, A)$ such that

$$f(\varphi_X(\text{Id}_X)) = \varphi_A(f) = y.$$

- (1) Let k be a positive integer. Prove that for any group homomorphism $f: X \rightarrow A$,

$$f(\varphi_X(k \cdot \text{Id}_X)) = \varphi_A(k \cdot f) = k \cdot f(\varphi_X(\text{Id}_X)).$$

Hint: Use the naturality of φ (the commutative diagram above) with f replaced by $k \cdot f$, and the fact that group homomorphisms preserve scalar multiplication.

- (2) Prove that there exists a positive integer n such that $n \cdot \varphi_X(\text{Id}_X) = 0_X$ (the zero element of X).

- (3) Let $A = \mathbb{Z}/(n+1)\mathbb{Z}$ (where n is the integer from Part (2)), and let $y = \bar{1} \in A$ denote the class of 1 in A . Using Parts (1) and (2), derive a contradiction, and thereby conclude that the functor T is not representable.

Hint: Note that $n \cdot y \neq 0_A \in T(A)$.

19. Consider the forgetful functor

$$F: \mathbf{Ring} \longrightarrow \mathbf{Set}$$

that sends a unitary ring R to its underlying set, and sends a morphism of unitary rings $f: R \rightarrow S$ to itself as a morphism of sets. Prove that this functor is representable by the polynomial ring with integer coefficients $\mathbb{Z}[T]$.

20. Let n be a positive integer. Consider the functor

$$F: \mathbf{CRing} \longrightarrow \mathbf{Set}$$

that sends a commutative unitary ring R to the set of elements $r \in R$ satisfying $r^n = r$, and sends a morphism of commutative unitary rings $f: R \rightarrow S$ to the restriction of f to these elements. Prove that this functor is representable by the quotient ring $\mathbb{Z}[T]/(T^n - T)$.

21. Let R be a commutative unitary ring, S be a set of elements of R . Consider the functor

$$F: \mathbf{CRing} \longrightarrow \mathbf{Set}$$

that sends a commutative unitary ring R' to the set of ring homomorphisms $f \in \mathbf{CRing}(R, R')$ such that $f(r) = 0_{R'}$ for all $r \in S$. Prove that this functor is representable by the quotient ring R/I_S , where I_S is the ideal of R generated by S .

22. Let R be a commutative unitary ring. Consider the forgetful functor

$$F: \mathbf{Mod}_R \rightarrow \mathbf{Set}$$

that sends an R -module M to its underlying set (i.e., $F(M) = M$), and sends a morphism of R -modules $f: M \rightarrow N$ to itself as a morphism of sets. Prove that this functor is representable by R (viewed as an R -module).

23. Let R be a commutative unitary ring and I be a set. Consider the functor

$$F: \mathbf{Mod}_R \rightarrow \mathbf{Set}$$

that sends an R -module M to M^I , and sends a morphism of R -modules $f: M \rightarrow N$ to the morphism $F(f): M^I \rightarrow N^I$ defined by

$$F(f)((x_i)_{i \in I}) = (f(x_i))_{i \in I}.$$

Prove that this functor is representable by the free R -module $R^{\oplus I}$.

- 24.** Let R be a commutative unitary ring. Define the forgetful functor

$$F: \mathbf{Alg}_R \longrightarrow \mathbf{Set}$$

that sends a R -algebra A to its underlying set, and sends a morphism of R -algebras $f: A \rightarrow B$ to itself as a morphism of sets. Prove that this functor is representable by the R -algebra $R[T]$.

- 25.** Consider the forgetful functor

$$F: \mathbf{Top} \longrightarrow \mathbf{Set}$$

that sends a topological space X to its underlying set, and sends a morphism of topological spaces $f: X \rightarrow Y$ (i.e., a continuous map) to itself as a morphism of sets. Prove that this functor is representable by the singleton topological space $\{\emptyset\}$ (i.e., a one-point space with the trivial topology).

- 26.** Consider the functor

$$F: \mathbf{Top} \longrightarrow \mathbf{Set}$$

that sends a topological space X to the set of all continuous loops in X , that is

$$F(X) = \{f: [0, 1] \rightarrow X \mid f \text{ is continuous, } f(0) = f(1)\},$$

and sends a morphism of topological spaces $g: X \rightarrow Y$ to

$$F(g): F(X) \longrightarrow F(Y), \quad f \longmapsto g \circ f.$$

Prove that this functor is representable by the unit circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

with the Euclidean topology.

- 27.** Let X be a topological space and $\sim$ be an equivalence relation on X . Consider the functor

$$F: \mathbf{Top} \longrightarrow \mathbf{Set}$$

that sends a topological space Y to the set of all continuous maps $f \in \mathbf{Top}(X, Y)$ such that

$$\forall (x, y) \in X \times X, \quad x \sim y \implies f(x) = f(y).$$

Prove that this functor is representable by the quotient topological space $X/\sim$.

28. Define the functor

$$F: \mathbf{Top}^{\text{op}} \longrightarrow \mathbf{Set}$$

that sends a topological space X to the set of all open subsets of X , that is

$$F(X) = \{U \subseteq X \mid U \text{ is open in } X\},$$

and sends a continuous map $f: X \rightarrow Y$ (viewed as a morphism in $\mathbf{Top}^{\text{op}}(Y, X)$) to

$$F(f): F(Y) \longrightarrow F(X), \quad V \longmapsto f^{-1}(V).$$

Prove that this functor is representable by the Sierpiński space $S = \{0, 1\}$ with $\{\emptyset, \{1\}, S\}$ as its set of open subsets.

Chapter 2

Bilinear algebra

2.1 Linear forms

In this section, we fix a unitary ring K .

2.1.1 Definition. Let V be a left K -module. We denote by $V^\vee$ the set of left K -module homomorphisms from V to K . Note that $V^\vee$ is equipped with the following structure of right K -module:

$$K \times V^\vee \longrightarrow V^\vee, \quad (a, f) \longmapsto f(\cdot) a.$$

Recall that, in the particular case where

$$V := \{(a_1, \dots, a_n) \mid \forall i \in \{1, \dots, n\}, a_i \in K\}$$

is the set of row vectors, the set $V^\vee$ is composed by column vectors:

$$V^\vee := \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mid \forall i \in \{1, \dots, n\}, a_i \in K \right\}.$$

In other words, if we view K^n as a left K -module, then $(K^n)^\vee$ is isomorphic to K^n viewed as a right K -module.

2.1.2 Definition. Let V and W be left K -modules, $\varphi : V \rightarrow W$ be a homomorphism of K -modules. We denote by $\varphi^\vee : W^\vee \rightarrow V^\vee$ the homomorphism of right K -modules that sends $g : W \rightarrow K$ to $g \circ \varphi \in V^\vee$. Note that, for any $a \in K$, one has

$$(g \circ \varphi)(\cdot) a = g(\varphi(\cdot)) a = (g(\cdot) a) \circ \varphi.$$

2.1.3 Proposition. Let E , F and G be left K -modules, and $\varphi : E \rightarrow F$ and $\psi : F \rightarrow G$ be homomorphism of left K -modules. Then

$$(\psi \circ \varphi)^\vee = \varphi^\vee \circ \psi^\vee.$$

Proof. Let $g \in G^\vee$. By definition, one has $\psi^\vee(g) = g \circ \psi$. Hence

$$\varphi^\vee(\psi^\vee(g)) = \varphi^\vee(g \circ \psi) = (g \circ \psi) \circ \varphi = g \circ (\psi \circ \varphi) = (\psi \circ \varphi)^\vee(g).$$

□

2.1.4 Example. Let n and p be natural numbers. Let $\alpha : (K^n)^\vee \rightarrow K^n$ and $\beta : (K^p)^\vee \rightarrow K^p$ be the isomorphisms of right K -modules described in Definition 2.1.1. If $\varphi : K^n \rightarrow K^p$ is a homomorphism of left K -modules, we denote by $\varphi^\tau : K^p \rightarrow K^n$ the homomorphism of right K -modules

$$\alpha \circ \varphi^\vee \circ \beta^{-1} : K^p \longrightarrow K^n.$$

Note that the following diagram is commutative:

$$\begin{array}{ccc} (K^p)^\vee & \xrightarrow{\varphi^\vee} & (K^n)^\vee \\ \beta \downarrow & & \downarrow \alpha \\ K^p & \xrightarrow{\varphi^\tau} & K^n. \end{array}$$

Suppose that φ is given by a matrix

$$\begin{pmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,p} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n,1} & b_{n,2} & \cdots & b_{n,p} \end{pmatrix}$$

Then φ sends $(a_1, \dots, a_n) \in K^n$ to

$$\left(\sum_{i=1}^n a_i b_{i,1}, \dots, \sum_{i=1}^n a_i b_{i,p} \right).$$

Let

$$g = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix} \in (K^p)^\vee,$$

which sends $(\lambda_1, \dots, \lambda_p) \in K^p$ to

$$\lambda_1 y_1 + \cdots + \lambda_p y_p.$$

Then $g \circ \varphi$ sends $(a_1, \dots, a_n)$ to

$$\sum_{j=1}^p \sum_{i=1}^n a_i b_{i,j} y_j = \sum_{i=1}^n a_i \sum_{j=1}^p b_{i,j} y_j.$$

Therefore, $g \circ \varphi$ is represented by the column vector

$$\begin{pmatrix} b_{1,1}y_1 + \cdots + b_{1,p}y_p \\ \vdots \\ b_{n,1}y_1 + \cdots + b_{n,p}y_p \end{pmatrix}$$

Let K^{op} be the opposite ring of K . As a homomorphism of left K^{op} -modules, φ^τ sends $(y_1, \dots, y_p)$ to

$$\left(\sum_{j=1}^p y_j b_{1,j}, \dots, \sum_{j=1}^p y_j b_{n,j} \right).$$

Hence it is represented by the following matrix

$$\begin{pmatrix} b_{1,1} & b_{2,1} & \cdots & b_{n,1} \\ b_{1,2} & b_{2,2} & \cdots & b_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1,p} & b_{2,p} & \cdots & b_{n,p} \end{pmatrix} \in M_{p,n}(K^{\text{op}})$$

2.1.5 Proposition. *Let V be a free left K -module of finite rank r and $(e_i)_{i=1}^r$ be a basis of V . For any $i \in \{1, \dots, r\}$, let*

$$e_i^\vee : V \longrightarrow K, \quad (\lambda_1 e_1 + \cdots + \lambda_r e_r) \longmapsto \lambda_i.$$

Then $V^\vee$ is a free right K -module and $(e_i^\vee)_{i=1}^r$ is a basis of $V^\vee$, called the dual basis of $(e_i)_{i=1}^r$.

Proof. Let $f : V \rightarrow K$ be a linear form. Then

$$f(\lambda_1 e_1 + \cdots + \lambda_r e_r) = \lambda_1 f(e_1) + \cdots + \lambda_r f(e_r),$$

so that

$$f(\cdot) = e_1^\vee(\cdot)f(e_1) + \cdots + e_r^\vee(\cdot)f(e_r).$$

Moreover, if $(b_1, \dots, b_r) \in K^r$ is such that

$$f(\cdot) = e_1^\vee(\cdot)b_1 + \cdots + e_r^\vee(\cdot)b_r,$$

then, for any $i \in \{1, \dots, r\}$,

$$f(e_i) = e_1^\vee(e_i)b_1 + \cdots + e_r^\vee(e_i)b_r = b_i.$$

Hence $f(\cdot)$ can be written in a unique way as a linear combination of $e_1^\vee, \dots, e_r^\vee$, namely $V^\vee$ is a free right K -module and $(e_i^\vee)_{i=1}^r$ is a basis of $V^\vee$. $\square$

2.2 Bilinear forms

In this section, we fix a commutative unitary ring K .

2.2.1 Definition. Let V be a K -module.

We call *linear form* on V any homomorphism of K -modules from V to K . The set of linear forms on V forms a sub- K -module of K^V , which is denoted as $V^\vee$ according to the notation of Definition 2.1.1.

We call *bilinear form* on V any bilinear mapping from $V \times V$ to K . The set of bilinear forms on V forms a sub- K -module of $K^{V \times V}$. We denote it as $\text{Bil}_K(V)$. We say that a bilinear form $\varphi : V \times V \rightarrow K$ is *symmetric* if

$$\forall (x, y) \in V \times V, \quad \varphi(x, y) = \varphi(y, x).$$

We say that a bilinear form $\varphi : V \times V \rightarrow K$ is *alternating* if

$$\forall x \in V, \quad \varphi(x, x) = 0.$$

2.2.2 Example. Let n be a natural number. The mapping

$$K^n \times K^n \longrightarrow K, \quad ((x_1, \dots, x_n), (y_1, \dots, y_n)) \longmapsto \sum_{i=1}^n x_i y_i$$

is a symmetric bilinear form on K^n .

2.2.3 Definition. Let V be a K -module and φ be a symmetric bilinear form on V . We say that two elements x and y of V are φ -orthogonal if $\varphi(x, y) = 0$. We denote this condition as $x \perp_\varphi y$. If there is no ambiguity on the symmetric bilinear form φ , the symbol $\perp_\varphi$ is simplified as $\perp$.

Let A be a subset of V . We denote by $A^\perp$ the set of $y \in V$ that satisfies

$$\forall x \in A, \quad \varphi(x, y) = 0.$$

Namely $y \in A^\perp$ if and only if y is φ -orthogonal to all elements of A .

2.2.4 Proposition. Let V be a K -module and φ be a symmetric bilinear form on V .

- (1) For any $A \in \mathcal{P}(V)$, $A^\perp$ is a sub- K -module of V .
- (2) For any $A \in \mathcal{P}(V)$, the sub- K -module of V generated by A is contained in $(A^\perp)^\perp$.
- (3) Let A and B be subsets of V . If $A \subseteq B$, then $B^\perp \subseteq A^\perp$.

Proof. (1) Let $(x, y) \in A^\perp \times A^\perp$, $(a, b) \in K \times K$. For any $z \in A$ one has

$$\varphi(z, ax + by) = a\varphi(z, x) + b\varphi(z, y) = 0.$$

Hence $ax + by \in A^\perp$.

(2) By (1), it suffices to check that $A \subseteq (A^\perp)^\perp$. Let $x \in A$. For any $y \in A^\perp$, one has

$$\varphi(x, y) = 0.$$

Hence $x \in (A^\perp)^\perp$.

(3) Let $y \in B^\perp$. For any $x \in A$, one has $x \in B$. Hence $\varphi(x, y) = 0$. Therefore, $y \in A^\perp$. $\square$

2.2.5 Definition. Let V be a K -module and $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form. We denote by $\ell_\varphi : V \rightarrow V^\vee$ the K -linear mapping that sends $x \in V$ to $\varphi(x, \cdot)$. The kernel of this K -linear mapping is called the *kernel* of φ .

We say that the bilinear form φ is *non-degenerate* if the mapping ℓ_φ is an isomorphism.

2.2.6 Theorem. Let K be a field and V be a finite-dimensional vector space and φ be a symmetric bilinear form on V . Then φ is non-degenerate if and only if its kernel is $\{0\}$.

Proof. If ℓ_φ is an isomorphism, its kernel is necessarily $\{0\}$.

Suppose that the kernel of ℓ_φ is $\{0\}$. Then ℓ_φ is injective. Note that

$$\dim_K(V) = \dim_K \text{Ker}(\ell_\varphi) + \dim_K \text{Im}(\ell_\varphi).$$

Hence $\dim_K(V) = \dim_K \text{Im}(\ell_\varphi)$, which leads to

$$\text{Im}(\ell_\varphi) = V.$$

Hence ℓ_φ is surjective. $\square$

2.3 Quadratic forms

In this section, K denotes a commutative unitary ring.

2.3.1 Definition. Let V be a K -module. We call *quadratic form* over K any mapping $q : V \rightarrow K$ such that, there exists a bilinear form $\varphi : V \times V \rightarrow K$ satisfying

$$\forall x \in V, \quad q(x) = \varphi(x, x).$$

Note that the set of all quadratic forms on V is a sub- K -module of K^V .

2.3.2 Proposition. *Let V be a K -module and $q : V \rightarrow K$ be a quadratic form. Then, for any $x \in V$ and $a \in K$ one has*

$$q(ax) = a^2q(x).$$

Proof. Let $\varphi : V \times V \rightarrow K$ be a bilinear form such that

$$\forall y \in V, \quad q(y) = \varphi(y, y).$$

Then, for any $(a, x) \in K \times V$, one has

$$q(ax) = \varphi(ax, ax) = a^2\varphi(x, x) = a^2q(x).$$

□

2.3.3 Definition. Let V be a K -module and $q : V \rightarrow K$ be a quadratic form. The set

$$C(q) := \{x \in V \mid q(x) = 0\}$$

is a cone in V (in the sense that, for any $x \in C(q)$ and any $\lambda \in K$, one has $\lambda x \in C(q)$). We call it the *isotropic cone* of q . If there exists at least one non-zero element of $C(q)$, we say that the quadratic form q is *isotropic*.

2.3.4 Proposition. *Let V be a K -module and $q : V \rightarrow K$ be a quadratic form.*

(1) *The mapping $\psi : V \times V \rightarrow K$ defined as*

$$\psi(x, y) := q(x + y) - q(x) - q(y)$$

is a symmetric bilinear form on V .

(2) *Assume that 2 is invertible in K . Then*

$$\forall x \in V, \quad q(x) = \frac{1}{2}\psi(x, x).$$

Proof. Let $\varphi : V \times V \rightarrow K$ be a bilinear form such that

$$\forall x \in V, \quad q(x) = \varphi(x, x).$$

Then

$$q(x + y) - q(x) - q(y) = \varphi(x + y, x + y) - \varphi(x, x) - \varphi(y, y) = \varphi(x, y) + \varphi(y, x).$$

In particular,

If 2 is invertible in K , then

$$\frac{1}{2}\psi(x, x) = \frac{1}{2}(\varphi(x, x) + \varphi(x, x)) = \varphi(x, x) = q(x).$$

□

2.3.5 Remark. Assume that 2 is invertible in K . Proposition 2.3.4 shows that the correspondence

$$q \longleftrightarrow \left(\varphi_q : V \times V \rightarrow K, (x, y) \mapsto \frac{1}{2}(q(x+y) - q(x) - q(y)) \right)$$

defines an isomorphism between the K -module of all quadratic forms and that of all symmetric bilinear forms on V . The symmetric bilinear form φ_q is called the *polarization* of q . The kernel of φ_q , which is defined as

$$\{x \in V \mid \forall y \in V, \varphi_q(x, y) = 0\}$$

is also called the *kernel* of q , denoted as $\text{Ker}(q)$. Note that the kernel of q is contained in the isotropic cone of q .

2.3.6 Definition. Let K be a field, V be a vector space over K and $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form. Let $(e_i)_{i \in I}$ be a family of non-zero elements of V . If, for any $(i, j) \in I \times I$ such that $i \neq j$, one has $e_i \perp e_j$, then we say that the family $(e_i)_{i \in I}$ is *orthogonal* with respect to φ .

2.3.7 Proposition. Let K be a field, V be a vector space over K , $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form, and $q : V \rightarrow K$ be the quadratic form defined as $q(x) := \varphi(x, x)$. Assume that the quadratic form q is not isotropic. If $(e_i)_{i \in I}$ is an orthogonal family of elements of V , then it is linearly independent over K .

Proof. Without loss of generality, we may assume that $I = \{1, \dots, n\}$, where n is a positive integer. Let $(a_1, \dots, a_n) \in K^n$ such that $a_1 e_1 + \dots + a_n e_n = 0$. For any $i \in \{1, \dots, n\}$, one has

$$\varphi(a_1 e_1 + \dots + a_n e_n, e_i) = a_i \varphi(e_i, e_i) = 0.$$

Since φ is not isotropic, one has $\varphi(e_i, e_i) \neq 0$ and hence $a_i = 0$. □

2.3.8 Proposition. Let K be a field such that 2 is invertible in K . Let V be a finite-dimensional vector space over K and $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form. There exists a basis of V which is orthogonal with respect to φ .

Proof. We reason by induction on the rank r of V over K . The cases where $r = 0$ or $r = 1$ are trivial. In the following, we assume that $r \geq 2$.

If φ is the zero bilinear form, then any basis of V is orthogonal with respect to φ . Otherwise there exists an element $e_1 \in V$ such that $\varphi(e_1, e_1) \neq 0$. Consider the K -linear form $\ell_{e_1} : V \rightarrow K$ sending $x \in V$ to $\varphi(e_1, x)$. This mapping is non-zero and hence is surjective (since K is a field). Therefore, its kernel W has dimension $r - 1$ over K , and e_1 does not belong to W since $\varphi(e_1, e_1) \neq 0$.

Applying the induction hypothesis to the restriction of φ to $W \times W$, we obtain the existence of a basis $(e_i)_{i=2}^r$ of W which is orthogonal with respect to φ . Then the basis $(e_i)_{i=1}^r$ of V is orthogonal with respect to φ . □

2.3.9 Proposition. *Let K be a field such that 2 is invertible in K . Let V be a finite-dimensional vector space over K and $q : V \rightarrow K$ be a quadratic form. There exists a basis $(\alpha_i)_{i=1}^r$ of $V^\vee$ and an element $(\lambda_1, \dots, \lambda_r) \in K^r$ such that*

$$\forall x \in V, \quad q(x) = \lambda_1 \alpha_1(x)^2 + \dots + \lambda_r \alpha_r(x)^2.$$

Proof. Let $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form such that

$$\forall x \in V, \quad q(x) = \varphi(x, x)$$

and $(e_i)_{i=1}^r$ be a basis of V that is orthogonal with respect to φ . Let $(\alpha_i)_{i=1}^r$ be the dual basis of $(e_i)_{i=1}^r$. Recall that, for any $i \in \{1, \dots, r\}$,

$$\alpha_i(a_1 e_1 + \dots + a_r e_r) = a_i.$$

For any $x = a_1 e_1 + \dots + a_r e_r \in V$, one has

$$q(x) = \varphi(x, x) = \sum_{i=1}^r \sum_{j=1}^r a_i a_j \varphi(e_i, e_j) = \sum_{i=1}^r a_i^2 \varphi(e_i, e_i),$$

where the last equality comes from the hypothesis that the basis $(e_i)_{i=1}^r$ is orthogonal with respect to φ . Hence

$$q(\cdot) = \sum_{i=1}^r \varphi(e_i, e_i) \alpha_i(\cdot)^2,$$

as required. □

2.3.10 Remark. Let K be a field such that 2 is invertible in K . Let V be a finite dimensional vector space over K , $\varphi : V \times V \rightarrow K$ be a symmetric bilinear form and $q : V \rightarrow K$ be the quadratic form defined as $q(x) := \varphi(x, x)$. Assume that q can be written as

$$q(x) = \lambda_1 \alpha_1(x)^2 + \dots + \lambda_r \alpha_r(x)^2,$$

where $(\alpha_i)_{i=1}^r$ is a basis of $V^\vee$ and $(\lambda_1, \dots, \lambda_r) \in K^r$. Let $(e_i)_{i=1}^r$ be a basis of V such that $(\alpha_i)_{i=1}^r$ is the dual basis of $(e_i)_{i=1}^r$. Then one has

$$q(b_1 e_1 + \dots + b_r e_r) = \lambda_1 b_1^2 + \dots + \lambda_r b_r^2.$$

In particular, for i and j in $\{1, \dots, r\}$ such that $i \neq j$, one has

$$q(e_i + e_j) = \lambda_i + \lambda_j = q(e_i) + q(e_j),$$

which shows that $(e_i)_{i=1}^r$ is orthogonal with respect to φ .

2.3.11 Remark. Let K be a field of characteristic $\neq 2$ and $q : K^r \rightarrow K$ be a quadratic form. Then q can be written as

$$q(x_1, \dots, x_r) = \sum_{1 \leq i \leq j \leq r} a_{i,j} x_i x_j = (a_{1,1} x_1^2 + \dots + a_{r,r} x_r^2) + \sum_{1 \leq i < j \leq r} a_{i,j} x_i x_j.$$

We can use an algorithm to write q as a linear combination of squares of linear forms which are linearly independent. It suffices to consider the case where q is non-zero.

Consider first the case where there exists $i \in \{1, \dots, r\}$ such that $a_{i,i} \neq 0$. We then write q as

$$q(x_1, \dots, x_r) = a_{i,i} x_i^2 + \left(\sum_{j < i} a_{j,i} x_j + \sum_{j > i} a_{i,j} x_j \right) x_i + \tilde{q}(x_1, \dots, x_r),$$

where $\tilde{q}$ is a quadratic form on K^r , the expanded expression of which does not contain the occurrence of x_i . By completing the square, we obtain

$$q(x_1, \dots, x_r) = a_{i,i} \left(x_i + \frac{1}{2a_{i,i}} \left(\sum_{j < i} a_{j,i} x_j + \sum_{j > i} a_{i,j} x_j \right) \right)^2 + q^*(x_1, \dots, x_r),$$

where

$$q^*(x_1, \dots, x_r) = \tilde{q}(x_1, \dots, x_r) - \frac{1}{4a_{i,i}^2} \left(\sum_{j < i} a_{j,i} x_j + \sum_{j > i} a_{i,j} x_j \right)^2.$$

Again in the expression of q^* there is no occurrence of x_i . The algorithm can then be iterated to q^* to write it as a K -linear combination of squares of linear forms that are linearly independent.

Consider now the case where there are only crossing terms in the expression of q . In this case there exist two different indices i and j such that $i < j$ and $a_{i,j} \neq 0$. We then write q in the form

$$q(x_1, \dots, x_r) = a_{i,j} x_i x_j + \varphi(x_1, \dots, x_r) x_i + \psi(x_1, \dots, x_r) x_j + q_1(x_1, \dots, x_r),$$

where φ and ψ are linear forms, q_1 is a quadratic form. Moreover, in the expanded expression of φ , ψ and q_1 , the terms x_i and x_j do not occur. We then write q as

$$q(x_1, \dots, x_r) = a_{i,j} \left(x_i + \frac{1}{a_{i,j}} \psi(x_1, \dots, x_r) \right) \left(x_j + \frac{1}{a_{i,j}} \varphi(x_1, \dots, x_r) \right) + q_2(x_1, \dots, x_r),$$

where

$$q_2(x_1, \dots, x_r) = q_1(x_1, \dots, x_r) - \frac{1}{a_{i,j}} \varphi(x_1, \dots, x_r) \psi(x_1, \dots, x_r).$$

Let f_1 and f_2 be elements of $(K^r)^\vee$ defined as

$$\begin{aligned} f_1(x_1, \dots, x_r) &:= x_i + \frac{1}{a_{i,j}}\psi(x_1, \dots, x_r) + x_j + \frac{1}{a_{i,j}}\varphi(x_1, \dots, x_r), \\ f_2(x_1, \dots, x_r) &:= x_i + \frac{1}{a_{i,j}}\psi(x_1, \dots, x_r) - x_j - \frac{1}{a_{i,j}}\varphi(x_1, \dots, x_r). \end{aligned}$$

Then one has

$$q(x_1, \dots, x_r) = \frac{a_{i,j}}{4}f_1(x_1, \dots, x_r)^2 - \frac{a_{i,j}}{4}f_2(x_1, \dots, x_r)^2 + q_2(x_1, \dots, x_r).$$

One can then iterate the algorithm to q_2 to write it as a K -linear combination of squares of linear forms which are linearly independent.

2.4 Quadratic form over a real vector space

2.4.1 Definition. Let V be a vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form. If q takes values in $\mathbb{R}_{\geq 0}$ (resp. $\mathbb{R}_{\leq 0}$) we say that q is *positive semi-definite* (resp. *negative semi-definite*). If for any $x \in V \setminus \{0\}$ one has $q(x) > 0$ (resp. $q(x) < 0$), we say that q is *positive definite* (resp. *negative definite*).

By definition, if a quadratic form q is positive semi-definite (resp. negative semi-definite, positive definite, negative definite), then the restriction of q to any vector subspace of V is positive semi-definite (resp. negative semi-definite, positive definite, negative definite).

2.4.2 Definition. Let V be a vector space over $\mathbb{R}$ and $\varphi : V \times V \rightarrow \mathbb{R}$ be a symmetric bilinear form. If the quadratic form $x \mapsto \varphi(x, x)$ is positive semi-definite (resp. negative semi-definite, positive definite, negative definite), we say that the symmetric bilinear form φ is positive semi-definite (resp. negative semi-definite, positive definite, negative definite).

2.4.3 Lemma. Let V be a vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form which is semi-definite. Then the kernel of q is equal to the isotropic cone of q .

Proof. Without loss of generality, we may assume that q is positive semidefinite (if q is negative semidefinite, then $-q$ is positive semidefinite). By definition, the kernel of q is contained in the isotropic cone.

Let φ be the polarisation of q . Let x be an element of the isotropic cone of q . Let y be an arbitrary element of V . For any $t \in \mathbb{R}$, one has

$$0 \leq q(y + tx) = q(y) + 2t\varphi(x, y) + t^2q(x) = q(y) + 2t\varphi(x, y).$$

This is possible only when $\varphi(x, y) = 0$. Therefore, x belongs to the kernel of q . $\square$

2.4.4 Theorem (Cauchy Schwarz). *Let V be a vector space over $\mathbb{R}$ and $\varphi : V \times V \rightarrow \mathbb{R}$ be a symmetric bilinear form. Let $q : V \rightarrow \mathbb{R}$ be the quadratic form defined as $q(x) := \varphi(x, x)$. Assume that q is positive semi-definite. For any $(x, y) \in V \times V$ one has*

$$\varphi(x, y)^2 \leq q(x)q(y).$$

The equality holds when the classes of x and y in $V/\text{Ker}(\varphi)$ are colinear.

Proof. In the case where x or y belongs to $\text{Ker}(\varphi)$, one has

$$\varphi(x, y)^2 = q(x)q(y) = 0,$$

and the classes of x and y in $V/\text{Ker}(\varphi)$ are colinear.

In what follows, we assume that x does not belong to the kernel of φ . By Lemma 2.4.3, we see that x does not belong to the isotropic cone of q , that is, $q(x) \neq 0$. Let

$$t = -\frac{\varphi(x, y)}{q(x)}.$$

One has

$$0 \leq q(y + tx) = q(y) + 2t\varphi(x, y) + t^2q(x) = q(y) - \frac{2\varphi(x, y)^2}{q(x)} + \frac{\varphi(x, y)^2}{q(x)},$$

which leads to

$$\frac{\varphi(x, y)^2}{q(x)} \leq q(y),$$

or equivalently $\varphi(x, y)^2 \leq q(x)q(y)$.

The equality holds if and only if $y + tx$ belongs to the isotropic cone (which is also the kernel) of φ , namely the classes of x and y in $V/\text{Ker}(\varphi)$ are colinear. $\square$

2.4.5 Definition. Let K be a commutative unitary ring and V be a K -module. Let $(V_i)_{i \in I}$ be a family of sub- K -modules of V . We say that V is *generated by* $(V_i)_{i \in I}$ (resp. a *direct sum* of $(V_i)_{i \in I}$) if the homomorphism of K -modules

$$\bigoplus_{i \in I} V_i \longrightarrow V, \quad (x_i)_{i \in I} \longmapsto \sum_{i \in I} x_i,$$

is surjective (resp. an isomorphism). Note that, V is a direct sum of $(V_i)_{i \in I}$ if and only if each $x \in V$ can be decomposed in a unique way as a sum

$$x = \sum_{i \in I} x_i$$

with $x_i \in V_i$ for any $i \in I$, and $x_i = 0$ for all but finitely many $i \in I$.

2.4.6 Proposition. *Let K be a commutative unitary ring and V be a K -module. Let W_1 and W_2 be sub- K -modules of V such that V is generated by W_1 and W_2 . Then V is a direct sum of W_1 and W_2 if and only if $W_1 \cap W_2 = \{0\}$.*

Proof. Assume that V is a direct sum of W_1 and W_2 . Let

$$f : W_1 \oplus W_2 \longrightarrow V, \quad (x, y) \longmapsto x + y.$$

If x belongs to $W_1 \cap W_2$, then $f(x, -x) = 0$. Hence $(x, -x) = 0$, which leads to $x = 0$.

Conversely, if $W_1 \cap W_2 = \{0\}$, for any $(x, y) \in \text{Ker}(f)$, one has $x = -y \in W_1 \cap W_2$. Hence $x = y = 0$. $\square$

2.4.7 Definition. Let V be a vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form. We call a *Sylvester decomposition* of q any decomposition of V into the form of the direct sum of three vector subspaces

$$V = V_+ + V_- + V_0,$$

where the restriction of q to V_+ is positive definite, the restriction of q to V_- is negative definite and the restriction of q to V_0 vanishes.

2.4.8 Proposition. *Let V be a finite-dimensional vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form. Let*

$$V = V_+ + V_- + V_0$$

be a Sylvester decomposition of V . Then V_0 identifies with the kernel of q .

Proof. By definition, V_0 is contained in the isotropic cone of $q|_{V_+ + V_0}$, which identifies with the kernel of $q|_{V_+ + V_0}$ by Lemma 2.4.3. So any element of V_0 is orthogonal to $V_+ + V_0$. For the same reason, any element of V_0 is also orthogonal to $V_- + V_0$. Therefore, V_0 is contained in the kernel of q .

Let φ be the polarisation of q . Let x be an element of the kernel of q . We decompose x as

$$x = x_+ + x_- + x_0,$$

where $(x_+, x_-, x_0) \in V_+ \times V_- \times V_0$. Since V_0 is contained in the kernel of q , we obtain that $x_+ + x_-$ also belongs to the kernel of q . In particular, one has

$$\varphi(x_+ + x_-, x_+) = q(x_+) + \varphi(x_-, x_+) = 0$$

Similarly

$$\varphi(x_+ + x_-, x_-) = \varphi(x_+, x_-) + q(x_-) = 0.$$

Therefore,

$$\varphi(x_+, x_-) = -q(x_+) \leq 0, \quad \varphi(x_+, x_-) = -q(x_-) \geq 0,$$

which leads to

$$\varphi(x_+, x_-) = q(x_+) = q(x_-) = 0.$$

By Lemma 2.4.3, we obtain that both x_+ and x_- belong to the kernel of q . $\square$

2.4.9 Proposition. *Let K be a unitary ring and M be a left K -module. If E and F are left sub- K -modules of M , then the mapping*

$$E/E \cap F \longrightarrow (E + F)/F,$$

sending the class of $x \in E/E \cap F$ to that of x in $(E + F)/F$, is an isomorphism. In particular, in the case where K is a field and M is a finite-dimensional vector space over K , one has

$$\dim(E \cap F) + \dim(E + F) = \dim(E) + \dim(F).$$

Proof. We consider the homomorphism of left R -modules

$$f : E \longrightarrow (E + F)/F$$

that sends $x \in E$ to the class of x in $(E + F)/F$. Note that, for any $(x, y) \in E \times F$, the class of $x + y$ in $(E + F)/F$ equals to the class of x . Hence f is surjective. Moreover, $x \in E$ belongs to the kernel of f if and only if it belongs to F . Hence the kernel of f is equal to $E \cap F$. Therefore, f induces an isomorphism between $E/E \cap F$ and $(E + F)/F$. $\square$

2.4.10 Theorem (Sylvester's law of inertia). *Let V be a finite-dimensional vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form on V .*

(1) *The quadratic form q admits a Sylvester decomposition.*

(2) *If*

$$V = V_+ + V_- + V_0 = W_+ + W_- + V_0$$

be two Sylvester decompositions of q . Then V_+ and W_+ have the same dimension over $\mathbb{R}$, V_- and W_- have the same dimension over $\mathbb{R}$.

Proof. (1) Let φ be the polarisation of q , $(e_i)_{i=1}^r$ be a basis of V which is orthogonal with respect to φ and $(\alpha_i)_{i=1}^r$ be its dual basis. Then one can decompose q as

$$q(x) = \sum_{i=1}^r \lambda_i \alpha_i(x)^2,$$

where $(\lambda_1, \dots, \lambda_r) \in \mathbb{R}^r$. Let

$$V_+ = \sum_{\substack{i \in \{1, \dots, r\}, \\ \lambda_i > 0}} \mathbb{R}e_i, \quad V_- = \sum_{i \in \{1, \dots, r\}, \lambda_i < 0} \mathbb{R}e_i, \quad V_0 = \sum_{i \in \{1, \dots, r\}, \lambda_i = 0} \mathbb{R}e_i.$$

Then $V = V_+ + V_- + V_0$ is a Sylvester decomposition of q .

(2) Note that the restriction of q to $V_- + V_0$ is negative semidefinite, the restriction of q to W_+ is positive definite. Hence

$$(V_- + V_0) \cap W_+ = \{0\}.$$

Therefore,

$$\dim(V_- + V_0) + \dim(W_+) \leq \dim(V),$$

which leads to

$$\dim(W_+) \leq \dim(V) - \dim(V_- + V_0) = \dim(V_+).$$

By the same argument, we obtain $\dim(V_+) \leq \dim(W_+)$. Therefore, the equality $\dim(W_+) = \dim(V_+)$ is satisfied, which also implies that $\dim(W_-) = \dim(V_-)$. $\square$

2.4.11 Definition. Let V be a finite-dimensional vector space over $\mathbb{R}$ and $q : V \rightarrow \mathbb{R}$ be a quadratic form on V . If $V = V_+ + V_- + V_0$ is a Sylvester decomposition of q , then the triplet $(\dim(V_+), \dim(V_-), \dim(V_0))$ is called the *Sylvester inertia* of q .

Exercises

- Let $V = \mathbb{R}^3$ and $(e_i)_{i=1}^3$ be the canonical basis of V . Denote by $(e_i^\vee)_{i=1}^3$ the dual basis of $(e_i)_{i=1}^3$. Write the following linear form as a linear combination of vectors in the dual basis.

(a) $\alpha(x, y, z) = 3x + 4y + 5z$.

(b) The differential at $(0, 0, 0)$ of the function $f(x, y, z) = \exp(x + y + z)$.

(c)

$$\beta(x, y, z) = (1, 1, 1)A \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \text{ where } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

- Let V be the vector space of all polynomials in $\mathbb{R}[T]$ of degree ≤ 2 . Let $e_1 = 1$, $e_2 = 1 + T$ and $e_3 = 1 + T + T^2$.

(a) Prove that $(e_i)_{i=1}^3$ is a basis of the vector space V .

(b) Prove that $I : V \rightarrow \mathbb{R}$, $I(f) = \int_0^1 f(t) dt$ is a linear form on V .

(c) Write I as a linear combination of the vectors in the dual basis of $(e_i)_{i=1}^3$.

3. Let V be a vector space over a field K and $U \subseteq V$ be a subspace. Define its annihilator

$$U^0 = \{ f \in V^\vee \mid f(u) = 0 \text{ for all } u \in U \}.$$

- (a) Prove that U^0 is a subspace of $V^\vee$.
 (b) If V is finite dimensional, prove that $\dim U + \dim U^0 = \dim V$.
4. Let $T : V \rightarrow W$ be a linear map between finite dimensional vector spaces. Prove that:

$$\ker(T^\vee) = (\operatorname{Im} T)^0, \quad \operatorname{Im}(T^\vee) = (\ker T)^0.$$

5. Let V be a finite dimensional vector space over $K = (\mathbb{R} \text{ or } \mathbb{C})$. Prove that the map from V to $V^{\vee\vee}$, which sends $x \in V$ to the linear form $\alpha \mapsto \alpha(x)$ on $V^\vee$, is a K -linear isomorphism.
6. Let V be a vector space over $K = (\mathbb{R} \text{ or } \mathbb{C})$ and W be a vector subspace of V .

- (a) Denote by $\sim$ the binary relation on V defined as

$$x \sim y \iff x - y \in W.$$

Prove that it is an equivalence relation.

- (b) For any element $x \in V$, denote by $[x]$ the equivalence class of x with respect to the equivalence relation $\sim$. Let V/W be the quotient set $V/\sim$. Prove that the maps

$$\begin{aligned} (V/W) \times (V/W) &\longrightarrow V/W \\ ([x], [y]) &\longmapsto [x + y] \end{aligned}$$

and

$$\begin{aligned} K \times (V/W) &\longrightarrow V/W \\ (a, [x]) &\longmapsto [ax] \end{aligned}$$

are well defined.

- (c) Prove that the set V/W equipped with the above maps forms a vector space over K and that the projection $\pi : V \rightarrow V/W$, $\pi(x) = [x]$ is a K -linear map.
- (d) Determine the kernel of π .
- (e) Prove that $\pi^\vee : (V/W)^\vee \rightarrow V^\vee$ is an injective map.
- (f) Let $\eta : W \rightarrow V$ be the inclusion map of W into V . Prove that $\eta^\vee$ is a surjective map.

- (g) Let $\Delta : V \times V^\vee \rightarrow \mathbb{R}$ be the bilinear map defined as $\Delta(x, \alpha) = \alpha(x)$. Denote by $W_\Delta^\perp$ the set of all $\alpha \in V^\vee$ such that $\alpha(x) = 0$ for any $x \in W$. Prove that $W_\Delta^\perp$ identifies with the image of $\pi^\vee$ and the kernel of $\eta^\vee$.
- (h) Assume that V is finite dimensional. Prove that $\dim(W_\Delta^\perp) = \dim(V) - \dim(W)$.
7. Determine the matrix of the following bilinear forms with respect to the basis assigned.
- (a) $\varphi : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$, $\varphi(x, y) = x_1y_2 + x_2y_1 + x_1y_1 - x_2y_2$, canonical basis.
- (b) $\varphi : \mathbb{R}[T]_2 \times \mathbb{R}[T]_2 \rightarrow \mathbb{R}$, $\varphi(P, Q) = \int_0^1 P(t)Q(t)dt$, the basis $\{1, T, T^2\}$.
8. Let $\varphi : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ be the bilinear form defined as

$$\varphi(x, y) = x_1y_1 - \frac{1}{2}x_2y_1 + \frac{3}{2}x_1y_2 - x_2y_2.$$

Is this bilinear form symmetric? If it is not the case, please decompose φ into the sum of a symmetric bilinear form and a anti-symmetric bilinear form.

9. Transform the following quadratic forms into the sum of squares of independent linear forms. Determiner an orthogonal basis and the signature of each quadratic form.
- (a) $q : \mathbb{R}^2 \rightarrow \mathbb{R}$, $q(x) = 5x_1^2 + 2x_1x_2 + 5x_2^2$.
- (b) $q : \mathbb{R}^3 \rightarrow \mathbb{R}$, $q(x) = (x_1 + x_2 + x_3)^2 + (x_1 + 2x_2 + x_3)^2 + (x_1 + x_3)^2$.
- (c) $q : \mathbb{R}^3 \rightarrow \mathbb{R}$, $q(x) = 7x_1x_2 + 8x_1x_3 + 4x_2x_3$.
- (d) $q : \mathbb{R}[T]_2 \rightarrow \mathbb{R}$, $q(P) = P(2)P(1) + P(1)P(0)$.
10. Let $E = \mathbb{R}[T]_n$. We consider the map $q : E \rightarrow \mathbb{R}$ defined as

$$q(P) = \int_0^1 P(t)P'(t) dt$$

- (a) Prove that q is a quadratic form.
- (b) Write q as the difference of squares of two linearly independent linear forms.
- (c) Compute the kernel and the rank of the polar form of q (namely the symmetric bilinear form associated to q).
- (d) Determine an orthogonal basis for the polar form of q .
- (e) Determine the isotopic cone of q .

- 11.** Let q be a quadratic form on a finite dimensional vector space V over $\mathbb{R}$, $f_1, \dots, f_p$ be linear forms on V and $\alpha_1, \dots, \alpha_p$ be real numbers. Assume that q can be written as

$$q = \alpha_1 f_1^2 + \dots + \alpha_p f_p^2.$$

- (a) Prove that $\text{rk}(f_1, \dots, f_p) \geq \text{rk}(q)$.
 (b) Prove that, if $f_1, \dots, f_p$ are linearly independent and if $\alpha_1, \dots, \alpha_p$ are non-zero, then the rank of the quadratic form q is equal to p .
- 12.** Let n and p be two integers in $\mathbb{Z}_{\geq 1}$. Let A be a matrix in $M_{n,p}(\mathbb{R})$ and B be the matrix AA^T .

- (a) Prove that the matrix B is symmetric.
 (b) Let $q: \mathbb{R}^n \rightarrow \mathbb{R}$ be the quadratic form defined as

$$q(x) = (x_1, \dots, x_n) B \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Prove that q is positive.

- (c) Determine the kernel of the quadratic form q .
 (d) Determine the signature of q in terms of the rank of A .
 (e) Determine the signature of the quadratic form

$$q_1: \mathbb{R}^n \rightarrow \mathbb{R}, \quad q_1(x) = \sum_{1 \leq i, j \leq n} x_i x_j.$$

- (f) Determine the signature of the quadratic form

$$q_2: \mathbb{R}^n \rightarrow \mathbb{R}, \quad q_2(x) = \sum_{1 \leq i, j \leq n} ij x_i x_j.$$

- 13.** Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

This matrix is indefinite (its eigenvalues are 3 and -1). Take $\mathbf{x} = (1, 1)^T$ and $\mathbf{y} = (1, -1)^T$.

- (a) Compute $\mathbf{x}^T A \mathbf{x}$, $\mathbf{y}^T A \mathbf{y}$, and $\mathbf{x}^T A \mathbf{y}$.
 (b) Does the inequality $(\mathbf{x}^T A \mathbf{y})^2 \leq (\mathbf{x}^T A \mathbf{x})(\mathbf{y}^T A \mathbf{y})$ hold?
 (c) Explain why the standard proof using $Q(\mathbf{x} + t\mathbf{y}) \geq 0$ fails here. (Hint: what is the sign of Q for some vectors?)

14. In $\mathbb{R}^4$, let U be the subspace spanned by $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and let W be the subspace spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$. Find $\dim(U \cap W)$, $\dim(U + W)$, and verify the dimension formula

$$\dim(E \cap F) + \dim(E + F) = \dim E + \dim F.$$

Chapter 3

Self adjoint operators

In this chapter, K denotes $\mathbb{R}$ or $\mathbb{C}$. We equip K with the usual absolute value $|\cdot|$. Recall that, in the case where $K = \mathbb{R}$, then the absolute value $|\cdot|$ sends $a \in \mathbb{R}$ to $\max\{a, -a\}$; in the case where $K = \mathbb{C}$, the absolute value $|\cdot|$ sends $x + yi$ to

$$\sqrt{x^2 + y^2},$$

where $(x, y) \in \mathbb{R} \times \mathbb{R}$.

3.1 Inner product space

3.1.1 Definition. Let V be a vector space over K , and $\langle \cdot, \cdot \rangle$ be a mapping from $V \times V$ to K . We say that $\langle \cdot, \cdot \rangle$ is a *semi-inner product* if the following condition holds:

- (1) $\langle \cdot, \cdot \rangle$ is linear with respect to the second variable, namely, for any $x \in V$, $(y, z) \in V \times V$, $(a, b) \in K \times K$, one has

$$\langle x, ay + bz \rangle = a\langle x, y \rangle + b\langle x, z \rangle.$$

- (2) $\langle \cdot, \cdot \rangle$ is *conjugate symmetric*, namely, for any $(x, y) \in V \times V$, one has

$$\langle y, x \rangle = \overline{\langle x, y \rangle}.$$

- (3) $\langle \cdot, \cdot \rangle$ is semi-positive, namely, for any $x \in V$,

$$\langle x, x \rangle \geq 0.$$

The pair $(V, \langle \cdot, \cdot \rangle)$ is called a *semi-inner product space*. Note that, in the particular case where $K = \mathbb{R}$, a semi-inner product on V is just a positive semi-definite bilinear form.

If in addition $\langle x, x \rangle > 0$ for any $x \in V \setminus \{0\}$, we say that $\langle \cdot, \cdot \rangle$ is an *inner product* and that $(V, \langle \cdot, \cdot \rangle)$ is an *inner product space*.

3.1.2 Remark. Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space. Let $(x, y, z) \in V \times V \times V$, and $(a, b) \in K \times K$, one has

$$\begin{aligned} \langle ax + by, z \rangle &= \overline{\langle z, ax + by \rangle} = \overline{a\langle z, x \rangle + b\langle z, y \rangle} \\ &= \bar{a}\langle x, z \rangle + \bar{b}\langle y, z \rangle. \end{aligned}$$

3.1.3 Example. Let n be a positive integer. Then the mapping $\langle \cdot, \cdot \rangle : K^n \times K^n \rightarrow K$ defined as

$$\langle (a_1, \dots, a_n), (b_1, \dots, b_n) \rangle = \bar{a}_1 b_1 + \dots + \bar{a}_n b_n$$

is an inner product. We call it the *canonical inner product*.

3.1.4 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space. If x and y are elements of V such that $\langle x, y \rangle = 0$ (in this case, one has $\langle y, x \rangle = 0$ too), then we say that x and y are *orthogonal* and we write $x \perp y$. Note that, if x and y are orthogonal, then, for any $(a, b) \in K \times K$, one has

$$\langle ax, by \rangle = \bar{a}\bar{b}\langle x, y \rangle = 0,$$

and hence ax and by are orthogonal. Moreover,

$$\langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle.$$

For any subset A of V , we denote by $A^\perp$ the set of $x \in V$ such that

$$\forall y \in A, \quad \langle x, y \rangle = 0.$$

Moreover, $V^\perp$ is called the kernel of the semi-inner product $\langle \cdot, \cdot \rangle$.

3.1.5 Proposition. Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space.

- (1) For any subset A of V , $A^\perp$ is a vector subspace of V .
- (2) For any subset A of V , the vector subspace of V generated by A is contained in $(A^\perp)^\perp$.
- (3) Let A and B be subsets of V . If $A \subseteq B$, then $B^\perp \subseteq A^\perp$.
- (4) Let A be a subset of V and W be the vector subspace of V generated by A . Then one has $A^\perp = W^\perp$.

Proof. (1) Let $(x, y) \in A^\perp \times A^\perp$ and $(a, b) \in K \times K$. For any $z \in A$, one has

$$\langle ax + by, z \rangle = \bar{a}\langle x, z \rangle + \bar{b}\langle y, z \rangle = 0.$$

Hence $ax + by \in A^\perp$.

(2) By (1), it suffices to check that $A \subseteq (A^\perp)^\perp$. For any $x \in A$ and any $y \in A^\perp$, one has

$$\langle x, y \rangle = \overline{\langle y, x \rangle} = 0.$$

Hence $x \in (A^\perp)^\perp$.

(3) Let $y \in B^\perp$. For any $x \in A$, one has $x \in B$ and hence $\langle y, x \rangle = 0$. Therefore, $y \in B^\perp$.

(4) By (3), one has $W^\perp \subseteq A^\perp$ since $A \subseteq W$. Moreover, by (2), one has $W \subseteq (A^\perp)^\perp$. Therefore, by (2) and (3) we get

$$A^\perp \subseteq ((A^\perp)^\perp)^\perp \subseteq W^\perp.$$

□

3.1.6 Theorem (Cauchy-Schwarz). *Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space. For any $(x, y) \in V \times V$, one has*

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

Proof. Let $u \in K$ with $|u| = 1$ such that

$$u\langle x, y \rangle = |\langle x, y \rangle|$$

For any real number t , one has

$$0 \leq \langle x + tuy, x + tuy \rangle = \langle x, x \rangle + 2t|\langle x, y \rangle| + t^2\langle y, y \rangle.$$

This implies that the discriminant of the quadratic function

$$t \longmapsto \langle x, x \rangle + 2t|\langle x, y \rangle| + t^2\langle y, y \rangle$$

is ≤ 0 (this is also true when $\langle y, y \rangle = 0$). Hence we obtain

$$4|\langle x, y \rangle|^2 \leq 4\langle x, x \rangle \langle y, y \rangle,$$

as required. □

3.1.7 Proposition. *Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space.*

(1) *The mapping*

$$\|\cdot\| : V \longrightarrow \mathbb{R}_{\geq 0}, \quad x \longmapsto \langle x, x \rangle^{1/2}$$

is a seminorm on V , called the seminorm associated with the inner product $\langle \cdot, \cdot \rangle$. It is a norm when $\langle \cdot, \cdot \rangle$ is an inner product.

(2) *The kernel of $\langle \cdot, \cdot \rangle$ is equal to*

$$\{x \in V \mid \|x\| = 0\}.$$

(3) *Let W be a vector subspace of V , which is contained in the kernel of $\langle \cdot, \cdot \rangle$. Then the semi-inner product $\langle \cdot, \cdot \rangle$ induces by passing to quotient a semi-inner product*

$$\langle \cdot, \cdot \rangle^\sim : (V/W) \times (V/W) \longrightarrow K$$

such that

$$\langle [x], [y] \rangle^\sim = \langle x, y \rangle$$

for any $(x, y) \in V \times V$. Moreover, this semi-inner product is an inner product when W is equal to the kernel of $\langle \cdot, \cdot \rangle$.

Proof. (1) For any $(a, x) \in K \times V$, one has

$$\|ax\| = \langle ax, ax \rangle^{1/2} = (a\bar{a} \cdot \langle x, x \rangle)^{1/2} = |a| \cdot \|(x, x)\|^{1/2} = |a| \cdot \|x\|.$$

For any $(x, y) \in V \times V$, one has

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + 2\operatorname{Re}\langle x, y \rangle + \langle y, y \rangle.$$

By Cauchy-Schwarz inequality, one has

$$\operatorname{Re}\langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \cdot \|y\|.$$

Hence

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \cdot \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

which leads to the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$.

If $\langle \cdot, \cdot \rangle$ is an inner product, then $\|x\| = \langle x, x \rangle^{1/2} = 0$ implies $x = 0$, namely $\|\cdot\|$ is a norm.

(2) Clearly the kernel of $\langle \cdot, \cdot \rangle$ is contained in $\{x \in V \mid \|x\| = 0\}$. Conversely, suppose that $x \in V$ is such that $\|x\| = 0$. For any $y \in V$, by Cauchy-Schwarz inequality one has

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\| = 0,$$

so that $\langle x, y \rangle = 0$.

(3) Let $(x, y) \in V \times V$ and $(u, v) \in W \times W$. One has

$$\langle x + u, y + v \rangle = \langle x, y \rangle + \langle x, v \rangle + \langle u, y \rangle + \langle u, v \rangle = \langle x, y \rangle.$$

Therefore, $\langle \cdot, \cdot \rangle$ induces by passing to quotient a mapping

$$\langle \cdot, \cdot \rangle^\sim : (V/W) \times (V/W) \longrightarrow K$$

such that

$$\langle [x], [y] \rangle^\sim = \langle x, y \rangle$$

for any $(x, y) \in V \times V$.

For any $(x, y, z) \in V^3$ and $(a, b) \in K \times K$, one has

$$\begin{aligned} \langle [x], a[y] + b[z] \rangle^\sim &= \langle [x], [ay + bz] \rangle^\sim = \langle x, ay + bz \rangle = a\langle x, y \rangle + b\langle x, z \rangle \\ &= a\langle [x], [y] \rangle^\sim + b\langle [x], [z] \rangle^\sim, \end{aligned}$$

$$\langle [y], [x] \rangle^\sim = \langle y, x \rangle = \overline{\langle x, y \rangle} = \overline{\langle [x], [y] \rangle^\sim},$$

$$\langle [x], [x] \rangle^\sim = \langle x, x \rangle \geq 0,$$

so $\langle \cdot, \cdot \rangle^\sim$ is a semi-inner product.

Finally, if

$$\langle [x], [x] \rangle^\sim = \langle x, x \rangle = 0,$$

then x belongs to the kernel of $\langle \cdot, \cdot \rangle$. In the case where W is the kernel of $\langle \cdot, \cdot \rangle$, this means that $[x] = 0$ in V/W . $\square$

3.1.8 Proposition. *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ be the norm on V defined as $\|x\| := \langle x, x \rangle^{1/2}$.*

(1) *For any $x \in V$, the mapping*

$$\ell_x : V \longrightarrow K, \quad z \longmapsto \langle x, z \rangle$$

is a bounded K -linear mapping, whose operator norm is equal to $\|x\|$.

(2) *For any $(x, y) \in V \times V$, $\ell_x = \ell_y$ if and only if $x = y$.*

Proof. (1) For any $z \in V$, by Cauchy-Schwarz inequality one has

$$|\langle x, z \rangle| \leq \|x\| \cdot \|z\|,$$

so that ℓ_x is a bounded K -linear mapping with $\|\ell_x\| \leq \|x\|$. Moreover,

$$\ell_x(x) = \langle x, x \rangle = \|x\|^2,$$

so $\|\ell_x\| = \|x\|$.

(2) One has $\ell_x - \ell_y = \ell_{x-y}$, so that

$$\|\ell_x - \ell_y\| = \|x - y\|.$$

Therefore, $\ell_x - \ell_y = 0$ if and only if $x - y = 0$. $\square$

3.1.9 Proposition. *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. We equip V with the norm $\|\cdot\|$ defined as $\|x\| := \langle x, x \rangle^{1/2}$.*

(1) *For any subset A of V , $A^\perp$ is a closed subset of V .*

(2) *Let A be a subset of V and B be its closure. Then one has $A^\perp = B^\perp$.*

Proof. (1) For any $x \in V$, let $\ell_x : V \rightarrow K$ be the linear form sending $y \in V$ to $\langle x, y \rangle$. It is bounded and thus continuous. Therefore,

$$A^\perp = \bigcap_{x \in A} \text{Ker}(\ell_x)$$

is closed.

(2) By Proposition, one has $B^\perp \subseteq A^\perp$. Let $y \in A^\perp$. Then, $\text{Ker}(\ell_y) \supseteq A$. Since ℓ_y is continuous, we obtain that $\text{Ker}(\ell_y) = \ell_y^{-1}(\{0\})$ is closed. Hence $B \subseteq \text{Ker}(\ell_y)$. $\square$

3.1.10 Proposition. *Let $(V, \|\cdot\|)$ be a normed vector space over K . The norm $\|\cdot\|$ is associated with some inner product if and only if, for any $(x, y) \in V \times V$,*

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2).$$

Proof. Suppose that there exists a semi-inner product $\langle \cdot, \cdot \rangle$ such that

$$\forall x \in V, \quad \|x\|^2 = \langle x, x \rangle.$$

Then

$$\begin{aligned} & \|x + y\|^2 + \|x - y\|^2 \\ &= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 + \|x\|^2 - \langle x, y \rangle - \langle y, x \rangle + \|y\|^2 \\ &= 2\|x\|^2 + 2\|y\|^2. \end{aligned}$$

(2) We first treat the real case (in Steps 1-3) in proving that the mapping

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{R}, \quad (x, y) \longmapsto \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$$

is an inner product.

Step 1. First, by definition, for any $(x, y) \in V \times V$ one has

$$\langle x, y \rangle = \langle y, x \rangle.$$

Moreover, the mapping $\langle \cdot, \cdot \rangle$ is continuous with respect to the product topology on V since the addition and the subtraction are continuous, and the mapping of taking the norm is also continuous.

Step 2. We now prove that the mapping $\langle \cdot, \cdot \rangle$ is bi-additive. By symmetry it suffices to check that $\langle \cdot, \cdot \rangle$ is linear with respect to the second coordinate. Let x, y and z be elements of V . One has

$$\begin{aligned}\|x + y + z\|^2 + \|x - y + z\|^2 &= 2\|x + z\|^2 + 2\|y\|^2, \\ \|x + y + z\|^2 + \|y - x + z\|^2 &= 2\|y + z\|^2 + 2\|x\|^2.\end{aligned}$$

Taking the average of these two equalities, we get

$$\|x + y + z\|^2 = \|x + z\|^2 + \|y + z\|^2 + \|x\|^2 + \|y\|^2 - \frac{1}{2}(\|x - y + z\|^2 + \|y - x + z\|^2).$$

Replacing z by $-z$, we obtain

$$\|x + y - z\|^2 = \|x - z\|^2 + \|y - z\|^2 + \|x\|^2 + \|y\|^2 - \frac{1}{2}(\|x - y - z\|^2 + \|y - x - z\|^2).$$

Taking the difference, we get that

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle.$$

Step 3. It remains to prove that, for any $(x, y) \in V \times V$, $\lambda \in \mathbb{R}$, one has

$$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle.$$

By induction, we deduce from the previous step that

$$\langle nx, y \rangle = n \langle x, y \rangle$$

for any $n \in \mathbb{N}$. Similarly, since

$$\langle -x, y \rangle = \frac{1}{4}(\| -x + y \|^2 - \| -x - y \|^2) = -\langle x, y \rangle.$$

Again by induction and the result of the previous step, we get

$$\forall n \in \mathbb{N}, \quad \langle -nx, y \rangle = -n \langle x, y \rangle.$$

Now let λ be a rational number, which is written as p/q , where $p \in \mathbb{Z}$ and $q \in \mathbb{N} \setminus \{0\}$. One has

$$\langle x, y \rangle = \langle q \cdot q^{-1}x, y \rangle = q \langle q^{-1}x, y \rangle.$$

Hence

$$\langle \lambda x, y \rangle = \langle pq^{-1}x, y \rangle = p \langle q^{-1}x, y \rangle = \lambda \langle x, y \rangle.$$

Finally, by the continuity of the mapping $\langle \cdot, \cdot \rangle$ and the density of $\mathbb{Q}$ in $\mathbb{R}$, we deduce that the equality

$$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$$

holds for any $\lambda \in \mathbb{R}$.

Step 4. We now assume that $K = \mathbb{C}$. For $(x, y) \in V \times V$ we define

$$\langle x, y \rangle_{\mathbb{C}} := \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) + \frac{1}{4}(\|x + iy\|^2 - \|x - iy\|^2)i.$$

Let

$$\langle x, y \rangle_1 := \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2), \quad \langle x, y \rangle_2 = \frac{1}{4}(\|x + iy\|^2 - \|x - iy\|^2).$$

By the real case proved above we obtain that $\langle \cdot, \cdot \rangle_1$ is a real bilinear form. Since

$$\forall (x, y) \in V \times V, \quad \langle x, y \rangle_2 = \langle x, iy \rangle_1,$$

we deduce that $\langle \cdot, \cdot \rangle_2$ is also a real bilinear form. Thus we deduce that $\langle \cdot, \cdot \rangle_{\mathbb{C}}$ is a real bilinear mapping. Moreover, by definition, one has

$$\forall (x, y) \in V \times V, \quad \langle x, iy \rangle_{\mathbb{C}} = i\langle x, y \rangle.$$

Therefore, $\langle \cdot, \cdot \rangle_{\mathbb{C}}$ is $\mathbb{C}$ -linear with respect to the second coordinate. Still by definition, one has

$$\forall (x, y) \in V \times V, \quad \langle y, x \rangle_{\mathbb{C}} = \overline{\langle x, y \rangle_{\mathbb{C}}}.$$

Finally, for any $x \in V \setminus \{0\}$, one has

$$\langle x, x \rangle_{\mathbb{C}} = \frac{1}{4}\|2x\|^2 + \frac{1}{4}(|1 + i|^2 - |1 - i|^2)\|x\|^2 = \|x\|^2 > 0.$$

Hence $\langle \cdot, \cdot \rangle_{\mathbb{C}}$ is a complex inner product such that □

3.2 Orthogonal projection

3.2.1 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. If V is complete with respect to the norm

$$\|\cdot\| : V \longrightarrow \mathbb{R}_{\geq 0}, \quad x \longmapsto \sqrt{\langle x, x \rangle},$$

then we say that $(V, \langle \cdot, \cdot \rangle)$ is a *Hilbert space*. In particular, any finite-dimensional inner product space is a Hilbert space.

3.2.2 Theorem. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and W be a vector subspace of V . We assume that, the vector space W , equipped with the restriction of $\langle \cdot, \cdot \rangle$, forms a Hilbert space. For any $x \in V$, there is a unique element $p_W(x) \in W$ such that $\|x - p_W(x)\|$ minimizes the function

$$(y \in W) \longmapsto \|x - y\|.$$

Moreover, $p_W(x)$ is also the only element of W such that $x - p_W(x) \in W^\perp$.

Proof. We first prove that there exists an element $p_W(x) \in W$ that minimizes the function

$$(y \in W) \longrightarrow \|x - y\|.$$

Let

$$c := \inf_{y \in W} \|x - y\|$$

and take a sequence $(y_n)_{n \in \mathbb{N}}$ in W such that

$$\lim_{n \rightarrow +\infty} \|x - y_n\| = c.$$

Note that, for any $(n, m) \in \mathbb{N}^2$,

$$\begin{aligned} \|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - \|y_n + y_m - 2x\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4c^2, \end{aligned}$$

which leads to

$$\lim_{N \rightarrow +\infty} \sup_{\substack{(n, m) \in \mathbb{N}^2 \\ n \geq N, m \geq N}} \|y_n - y_m\|^2 = 0.$$

Therefore, $(y_n)_{n \in \mathbb{N}}$ is a Cauchy sequence, which converges in W since $(W, \|\cdot\|)$ is a Hilbert space. Hence there exists $p_W(x) \in W$ such that

$$\|x - p_W(x)\| = \inf_{y \in W} \|x - y\|.$$

We now prove that $x - p_W(x)$ belongs to $W^\perp$. Let y be an element of W and pick $a \in K$ such that $|a| = 1$ and $a\langle x - p_W(x), y \rangle = |\langle x - p_W(x), y \rangle|$. For any $t \in \mathbb{R}$, one has

$$\|x - p_W(x) + tay\|^2 = \|x - p_W(x)\|^2 + 2t|\langle x - p_W(x), y \rangle| + t^2\|y\|^2 \geq \|x - p_W(x)\|^2.$$

Therefore,

$$\forall t \in \mathbb{R}, 2t|\langle x - p_W(x), y \rangle| + t^2\|y\|^2 \geq 0,$$

which leads to $\langle x - p_W(x), y \rangle = 0$.

Finally, if z is an element of W such that $z \neq p_W(x)$, then one has

$$\begin{aligned} \|x - z\|^2 &= \|(x - p_W(x)) + (p_W(x) - z)\|^2 \\ &= \|x - p_W(x)\|^2 + \|p_W(x) - z\|^2 > \|x - p_W(x)\|^2. \end{aligned}$$

Therefore, $p_W(x)$ is the unique minimum point of the function

$$(y \in W) \longmapsto \|x - y\|.$$

Similarly, if $x - z$ belongs to $W^\perp$, then

$$\|x - p_W(x)\|^2 = \|x - z\|^2 + \|z - p_W(x)\|^2 > \|x - z\|^2,$$

which is absurd. Therefore, $p_W(x)$ is also the unique element of W such that $x - p_W(x) \in W^\perp$. $\square$

3.2.3 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and W be a vector subspace of V such that W equipped with the restriction of $\langle \cdot, \cdot \rangle$ forms a Hilbert space. For any $x \in V$, the element $p_W(x)$ constructed in Theorem 3.2.2 is called the *orthogonal projection* of x to W .

3.2.4 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be a semi-inner product space. We say that a family $(x_i)_{i \in I}$ of non-zero elements of V is *orthogonal* if, for any $(i, j) \in I \times I$, one has $\langle x_i, x_j \rangle = 0$. If furthermore $\langle \cdot, \cdot \rangle$ is an inner product and $\|x_i\| = 1$ for any $i \in I$, we say that the family $(x_i)_{i \in I}$ is *orthonormal*. If $(x_i)_{i \in I}$ is an orthogonal (resp. orthonormal) family such that

$$\text{Vect}_K(\{x_i \mid i \in I\})$$

is dense in V , then we say that $(x_i)_{i \in I}$ is an *orthogonal* (resp. *orthonormal*) *basis* of V .

3.2.5 Proposition. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Any orthogonal family in V is linearly independent.

Proof. Without loss of generality, it suffices to check that, any orthogonal family of the form $(x_i)_{i=1}^n$ is linearly independent. Let $(a_1, \dots, a_n) \in K^n$ such that $a_1x_1 + \dots + a_nx_n = 0$. For any $i \in \{1, \dots, n\}$, one has

$$0 = \langle a_1x_1 + \dots + a_nx_n, x_i \rangle = a_i \langle x_i, x_i \rangle.$$

Since $\langle x_i, x_i \rangle \neq 0$, we obtain $a_i = 0$. □

3.2.6 Proposition. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $(x_i)_{i \in I}$ be an orthogonal family in V . Let $(a_i)_{i \in I} \in K^I$. Suppose that

$$\sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2 := \sup_{\substack{J \subset I \\ J \text{ finite}}} \sum_{i \in J} |a_i|^2 \cdot \|x_i\|^2 < +\infty.$$

- (1) The set $I' := \{i \in I \mid a_i \neq 0\}$ is countable.
- (2) In the case where I' is infinite, for any bijection $\varphi : \mathbb{N} \rightarrow I'$, the series

$$\sum_{n \in \mathbb{N}} a_{\varphi(n)} x_{\varphi(n)}$$

converges in V . Moreover, its limit does not depend on the choice of φ . We denote by

$$\sum_{i \in I} a_i x_i$$

this limit.

(3) One has

$$\left\| \sum_{i \in I} a_i x_i \right\|^2 = \sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2.$$

Moreover, for any bijection $\varphi : \mathbb{N} \rightarrow I'$ and any $y \in V$, the series

$$\sum_{n \in \mathbb{N}} \langle a_{\varphi(n)} x_{\varphi(n)}, y \rangle$$

converges absolutely to

$$\left\langle \sum_{i \in I} a_i x_i, y \right\rangle.$$

Proof. (1) For any $\varepsilon \in \mathbb{Q}_{>0}$, the set

$$\{i \in I \mid |a_i| \cdot \|x_i\| > \varepsilon\}$$

is finite since otherwise

$$\sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2 = +\infty.$$

Therefore,

$$I' = \bigcup_{\varepsilon \in \mathbb{Q}_{>0}} \{i \in I \mid |a_i| \cdot \|x_i\| > \varepsilon\}$$

is countable.

(2) We prove that the series is a Cauchy sequence. Since $(x_i)_{i \in I}$ forms an orthogonal family, for any natural number N , n and m such that $n > m \geq N$, one has

$$\left\| \sum_{k=m+1}^n a_{\varphi(k)} x_{\varphi(k)} \right\|^2 = \sum_{k=m+1}^n |a_{\varphi(k)}|^2 \cdot \|x_{\varphi(k)}\|^2.$$

Note that

$$\sum_{n \in \mathbb{N}} |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2 = \sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2 < +\infty.$$

Hence

$$\lim_{N \rightarrow +\infty} \sup_{\substack{(n,m) \in \mathbb{N}^2 \\ n > m \geq N}} \left\| \sum_{k=m+1}^n a_{\varphi(k)} x_{\varphi(k)} \right\| = 0,$$

namely the series

$$\sum_{n \in \mathbb{N}} a_{\varphi(n)} x_{\varphi(n)}$$

is a Cauchy sequence, that converges to some element $x \in V$ since V is a Hilbert space. Moreover, for any $N \in \mathbb{N}$, one has

$$\begin{aligned} \left\| x - \sum_{n=0}^N a_{\varphi(n)} x_{\varphi(n)} \right\|^2 &= \lim_{m \rightarrow +\infty} \left\| \sum_{n=0}^{N+m} a_{\varphi(n)} x_{\varphi(n)} - \sum_{n=0}^N a_{\varphi(n)} x_{\varphi(n)} \right\|^2 \\ &= \lim_{m \rightarrow +\infty} \sum_{n=N+1}^{N+m} |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2 = \sum_{n=N+1}^{+\infty} |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2. \end{aligned}$$

It remains to check that x does not depend on the choice of φ . Let $\psi : \mathbb{N} \rightarrow I'$ be another bijection. For any $N \in \mathbb{N}$, there exists $M \in \mathbb{N}$ such that

$$\{\varphi(0), \dots, \varphi(N)\} \subseteq \{\psi(0), \dots, \psi(M)\},$$

or equivalently

$$\{\psi(n) \mid n \geq M+1\} \subseteq \{\varphi(n) \mid n \geq N+1\}.$$

Therefore,

$$\left\| x - \sum_{n=0}^M a_{\psi(n)} x_{\psi(n)} \right\|^2 = \sum_{n=M+1}^{+\infty} |a_{\psi(n)}|^2 \cdot \|x_{\psi(n)}\|^2 \leq \sum_{n=N+1}^{+\infty} |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2,$$

which leads to

$$\lim_{N \rightarrow +\infty} \left\| x - \sum_{n=0}^M a_{\psi(n)} x_{\psi(n)} \right\|^2 = 0.$$

(3) Pick an arbitrary bijection $\varphi : \mathbb{N} \rightarrow I'$. Since the mapping $\|\cdot\|^2$ is continuous, one has

$$\begin{aligned} \left\| \sum_{i \in I} a_i x_i \right\|^2 &= \lim_{N \rightarrow +\infty} \left\| \sum_{n=0}^N a_{\varphi(n)} x_{\varphi(n)} \right\|^2 \\ &= \lim_{N \rightarrow +\infty} \sum_{n=0}^N |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2 \\ &= \sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2. \end{aligned}$$

Moreover, by Cauchy-Schwarz inequality,

$$\begin{aligned} &\left| \left\langle \sum_{i \in I} a_i x_i, y \right\rangle - \sum_{n=0}^N \langle a_{\varphi(n)} x_{\varphi(n)}, y \rangle \right| \\ &= \left| \left\langle \sum_{n=N+1}^{+\infty} a_{\varphi(n)} x_{\varphi(n)}, y \right\rangle \right| \\ &\leq \left(\sum_{n=N+1}^{+\infty} |a_{\varphi(n)}|^2 \cdot \|x_{\varphi(n)}\|^2 \right)^{1/2} \cdot \|y\|. \end{aligned}$$

Hence

$$\lim_{N \rightarrow +\infty} \sum_{n=0}^N \langle a_{\varphi(n)} x_{\varphi(n)}, y \rangle = \left\langle \sum_{i \in I} a_i x_i, y \right\rangle.$$

Finally, it remains to check that

$$\sum_{i \in I} |\langle a_i x_i, y_i \rangle| < +\infty.$$

For this purpose, we pick, for any $i \in I$, an element $b_i \in K$ such that $|b_i| = 1$ and

$$\langle b_i a_i x_i, y \rangle = |\langle a_i x_i, y \rangle|.$$

Note that

$$\sum_{i \in I} |b_i a_i|^2 \cdot \|x_i\|^2 = \sum_{i \in I} |a_i|^2 \cdot \|x_i\|^2 < +\infty,$$

we obtain that

$$\sum_{i \in I} |\langle a_i x_i, y \rangle| = \sum_{n \in \mathbb{N}} \langle b_{\varphi(n)} a_{\varphi(n)} x_{\varphi(n)}, y \rangle < +\infty,$$

as required. $\square$

3.2.7 Proposition (Parseval theorem). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and W be a vector subspace of V such that W equipped with the restriction of $\langle \cdot, \cdot \rangle$ forms a Hilbert space.*

- (1) *The mapping of orthogonal projection $p_W : V \rightarrow W$ is a K -linear mapping.*
- (2) *For any $x \in W$, one has $p_W(x) = x$. In particular, one has $p_W \circ p_W = p_W$.*
- (3) *If $(e_i)_{i \in I}$ is an orthonormal basis of W , then, for any $x \in V$, one has*

$$\sum_{i \in I} |\langle e_i, x \rangle|^2 \leq \|x\|^2$$

and

$$p_W(x) = \sum_{i \in I} \langle e_i, x \rangle e_i.$$

Proof. (1) Let $(x, y) \in V \times V$ and $(a, b) \in K \times K$. For any $z \in W$, one has

$$\langle z, (ax + by) - (ap_W(x) + bp_W(y)) \rangle = a \langle z, x - p_W(x) \rangle + b \langle z, y - p_W(y) \rangle = 0.$$

By the uniqueness part of Theorem 3.2.2, we obtain

$$p_W(ax + by) = ap_W(x) + bp_W(y).$$

(2) For any $z \in W$, $0 = z - z$ belongs to $W^\perp$. Hence $p_W(z) = z$. In particular, for any $x \in V$, one has $p_W(p_W(x)) = p_W(x)$.

(3) We first consider the case where I is a finite set. In this case, without loss of generality, we may assume that $I = \{1, \dots, n\}$. Let

$$y = x - \sum_{i=1}^n \langle e_i, x \rangle e_i.$$

For any $j \in \{1, \dots, n\}$, one has

$$\langle e_j, y \rangle = \langle e_j, x \rangle - \langle e_j, x \rangle \langle e_j, e_j \rangle = 0.$$

Hence $y \in \{e_1, \dots, e_n\}^\perp = W^\perp$. Therefore,

$$p_W(x) = \sum_{i=1}^n \langle e_i, x \rangle e_i.$$

In particular,

$$\|p_W(x)\|^2 = \sum_{i=1}^n |\langle e_i, x \rangle|^2 \leq \|x\|^2$$

by Cauchy-Schwarz.

We now consider the general case. By the particular case above, we obtain that, for any finite subset J of I , one has

$$\sum_{i \in J} |\langle e_i, x \rangle|^2 \leq \|x\|^2.$$

Taking the supremum with respect to J , we get

$$\sum_{i \in I} |\langle e_i, x \rangle|^2 \leq \|x\|^2.$$

In particular,

$$\sum_{i \in I} \langle e_i, x \rangle e_i$$

is well defined. Note that, for any $j \in I$, one has

$$\left\langle e_j, \sum_{i \in I} \langle e_i, x \rangle e_i \right\rangle = \langle e_j, x \rangle.$$

Therefore,

$$x - \sum_{i \in I} \langle e_i, x \rangle e_i \in \{e_i \mid i \in I\}^\perp = W^\perp,$$

which leads to

$$p_W(x) = \sum_{i \in I} \langle e_i, x \rangle e_i.$$

□

3.2.8 Theorem (Gram–Schmidt). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and*

$$\{0\} = V_0 \subseteq V_1 \subseteq \dots \subseteq V_n$$

be a sequence of vector subspaces of V such that

$$\forall i \in \{1, \dots, n\}, \dim(V_i) = i.$$

Then there exists an orthonormal family $(x_i)_{i=1}^n$ in V such that $x_i \in V_i \setminus V_{i-1}$ for any $i \in \{1, \dots, n\}$.

Proof. For any $i \in \{1, \dots, n\}$, we pick an arbitrary element $y_i \in V_i \setminus V_{i-1}$ and let

$$x_i = \frac{y_i - p_{V_{i-1}}(y_i)}{\|y_i - p_{V_{i-1}}(y_i)\|}.$$

Then one has $x_i \in (V_i \setminus V_{i-1}) \cap V_{i-1}^\perp$. Therefore, $(x_i)_{i=1}^n$ is an orthonormal family. $\square$

3.2.9 Remark. In the proof of Theorem 3.2.8, $(x_j)_{j=1}^{i-1}$ forms an orthonormal basis of V_{i-1} . Hence the vector x_i can be constructed in a recursive way as

$$z_i := y_i - \sum_{j=1}^{i-1} \langle x_j, y_i \rangle x_j = y_i - \frac{1}{\|z_j\|^2} \sum_{j=1}^{i-1} \langle z_j, y_j \rangle z_j, \quad x_i = \frac{1}{\|z_i\|} z_i.$$

Note that

$$\|z_i\|^2 = \|y_i\|^2 - \sum_{j=1}^{i-1} |\langle x_j, y_i \rangle|^2 = \|y_i\|^2 - \sum_{j=1}^{i-1} \frac{|\langle z_j, y_i \rangle|^2}{\|z_j\|^2}.$$

3.3 Adjoint of a bounded linear mapping

3.3.1 Lemma. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and W be a vector subspace of V . If W is not dense in V , then $W^\perp$ is non-zero.

Proof. By Proposition 3.1.9, if W' denotes the closure of W in V , then $W'^\perp = W^\perp$. Therefore, without loss of generality we may assume that W is closed (hence is a Hilbert space) and $W \neq V$. Let $x \in V \setminus W$. Then $p_W(x) \neq x$. Hence

$$x - p_W(x) \in W^\perp \setminus \{0\}.$$

$\square$

3.3.2 Definition. Let V and W be vector spaces over K . We call *conjugate homomorphism* from V to W any mapping $f : V \rightarrow W$ such that, for any $(a, b) \in K \times K$ and any $(x, y) \in V \times V$, one has

$$f(ax + by) = \bar{a}f(x) + \bar{b}f(y).$$

If in addition f is a bijection, we say that f is a conjugate isomorphism.

3.3.3 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space. We say that a K -linear mapping $f : V \rightarrow V$ is *isometric* or an *isometry* if $\|f(x)\| = \|x\|$ for any $x \in V$. Note that such that an isometric K -linear mapping is necessarily bounded and injective.

3.3.4 Theorem (Riesz representation theorem). Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space. For any bounded K -linear mapping $f : V \rightarrow K$, there exists a unique $x \in V$ such that $f = \ell_x := \langle x, \cdot \rangle$. Moreover, the mapping from V to $\mathcal{L}(V, K)$, that sends $x \in V$ to ℓ_x , is a conjugate isomorphism of vector spaces which is also an isometry, where we consider the operator norm on $\mathcal{L}(V, K)$.

Proof. By Proposition 3.1.8, the mapping

$$V \longrightarrow \mathcal{L}(V, K)$$

is well defined and is injective. Moreover, for any $x \in V$, the operator norm of ℓ_x identifies with $\|x\|$.

Let $(x, y) \in V \times V$, $(a, b) \in K \times K$, one has

$$\forall z \in V, \quad \ell_{ax+by}(z) = \langle ax + by, z \rangle = \bar{a}\langle x, z \rangle + \bar{b}\langle y, z \rangle.$$

Hence the mapping $(x \in V) \mapsto \ell_x$ is a conjugate homomorphism.

We then prove that the mapping $(x \in V) \mapsto \ell_x$ is surjective. Since f is a bounded linear mapping, its kernel W is a closed vector subspace of V . If f is the zero mapping, then $x = 0 \in V$ is the only element such that $f = \ell_x$. In what follows, we assume that f is non-zero. In this case W is not dense in V . By Lemma 3.3.1, there exists $y \in W^\perp$ such that $\|y\| = 1$. Let $x = \overline{f(y)}y$. Since $V/W \cong K$, for any $z \in V$, there exists $\lambda \in K$ such that $z - \lambda y \in W$. Hence

$$\ell_x(z) = \langle \overline{f(y)}y, z \rangle = f(y)\langle y, z \rangle = f(y)\langle y, \lambda y \rangle = \lambda f(y) = f(\lambda y) = f(z).$$

Therefore, the mapping

$$V \longrightarrow \mathcal{L}(V, K)$$

is a conjugate isomorphism, which is also an isometry. $\square$

3.3.5 Corollary. *Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be Hilbert spaces and $f : V \rightarrow W$ be a bounded linear mapping. There exists a unique bounded linear mapping $f^* : W \rightarrow V$ such that*

$$\forall (x, y) \in W \times V, \quad \langle f^*(x), y \rangle_V = \langle x, f(y) \rangle_W.$$

The bounded linear mapping f^ is called the adjoint of f . Moreover, the operator norm of f^* is equal to that of f , and one has $f^{**} = f$.*

Proof. For any $x \in W$, the mapping $\ell_x \circ f$ is a bounded linear mapping. By Theorem 3.3.4, there exists a unique element of V , which we denote by $f^*(x)$, such that $\ell_x \circ f = \ell_{f^*(x)}$. Moreover, for any $(x, x') \in W \times W$ and $(a, b) \in K \times K$, one has

$$\begin{aligned} \ell_{f^*(ax+bx')} &= \ell_{ax+bx'} \circ f = (\bar{a}\ell_x + \bar{b}\ell_{x'}) \circ f = \bar{a}(\ell_x \circ f) + \bar{b}(\ell_{x'} \circ f) \\ &= \bar{a}\ell_{f^*(x)} + \bar{b}\ell_{f^*(x')} = \ell_{af^*(x)+bf^*(x')}, \end{aligned}$$

Therefore, there exists a unique linear mapping $f^* : W \rightarrow V$ such that

$$\forall (x, y) \in W \times V, \quad \langle f^*(x), y \rangle = \langle x, f(y) \rangle.$$

It remains to check that f^* is bounded. For any $x \in W$, one has

$$\|f^*(x)\|_V = \|\ell_{f^*(x)}\| = \|\ell_x \circ f\| \leq \|\ell_x\| \cdot \|f\| = \|x\|_W \cdot \|f\|.$$

Hence $\|f^*\| \leq \|f\|$.

For any $(x, y) \in W \times V$

$$\langle y, f^*(x) \rangle_V = \overline{\langle f^*(x), y \rangle_V} = \overline{\langle x, f(y) \rangle_V} = \langle f(y), x \rangle_W.$$

Hence we get $f^{**} = f$, which also leads to

$$\|f\| = \|f^{**}\| \leq \|f^*\|.$$

Therefore, the equality $\|f^*\| = \|f\|$ holds. $\square$

3.3.6 Example. Let n and p be positive integers and $\langle \cdot, \cdot \rangle_n$ and $\langle \cdot, \cdot \rangle_p$ be the canonical inner products on K^n and K^p , respectively. Let $A : K^n \rightarrow K^p$ be a K -linear mapping. We write A in the form of an n by p matrix as

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,p} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,p} \end{pmatrix}.$$

Recall that A sends $(y_1, \dots, y_n) \in K^n$ to

$$\left(\sum_{j=1}^n y_j a_{j,1}, \dots, \sum_{j=1}^n y_j a_{j,p} \right).$$

For any $x = (x_1, \dots, x_p) \in K^p$, the mapping $\ell_x \circ A$ sends $(y_1, \dots, y_n)$ to

$$\sum_{i=1}^p \overline{x_i} \sum_{j=1}^n y_j a_{j,i} = \sum_{j=1}^n \left(\sum_{i=1}^p \overline{x_i} a_{j,i} \right) y_j = \sum_{j=1}^n \left(\sum_{i=1}^p x_i \overline{a_{j,i}} \right) y_j.$$

Therefore, A^* is given by the matrix

$$\begin{pmatrix} \overline{a_{1,1}} & \overline{a_{2,1}} & \cdots & \overline{a_{n,1}} \\ \overline{a_{1,2}} & \overline{a_{2,2}} & \cdots & \overline{a_{n,2}} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{1,p}} & \overline{a_{2,p}} & \cdots & \overline{a_{n,p}} \end{pmatrix},$$

namely $A^* = \overline{A}^T$.

3.3.7 Proposition. Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be Hilbert spaces. Then $f \mapsto f^*$ defines a conjugate homomorphism from $\mathcal{L}(V, W)$ to $\mathcal{L}(W, V)$.

Proof. Let $(f, g) \in \mathcal{L}(V, W) \times \mathcal{L}(V, W)$, and $(a, b) \in K \times K$. For any $(x, y) \in W \times V$, one has

$$\begin{aligned}\langle x, (af + bg)(y) \rangle_W &= a\langle x, f(y) \rangle_W + b\langle x, g(y) \rangle_W = a\langle f^*(x), y \rangle_V + b\langle g^*(x), y \rangle_V \\ &= \langle \bar{a}f^*(x) + \bar{b}g^*(x), y \rangle_V = \langle (\bar{a}f^* + \bar{b}g^*)(x), y \rangle_V.\end{aligned}$$

Hence $(af + bg)^* = \bar{a}f^* + \bar{b}g^*$. $\square$

3.3.8 Proposition. Let $(V_1, \langle \cdot, \cdot \rangle_1)$, $(V_2, \langle \cdot, \cdot \rangle_2)$ and $(V_3, \langle \cdot, \cdot \rangle_3)$ be Hilbert spaces, $f : V_1 \rightarrow V_2$ and $g : V_2 \rightarrow V_3$ be bounded linear mappings. Then

$$(g \circ f)^* = f^* \circ g^*.$$

Proof. For $(x, y) \in V_3 \times V_1$, let

$$\langle x, (g \circ f)(y) \rangle_{V_3} = \langle x, g(f(y)) \rangle_{V_3} = \langle g^*(x), f(y) \rangle_{V_2} = \langle f^*(g^*(x)), y \rangle_{V_1}.$$

Therefore, we get $(g \circ f) = f^* \circ g^*$. $\square$

3.3.9 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space. We say that a bounded endomorphism $f : V \rightarrow V$ is *self-adjoint* if $f^* = f$, namely

$$\forall (x, y) \in V \times V, \quad \langle f(x), y \rangle = \langle x, f(y) \rangle.$$

3.3.10 Remark. Let n be a positive integer. We equip K^n with the canonical inner product. Let $A : K^n \rightarrow K^n$ be a K -linear mapping. Then one has $A^* = \bar{A}^T$. In the case where $K = \mathbb{R}$ and one has $A^* = A^T$, we say that A is a *symmetric matrix*. In the case where $K = \mathbb{C}$, and $A = \bar{A}^T$ we say that A is a *Hermitian matrix*.

3.3.11 Example. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and W be a closed vector subspace of V . Then the mapping of orthogonal projection $p_W : V \rightarrow W$ is self-adjoint. In fact, for any $(x, y) \in V \times V$, one has

$$\langle x, p_W(y) \rangle = \langle p_W(x), p_W(y) \rangle + \langle x - p_W(x), p_W(y) \rangle = \langle p_W(x), p_W(y) \rangle$$

since $x - p_W(x)$ belongs to $W^\perp$ and $p_W(y) \in W$. Similarly,

$$\langle p_W(x), y \rangle = \langle p_W(x), p_W(y) \rangle + \langle p_W(x), y - p_W(y) \rangle = \langle p_W(x), p_W(y) \rangle.$$

Therefore,

$$\langle p_W(x), y \rangle = \langle x, p_W(y) \rangle,$$

namely $p_W^* = p_W$.

3.4 Reduction of self-adjoint mappings

3.4.1 Definition. Let F be a field, V be a vector space over F and $f : V \rightarrow V$ be an endomorphism of vector spaces over F . Let $\lambda \in F$. If there exists $x \in V \setminus \{0\}$ such that $f(x) = \lambda x$, then we say that λ is an *eigenvalue* of f and that x is an *eigenvector* of f associated with the eigenvalue λ . Note that $\lambda \in K$ is an eigenvalue of f if and only if $\lambda \text{Id}_V - f$ is not injective.

Suppose that V is a finite-dimensional vector space over F . We say that an endomorphism $f : V \rightarrow V$ is *diagonalizable* if there exists a basis of V which is composed by eigenvectors of f .

3.4.2 Remark. Let F be a field, V be a finite-dimensional vector space over F and $f : V \rightarrow V$ be an endomorphism. Let $(e_i)_{i=1}^n$ be a basis of f which is composed by eigenvectors of f . For any $i \in I$, let $\lambda_i \in F$ be the eigenvalue of e_i , namely $f(e_i) = \lambda_i e_i$. Let $\varphi : K^n \rightarrow V$ the mapping that sends $(a_1, \dots, a_n)$ to $a_1 e_1 + \dots + a_n e_n$, then $\varphi^{-1} \circ f \circ \varphi$ sends $(a_1, \dots, a_n)$ to

$$\varphi^{-1}(\lambda_1 a_1 e_1 + \dots + \lambda_n a_n e_n) = (\lambda_1 a_1, \dots, \lambda_n a_n),$$

namely

$$\varphi^{-1} \circ f \circ \varphi := \text{diag}(\lambda_1, \dots, \lambda_n).$$

3.4.3 Definition. Let $(V, \|\cdot\|)$ be a Banach space over K (which is $\mathbb{R}$ or $\mathbb{C}$) and $f : V \rightarrow V$ be a bounded linear mapping. We call $\lambda \in K$ a *regular value* of f if $\lambda \text{Id}_V - f$ is an isomorphism of K -vector spaces. Note that the open mapping theorem shows that, if $\lambda \text{Id}_V - f$ is a bijection, then $(\lambda \text{Id}_V - f)^{-1}$ is a bounded linear mapping. We denote by $\sigma(f)$ the set of $\lambda \in K$ such that $\lambda \text{Id}_V - f$ is invertible, and we call this set the *resolvent* of f . The set $\rho(f) := K \setminus \sigma(f)$ is called the *spectrum* of f . In the case where V is a finite-dimensional vector space over K , a K -linear mapping is an isomorphism if and only if it is injective. Therefore, the spectrum of f is equal to the set of eigenvalues of f . In the infinite dimensional case, the points of the spectrum can be divided in two classes:

- I. $\lambda \in \rho(f)$ such that $\lambda \text{Id}_V - f$ is not injective,
- II. $\lambda \in \rho(f)$ such that $\lambda \text{Id}_V - f$ is injective but not surjective.

The points of type I. correspond to eigenvalues of f , but, in general, the set of points of type II. might be non-empty.

3.4.4 Example. Consider a Hilbert space of countable dimension over K with an orthonormal basis $\{e_n\}_{n \in \mathbb{N}}$, and let f be such that $f(e_n) = \frac{1}{n} e_n$ for all $n \in \mathbb{N}$. The set of eigenvalues of f is exactly $\{\frac{1}{n} | n \in \mathbb{N}\}$, while the spectrum contains also the

value $0 \in K$. Indeed, f^{-1} sends e_n to ne_n and therefore is not bounded. Using the open mapping theorem, we conclude, that f cannot be surjective. This implies, that 0 is an element of the spectrum, that is not an eigenvalue.

3.4.5 Theorem. *Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space. Let f be a self-adjoint bounded K -linear mapping. Then there exists an orthonormal basis $(e_i)_{i=1}^n$ of V that is composed of eigenvectors of f . In particular, f is diagonalizable. Moreover, the eigenvalues of f are real numbers.*

Proof. If λ is an eigenvalue of f and v is an eigenvector of eigenvalue λ , then

$$\lambda \langle v, v \rangle = \langle v, \lambda v \rangle = \langle v, f(v) \rangle = \langle f(v), v \rangle$$

We reason by induction on the dimension n of V over K . The case where $n = 0$ is trivial.

In the following, we suppose that $\dim(V) > 0$. Since f is self-adjoint, for any $x \in V$ one has

$$\langle x, f(x) \rangle = \langle f(x), x \rangle = \overline{\langle x, f(x) \rangle}.$$

Hence $\langle x, f(x) \rangle \in \mathbb{R}$. We denote by $q : V \rightarrow \mathbb{R}$ the mapping that sends $x \in V$ to $\langle x, f(x) \rangle$. It is a continuous mapping. Let S be the unit circle of V , namely

$$S := \{x \in V \mid \|x\| = 1\}.$$

It is a bounded and closed subset of V . Since V is finite-dimensional, we obtain that S is compact. Hence the mapping $q(\cdot)$ attains its maximum λ . Note that, for any $y \in V$, there exists $a \in K$ and $x \in S$ such that $y = ax$, so that

$$q(y) = \langle ax, f(ax) \rangle = |a|^2 q(x) \leq |a|^2 \lambda = \lambda \|y\|^2.$$

Let e_1 be an element of S such that $q(e_1) = \lambda$. Let $y \in S$ and $\theta \in K$ be such that $|\theta| = 1$. For any $t > 0$, one has

$$\begin{aligned} q(\theta e_1 + ty) &= \langle \theta e_1 + ty, f(\theta e_1 + ty) \rangle = q(e_1) + t^2 q(y) + 2t \operatorname{Re}(\theta \langle y, f(e_1) \rangle) \\ &\leq \lambda \|\theta e_1 + ty\|^2 = \lambda (1 + t^2 + 2t \operatorname{Re}(\theta \langle y, e_1 \rangle)). \end{aligned}$$

Therefore,

$$tq(y) + 2 \operatorname{Re}(\theta \langle y, f(e_1) \rangle) \leq \lambda t + 2\lambda \operatorname{Re}(\theta \langle y, e_1 \rangle).$$

Taking the limit when $t \rightarrow 0$, we get

$$\operatorname{Re}(\theta \langle y, f(e_1) \rangle) \leq \lambda \operatorname{Re}(\theta \langle y, e_1 \rangle).$$

Replacing y by $-y$, we get the inequality with reversed direction. Hence

$$\operatorname{Re}(\theta \langle y, f(e_1) \rangle) = \lambda \operatorname{Re}(\theta \langle y, e_1 \rangle).$$

which leads to

$$\operatorname{Re}(\theta\langle y, f(e_1) - \lambda e_1 \rangle) = 0.$$

Since θ is arbitrary, we may choose θ such that

$$\operatorname{Re}(\theta\langle y, f(e_1) - \lambda e_1 \rangle) = |\langle y, f(e_1) - \lambda v \rangle|.$$

Thus we get $f(e_1) = \lambda e_1$, namely, λ is an eigenvalue of f and e_1 is an eigenvector of f of eigenvalue λ .

Let $W = \{e_1\}^\perp$. For any $y \in W$, one has

$$\langle e_1, f(y) \rangle = \langle f(e_1), y \rangle = \langle \lambda e_1, y \rangle = \bar{\lambda} \langle e_1, y \rangle = 0.$$

Hence the restriction of f to W gives a self-adjoint endomorphism of W . Since $W = \operatorname{Ker}(\ell_{e_1})$, its dimension is $n-1$. We apply the induction hypothesis to $f|_W : W \rightarrow W$ to obtain that there exists an orthonormal basis $(e_i)_{i=2}^n$ of W which is composed of eigenvectors of f . Since $e_1 \in W^\perp$, we obtain that $(e_i)_{i=1}^n$ is an orthogonal basis of V . The theorem is thus proved. $\square$

3.4.6 Corollary. *Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space and $f : V \rightarrow V$ be a self-adjoint bounded K -linear mapping. Then the operator norm of f is equal to the maximal absolute value of its eigenvalues.*

Proof. Let λ be an eigenvalue of f and v be an eigenvector of eigenvalue λ such that $\|v\| = 1$. By definition, one has $f(v) = \lambda v$. Hence, by Cauchy-Schwarz inequality,

$$|\lambda| = |\langle v, f(v) \rangle| \leq \|v\| \cdot \|f(v)\| \leq \|f\|.$$

Let $(e_i)_{i=1}^n$ be an orthonormal basis of V which is composed of eigenvectors of V . For each $i \in \{1, \dots, n\}$, let λ_i be the eigenvalue corresponding to e_i . Let $x = a_1 e_1 + \dots + a_n e_n$ be an element of V . One has

$$f(x) = a_1 \lambda_1 e_1 + \dots + \lambda_n a_n e_n.$$

Hence

$$\begin{aligned} \|f(x)\|^2 &= |a_1 \lambda_1|^2 + \dots + |a_n \lambda_n|^2 \\ &\leq \left(\max_{i \in \{1, \dots, n\}} |\lambda_i|^2 \right) (|a_1|^2 + \dots + |a_n|^2) \\ &= \left(\max_{i \in \{1, \dots, n\}} |\lambda_i|^2 \right) \|x\|^2, \end{aligned}$$

which leads to

$$\|f\|^2 \leq \max_{i \in \{1, \dots, n\}} |\lambda_i|^2.$$

$\square$

3.4.7 Definition. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space, and $f : V \rightarrow V$ be a self-adjoint bounded K -linear mapping.

(1) We say that f is *positive semi-definite* if

$$\forall x \in V, \quad \langle x, f(x) \rangle \geq 0.$$

(2) We say that f is *negative semi-definite* if

$$\forall x \in V, \quad \langle x, f(x) \rangle \leq 0.$$

(3) We say that f is *positive definite* if

$$\forall x \in V \setminus \{0\}, \quad \langle x, f(x) \rangle > 0.$$

(4) We say that f is *negative definite* if

$$\forall x \in V \setminus \{0\}, \quad \langle x, f(x) \rangle < 0.$$

3.4.8 Proposition. Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional non-zero inner product space, and $f : V \rightarrow V$ be a self-adjoint K -linear mapping.

(1) f is positive semidefinite if and only if all eigenvalues of f are ≥ 0 ,

(2) f is negative semidefinite if and only if all eigenvalues of f are ≤ 0 ,

(3) f is positive definite if and only if all eigenvalues of f are > 0 .

(4) f is negative definite if and only if all eigenvalues of f are < 0 .

Proof. Let $(e_i)_{i=1}^n$ be an orthonormal basis of V which is composed of eigenvectors of f . For any $i \in \{1, \dots, n\}$, let λ_i be the eigenvalue of e_i .

If f is positive semidefinite (resp. negative semidefinite, positive definite, negative definite), then, for any $i \in \{1, \dots, n\}$, one has

$$\lambda_i = \langle e_i, f(e_i) \rangle \geq 0 \quad (\text{resp. } \leq 0, > 0, < 0).$$

Conversely, for any $x = a_1 e_1 + \dots + a_n e_n \in V \setminus \{0\}$, one has

$$\langle x, f(x) \rangle = \lambda_1 |a_1|^2 + \dots + \lambda_n |a_n|^2.$$

If, for any $i \in \{1, \dots, n\}$, one has $\lambda_i \geq 0$ (resp. $\leq 0, > 0, < 0$), then, one has $\langle x, f(x) \rangle \geq 0$ (resp. $\leq 0, > 0, < 0$), as required. $\square$

3.5 Linear isometry

3.5.1 Proposition. *Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space. A K -linear mapping $f : V \rightarrow V$ is isometric if and only if $f^* \circ f = \text{Id}_V$, or equivalently,*

$$\forall (x, y) \in V \times V, \quad \langle f(x), f(y) \rangle = \langle x, y \rangle.$$

Proof. For any $(x, y) \in V \times V$, one has

$$\langle f(x), f(y) \rangle = \langle f^*(f(x)), y \rangle.$$

Suppose that $f^* \circ f = \text{Id}_V$, then, for any $(x, y) \in V \times V$, one has

$$\langle f(x), f(y) \rangle = \langle f^*(f(x)), y \rangle = \langle x, y \rangle.$$

In particular, for any $x \in V$, one has

$$\|f(x)\|^2 = \langle f(x), f(x) \rangle = \langle x, x \rangle^2 = \|x\|^2.$$

Conversly, suppose that f is isometric. Let $(x, y) \in V \times V$ and $\theta \in K$ such that $|\theta| = 1$. One has

$$\begin{aligned} \|f(x + \theta y)\|^2 &= \|f(x)\|^2 + \|f(y)\|^2 + 2\text{Re}(\theta \langle f(x), f(y) \rangle), \\ \|x + \theta y\|^2 &= \|x\|^2 + \|y\|^2 + 2\text{Re}(\theta \langle x, y \rangle), \end{aligned}$$

so that

$$\text{Re}(\theta \langle f^*(f(x)), y \rangle) = \text{Re}(\theta \langle x, y \rangle),$$

or equivalently

$$\text{Re}(\theta \langle f^*(f(x)) - x, y \rangle) = 0.$$

Choose $\theta \in K$ such that

$$\theta \langle f^*(f(x)) - x, y \rangle \in \mathbb{R},$$

we deduce that

$$f^*(f(x)) = x.$$

□

3.5.2 Remark. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $f : V \rightarrow V$ be an isometric K -linear mapping. Note that f is necessarily injective. In the case where V is finite-dimensional, then f is actually an isomorphism. In general, a isometric K -linear mapping $V \rightarrow V$ which is an isomorphism of K -vector spaces is called an *automorphism* of the Hilbert space $(V, \langle \cdot, \cdot \rangle)$. The set of automorphisms of $(V, \langle \cdot, \cdot \rangle)$ form a group under the composition of mappings, which is denoted as $\text{Aut}(V, \langle \cdot, \cdot \rangle)$. In the case where $K = \mathbb{R}$, the group of automorphisms $\text{Aut}(V, \langle \cdot, \cdot \rangle)$ is also called the *orthogonal group* of $(V, \langle \cdot, \cdot \rangle)$ and is denoted as $O(V, \langle \cdot, \cdot \rangle)$. In the case where $K = \mathbb{C}$, the group of automorphisms $\text{Aut}(V, \langle \cdot, \cdot \rangle)$ is also called the *unitary group* of $(V, \langle \cdot, \cdot \rangle)$ and is denoted as $U(V, \langle \cdot, \cdot \rangle)$.

3.5.3 Example. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and W be a closed subspace of $(V, \langle \cdot, \cdot \rangle)$. We call orthogonal symmetry with respect to W the following mapping

$$s_W : V \longrightarrow V, \quad x \longmapsto 2p_W(x) - x.$$

Clearly s_W is a K -linear mapping. Moreover, for any $x \in V$, one has

$$s_W(s_W(x)) = 2p_W(2p_W(x) - x) - (2p_W(x) - x) = 4p_W(x) - 2p_W(x) - 2p_W(x) + x = x.$$

Therefore, s_W is an *involution* of V (namely $s_W \circ s_W = \text{Id}_V$). Moreover, one has

$$s_W^* = 2p_W^* - \text{Id}_V^* = 2p_W - \text{Id}_V.$$

Hence s_W is in the same time an automorphism of Hilbert space and a self-adjoint endomorphism.

Exercises

1. Consider $L^2([-1, 1])$, the space of square integrable real functions on $[-1, 1]$. As usual, two functions that differ at a Lebesgue null set are identified.

a) Prove that

$$(f, g) \mapsto \langle f, g \rangle := \int_{-1}^1 f(x)g(x) \, dx$$

is an inner product.

It is a standard fact in real analysis that the inner product above induces a norm $\|\cdot\|_{L^2}$ with respect to which $L^2([-1, 1])$ is complete. Equivalently, $L^2([-1, 1])$ is a Hilbert space. In fact, the L^2 space on any measure space is complete.

b) Let V be a Hilbert space and $W \subset V$ be a subspace. Prove that W is closed if and only if it is a Hilbert space with respect to the inner product on V .

c) Determine whether $C([-1, 1])$, $C^1([-1, 1])$, and $C^\infty([-1, 1])$ are Hilbert spaces under $\langle \cdot, \cdot \rangle$.

The *Legendre polynomials* $\{P_n\}_{n \geq 0}$ are defined as the Gram–Schmidt orthonormalisation of $\{1, x, x^2, x^3, \dots\}$.

d) Calculate P_0, P_1, P_2 .

e) Determine whether $\{P_n\}$ is complete in $L^2([-1, 1])$.

2. Let $\ell^2(\mathbb{Z}) \subset \mathbb{C}^{\mathbb{Z}}$ denote the space of complex sequences

$$\{a = (a_i)_{i \in \mathbb{Z}}; |a| := \sum_{i \in \mathbb{Z}} |a_i|^2 < +\infty\}.$$

In fact, $\ell^2(\mathbb{Z})$ is nothing but the L^2 space of functions $f : \mathbb{Z} \mapsto \mathbb{C}$ with respect to the counting measure, but we can directly

- a) Prove that $|\cdot|$ is a norm on $\ell^2(\mathbb{Z})$.
- b) Prove that $\ell^2(\mathbb{Z})$ has the structure of an inner product space, i.e., find an inner product associated with the norm $|\cdot|$.
- c) Prove that $\ell^2(\mathbb{Z})$ is complete with respect to the norm $|\cdot|$. In particular, $\ell^2(\mathbb{Z})$ is a Hilbert space.
- d) Find a complete orthonormal family for $\ell^2(\mathbb{Z})$.
- e) Define the maps

$$\sigma_L : (a_i)_{i \in \mathbb{Z}} \mapsto (a_{i+1})_{i \in \mathbb{Z}},$$

to be the shift to the left. Find the adjoint of σ_L

3. Let V be an inner product space and let A, B be linear operators on V . We denote by $*$ the adjoint. Prove that
- a) $(A + B)^* = A^* + B^*$.
 - b) $(AB)^* = B^* A^*$.
 - c) $(A^*)^* = A$.
 - d) For $\lambda \in \mathbb{C}$, $(\lambda A + B)^* = \bar{\lambda} A^* + B^*$.
 - e) Suppose A has an inverse A^{-1} , then A^* is invertible, with $(A^{-1})^* = (A^*)^{-1}$.
4. We study the *Dirichlet problem for the Poisson equation* on the open unit disk $D \subset \mathbb{R}^2$. Define the differential operators

$$\nabla := \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2} \right)$$

$$\Delta := \left(\frac{\partial}{\partial x_1} \right)^2 + \left(\frac{\partial}{\partial x_2} \right)^2.$$

For a fixed real function $f \in L^2(D)$, we wish to solve for twice differentiable functions u in the partial differential equation

$$-\Delta u + f = 0 \text{ on } D, u = 0 \text{ on } \partial D.$$

a) Let $C_c^\infty(D)$ denote the space of real valued compactly supported smooth functions on D . Prove that

$$\|\cdot\|_{H^1} : u \mapsto \|u\|_{L^2(D)} + \|\nabla u\|_{L^2(D)}$$

is a norm, with respect to which $C_c^\infty(D)$ is not complete.

We write $H_0^1(D)$ for the completion of $C_c^\infty(D)$ with respect to $\|\cdot\|_{H^1}$. It is one of what are called *zero-trace Sobolev spaces*, and we have $C_0^k(D) \subset H_0^1(D)$ for all $k = 0, 1, \dots, \infty$, where $C_0^k(D)$ denotes the space of k -times differentiable functions that vanishes on ∂D .

b) Prove that any function in $H_0^1(D)$ vanishes on ∂D .

The *Poincaré inequality* implies that $\|\cdot\|_{H^1}$ is equivalent to the norm

$$\|\cdot\|_{H_0^1} : u \mapsto \|\nabla u\|_{L^2(D)},$$

that is, they define the same Banach space structure.

c) Prove that

$$\langle \cdot, \cdot \rangle_{H_0^1} : (u, v) \mapsto \int_D \nabla u \cdot \nabla v$$

equips $H_0^1(D)$ with the structure of a Hilbert space.

Suppose u is an actual solution of the Dirichlet problem, then for all differentiable v , the integration-by-part formula asserts that

$$-\int_D f v = \int_D v \Delta u = \int_D \nabla u \cdot \nabla v.$$

Motivated by this, we say $u \in H_0^1(D)$ is a *weak solution* to the Dirichlet problem if for all $v \in H_0^1(D)$ we have

$$\int_D \nabla u \cdot \nabla v = - \int_D f v.$$

d) Prove that $v \mapsto - \int_D f v$ is a continuous linear functional on $H_0^1(D)$.

e) Apply the Riesz theorem and show that the weak solution to the Dirichlet problems exists and is unique.

A C^2 function u is called *harmonic* if $\Delta u = 0$.

f) Prove that the only harmonic function on D that vanishes on ∂D is 0.

5. We build a primer on *Fourier analysis*.

Let $L^2(S^1)$ be the space of complex-valued square-integrable periodic functions on $\mathbb{R}$ with period 2π . Equivalently,

$$L^2(S^1) = \left\{ f : [0, 2\pi] \rightarrow \mathbb{C} : \int_0^{2\pi} |f(\theta)|^2 d\theta < \infty \right\}.$$

a) Prove that $L^2(S^1)$ has the structure of a Hilbert space with respect to the usual L^2 inner product $\langle \cdot, \cdot \rangle$ and L^2 norm $\| \cdot \|_{L^2}$. You may use the fact that $L^2(\mathbb{R})$ is a Hilbert space.

b) Prove that the functions

$$\left\{ e_n(\theta) := \frac{\exp(in\theta)}{\sqrt{2\pi}} \right\}_{n \in \mathbb{Z}}$$

is an orthonormal family.

c) Prove the Parseval identity for $\{e_n\}$:

$$\|f\|_{L^2}^2 = \sum_{n \in \mathbb{Z}} \langle f, e_n \rangle^2,$$

and conclude that $\{e_n\}_{n \in \mathbb{Z}}$ is a complete orthonormal family.

Write $\hat{f}(n) = \langle f, e_n \rangle$, it is called the n -th *Fourier coefficient* of f . The *Fourier transform* is the map that sends f to the function $\hat{f} : \mathbb{Z} \rightarrow \mathbb{C}$.

d) Prove the *Plancherel theorem*: the Fourier transform is an isometry between the Hilbert spaces $L^2(S^1)$ and $\ell^2(\mathbb{Z})$.

e) Try to determine the inverse map of the Fourier transform.

f) Suppose $f \in L^2(S^1)$ is twice differentiable, calculate the Fourier transform of $\frac{d}{dx}f$ and $(\frac{d}{dx})^2f$.

g) Prove that if f is smooth then $\{\hat{f}(n)\}$ is fast decaying in the sense that

$$\lim_{|n| \rightarrow \infty} \hat{f}(n)P(n) = 0$$

for any polynomial P .

h) Prove the converse of f). This is one test for regularity of L^2 functions.

6. Let V be a Hilbert space. V is called *seperable* if there exists a countable subset in V that is dense.

a) Suppose the dimension of V is countable, i.e., V has a countable complete orthonormal family, prove that V is seperable.

b) Suppose V is seperable, prove that dimension of V is countable.

c) Prove that any complex seperable Hilbert space of dimension infinity is isometric to $\ell^2(\mathbb{Z})$.

7. Let V be a Hilbert space and let $W \subset V$ be a linear subspace.

a) Do we have

$$(W^\perp)^\perp = W?$$

b) Prove that the equality in a) holds when W is a closed subspace.

8. Let V be a Hilbert space and $P : V \rightarrow V$ a linear map with

$$P(v) = P^2(v), \quad |P(v)| \leq |v|, \quad \forall v \in V.$$

Prove that

a) P is self-adjoint.

b) There exists a closed subspace $W \subset V$ such that $P = P_W$, the orthogonal projection onto W . In particular, $\ker P = \operatorname{im}(P)^\perp$.

c) $\operatorname{id} - P = P_{W^\perp}$.

9. Let V be a Hilbert space, prove that:

a) Suppose A is a self-adjoint operator on A . If $\langle Av, v \rangle = 0$ for all $v \in V$, then $A = 0$.

b) a) fails for operators that are not self-adjoint.

c) Let $A, B : V \rightarrow V$ be self-adjoint linear maps, then $A + B$ is self-adjoint, and AB is self-adjoint if and only if $AB = BA$.

d) Let f be a polynomial. If A is self-adjoint, then so is $f(A)$.

10. a) Let $V = L^2([0, 1])$ be the space of square integrable functions on the unit interval. Let M be the operator defined by

$$Mf : x \mapsto xf(x).$$

Prove that each $\lambda \in [0, 1]$ is a spectral value of M . Are they eigenvalues?

A is called a *compact operator* if the closure of the image of the unit ball in V is compact.

b) Prove that a linear operator on a finite-dimensional space is compact.

The *Riesz-Schauder theorem* asserts that for a compact operator A ,

$$I - A \text{ injective} \iff I - A \text{ surjective} \iff I - A \text{ invertible}.$$

c) Prove that nonzero spectral values of a compact operator A are eigenvalues.

11. a) Prove that the shift σ_L in Exercise 2 has no eigenvalue.

Define $T : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ by

$$T : (a_i)_{i \in \mathbb{Z}} \mapsto \left(\frac{a_i}{1 + n^2} \right)_{i \in \mathbb{Z}}.$$

b)★ Prove that T is a compact operator.

c) Find the *spectrum* of T , i.e., the set of all spectral values.

The pullback of T by the Plancherel isomorphism is also a compact operator on $L^2(S^1)$ with the same spectrum. It is called the *Sobolev smoothing operator* and can be explicitly given by convolution.

12. Consider the matrix

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

a) Prove that J has eigenvalues $\pm i$.

b) Find a complex matrix that diagonalizes J can be diagonalized by a complex matrix.

c) Is J diagonalisable by a real matrix?

d) Let A be an $n \times n$ matrix whose eigenvalues are all real. Prove that if A is diagonalized by a complex matrix, then it can be diagonalized by a real one.

13. Let A be a complex $n \times n$ matrix and $V \subset \mathbb{C}^n$ be a linear subspace. V is called an *invariant subspace* if $A(V) \subset V$.

a) Prove that V is an invariant subspace if and only if $V^\perp$ is an invariant subspace of A^* .

A complex matrix is called *normal* if $A^*A = AA^*$, where $*$ denotes the adjoint. We say a complex matrix U is *unitary* if $UU^* = U^*U = I$. We pick this name to distinguish real and complex orthogonal matrices. In this exercise we prove that a matrix is unitarily diagonalisable if and only if it is normal.

b) Suppose A is diagonalisable by an orthogonal matrix P , then A is normal.

Conversely, suppose A is normal, prove that

c) Suppose λ is an eigenvalue of A , then $\bar{\lambda}$ is an eigenvalue of A^* with the same eigenvectors.

d) Let $\lambda_1 \neq \lambda_2$ be eigenvalues of A and v_1, v_2 eigenvectors of λ_1, λ_2 respectively, then v_1 and v_2 are perpendicular.

e) A is diagonalisable by a unitary matrix. Prove by induction on the size of A .

It is in fact a direct corollary, but we can mimick the steps above to prove:

f) A real matrix A is orthogonally diagonalisable if and only if it is symmetric.

14. We describe how to complete squares for quadratic forms and calculate orthogonal diagonalisation bare-handed. For simplicity we work with the real field.

Let A be a matrix, recall that we defined the *elementary row and column transforms*

- R_{ij} : Interchanging the i -th and j -th rows (columns).
- R_{i+j} : Add the j -th row (column) to the i -th.
- $R_{\lambda i}$: Multiplying the i -th row (column) with a real scalar $\lambda \neq 0$.

a) Prove that performing an elementary row transform R on A is equivalent to multiplying a matrix T_R to the left of A .

b) Prove that performing the same column transform R on A is equivalent to multiplying the matrix T_R^T to the right of A , where T denotes the transpose.

c) Prove that $T_R T_R^T$ is either the identity, or an invertible diagonal matrix, with at most one diagonal element not equal to one.

The *Gauss elimination theorem* implies that each invertible matrices can be transformed into the identity using only elementary row transforms or column transforms.

d) Let T_{ij} , T_{i+j} , $T_{\lambda i}$ denote the matrices found in a). Prove that such matrices generate the group GL_n of invertible $n \times n$ matrices, called the *general linear group*.

An *elementary transform pair* of A is obtained by an elementary row transformation followed by the same elementary transformation on the column.

e) Prove that any symmetric matrix is reduced to a diagonal matrix by finitely many elementary transform pairs. This in fact holds for normal matrices allowing complex λ .

f) Apply elementary transform pairs to complete the squares and find the signature of the quadratic form

$$q_E(x, y) = \sum_{1 \leq i \leq j \leq n} x_i y_i.$$

- 15.** We recall a minitest problem on 21th Mar. Consider $C([0, 1])$, the space of continuous real functions on the unit interval $[0, 1]$. Pick $p \in C([0, 1])$ fixed. We claim that the symmetric bilinear function

$$(f, g) \mapsto \int_0^1 f(x)g(x)p(x) \, dx$$

is an inner product on $C([0, 1])$, i.e., it is positive definite, if and only if $p \geq 0$ on $[0, 1]$, and the set where $p > 0$ is dense in $[0, 1]$.

Chapter 4

Multilinear algebra

4.1 Tensor product

In this section, we fix a commutative unitary ring K . If n is a natural number, $V_1, \dots, V_n$ and W are K -modules, we denote by $\text{Hom}_K^{(n)}(V_1, \dots, V_n, W)$ the set of n -linear mappings from $V_1 \times \dots \times V_n$ to W .

4.1.1 Theorem. *Let n be natural number and $V_1, \dots, V_n$ be K -modules. The functor F from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends any K -module W to the set of n -linear mappings from $V_1 \times \dots \times V_n$ to W is representable.*

Proof. We have seen that, for any K -module W , the mapping

$$W^{V_1 \times \dots \times V_n} \longrightarrow \mathbf{Mod}_K(K^{\oplus(V_1 \times \dots \times V_n)}, W)$$

sending any

$$s \in W^{V_1 \times \dots \times V_n}$$

to

$$f_s : K^{\oplus(V_1 \times \dots \times V_n)} \longrightarrow W, \quad (a_{\mathbf{x}})_{\mathbf{x} \in V_1 \times \dots \times V_n} \longmapsto \sum_{\mathbf{x} \in V_1 \times \dots \times V_n} a_{\mathbf{x}} s(\mathbf{x}),$$

is a bijection.

Let $(e_{\mathbf{x}})_{\mathbf{x} \in V_1 \times \dots \times V_n}$ be the canonical basis of $K^{\oplus(V_1 \times \dots \times V_n)}$. Recall that, as a mapping from

$$V_1 \times \dots \times V_n \longrightarrow K,$$

$e_{\mathbf{x}}$ sends $\mathbf{x}$ to 1 and any

$$\mathbf{y} \in (V_1 \times \dots \times V_n) \setminus \{\mathbf{x}\}$$

to 0. Let M be the sub- K -module of $K^{\oplus(V_1 \times \dots \times V_n)}$ generated by the elements of the form

$$e_{(x_1, \dots, x_{i-1}, ax'_i + bx''_i, x_{i+1}, \dots, x_n)} - ae_{(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)} - be_{(x_1, \dots, x_{i-1}, x''_i, x_{i+1}, \dots, x_n)}, \quad (4.1)$$

where a and b are elements of K . Note that, for any element $s \in W^{V_1 \times \cdots \times V_n}$, the image of the element (4.1) by f_s is

$$\begin{aligned} s(x_1, \dots, x_{i-1}, ax'_i + bx''_i, x_{i+1}, \dots, x_n) - as(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \\ - bs(x_1, \dots, x_{i-1}, x''_i, x_{i+1}, \dots, x_n). \end{aligned}$$

Therefore, s is an n -linear mapping if and only if f_s vanishes on M and thus induces a homomorphism from

$$V_1 \otimes \cdots \otimes V_n := K^{\oplus(V_1 \times \cdots \times V_n)} / M$$

to W . Therefore, we obtain that the functor F is representable by the quotient K -module $V_1 \otimes \cdots \otimes V_n$. □

4.1.2 Definition. Let n be a natural number, and $V_1, \dots, V_n$ be K -modules. The K -module $V_1 \otimes \cdots \otimes V_n$ constructed in the proof of Theorem 4.1.1 is called the *tensor product* of $V_1, \dots, V_n$. For any $(x_1, \dots, x_n) \in V_1 \times \cdots \times V_n$, we denote by $x_1 \otimes \cdots \otimes x_n$ the image of $(x_1, \dots, x_n)$ by the universal n -linear mapping

$$V_1 \times \cdots \times V_n \longrightarrow V_1 \otimes \cdots \otimes V_n.$$

Note that, for any K -module W and any n -linear mapping $f : V_1 \times \cdots \times V_n \rightarrow W$, there is a unique homomorphism of K -modules

$$\tilde{f} : V_1 \otimes \cdots \otimes V_n \longrightarrow W$$

such that

$$\tilde{f}(x_1 \otimes \cdots \otimes x_n) = f(x_1, \dots, x_n).$$

Sometimes, the tensor product $V_1 \otimes \cdots \otimes V_n$ is also denoted as $V_1 \otimes_K \cdots \otimes_K V_n$ in order to emphasize the commutative unitary ring K over which we consider modules.

4.1.3 Proposition. Let n be an integer such that $n \geq 2$, and $V_1, \dots, V_n$ be K -modules. Let $i \in \{1, \dots, n-1\}$ and $j \in \{0, \dots, n-i\}$. Then there is an isomorphism of K -modules

$$V_1 \otimes \cdots \otimes V_n \longrightarrow V_1 \otimes \cdots \otimes V_{i-1} \otimes (V_i \otimes \cdots \otimes V_{i+j}) \otimes V_{i+j+1} \otimes \cdots \otimes V_n,$$

that sends $x_1 \otimes \cdots \otimes x_n$ to

$$x_1 \otimes \cdots \otimes x_{i-1} \otimes (x_i \otimes \cdots \otimes x_{i+j}) \otimes x_{i+j+1} \otimes \cdots \otimes x_n.$$

Proof. First, the mapping

$$V_1 \times \cdots \times V_n \longrightarrow V_1 \otimes \cdots \otimes V_{i-1} \otimes (V_i \otimes \cdots \otimes V_{i+j}) \otimes V_{i+j+1} \otimes \cdots \otimes V_n$$

that sends $(x_1, \dots, x_n)$ to

$$x_1 \otimes \cdots \otimes x_{i-1} \otimes (x_i \otimes \cdots \otimes x_{i+j}) \otimes x_{i+j+1} \otimes \cdots \otimes x_n$$

is n -linear, hence it induces a homomorphism of K -modules

$$V_1 \otimes \cdots \otimes V_n \longrightarrow V_1 \otimes \cdots \otimes V_{i-1} \otimes (V_i \otimes \cdots \otimes V_{i+j}) \otimes V_{i+j+1} \otimes \cdots \otimes V_n,$$

that sends $x_1 \otimes \cdots \otimes x_n$ to

$$x_1 \otimes \cdots \otimes x_{i-1} \otimes (x_i \otimes \cdots \otimes x_{i+j}) \otimes x_{i+j+1} \otimes \cdots \otimes x_n.$$

To show that this homomorphism is an isomorphism, it suffices to check that both sides represent the same functor from $\mathbf{Mod}_K$ to $\mathbf{Set}$ that sends a K -module to the set of all n -linear mappings from $V_1 \times \cdots \times V_n$ to W . By Theorem 4.1.1, one has the following bijections that are functorial on W :

$$\begin{aligned} & \mathrm{Hom}_K(V_1 \otimes \cdots \otimes V_n, W) \\ & \quad \downarrow \\ & \mathrm{Hom}_K^{(n)}(V_1, \dots, V_n, W) \\ & \quad \downarrow \\ & \mathrm{Hom}_K^{(n-j)}(V_1, \dots, V_{i-1}, V_i \otimes \cdots \otimes V_{i+j}, V_{i+j+1}, \dots, V_n, W) \\ & \quad \downarrow \\ & \mathrm{Hom}_K(V_1 \otimes \cdots \otimes V_{i-1} \otimes (V_i \otimes \cdots \otimes V_{i+j}) \otimes V_{i+j+1} \otimes \cdots \otimes V_n, W). \end{aligned}$$

Therefore, we get the desired result. $\square$

4.2 Tensor algebra

4.2.1 Definition. Let K be a commutative unitary ring and E be a K -module. For any $n \in \mathbb{N}_{\geq 1}$, we denote by $E^{\otimes n}$ the K -module

$$\underbrace{E \otimes \cdots \otimes E}_{n \text{ copies}}.$$

By convention $E^{\otimes 0}$ denotes K . Let $T(E)$ be the direct sum

$$\bigoplus_{n \in \mathbb{N}} E^{\otimes n}.$$

For any $n \in \mathbb{N}$, one has a homomorphism of K -modules from $E^{\otimes n}$ to $T(E)$ that sends $\alpha_n \in E^{\otimes n}$ to the element of $T(E)$ whose n -th coordinate is α_n and other coordinates are zero. By this mapping we could consider $E^{\otimes n}$ as a sub- K -module. Thus, by abuse of notation, an element

$$(\alpha_n)_{n \in \mathbb{N}} \in T(E)$$

is written as

$$\sum_{n \in \mathbb{N}} \alpha_n.$$

We equip $T(E)$ with the bilinear mapping

$$\otimes : T(E) \times T(E) \longrightarrow T(E)$$

that sends

$$(x_1 \otimes \cdots \otimes x_n, x_{n+1} \otimes \cdots \otimes x_{n+m}) \in E^{\otimes n} \times E^{\otimes m}$$

to

$$x_1 \otimes \cdots \otimes x_n \otimes x_{n+1} \otimes \cdots \otimes x_{n+m} \in E^{\otimes(n+m)},$$

so that $T(E)$ equipped with this composition law forms a K -algebra. We call it the *tensor algebra* of the K -module E . Sometimes we use the expression $T_K(E)$ to denote the tensor algebra of the K -module E in order to emphasize the commutative unitary ring K over which we consider modules and algebras.

4.2.2 Theorem. *Let K be a commutative unitary ring and E be a K -module. The functor from $\mathbf{Alg}_K$ to $\mathbf{Set}$, that sends any K -algebra A to the set of K -module homomorphisms from E to A , is representable by the tensor algebra $T(E)$.*

Proof. It suffices to check that, for any K -algebra A , any K -module homomorphism $f : E \rightarrow A$ extends in a unique way to a homomorphism λ_f of K -algebras from $T(E)$ to A .

Assume that such a homomorphism λ_f exists. Then, for any $x_1 \otimes \cdots \otimes x_n \in E^{\otimes n}$, one has

$$\lambda_f(x_1 \otimes \cdots \otimes x_n) = f(x_1) \cdots f(x_n),$$

which proves the uniqueness of λ_f since $T(E)$ is generated by elements of the form $x_1 \otimes \cdots \otimes x_n$.

It remains to prove the existence of λ_f . First, the mapping

$$E^n \longrightarrow A, \quad (x_1, \dots, x_n) \longmapsto f(x_1) \cdots f(x_n)$$

is n -linear, hence it induces a K -linear mapping $E^{\otimes n} \rightarrow A$. These K -linear mappings induce then a K -linear mapping from $\lambda_f : T(E) \rightarrow A$ that sends $x_1 \otimes \cdots \otimes x_n$ to $f(x_1) \cdots f(x_n)$. Moreover, one has

$$\begin{aligned} \lambda_f(x_1 \otimes \cdots \otimes x_n \otimes x_{n+1} \otimes \cdots \otimes x_{n+m}) &= \prod_{i=1}^{n+m} f(x_i) = \left(\prod_{i=1}^n f(x_i) \right) \left(\prod_{j=1}^m f(x_{n+j}) \right) \\ &= \lambda_f(x_1 \otimes \cdots \otimes x_n) \lambda_f(x_{n+1} \otimes \cdots \otimes x_{n+m}). \end{aligned}$$

Therefore, λ_f is a homomorphism of K -algebras. $\square$

4.2.3 Corollary. *Let V and W be K -modules and $f : V \rightarrow W$ be a homomorphism of K -modules. Then f induces a homomorphism of K -algebras*

$$T(f) : T(V) \longrightarrow T(W)$$

such that

$$T(f)(x_1 \otimes \cdots \otimes x_n) = f(x_1) \otimes \cdots \otimes f(x_n).$$

4.2.4 Remark. We call *graded K -algebra* any K -algebra A that can be written as a K -module in the form

$$\bigoplus_{n \in \mathbb{N}} A_n,$$

such that, for any $(n, m) \in \mathbb{N} \times \mathbb{N}$,

$$\forall (a_n, a_m) \in A_n \times A_m, \quad a_n a_m \in A_{n+m}.$$

For example, if V is a K -module, then the tensor algebra $T(V)$ is a graded K -algebra.

Given a graded K -algebra

$$A = \bigoplus_{n \in \mathbb{N}} A_n,$$

for any $n \in \mathbb{N}$, an element of A_n is said to be *homogeneous of degree n* (thus the zero element of A is homogeneous of degree n for any $n \in \mathbb{N}$). Note that any element $a \in A$ can be written in a unique way as

$$a = \sum_{n \in \mathbb{N}} a_n$$

with $a_n \in A_n$ (here we consider A_n as a sub- K -module of A). The element a_n is called the *homogeneous component* of degree n of a .

Let $A = \bigoplus_{n \in \mathbb{N}} A_n$ be a graded K -algebra and I be an ideal of A . We say that I is a *homogeneous ideal* if, for any element $a \in I$, all homogeneous components

of a belong to I (this is equivalent to the condition that the ideal is generated by homogeneous elements). Note that a homogeneous ideal I of A can be written as

$$I = \bigoplus_{n \in \mathbb{N}} I_n,$$

where I_n is the sub- K -module of I which consists of homogeneous elements of degree n of I . Note that the quotient ring A/I admits a graded- K -algebra structure as

$$A/I = \bigoplus_{n \in \mathbb{N}} A_n/I_n.$$

4.3 Symmetric algebra

4.3.1 Definition. For any set X , we denote by S_X the set of all bijections from X to X . Recall that, the set S_X equipped with the composition of mappings, forms a group, called the *symmetric group* of X . The elements of S_X are called *permutations*. If $x_1, \dots, x_n$ are distinct elements X , we denote by

$$(x_1 \ x_2 \ \cdots \ x_n)$$

the mapping that sends x_i to x_{i+1} for $i \in \{1, \dots, n-1\}$, x_n to x_1 and any $y \in X \setminus \{x_1, \dots, x_n\}$ to y itself. This element is called an n -cycle. A 2-cycle is also called a *transposition*.

During the first semester, we have proved that, when X is finite, then for any permutation $\sigma \in S_X$, there exists pairwise disjoint subsets $X_1, \dots, X_d$ of cardinality $n_1, \dots, n_d$ of X , together with n_i -cycles τ_i of elements of X_i , such that

$$\sigma = \tau_1 \circ \cdots \circ \tau_d.$$

Moreover, any permutation can be written as a composite of transpositions.

4.3.2 Remark. Let n be an integer such that $n \geq 2$. The symmetric group $S_{\{1, \dots, n\}}$ is also denoted as S_n . A transposition of the form $(i \ i+1)$ is called an adjacent transposition.

4.3.3 Theorem. Let V be a K -module. The functor from $\mathbf{CAlg}_K$ to $\mathbf{Set}$, that sends a commutative K -algebra A to the set of K -module homomorphisms from V to A , is representable.

Proof. Let I be the ideal of $T(V)$ generated by elements of the form $x \otimes y - y \otimes x$, where $(x, y) \in V \times V$. Since any permutation in the symmetric group S_n can be written as a composition of adjacent transpositions, for any $(x_1, \dots, x_n) \in V^n$ and any $\sigma \in S_n$, one has

$$x_1 \otimes \cdots \otimes x_n - x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)} \in I.$$

Hence the equivalence classes of $x_1 \otimes \cdots \otimes x_n$ and $x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}$ in $S(V)$ are equal. In particular, the quotient ring $S(V) := T(V)/I$ is commutative.

Let $f : V \rightarrow A$ be a homomorphism of K -modules. It induces in a unique way a homomorphism of K -algebras from $\lambda_f : T(V) \rightarrow A$. Note that, for any $(x, y) \in V \times V$, one has

$$\lambda_f(x \otimes y - y \otimes x) = f(x)f(y) - f(y)f(x) = 0$$

since A is commutative. Hence λ_f vanishes on I and hence induces in a unique way a homomorphism of K -algebras from $S(V)$ to A that extends f . Therefore, $S(V)$ represents the functor defined in the Theorem. $\square$

4.3.4 Definition. Let V be a K -module. The commutative K -algebra $S(V)$ constructed in the proof of Theorem 4.3.3 is called the *symmetric algebra* of V .

4.3.5 Remark. In the proof of Theorem 4.3.3, the ideal I is generated by homogeneous elements. Therefore $S(V)$ is a graded K -algebra. We denote by $S^n(V)$ its homogeneous component of degree n . The composition law of $S(V)$ is written multiplicatively. For any $n \in \mathbb{N}$, one has a natural surjective mapping $V^{\otimes n} \rightarrow S^n(V)$ that sends $x_1 \otimes \cdots \otimes x_n$ to $x_1 \cdots x_n$ for any $(x_1, \dots, x_n) \in V^n$. Moreover, since I_n is generated as a K -module by elements of the form

$$x_1 \otimes \cdots \otimes x_n - x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)},$$

we obtain that $S^n(V)$ is actually the quotient of $V^{\otimes n}$ by the linear action of the symmetric group S_n as follows:

$$\forall \sigma \in S_n, \quad \sigma(x_1 \otimes \cdots \otimes x_n) := x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}.$$

4.3.6 Example. Let K be a commutative unitary ring and V be a free K -module of finite type, and $(T_i)_{i=1}^d$ be a basis of V over K . For any commutative K -algebra A , the set of homomorphism of K -modules from V to A is in functorial bijection with $A^{\{T_1, \dots, T_d\}}$. We denote by $K[T_1, \dots, T_d]$ the graded K -algebra $S(V)$, called the polynomial algebra of variables $T_1, \dots, T_d$.

4.4 Exterior algebra

In this section, K denotes a commutative unitary ring.

4.4.1 Definition. Let V be a K -module. We denote by $\Lambda(V)$ the quotient K -algebra of $T(V)$ modulo the ideal generated by homogeneous elements (of degree 2) of the form $x \otimes x$, where $x \in V$. The homogeneous component of degree n of $\Lambda(V)$ is denoted as $\Lambda^n(V)$. The composition law of $\Lambda(V)$ is denoted as $\wedge$. If n is a positive integer and $(x_1, \dots, x_n) \in V^n$, then the image of $x_1 \otimes \cdots \otimes x_n \in V^{\otimes n}$ by the quotient mapping $V^{\otimes n} \rightarrow \Lambda^n(V)$ is denoted as $x_1 \wedge \cdots \wedge x_n$.

4.4.2 Proposition. *Let n be a natural number. There is a homomorphism of groups sgn from S_n to the multiplicative group $\{1, -1\}$ that sends any transposition to -1 .*

Proof. Consider the mapping

$$\text{sgn} : S_n \longrightarrow \mathbb{Q}^\times$$

defined as

$$\text{sgn}(\sigma) := \prod_{\substack{(i,j) \in \mathbb{N}^2 \\ 1 \leq i < j \leq n}} \frac{\sigma(i) - \sigma(j)}{i - j}.$$

Let Θ be the set of all $A \in \mathcal{P}(\{1, \dots, n\})$ of cardinality 2. We can rewrite $\text{sgn}(\sigma)$ as

$$\text{sgn}(\sigma) = \prod_{\{i,j\} \in \Theta} \frac{\sigma(i) - \sigma(j)}{i - j}.$$

Thus, for any elements σ and τ in S_n , one has

$$\text{sgn}(\tau \circ \sigma) = \prod_{\{i,j\} \in \Theta} \frac{\tau(\sigma(i)) - \tau(\sigma(j))}{\sigma(i) - \sigma(j)} \cdot \frac{\sigma(i) - \sigma(j)}{i - j} = \text{sgn}(\sigma) \text{sgn}(\tau).$$

Therefore sgn is a homomorphism of groups.

It remains to check that, if τ is a transposition, then $\text{sgn}(\tau) = -1$. Suppose that τ is of the form $(a \ b)$. Then

$$\text{sgn}(\tau) = \frac{b-a}{a-b} \cdot \left(\prod_{i \in \{1, \dots, n\} \setminus \{a, b\}} \frac{i-a}{i-b} \cdot \frac{i-b}{i-a} \right) \cdot \left(\prod_{\{i,j\} \subseteq \{1, \dots, n\} \setminus \{a, b\}} \frac{i-j}{i-j} \right) = -1.$$

□

4.4.3 Proposition. *Let V be a K -module and n be a positive integer. Let $(x_1, \dots, x_n) \in V^n$, one has*

$$x_{\sigma(1)} \wedge \dots \wedge x_{\sigma(n)} = \text{sgn}(\sigma) x_1 \wedge \dots \wedge x_n.$$

Proof. The case where $n = 1$ is trivial. We first treat the case where $n = 2$. Let x and y be elements of V . One has

$$0 = (x + y) \wedge (x + y) = x \wedge x + x \wedge y + y \wedge x + y \wedge y = x \wedge y + y \wedge x,$$

which leads to $y \wedge x = -x \wedge y$.

For the general case, since any permutation can be written as the composition of several adjacent transpositions, we may assume that σ is of the form $(i \ i+1)$, where $i \in \{1, \dots, n-1\}$. By the particular case proved above, one has

$$x_{i+1} \wedge x_i = -x_i \wedge x_{i+1}.$$

Therefore,

$$x_1 \wedge \cdots \wedge x_{i-1} \wedge x_{i+1} \wedge x_i \wedge x_{i+2} \wedge \cdots \wedge x_n = -x_1 \wedge \cdots \wedge x_n.$$

□

4.4.4 Remark. By Proposition 4.4.3, we obtain that the kernel of the K -module homomorphism $V^{\otimes n} \rightarrow \Lambda^n(V)$ is generated by elements of the form $x_1 \otimes \cdots \otimes x_n$, where at least two elements among $x_1, \dots, x_n$ are identical.

4.4.5 Definition. Let V and M be K -modules, n be a positive integer, and $f : V^n \rightarrow M$ be an n -linear mapping. If, for any $(x_1, \dots, x_n) \in V^n$ such that $x_i = x_{i+1}$ holds for a certain $i \in \{1, \dots, n-1\}$, one has $f(x_1, \dots, x_n) = 0$, then we say that f is an *alternating* n -linear mapping.

4.4.6 Proposition. Let V and M be K -modules, n be a positive integer, $f : V^n \rightarrow M$ be an alternating n -linear mapping. There exists a unique K -linear mapping $\widehat{f} : \Lambda^n V \rightarrow M$ such that

$$\forall (x_1, \dots, x_n) \in V^n, \quad \widehat{f}(x_1 \wedge \cdots \wedge x_n) = f(x_1, \dots, x_n). \quad (4.2)$$

In particular, if $(x_1, \dots, x_n) \in V^n$ and if there exists an alternating n -linear mapping $g : V^n \rightarrow M$ such that $g(x_1, \dots, x_n) \neq 0$, then $x_1 \wedge \cdots \wedge x_n \neq 0$.

Proof. The uniqueness follows from (4.2). In the following, we prove the existence. By Theorem 4.1.1, there exists a unique K -linear mapping $\widetilde{f} : V^{\otimes n} \rightarrow M$ such that

$$\forall (x_1, \dots, x_n) \in V^{\otimes n}, \quad \widetilde{f}(x_1 \otimes \cdots \otimes x_n) = f(x_1, \dots, x_n).$$

Moreover, since f is alternating, we obtain that, if $x_1, \dots, x_n$ are elements of V such that at least two of them are identical, then

$$\widetilde{f}(x_1 \otimes \cdots \otimes x_n) = f(x_1, \dots, x_n) = 0.$$

Therefore, $\widetilde{f}$ induces by quotient a K -linear mapping $\widehat{f} : \Lambda^n(V) \rightarrow M$ such that

$$\widehat{f}(x_1 \wedge \cdots \wedge x_n) = \widetilde{f}(x_1 \otimes \cdots \otimes x_n) = f(x_1, \dots, x_n).$$

□

4.4.7 Example. Let V be a K -module and n be a positive integer. Let $f_1, \dots, f_n$ be K -linear forms on V (that is K -linear mapping from V to K). Consider the n -linear mapping

$$V^n \longrightarrow K, \quad (x_1, \dots, x_n) \longmapsto \sum_{\sigma \in S_n} \text{sgn}(\sigma) f_{\sigma(1)}(x_1) \cdots f_{\sigma(n)}(x_n). \quad (4.3)$$

If i and j are indices in $\{1, \dots, n\}$ and $x \in V$ such that $i < j$ and $x_i = x_j = x$, then

$$\begin{aligned}
& \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) f_{\sigma(1)}(x_1) \cdots f_{\sigma(n)}(x_n) \\
&= \sum_{\substack{(k, \ell) \in \{1, \dots, n\}^2 \\ k < \ell}} \left(\sum_{\substack{\sigma \in S_n \\ \sigma(k)=i, \sigma(\ell)=j}} \operatorname{sgn}(\sigma) f_{\sigma(1)}(x_1) \cdots f_{\sigma(n)}(x_n) \right. \\
&\quad \left. + \sum_{\substack{\sigma \in S_n \\ \sigma(k)=j, \sigma(\ell)=i}} \operatorname{sgn}(\sigma) f_{\sigma(1)}(x_1) \cdots f_{\sigma(n)}(x_n) \right) \\
&= 0.
\end{aligned}$$

Therefore, (4.3) defines an alternating linear form on $V^{\otimes n}$, and hence induces by passing to quotient a K -linear form on $\Lambda^n(V)$, which we denote as $f_1 \wedge \cdots \wedge f_n$.

Consider the particular case where V is a free K -module of finite type and $(e_i)_{i=1}^n$ is a basis of V . Let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$, namely

$$e_i^\vee(a_1 e_1 + \cdots + a_n e_n) := a_i.$$

Then the following equality holds:

$$(e_1^\vee \wedge \cdots \wedge e_n^\vee)(e_1 \wedge \cdots \wedge e_n) = \sum_{\sigma \in S_n} e_{\sigma(1)}^\vee(e_1) \cdots e_{\sigma(n)}^\vee(e_n) = 1.$$

4.5 Linear independence criterion

4.5.1 Lemma. *Let V be a K -module, n be a positive integer. Let $(e_i)_{i=1}^n$ be a family of elements of V . For any $i \in \{1, \dots, n\}$, let $(a_{i,1}, \dots, a_{i,n})_{j=1}^n \in K^n$ and*

$$x_i = a_{i,1} e_1 + \cdots + a_{i,n} e_n.$$

Then the following equality holds

$$x_1 \wedge \cdots \wedge x_n = \left(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)} \right) e_1 \wedge \cdots \wedge e_n. \quad (4.4)$$

Proof. By the multilinearity of wedge product, we obtain

$$\begin{aligned}
x_1 \wedge \cdots \wedge x_n &= \sum_{(j_1, \dots, j_n) \in \{1, \dots, n\}^n} a_{1,j_1} \cdots a_{n,j_n} e_{j_1} \wedge \cdots \wedge e_{j_n} \\
&= \sum_{\sigma \in S_n} a_{1,\sigma(1)} \cdots a_{n,\sigma(n)} e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)},
\end{aligned}$$

where the second equality comes from the fact that, if there are distinct indices k and ℓ such that $j_k = j_\ell$, then

$$e_{j_1} \wedge \cdots \wedge e_{j_n} = 0.$$

By Corollary, for any $\sigma \in S_n$, one has

$$e_{\sigma(1)} \wedge \cdots \wedge e_{\sigma(n)} = \text{sgn}(\sigma) e_1 \wedge \cdots \wedge e_n.$$

Hence the equality (4.4) holds. □

4.5.2 Remark. Using the notation of Lemma 4.5.1 we can also calculate

$$\begin{aligned} x_1 \wedge \cdots \wedge x_d &= \sum_{(j_1, \dots, j_d) \in \{1, \dots, n\}^d} a_{1,j_1} \cdots a_{d,j_d} e_{j_1} \wedge \cdots \wedge e_{j_d} \\ &= \sum_{\{j_1, \dots, j_d\} \subseteq \{1, \dots, n\}} a_{1,j_1} \cdots a_{d,j_d} e_{j_1} \wedge \cdots \wedge e_{j_d}, \end{aligned}$$

where the second sum is taken over d -tuples in which $j_1, \dots, j_d$ are pairwise-different, i.e. there are no repetitions.

4.5.3 Theorem. Let V be a K -module, n be a positive integer, and $(e_i)_{i=1}^n$ be a family of elements of V .

- (1) Assume that, for any $\lambda \in K \setminus \{0\}$, one has

$$\lambda e_1 \wedge \cdots \wedge e_n \neq 0.$$

Then $(e_i)_{i=1}^n$ is K -linearly independent.

- (2) Assume that $(e_i)_{i=1}^n$ is a system of generators of V . For any $d \in \mathbb{N}$, the family

$$(e_{i_1} \wedge \cdots \wedge e_{i_d})_{1 \leq i_1 < \dots < i_d \leq n}$$

generates $\Lambda^d(V)$. In particular, $\Lambda^d(V) = \{0\}$ when $d > n$.

- (3) If V is a free K -module and $(e_i)_{i=1}^n$ is a basis of V , then, for any $d \in \mathbb{N}$, $(e_{i_1} \wedge \cdots \wedge e_{i_d})_{1 \leq i_1 < \dots < i_d \leq n}$ forms a basis of $\Lambda^d(V)$.

Proof. (1) Assume that $(e_i)_{i=1}^n$ is K -linearly independent. Without loss of generality, we may assume that there exists $\lambda \in K \setminus \{0\}$ and $(a_2, \dots, a_n) \in K^{n-1}$ such that λe_1 is a K -linear combination of $e_2, \dots, e_n$. Then one has

$$\lambda e_1 \wedge \cdots \wedge e_n = (a_2 e_2 + \cdots + a_n e_n) \wedge e_2 \wedge \cdots \wedge e_n = 0.$$

(2) Let $(x_1, \dots, x_d) \in V^d$. For any $i \in \{1, \dots, n\}$, one can express x_i as a linear combination

$$x_i = \sum_{j \in \{1, \dots, n\}} a_{i,j} e_j.$$

Therefore, $x_1 \wedge \dots \wedge x_d$ can be expressed as a K -linear combination of elements of the form $e_{j_1} \wedge \dots \wedge e_{j_d}$, where $j_1, \dots, j_d$ are distinct elements of $\{1, \dots, n\}$. Moreover, by Remark 4.5.2, we deduce that $x_1 \wedge \dots \wedge x_d$ can be written as a linear combination of elements of the form $e_{i_1} \wedge \dots \wedge e_{i_d}$ with $1 \leq i_1 < \dots < i_d \leq n$.

(3) We have seen in (2) that $\Lambda^d(V)$ is generated by

$$(e_{i_1} \wedge \dots \wedge e_{i_d})_{1 \leq i_1 < \dots < i_d \leq n}.$$

Let $(\lambda_{i_1, \dots, i_d})_{1 \leq i_1 < \dots < i_d \leq n}$ be a family of elements of K such that

$$\alpha := \sum_{1 \leq i_1 < \dots < i_d \leq n} \lambda_{i_1, \dots, i_d} e_{i_1} \wedge \dots \wedge e_{i_d} = 0.$$

Let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. For any $(i_1, \dots, i_d) \in \{1, \dots, n\}^d$ such that $i_1 < \dots < i_d$, one has

$$(e_{i_1}^\vee \wedge \dots \wedge e_{i_d}^\vee)(\alpha) = \lambda_{i_1, \dots, i_d} = 0.$$

Hence the family $(e_{i_1} \wedge \dots \wedge e_{i_d})_{1 \leq i_1 < \dots < i_d \leq n}$ is K -linearly independent. $\square$

4.5.4 Corollary. *Let V be a free K -module of finite rank n . Let $d \in \{1, \dots, n\}$ and*

$$F : \Lambda^{n-d}(V) \rightarrow \Lambda^n(V)$$

be a K -linear mapping. Then there exists a unique element $\alpha \in \Lambda^d(V)$ such that $F(\beta) = \alpha \wedge \beta$ for any $\beta \in \Lambda^{n-d}(V)$.

Proof. Let $(e_i)_{i=1}^n$ be a basis of V and $\eta = e_1 \wedge \dots \wedge e_n$. For any $m \in \{1, \dots, n\}$, let S_m be the set of subsets of m elements of $\{1, \dots, n\}$. For any $I = \{i_1, \dots, i_m\} \in S_m$ with $i_1 < \dots < i_m$, let $\beta_I := e_{i_1} \wedge \dots \wedge e_{i_m}$. By Theorem 4.5.3,

$$(\beta_{\{1, \dots, n\} \setminus J})_{J \in S_{n-m}} = (\beta_I)_{I \in S_m}$$

forms a basis of $\Lambda^m(V)$. For any $\beta \in \Lambda^{n-d}(V)$, let $f(\beta)$ be the element of K such that

$$F(\beta) = f(\beta)\eta.$$

We let

$$\alpha = \sum_{J \in S_{n-d}} \text{sgn}(J) f(\beta_J) \beta_{\{1, \dots, n\} \setminus J},$$

where $\text{sgn}(J) \in \{1, -1\}$ is such that

$$\beta_{\{1, \dots, n\} \setminus J} \wedge \beta_J = \text{sgn}(J)\eta.$$

Note that, for any $I \in S_{n-d}$, one has

$$\begin{aligned}\alpha \wedge \beta_I &= \sum_{J \in S_{n-d}} \operatorname{sgn}(J) f(\beta_J) \beta_{\{1, \dots, n\} \setminus J} \wedge \beta_I \\ &= \operatorname{sgn}(I) f(\beta_I) \beta_{\{1, \dots, n\} \setminus I} \wedge \beta_I = f(\beta_I) \eta = F(\beta_I).\end{aligned}$$

By the linearity, we obtain that $\alpha \wedge \beta = F(\beta)$ for any $\beta \in \Lambda^{n-d}(V)$.

If α and α' are elements of $\Lambda^d(V)$ such that $\alpha \wedge \beta = \alpha' \wedge \beta = F(\beta)$ holds for any $\beta \in \Lambda^{n-d}E$, then one has

$$(\alpha - \alpha') \wedge \beta_I = 0$$

for any $I \in S_{n-d}$. Hence $\alpha - \alpha' = 0$. The uniqueness is thus proved. $\square$

4.6 Determinant

In this section, we fix a commutative unitary ring.

4.6.1 Proposition. *Let V be a free K -module that is generated by one element. For any K -endomorphism $\varphi : V \rightarrow V$, there exists a unique $\lambda_\varphi \in K$ such that*

$$\forall x \in V, \quad \varphi(x) = \lambda_\varphi x.$$

Proof. Let x_0 be a generator of V . We write $\varphi(x_0)$ as $\lambda_\varphi x_0$, where $\lambda_\varphi \in K$. For any $x \in V$, there exists $b \in K$ such that $x = bx_0$. Hence

$$\varphi(x) = b\varphi(x_0) = b\lambda_\varphi x_0 = \lambda_\varphi x.$$

$\square$

4.6.2 Definition. Let V be a free K -module of rank n . Let $\varphi : V \rightarrow V$ be a K -linear mapping (we often called it a K -linear endomorphism), we denote by $\det(\varphi)$ the unique element of K such that, for any $(x_1, \dots, x_n) \in V^n$

$$\varphi(x_1) \wedge \dots \wedge \varphi(x_n) = \det(\varphi) x_1 \wedge \dots \wedge x_n.$$

The element $\det(\varphi)$ is called the *determinant* of φ .

4.6.3 Proposition. *Let V be a free K -module of rank n , and $\varphi : V \rightarrow V$ and $\psi : V \rightarrow V$ be K -linear endomorphisms. Then*

$$\det(\psi \circ \varphi) = \det(\psi) \det(\varphi).$$

Moreover, $\det(\operatorname{Id}_V) = 1$.

Proof. For any $(x_1, \dots, x_n) \in V^n$, one has

$$\psi(\varphi(x_1)) \wedge \dots \wedge \psi(\varphi(x_n)) = \det(\psi)\varphi(x_1) \wedge \dots \wedge \varphi(x_n) = \det(\psi)\det(\varphi)x_1 \wedge \dots \wedge x_n.$$

Therefore, one has $\det(\psi \circ \varphi) = \det(\psi)\det(\varphi)$.

Moreover, one has

$$\text{Id}_V(x_1) \wedge \dots \wedge \text{Id}_V(x_n) = x_1 \wedge \dots \wedge x_n.$$

Hence $\det(\text{Id}_V) = 1$. □

4.6.4 Theorem. *Let V be a free K -module of rank n and $\varphi : V \rightarrow V$ be a K -linear endomorphism. Then φ is a bijection (in this case we say that φ is a K -linear automorphism) if and only if $\det(\varphi)$ is invertible.*

Proof. Suppose that φ is invertible. By Proposition 4.6.3, we obtain that

$$\det(\varphi)\det(\varphi^{-1}) = \det(\text{Id}_V) = 1.$$

Hence $\det(\varphi)$ is invertible.

Conversely, suppose that $\det(\varphi)$ is invertible. Let $(e_i)_{i=1}^n$ be a basis of V over K . For any $i \in \{1, \dots, n\}$, let $x_i = \varphi(e_i)$. By (1) of Theorem 4.5.3, we obtain that $(x_i)_{i=1}^n$ is K -linear independent. It suffices to check that it generates V . Let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. Let

$$g = \det(\varphi)^{-1}(e_1^\vee \wedge \dots \wedge e_n^\vee).$$

Then $g(x_1, \dots, x_n) = 1$. The mapping $f : V^{n+1} \rightarrow V$ defined as

$$(y_0, \dots, y_n) \mapsto \sum_{i=0}^n (-1)^i g(y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_n) y_i$$

is an alternating K -linear mapping. In fact, if $(y_0, \dots, y_n)$ are elements of V^{n+1} and if $i \in \{1, \dots, n-1\}$ such that $y_i = y_{i+1}$, then

$$\begin{aligned} f(y_0, \dots, y_n) &= (-1)^i g(y_0, \dots, y_{i-1}, y_{i+1}, y_{i+2}, \dots, y_n) y_i \\ &\quad + (-1)^{i+1} g(y_0, \dots, y_{i-1}, y_i, y_{i+2}, \dots, y_n) y_{i+1} = 0. \end{aligned}$$

Let x be an element of V . One has

$$x \wedge x_1 \wedge \dots \wedge x_n = 0.$$

Hence $f(x, x_1, \dots, x_n) = 0$, which leads to

$$x + \sum_{i=1}^n g(x, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) x_i = 0.$$

Therefore, x is a K -linear combination of $x_1, \dots, x_n$. □

4.6.5 Proposition. *Let n be a positive integer and*

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix}.$$

Then

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)}.$$

Proof. Let $(e_i)_{i=1}^n$ be the canonical basis of K^n . For any $i \in \{1, \dots, n\}$, one has

$$A(e_i) = a_{i,1}e_1 + \cdots + a_{i,n}e_n.$$

By Lemma 4.5.1, one has

$$A(e_1) \wedge \cdots \wedge A(e_n) = \left(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{i,\sigma(i)} \right) e_1 \wedge \cdots \wedge e_n.$$

□

4.6.6 Definition. Let V and W be K -modules, and $\varphi : V \rightarrow W$ be a K -linear mapping. We denote by

$$\varphi^\vee : W^\vee \longrightarrow V^\vee$$

the K -linear mapping that sends $\alpha \in W^\vee$ to

$$\varphi^\vee(\alpha) := \alpha \circ \varphi.$$

4.6.7 Proposition. *Let V be a free K -module of finite rank and $\varphi : V \rightarrow V$ be an K -linear endomorphism of V . One has $\det(\varphi^\vee) = \det(\varphi)$.*

Proof. By definition, for any $\alpha \in V^\vee$, one has $\varphi^\vee(\alpha) = \alpha \circ \varphi$. Let $(e_i)_{i=1}^n$ be a basis of V and let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. One has

$$(e_1^\vee \circ \varphi) \wedge \cdots \wedge (e_n^\vee \circ \varphi)(e_1 \wedge \cdots \wedge e_n) = (e_1^\vee \wedge \cdots \wedge e_n^\vee)(\varphi(e_1) \wedge \cdots \wedge \varphi(e_n)) = \det(\varphi).$$

Therefore

$$(e_1^\vee \circ \varphi) \wedge \cdots \wedge (e_n^\vee \circ \varphi) = \det(\varphi) e_1^\vee \wedge \cdots \wedge e_n^\vee,$$

which leads to $\det(\varphi^\vee) = \det(\varphi)$. □

4.6.8 Corollary. *Let n be a positive integer and $A \in M_n(K)$ be an n by n matrix. One has*

$$\det(A^\tau) = \det(A).$$

4.7 Adjugate

In this section, we fix a free K -module E of finite rank n .

4.7.1 Theorem. *For any $\varphi \in \text{End}_K(E)$, there exists $\varphi^a \in \text{End}_K(E)$ such that, for any $(x_1, \dots, x_n) \in E^n$, one has*

$$x_1 \wedge \varphi(x_2) \wedge \cdots \wedge \varphi(x_n) = \varphi^a(x_1) \wedge x_2 \wedge \cdots \wedge x_n. \quad (4.5)$$

Moreover, one has

$$\varphi^a \circ \varphi = \det(\varphi) \text{Id}_E.$$

Proof. For any $x_1 \in E$, the mapping $\Lambda^{n-1}(E) \rightarrow \Lambda^n(E)$ that sends $x_2 \wedge \cdots \wedge x_n$ to $x_1 \wedge \varphi(x_2) \wedge \cdots \wedge \varphi(x_n)$ is K -linear. Therefore, there exists a unique element of E , that we denote as $\varphi^a(x_1)$, such that

$$x_1 \wedge \varphi(x_2) \wedge \cdots \wedge \varphi(x_n) = \varphi^a(x_1) \wedge x_2 \wedge \cdots \wedge x_n.$$

Moreover, the uniqueness also shows that $\varphi^a : E \rightarrow E$ is a K -linear mapping.

By (4.5), one has

$$\varphi^a(\varphi(x_1)) \wedge x_2 \wedge \cdots \wedge x_n = \varphi(x_1) \wedge \cdots \wedge \varphi(x_n) = \det(\varphi) x_1 \wedge x_2 \wedge \cdots \wedge x_n.$$

Hence we obtain $\varphi^a \circ \varphi = \det(\varphi) \text{Id}_E$. □

4.7.2 Definition. Let n be a positive integer and A be an n by n matrix. For $(i, j) \in \{1, \dots, n\}^2$, let $\widehat{A}_{i,j}$ be the $(n-1)$ by $(n-1)$ matrix obtained by removing the i -th row and the j -th column of the matrix A , and let

$$A_{i,j}^a := (-1)^{i+j} \det(\widehat{A}_{i,j}).$$

The matrix

$$\begin{pmatrix} A_{1,1}^a & A_{2,1}^a & \cdots & A_{n,1}^a \\ A_{1,2}^a & A_{2,2}^a & \cdots & A_{n,2}^a \\ \vdots & \vdots & \ddots & \vdots \\ A_{1,n}^a & A_{2,n}^a & \cdots & A_{n,n}^a \end{pmatrix}$$

is called the *adjugate matrix* of A , denoted as A^a . Note that, if we consider A as a K -endomorphism of K^n , then A^a identifies with the endomorphism constructed in Theorem 4.7.1.

4.7.3 Corollary. *Let n be a positive integer and*

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix}$$

be an element of $M_{n,n}(K)$. Then the following equality holds:

$$AA^a = \det(A)I_n.$$

In particular, for any $i \in I$, one has

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{i,j} A_{i,j}^a.$$

Exercises

In these exercise problems, R shall always denote a commutative ring with unity.

1. We build some easy properties of the tensor product. Let R be a commutative ring with unity and let A, B, C denote arbitrary R -modules. Prove:
 - a) Identity: $A \otimes_R R = A$ and $R \otimes_R A = A$.
 - b) Commutativity: $A \otimes_R B = B \otimes_R A$.
 - c) Associativity: Property 4.1.3 in lecture.
 - c) Distributivity: $A \otimes_R (B \oplus C) = A \otimes_R B \oplus A \otimes_R C$.
 - d) The above in fact generalizes as $A \otimes_R (\oplus_i B_i) = \oplus_i (A \otimes_R B_i)$ for arbitrary index $i \in I$.
 - e) Base extension: Let S be an R algebra, then

$$(A \otimes_R B) \otimes_R S = (A \otimes_R S) \otimes_S (B \otimes_R S).$$

2. Let R be a commutative ring with identity and let A, B be R -modules. Find examples to show that:
 - a) It's in general false that

$$A \otimes_R B = A \otimes_{\mathbb{Z}} B$$

when $R \neq \mathbb{Z}$.

- b) There may exist elements in $A \otimes_R B$ that cannot be written as $a \otimes b$ with $a \in A, b \in B$.
- c) $a \otimes b = a' \otimes b'$ does not imply $a = a', b = b'$.

3. a) Let S be a commutative R -algebra and A an R -module. Prove that $A \otimes_R S$ has the structure of an S -module as well.

b) Let $n \geq 1$. Prove that $\mathbb{R}^n \otimes_{\mathbb{C}} \mathbb{C} \cong \mathbb{C}^n$ as complex vector spaces.

4. In this exercise $\otimes$ just means tensor over $\mathbb{Z}$.

a) Find

$$\mathbb{Z}[x] \otimes \mathbb{C}.$$

b) Let A be a *torsion* abelian group, i.e., where every element has finite order, then

$$A \otimes \mathbb{Q} = 0.$$

c) Let $m > 0$. For any abelian group A we have

$$A \otimes (\mathbb{Z}/m\mathbb{Z}) = A/mA.$$

d) Let m, n be positive integers, then

$$\mathbb{Z}/m\mathbb{Z} \otimes \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/(\gcd(m, n)\mathbb{Z})$$

as abelian groups.

e) Let F, F' be fields of characteristic p, q respectively. If $p \neq q$, then $F \otimes F' = 0$.

5. a) Show that as $\mathbb{R}$ -modules, $\mathbb{R}[T_1, T_2] \cong (\mathbb{R}[T])^{\otimes 2}$.

b) Is it true that $L^2(\mathbb{R}^2) \cong (L^2(\mathbb{R}))^{\otimes 2}$?

6. a) Let M, N be flat R -modules and assume R is non-torsion. Prove that we have the decompositions

$$S^n(M \oplus N) = \bigoplus_{i=0}^n [S^i(M) \otimes S^{n-i}(N)],$$

$$\Lambda^n(M \oplus N) = \bigoplus_{i=0}^n [\Lambda^i(M) \otimes \Lambda^{n-i}(N)].$$

b) The above identity may fail for modules over rings with torsion. Find an example.

7. Let M be a free R -module of rank $r \in \mathbb{N}_+$. Then $T^n(M)$, $S^n(M)$, and $\Lambda^n(M)$ are free R modules as well. Try to determine their rank.

8. Let K be ring with unity.

a) Let $I_1 = \langle x^2, y \rangle$ and $I_2 = \langle x^2 + y \rangle$ be ideals of the polynomial ring $K[x, y]$. Describe $K[x, y]/I_1$ and $K[x, y]/I_2$ in terms of generators and relations.

Let $S = \oplus_{d \geq 0} S_d$ be a graded K -algebra. Recall that an ideal I is a *homogeneous* if I itself is graded by

$$I = \oplus_{d \geq 0} I_d,$$

where $I_d = I \cap S_d$. We will prove that S/I is graded if and only if I is homogeneous.

b) Let $I = \oplus_{d \geq 0} I_d$ be a graded ideal, prove that the quotient S/I is graded by

$$S/I = \oplus_{d \geq 0} (S_d/I_d).$$

This is Remark 4.2.4 in lecture.

c) Conversely, Let I be an ideal of S . Suppose the quotient $\pi : S \rightarrow S/I$ is graded by

$$S/I = \oplus_{d \geq 0} A_d$$

such that π maps S_d onto A_d . Denote the kernel of the map $\pi|_{S_d}$ by $I_d \subset I_d$. Prove that $I = \oplus_{d \geq 0} I_d$, i.e., I is a homogeneous ideal.

9. a) Let V be a vector space equipped with an inner product $\langle \cdot, \cdot \rangle$. Prove that

$$\left(\sum_i a_{1i} \otimes \cdots \otimes a_{ni}, \sum_j b_{1j} \otimes \cdots \otimes b_{nj} \right) \mapsto \sum_{i,j} \prod_{k=1}^n \langle a_{ki}, b_{kj} \rangle,$$

defines an inner product on $T^n(V)$ via linear extension.

b) Spell out the inner product on $L^2(\mathbb{R}) \otimes L^2(\mathbb{R})$. Is this inner product complete?

c) Prove that $L^2(\mathbb{R}) \otimes L^2(\mathbb{R})$ is isomorphic to a dense subset of $L^2(\mathbb{R}^2)$. Equivalently, the completion $L^2(\mathbb{R}) \hat{\otimes} L^2(\mathbb{R})$ of $L^2(\mathbb{R}) \otimes L^2(\mathbb{R})$ with respect to the inner product in b) is isomorphic to $L^2(\mathbb{R}^2)$ as Hilbert spaces.

10. a) Let $V = \mathbb{R}^2$, prove that

$$T^2(V) = S^2(V) \oplus \Lambda^2(V).$$

b) Let $V = \mathbb{R}^d$ with $d \geq 3$ instead, prove that

$$S^d(V) \oplus \Lambda^d(V) \subsetneq T^d(V),$$

and determine the missing part when $d = 3$.

c) Let M be a free module over R of rank d . For simplicity we assume R is non-torsion w.r.t. the addition. $\sigma \in S_n$ act on $T^n(V)$ by permuting order of taking tensor power, i.e., as the linear extension of the map defined by

$$\sigma \cdot (v_1 \otimes \cdots \otimes v_d) = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(d)}.$$

We define a map $s : T^n(M) \rightarrow T^n(M)$ by

$$u \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \sigma \cdot u.$$

Prove that

- c.1) The image of s is equal to $S^n(V)$.
 - c.2) For all $n \geq 2$, we have $\sum_{\sigma \in S_n} \text{sgn}(\sigma) = 0$.
 - c.3) The kernel of s contains $\Lambda^n(V)$ properly when $n \geq 3$.
- 11.** From now on, we work with square matrices over fields of characteristic zero. Prove, if it is not obvious for you from definition that: Consider an $n \times n$ matrix A and let A' be the matrices obtained by the following *elementary row transforms*
- a) A' is obtained by interchanging two rows of A , then $\det A' = -\det A$.
 - b) A' is obtained by multiplying a row with a constant λ . Then $\det A' = \lambda \det A$.
 - c) A' is obtained by adding one row of A to another. Then $\det A' = \det A$.
 - d) Let A^T denote the transpose of A . Prove that $\det A = \det A^T$, and try to build a)-c) for *elementary column transforms*.
- 12.** Let $a = (a_1, a_2)$ and $b = (b_1, b_2)$ be nonzero vectors on the real plane. Prove that the parallelogram spanned by a and b has area

$$|a_1 b_2 - a_2 b_1| = \left| \det \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix} \right|.$$

- 13.** Let $A = (a_{ij})_{n \times n}$ be a matrix. Let $1 \leq i_1 < i_2 < \cdots < i_m \leq n$ and $1 \leq j_1 < j_2 < \cdots < j_m \leq n$. Denote by $A_{[j_1, \dots, j_m]}^{[i_1, \dots, i_m]}$ the matrix whose entries are those of A that appears in the $i_1, \dots, i_m$ -th rows and $j_1, \dots, j_m$ -th columns, and by $A_{(j_1, \dots, j_m)}^{(i_1, \dots, i_m)}$ the matrix obtained from A by discarding the $i_1, \dots, i_m$ -th rows and $j_1, \dots, j_m$ -th columns. Prove the *Laplace expansion formula*

$$\det(A) = (-1)^{i_1 + \dots + i_m} \sum_{1 \leq j_1 < j_2 < \dots < j_m \leq n} (-1)^{j_1 + \dots + j_m} \det A_{[j_1, \dots, j_m]}^{[i_1, \dots, i_m]} \det A_{(j_1, \dots, j_m)}^{(i_1, \dots, i_m)}.$$

Note that when $m = 1$, this is just 4.7.3 in lecture.

14. Calculate the determinant of the matrix

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix}$$

using

- a) The permutation definition.
 b) Elementary column transforms.
 c) Expansion formulae.
15. a) Suppose $A = (a_{ij})_{n \times n}$ is an upper-triangular matrix, i.e., $a_{ij} = 0$ when $i < j$. Then

$$\det A = \prod_{i=1}^n a_{ii}.$$

Suppose an $(m+n) \times (m+n)$ matrix A can be written as the block form

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where A_{11} , A_{12} , A_{21} , and A_{22} have size $n \times n$, $n \times m$, $m \times n$ and $m \times m$ respectively. Prove that

- b) If $A_{21} = 0$, then $\det A = \det A_{11} \cdot \det A_{22}$.
 c) If A_{11} is invertible, then

$$\det A = \det A_{11} \cdot \det(A_{22} - A_{21}A_{11}^{-1}A_{12}).$$

- d) If A_{11} and A_{22} are identity matrices, then

$$\det A = \det(I_{n \times n} - A_{12}A_{21}) = \det(I_{m \times m} - A_{21}A_{12}).$$

- e) Suppose $m = n \geq 2$. Do we have

$$\det A = \det(A_{11}A_{22} - A_{12}A_{21})?$$

- d) Prove the equality in e) in the case where A_{11} and A_{21} commutes.

16. Calculate the *Vandermonde determinant* $\det V$, where

$$V = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ a_1^2 & a_2^2 & \cdots & a_n^2 \\ \vdots & \vdots & \cdots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \cdots & a_n^{n-1} \end{pmatrix}.$$

17. Calculate the determinant of the cyclic matrix

$$A = \begin{pmatrix} a_1 & a_2 & \cdots & a_{n-1} & a_n \\ a_2 & a_3 & \cdots & a_n & a_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_n & a_1 & \cdots & a_{n-2} & a_{n-1} \end{pmatrix}.$$

18. Calculate the determinant of the $n \times n$ matrix

$$A = \begin{pmatrix} x+y & xy & & & & & & \\ & 1 & x+y & xy & & & & \\ & & 1 & x+y & xy & & & \\ & & & & \cdots & \cdots & \cdots & \\ & & & & & & 1 & x+y & xy \\ & & & & & & & 1 & x+y \end{pmatrix},$$

where all blank entries are zero.

19. Let A be an invertible $n \times n$ matrix and b an $n \times 1$ vector.

Prove that the linear system $Ax = b$ has a unique solution $x = (x_1, \dots, x_n)^T$ given by *Cramer's formula*

$$x_i = \frac{\det A_i}{\det A}, \quad i = 1, \dots, n,$$

where A_i is the matrix obtained by replacing the i -th column of A by b .

20. Let A be an $n \times n$ invertible matrix. Suppose A has integer entries, prove that A^{-1} has integer entries if and only if $\det A = \pm 1$.
21. Let A be an $n \times n$ matrix. The *characteristic polynomial* of A is by definition the degree n polynomial

$$f_A(\lambda) = \det(\lambda I_n - A).$$

- Calculate the coefficient of the term λ^{n-1} and the constant term of f_A .
- If B is another matrix similar to A , then A and B has the same characteristic polynomial. In particular, A and B have the same trace.
- Let λ be a root of f_A . Prove that λ is an eigenvalue of A .
- Let A be an $m \times n$ matrix and B an $n \times m$ matrix. Prove that AB and BA has the same collection of eigenvalues. In particular, $\text{tr } AB = \text{tr } BA$.

22. Let

$$\begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 0 & 1 \end{pmatrix}.$$

- a) Find the characteristic polynomial f_A .
- b) Find the eigenvalues of A and the corresponding eigenvectors.
- c) Diagonalize A .

23. Let

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}.$$

- a) Find the characteristic polynomial f_A .
- b) Find the eigenvalues of A and the corresponding eigenvectors. Find an orthonormal basis consisting of eigenvectors.
- c) Diagonalize A by an orthogonal matrix.

Chapter 5

Haar measures

5.1 Reminder on general topology

5.1.1 Definition. Let X and Y be topological spaces and $f : X \rightarrow Y$ be a mapping. If, for any open subset U of X , the image of U by f is an open subset of Y , then we say that f is an open mapping.

5.1.2 Definition. Let X and Y be sets and $\pi : X \rightarrow Y$ be a surjective mapping. If $\mathcal{T}$ is a topology on X , we call *quotient topology* of $\mathcal{T}$ on Y the family of $V \subseteq Y$ such that $\pi^{-1}(V)$ belongs to $\mathcal{T}$. Note that, if one equip Y with the quotient topology, then the mapping π is continuous.

5.1.3 Proposition. *Let X be a topological space and $\pi : X \rightarrow Y$ be a surjective mapping of sets. We equip Y with the quotient topology and assume that π is an open mapping.*

(1) *A subset U of Y is open if and only if $\pi^{-1}(U)$ is open.*

(2) *A subset F of Y is closed if and only if $\pi^{-1}(F)$ is closed.*

Proof. (1) Since π is continuous, if U is open, then $\pi^{-1}(U)$ is open. Conversely, if $\pi^{-1}(U)$ is open, since π is a surjective open mapping, $U = \pi(\pi^{-1}(U))$ is open.

(2) If F is closed, then $Y \setminus F$ is open. Hence $X \setminus \pi^{-1}(F) = \pi^{-1}(Y \setminus F)$ is open, which implies that $\pi^{-1}(F)$ is closed.

Suppose that π is an open mapping. If $\pi^{-1}(F)$ is closed, then $X \setminus \pi^{-1}(F) = \pi^{-1}(X \setminus F)$ is open. Hence $X \setminus F$ is open, which implies that F is closed. $\square$

5.1.4 Proposition. *Let I be a set, $(X_i)_{i \in I}$ and $(Y_i)_{i \in I}$ be families topological spaces. For any $i \in I$, let $f_i : X_i \rightarrow Y_i$ be a mapping. Let*

$$X = \prod_{i \in I} X_i, \quad Y = \prod_{i \in I} Y_i$$

and $f : X \rightarrow Y$ be the mapping that sends $(x_i)_{i \in I}$ to $(f(x_i))_{i \in I}$.

- (1) If, for any $i \in I$, f_i is continuous, then f is continuous.
- (2) Suppose that, for any $i \in I$, f_i is a open mapping. Assume in addition that f_i is surjective for all but finitely many $i \in I$. then f is an open mapping.

Proof. (1) For any $i \in I$, let $\pi_i : X \rightarrow X_i$ and $p_i : Y \rightarrow Y_i$ be projections. For any $i \in I$, one has

$$p_i \circ f = f_i \circ \pi_i.$$

Since f_i and π_i are continuous, so is $f_i \circ \pi_i$. Therefore, $p_i \circ f$ is continuous for any $i \in I$, which shows that continuity of f .

(2) Let J_0 be the set of $i \in I$ such that f_i is not surjective. A topological basis of X is given by subsets of the form

$$\bigcap_{j \in J} \pi_j^{-1}(U_j),$$

where J is a finite subset of I containing J_0 , and U_j is an open subset of X_j . Note that

$$f\left(\bigcap_{j \in J} \pi_j^{-1}(U_j)\right) = \bigcap_{j \in J} p_j^{-1}(f_j(U_j)).$$

Since each f_j is an open mapping, we obtain that $f_j(U_j)$ is open, and hence

$$f\left(\bigcap_{j \in J} \pi_j^{-1}(U_j)\right)$$

is also open. □

5.1.5 Proposition. Let I be a set and $(X_i)_{i \in I}$ be a family of topological spaces. Let X be the product $\prod_{i \in I} X_i$. For any $i \in I$, let A_i be a subset of X_i and C_i be the closure of A_i in X_i . Let $A = \prod_{i \in I} A_i$ and $C = \prod_{i \in I} C_i$. Then C identifies with the closure of A in X .

Proof. We first prove that, C is a closed subset of X . For any $i \in I$, let $\pi_i : X \rightarrow X_i$ be the mapping of projection to the i -th coordinate. Note that

$$X \setminus C = \bigcup_{i \in I} \pi_i^{-1}(X_i \setminus C_i).$$

Hence $X \setminus C$ is open, which means that C is closed.

Let $x = (x_i)_{i \in I}$ be an element of C . Let U be an open neighbourhood of x in X . We will prove that $U \cap A \neq \emptyset$. Without loss of generality we may assume that U is of the form $\prod_{i \in I} U_i$, where each U_i is an open neighbourhood of x_i , and $U_i = X_i$ for all but finitely many $i \in I$. Since C_i is the closure of A_i , we obtain that $U_i \cap C_i \neq \emptyset$. Therefore, $U \cap A \neq \emptyset$. □

5.1.6 Proposition. *Let X be a topological space and $\sim$ be an equivalence relation on X . Let Y be the quotient of X by the equivalence relation $\sim$, and Γ be the graph of $\sim$. If the quotient set Y equipped with the quotient topology is Hausdorff, then Γ is a closed subset of $X \times X$. The converse is true when π is an open mapping.*

Proof. Let

$$\Delta_Y := \{(y, y) \mid y \in Y\} \subseteq Y \times Y$$

be the diagonal subset of Y . Let $\pi \times \pi : X \times X \rightarrow Y \times Y$ be the mapping sending (x, x') to $(\pi(x), \pi(x'))$. By definition, one has

$$\Gamma = (\pi \times \pi)^{-1}(\Delta_Y).$$

If Y is Hausdorff, then Δ_Y is closed and hence Γ is also closed since $\pi \times \pi$ is continuous.

Assume that π is an open mapping. By Proposition 5.1.4, we obtain that $\pi \times \pi$ is also an open mapping. If Γ is a closed subset of $X \times X$, then

$$(X \times X) \setminus \Gamma = (\pi \times \pi)^{-1}((Y \times Y) \setminus \Delta_Y)$$

is an open subset of $X \times X$. Its image by $\pi \times \pi$, which identifies with $(Y \times Y) \setminus \Delta_Y$, is thus an open subset of $Y \times Y$. Therefore, Y is a Hausdorff space. $\square$

5.1.7 Definition. We say that a topological space $(X, \mathcal{T})$ is *disconnected* if there exists two non-empty open subsets U and V of X such that $X = U \cup V$ and that $U \cap V = \emptyset$. If $(X, \mathcal{T})$ is not disconnected, then we say that it is *connected*.

We say that $(X, \mathcal{T})$ is *path connected* if, for any $(x, y) \in X \times X$, there exists a continuous mapping $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

5.1.8 Proposition. *Let $(X, \mathcal{T})$ be a topological space. Then it is connected if and only if there does not exist any closed and open subset of X other than $\emptyset$ and X itself.*

Proof. Suppose that $X = U \cup V$, where U and V are disjoint non-empty open subsets of X . Then $U \notin \{\emptyset, X\}$, and U is open and closed (since it is equal to $X \setminus V$).

Conversely, if U is a closed and open subset of X , then $V := X \setminus U$ is also an open subset of X . Moreover, if $U \notin \{\emptyset, X\}$, then V is non-empty. Note that $U \cap V = \emptyset$ and $U \cup V = X$. Thus X can be written as the union of two non-empty and disjoint open subsets of X . $\square$

5.1.9 Proposition. *Let $(X, \mathcal{T})$ be a topological space. If $(X, \mathcal{T})$ is path connected, then it is connected.*

Proof. We write X as the union of two disjoint non-empty open subsets U and V . Pick $x \in U$, $y \in V$ and a continuous mapping $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$. Then $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are disjoint open subsets of $[0, 1]$ containing 0 and 1 respectively, such that $[0, 1] = \gamma^{-1}(U) \cup \gamma^{-1}(V)$. Let

$$a = \sup\{t \in [0, 1] \mid [0, t[\subseteq \gamma^{-1}(U)\}.$$

Since $\gamma^{-1}(U)$ is closed, one has $a \in \gamma^{-1}(U)$. Since $\gamma^{-1}(U)$ is open, there exists $\varepsilon > 0$ such that $]a - \varepsilon, a + \varepsilon[\subseteq \gamma^{-1}(U)$. Thus $[0, a + \varepsilon[\subseteq \gamma^{-1}(U)$, which leads to a contradiction. $\square$

5.1.10 Definition. Let $(X, \mathcal{T})$ be a topological space. We call *connected component* of X any non-empty maximal element of the set (equipped with the order of inclusion) of all connected subsets of X . Note that an empty topological space is connected but does not have any connected component.

5.1.11 Lemma. Let $(X, \mathcal{T})$ be a topological space, and $(A_i)_{i \in I}$ be a non-empty family of connected subsets of X , and $i_0 \in I$. If, for any $i \in I$, $A_i \cap A_{i_0}$ is not empty, then $\bigcup_{i \in I} A_i$ is a connected subset of X .

Proof. Suppose that U and V be open subsets of X such that

$$\bigcup_{i \in I} A_i \subseteq U \cup V$$

and that

$$\left(\bigcup_{i \in I} A_i \right) \cap (U \cap V) = \emptyset.$$

Since A_{i_0} is connected, either $A_{i_0} \subseteq U$ or $A_{i_0} \subseteq V$. Without loss of generality, we suppose that $A_{i_0} \subseteq U$. For any $i \in I$, since $A_{i_0} \cap A_i$ is not empty, we obtain that $A_i \cap U \neq \emptyset$. Since A_i is connected, we obtain that $A_i \subseteq U$. Thus

$$\bigcup_{i \in I} A_i \subseteq U.$$

$\square$

5.1.12 Proposition. Let $(X, \mathcal{T})$ be a topological space. Then the following binary relation $\sim$ is an equivalence relation:

$$x \sim y \iff \text{there exists a connected subset } A \text{ of } X \text{ such that } \{x, y\} \subset A.$$

Moreover, the equivalence classes of this equivalence relation are precisely the connected components of X . In particular, any $x \in X$ is contained in a unique connected component (called the connected component of x).

Proof. The binary relation $\sim$ is clearly reflexive and symmetric. We prove that it is transitive. Let x, y and z be elements of X such that $x \sim y$ and $y \sim z$. There exist connected subsets A and B of X such that $\{x, y\} \subseteq A$ and $\{y, z\} \subseteq B$. Thus $A \cap B$ is not empty. By Lemma 5.1.11 we obtain that $A \cup B$ is connected. Since $\{x, z\} \subseteq A \cup B$, we get $x \sim z$.

Let C be a connected component of X , and let x be an arbitrary element of C . By definition, for any $y \in C$, one has $x \sim y$. If z is an element of X such that $x \sim z$, there exists a connected subset A of X such that $\{x, z\} \subseteq A$. Thus, by Lemma 5.1.11 we obtain that $A \cup C$ is connected and hence $A \subseteq C$. Hence C is equal to the equivalence class of x .

Conversely, let x be an element of X and C be the equivalence class of x . By definition, any connected subset of X containing x is contained in C . It remains to check that C is itself connected (and hence C is a connected component). For any $y \in C$ there exists a connected subset A_y of X such that $\{x, y\} \subseteq A_y$. By Lemma 5.1.11, we obtain that

$$\bigcup_{y \in C} A_y$$

is connected and contains C . Therefore, one has

$$C = \bigcup_{y \in C} A_y,$$

which proves that C is connected. $\square$

5.1.13 Proposition. *Let $(X, \mathcal{T})$ be a topological space. If A is a connected subset of X , then its closure $\overline{A}$ is also a connected subset of X . In particular, any connected component in X is closed.*

Proof. Suppose by contradiction that U and V are open subsets of X such that

$$(U \cap V) \cap \overline{A} = \emptyset, \quad \overline{A} \subseteq (U \cup V), \quad U \cap \overline{A} \neq \emptyset \text{ and } V \cap \overline{A} \neq \emptyset.$$

By the definition of closure, we obtain that $U \cap A \neq \emptyset$ and $V \cap A \neq \emptyset$. Moreover, one has $(U \cap V) \cap A = \emptyset$ and $A \subseteq U \cup V$. This contradicts the assumption that A is connected. $\square$

5.2 Topological groups

5.2.1 Definition. Let $(G, *)$ be a group equipped with a topology $\mathcal{T}$. If the mapping

$$\varphi: G \times G \longrightarrow G, \quad (x, y) \longmapsto x * \iota(y)$$

is continuous, then we say that $(G, *, \mathcal{T})$ is a *topological group*.

5.2.2 Definition. Let $(G, *)$ be a group and X be a set. Recall that a mapping $\varphi : G \times X \rightarrow X$ is called a *left action* (resp. *right action*) if the following conditions are satisfied:

- (1) for any $x \in X$, one has $\varphi(e, x) = x$, where e is the neutral element of G ,
- (2) for any $(a, b) \in G \times G$ and any $x \in X$, one has

$$\varphi(a * b, x) = \varphi(a, \varphi(b, x)) \quad (\text{resp. } \varphi(a * b, x) = \varphi(b, \varphi(a, x)))$$

For any $a \in G$, we denote by $L_a : X \rightarrow X$ (resp. $R_a : X \rightarrow X$) the mapping that sends $x \in X$ to $\varphi(a, x)$.

If $(G, *, \mathcal{T})$ is a topological group and X is a topological space, we say that a left action (resp. right action) $\varphi : G \times X \rightarrow X$ is *continuous* if φ is continuous as a mapping, where we consider the product topology on $G \times X$.

5.2.3 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group and X be a topological space. Let $\varphi : G \times X \rightarrow X$ be a continuous left or right action of G on X . Then, for any $a \in G$, the mapping $L_a : X \rightarrow X$, $x \mapsto \varphi(a, x)$ is a homeomorphism.*

Proof. We only prove the case of left action. The case of right action is similar.

The mapping $L_a : X \rightarrow X$, $x \mapsto \varphi(a, x)$ can be written as a composite

$$X \xrightarrow{i_a} G \times X \xrightarrow{\varphi} X,$$

where $i_a : X \rightarrow G \times X$ sends $x \in X$ to (a, x) . Note that the composite of i_a with the two projections $G \times X \rightarrow G$ and $G \times X \rightarrow X$ are respectively the constant mapping and the identity mapping, which are both continuous. Hence i_a is continuous. Therefore L_a is also continuous.

By definition

$$L_a \circ L_{\iota(a)} = L_{\iota(a)} \circ L_a = \text{Id}_X.$$

Hence L_a is a bijection and the inverse of L_a is continuous, namely L_a is a homeomorphism. $\square$

5.2.4 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group.*

- (1) *The mapping $\iota : G \rightarrow G$ of taking the inverse is a homeomorphism.*
- (2) *The mapping $G \times G \rightarrow G$ sending (x, y) to $x * y$ is continuous. It hence defines a continuous left action of G on itself.*
- (3) *The mapping $G \times G \rightarrow G$ sending (x, y) to $y * x$ is continuous. It hence defines a continuous right action of G on itself.*

(4) For any $g \in G$, the mappings

$$L_g : G \longrightarrow G, \quad x \longmapsto g * x$$

and

$$R_g : G \longrightarrow G, \quad x \longmapsto x * g$$

are both homeomorphism of topological spaces.

Proof. Let e be the neutral element of G . We denote by $p_1 : G \times G \rightarrow G$ and $p_2 : G \times G \rightarrow G$ the two projections, namely $p_1(x, y) = x$ and $p_2(x, y) = y$.

(1) The mapping $\lambda : G \rightarrow G \times G$ sending $x \in G$ to (e, x) is continuous since its composite with the two projections from $G \times G$ to G , which are respectively the constant mapping $x \mapsto e$ and the identity mapping $x \mapsto x$, are continuous. Therefore, the mapping $\iota : G \rightarrow G$, which is the composite

$$G \xrightarrow{\lambda} G \times G \xrightarrow{\varphi} G \times G,$$

is continuous. Since $\iota \circ \iota = \text{Id}_G$, we obtain that ι is a homeomorphism.

(2) Let $\tau : G \times G \rightarrow G \times G$ be the mapping that sends (x, y) to $(x, \iota(y))$. One has $p_1 \circ \tau = p_1$ and $p_2 \circ \tau = \iota \circ p_2$. Since ι is continuous, we obtain that $p_1 \circ \tau$ and $p_2 \circ \tau$ are both continuous. Therefore, the mapping τ is continuous. Since the mapping $G \times G \rightarrow G$, $(x, y) \mapsto x * y$ is equal to the composite $\varphi \circ \tau$, we obtain that it is continuous.

(3) The mapping $G \times G \rightarrow G \times G$ sending (x, y) to (y, x) is continuous since its composites with both projections from $G \times G$ to G are continuous. Since the mapping $G \times G \rightarrow G$, $(x, y) \mapsto y * x$ is the composite of $G \times G \rightarrow G \times G$, $(x, y) \mapsto (y, x)$ with $G \times G \rightarrow G$, $(x, y) \mapsto x * y$, we obtain that it is continuous.

(4) follows from Proposition 5.2.3. □

5.2.5 Remark. Let $(G, *)$ be a group equipped with a topology. Then it is a topological group if and only if the mapping of composition $G \times G \rightarrow G$, $(x, y) \mapsto x * y$, and the mapping of inverse $\iota : G \rightarrow G$, are both continuous. The proof is quite similar to that of Proposition 5.2.4. We left it as an exercise.

5.2.6 Example. Let $(K, |\cdot|)$ be a valued field. The additive group $(K, +)$ and the multiplicative group $(K^\times, \cdot)$ are both topological groups.

5.2.7 Example. Let $(K, |\cdot|)$ be a complete valued field and $(V, \|\cdot\|)$ be a normed vector space over K . Then the additive group $(V, +)$ forms a topological group. Moreover, the mapping

$$K^\times \times V \longrightarrow V, \quad (a, x) \longmapsto ax$$

is a continuous left action.

5.2.8 Example. Let $(K, |\cdot|)$ be a complete valued field and $(V, \|\cdot\|)$ be a Banach space over K . We let $\mathcal{L}(V, V)$ be the vector space of bounded K -linear endomorphisms of V . We equip $\mathcal{L}(V, V)$ with the operator norm. Then the bilinear mapping

$$\mathcal{L}(V, V) \times \mathcal{L}(V, V) \longrightarrow \mathcal{L}(V, V), \quad (\varphi, \psi) \longmapsto \varphi \circ \psi$$

is bounded and hence is continuous. Let $\text{GL}(V)$ be the subset of $\mathcal{L}(V, V)$ which is composed of bounded K -linear isomorphisms. Note that, by Banach's open mapping theorem, for any $f \in \text{GL}(V)$, one has $f^{-1} \in \text{GL}(V)$. Therefore, $\text{GL}(V)$ forms a group under the composition. It has been shown in the first semester that the mapping of inversion $\iota : \text{GL}(V) \rightarrow \text{GL}(V)$ is of class C^∞ (and in particular continuous). Therefore, $\text{GL}(V)$ forms a topological group. Moreover, the mapping $\text{GL}(V) \times V \rightarrow V$, $(f, x) \mapsto f(x)$ is a continuous left action of $\text{GL}(V)$ on V . In fact, the mapping

$$\mathcal{L}(V, V) \times V \longrightarrow V, \quad (f, x) \longmapsto f(x)$$

is a bounded bilinear mapping and hence it is continuous.

5.2.9 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group and X be a topological space. Let $\varphi : G \times X \rightarrow X$ be a continuous left action. Let U be a subset of G and V be an open subset of X . Then*

$$\{\varphi(a, x) \mid (a, x) \in U \times V\}$$

is an open subset of X .

Proof. One has

$$\{\varphi(a, x) \mid (a, x) \in U \times V\} = \bigcup_{a \in U} L_a(V).$$

Since L_a is a homeomorphism, the set $L_a(V)$ is open. Therefore,

$$\{\varphi(a, x) \mid (a, x) \in U \times V\}$$

is also open. □

5.2.10 Proposition. *Let $(G, *, \mathcal{T})$ be a topological space, e be the neutral element of G and U be an open neighbourhood of e .*

- (1) *There exists an open neighbourhood V of e such that $V = \iota(V)$ and*

$$V * V := \{x * y \mid (x, y) \in V \times V\}$$

is contained in U .

- (2) For any $a \in U$, there exists an open neighbourhood V of e such that $V \subseteq U$ and

$$a * V := \{a * x \mid x \in V\} \subseteq U, \quad V * a := \{x * a \mid x \in V\} \subseteq U.$$

- (3) For any $b \in U$, there exists an open neighbourhood V of e such that $V \subseteq U$ and

$$b * V * \iota(b) := \{b * x * \iota(b) \mid x \in V\} \subseteq U.$$

Proof. (1) Let $\varphi : G \times G \rightarrow G$ be the mapping that sends (x, y) to $x * \iota(y)$. By definition, this mapping is continuous, and it sends (e, e) to e . Hence $\varphi^{-1}(U)$ is an open neighborhood of (e, e) . Thus there exist open neighborhoods V_1 and V_2 of e such that $V_1 \times V_2 \subseteq \varphi^{-1}(U)$. We take

$$V = (V_1 \cap V_2) \cap (\iota(V_1 \cap V_2)).$$

It is an open neighborhood of e such that $V = \iota(V)$, and

$$V \times V \subseteq \varphi^{-1}(U).$$

Therefore

$$V * V = V * \iota(V) \subseteq U.$$

- (2) We take

$$V := L_{\iota(a)}(U) \cap R_{\iota(a)}(U) \cap U.$$

This is an open neighborhood of e . Moreover, one has

$$a * V = L_a(V) \subseteq U, \quad V * a = R_a(V) \subseteq U.$$

- (3) We take

$$V = (\iota(b) * U * b) \cap U = R_b(L_{\iota(b)}(U)) \cap U$$

It is an open neighborhood of e since $\iota(b) * e * b = e$. Moreover,

$$b * V * \iota(b) \subseteq U.$$

□

5.2.11 Proposition. Let $(G, *, \mathcal{T})$ be a topological group, and e be the neutral element of G . It is a Hausdorff space if and only if the intersection of all open neighborhoods of e is equal to $\{e\}$.

Proof. “ $\implies$ ”: Suppose that $(G, \mathcal{T})$ is a Hausdorff space. For any $x \in G$ such that $x \neq e$, there exists an open neighborhood U_x of e such that $x \notin U_x$. Hence the intersection of all open neighborhoods of e , which is contained in $\bigcap_{x \in G \setminus \{e\}} U_x$, is equal to $\{e\}$.

“ $\impliedby$ ”: Let x and y be two elements of G such that $x \neq y$. Then $x * \iota(y) \neq e$. Hence there exists an open neighborhood U of e that does not contain $x * \iota(y)$. By Proposition 5.2.10, there exists an open neighborhood V of e such that $V * \iota(V) \subseteq U$. We claim that $L_x(V) \cap L_y(V) = \emptyset$. In fact, if $(a, b) \in V \times V$ is such that $x * a = y * b$. Then one has

$$a * \iota(b) = \iota(x) * y \in U,$$

which leads to a contradiction. $\square$

5.2.12 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group and H be a subgroup of G .*

- (1) *The closure $\overline{H}$ of H in G is also a subgroup.*
- (2) *If H is a normal subgroup of G , so is $\overline{H}$.*
- (3) *If H is open, then it is also closed.*

Proof. (1) Consider the mapping

$$\varphi : G \times G \longrightarrow G, \quad (x, y) \longmapsto x * \iota(y).$$

By definition, φ is a continuous mapping. Therefore, one has $\varphi^{-1}(\overline{H})$ is a closed subset of $G \times G$. Also

$$\varphi^{-1}(\overline{H}) \supseteq \varphi^{-1}(H) \supseteq H \times H$$

since H is a subgroup of G . Therefore, by Proposition 5.1.5 we obtain $\varphi^{-1}(\overline{H}) \supseteq \overline{H} \times \overline{H}$, which shows that $\overline{H}$ is a subgroup of G .

- (2) Suppose that H is a normal subgroup, namely, for any $g \in G$, one has

$$H \subseteq g * H * \iota(g).$$

Note that this condition is equivalent to

$$\forall g \in G, \quad H = g * H * \iota(g).$$

Let $\psi_g : G \rightarrow G$ the mapping that sends $x \in G$ to $g * x * \iota(g)$. Note that $\psi_g = R_{\iota(g)} \cdot L_g$. Hence ψ_g is continuous. Since H is a normal subgroup of G , one has

$$H \subseteq \psi_g^{-1}(H) \subseteq \psi_g^{-1}(\overline{H}).$$

Since $\psi_g^{-1}(\overline{H})$ is a closed subset of G , one has $\overline{H} \subseteq \psi_g^{-1}(\overline{H})$. Therefore, $\overline{H}$ is a normal subgroup of G .

(3) If H is open, then, for any $g \in G$, the set $g * H$ is open. Therefore,

$$H = G \setminus \bigcup_{g \in G, g * H \neq H} g * H$$

is closed. □

5.2.13 Definition. Let G be a group and X be a set. Let $\varphi : G \times X \rightarrow X$ be a left action of G on X (resp. right action of G on X).

- (1) If, for any element $g \in G$ that is *not* the neutral element, there exists $x \in X$ such that $\varphi(g, x) \neq x$, then we say that the action φ is *faithful*.
- (2) If, for any $(x, y) \in X \times X$, there exists $g \in G$ such that $\varphi(g, x) = y$, then we say that the action φ is *transitive* and that X is a *homogeneous space* of G .
- (3) For any $x \in X$, we denote by $\text{Stab}(x)$ the set of $g \in G$ such that $\varphi(g, x) = x$. This is a subgroup of G , called the *stabilizer* of the action φ .

5.2.14 Example. Let $(G, *, \mathcal{T})$ be a topological group, and H be a subgroup of G . We let G/H be the set of left cosets of G with respect to H . Let $\pi : G \rightarrow G/H$ be the mapping of projection. We equip G/H with the quotient topology, namely, a subset V of G/H is open if and only if $\pi^{-1}(V)$ is an open subset of G . Note that the mapping of projection π is open. In fact, if U is an open subset of G , then

$$\pi^{-1}(\pi(U)) = \bigcup_{h \in H} U * h$$

is an open subset of G .

The mapping

$$\psi : G \times (G/H) \longrightarrow G/H, \quad (g, x * H) \longmapsto (g * x) * H$$

defines a left action of G on the quotient set G/H . It turns out that this left action is continuous. In fact, if V is an open subset of G/H , then

$$\begin{aligned} \psi^{-1}(V) &= \{(g, x * H) \in G \times (G/H) \mid (g * x) * H \in V\} \\ &= \{(g, x * H) \in G \times (G/H) \mid g * x \in \pi^{-1}(V)\}. \end{aligned}$$

Let $m : G \times G \rightarrow G$, $(x, y) \mapsto x * y$ be the composition law of G . It is a continuous mapping. Hence $m^{-1}(\pi^{-1}(V))$ is an open subset of $G \times G$. Note that $\psi^{-1}(V)$ is the image of $m^{-1}(\pi^{-1}(V))$ by the mapping

$$G \times G \longrightarrow G \times (G/H), \quad (a, b) \longmapsto (a, \pi(b)).$$

Thus $\psi^{-1}(V)$ is an open subset of $G \times (G/H)$.

By definition, G acts transitively on G/H and hence G/H is a homogeneous space of G .

5.2.15 Example. Let $(G, *, \mathcal{T})$ be a topological group, and H be a normal subgroup of G . Let $\varphi : G \times G \rightarrow G$ be the mapping that sends (x, y) to $x * \iota(y)$. This mapping is continuous. We equip the quotient group G/H with the quotient topology. Let $\psi : (G/H) \times (G/H) \rightarrow G/H$ be the mapping that sends $(x * H, y * H)$ to $(x * \iota(y)) * H$. We claim that the mapping ψ is continuous. Let $\pi : G \rightarrow G/H$ be the mapping of projection. If V is an open subset of G/H , then $\psi^{-1}(V)$ is the image of $\varphi^{-1}(\pi^{-1}(V))$ by the open mapping

$$G \times G \longrightarrow (G/H) \times (G/H), \quad (x, y) \longmapsto (x * H, y * H),$$

hence is an open subset of $(G/H) \times (G/H)$.

5.2.16 Example. Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space over $\mathbb{R}$ and

$$S = \{x \in V \mid \langle x, x \rangle = 1\}$$

be the unit sphere of $(V, \langle \cdot, \cdot \rangle)$. Let $O(V, \langle \cdot, \cdot \rangle)$ be the orthogonal group of $(V, \langle \cdot, \cdot \rangle)$. Then $O(V, \langle \cdot, \cdot \rangle)$ acts continuously on S .

Let x and y be two elements of S . We choose two orthonormal basis $(e_i)_{i=1}^n$ and $(e'_i)_{i=1}^n$ on $(V, \langle \cdot, \cdot \rangle)$ such that $e_1 = x$ and $e'_1 = y$. Then the mapping

$$f : V \longrightarrow V, \quad \lambda_1 e_1 + \cdots + \lambda_n e_n \longmapsto \lambda_1 e'_1 + \cdots + \lambda_n e'_n$$

is an $\mathbb{R}$ -linear isometric that sends x to y . Hence the action of $O(V, \langle \cdot, \cdot \rangle)$ on S is transitive, namely S is a homogeneous space of the orthogonal group $O(V, \langle \cdot, \cdot \rangle)$.

5.2.17 Proposition. Let $(G, *, \mathcal{T})$ be a topological group, X be a topological space, and $\varphi : G \times X \rightarrow X$ be a continuous left action which is transitive. Let x_0 be an element of X and H be the stabilizer of x_0 . Then the mapping $f : G/H \rightarrow X$ sending $a * H$ to $\varphi(a, x_0)$ is bijective and continuous.

Proof. Let $\pi : G \rightarrow G/H$ be the mapping of projection and $g : G \rightarrow X$ be the mapping that sends $a \in G$ to $\varphi(a, x_0)$. Since φ and the mapping $G \rightarrow G \times X$, $a \mapsto (a, x_0)$ are both continuous, we obtain that the mapping g is continuous. Moreover, one has $g = f \circ \pi$. For any open subset U of X , one has

$$g^{-1}(U) = \pi^{-1}(f^{-1}(U))$$

is an open subset of G . Hence $f^{-1}(U)$ is an open subset of G/H .

It remains to check that f is a bijection. By definition, g is a surjective mapping, so that f is also surjective. If a and b are elements of G such that $\varphi(a, x_0) = \varphi(b, x_0)$, then one has

$$\varphi(\iota(a) * b, x_0) = x_0,$$

namely $\iota(a) * b \in H$. Thus $a * H = b * H$. □

5.2.18 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group, X be a topological space, and $\varphi : G \times X \rightarrow X$ be a continuous left action. If X is Hausdorff, then, for any $x_0 \in X$, the stabilizer of x_0 is a closed subgroup of G .*

Proof. The mapping f from G to X sending $a \in G$ to $\varphi(a, x_0)$ is continuous. The stabilizer of x_0 is equal to $f^{-1}(\{x_0\})$. Since X is a Hausdorff space, the set $\{x_0\}$ is closed. Hence $\text{Stab}(x_0) = f^{-1}(\{x_0\})$ is also closed. $\square$

5.2.19 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group. Let H be a subgroup of G and $\pi : G \rightarrow G/H$ be the mapping of projection. Then G/H equipped with the quotient topology is a Hausdorff space if and only if H is closed.*

Proof. Let e be the neutral element of G .

Suppose that G/H is Hausdorff. Then $\{e * H\}$ is closed in G/H . Hence

$$H = \pi^{-1}(e * H)$$

is a closed subset of G .

Conversely, suppose that H is closed. Let Γ be the graph of the following equivalence relation :

$$a \sim b \iff \iota(a) * b \in H.$$

It is easy to see, that Γ is closed. Indeed, the mapping f from $G \times G$ to G sending (x, y) to $\iota(x) * y$ is continuous and $\Gamma = f^{-1}(H)$. Let (a, b) be an element of $G \times G$ such that $\iota(a) * b \notin H$. There exist open neighbourhoods U_a and U_b of a and b respectively, such that $f(U_a \times U_b) \subseteq G \setminus H$. Thus

$$(a, b) \in U_a \times U_b \subseteq (G \times G) \setminus \Gamma.$$

We conclude that the projection $\pi : G \rightarrow G/H$ is open. By Proposition 5.1.6, we obtain that G/H is Hausdorff. $\square$

5.2.20 Definition. Let $(G, *, \mathcal{T})$ and $(G', *, \mathcal{T}')$ be topological groups. If a homomorphism of groups $f : G \rightarrow G'$ is continuous, then we say that it is a homomorphism of topological groups. If f is an isomorphism of groups which is also a homeomorphism, we say that f is an isomorphism of topological groups.

5.2.21 Proposition. *Let $(G, *, \mathcal{T})$ and $(G', *, \mathcal{T}')$ be topological groups and $f : G \rightarrow G'$ be a homomorphism of topological groups. Suppose that f is a surjective open mapping. Let N be the kernel of f and $\bar{f} : G/N \rightarrow G'$ be the isomorphism of groups that sends the equivalence class of x to $f(x)$. Then $\bar{f}$ is an isomorphism of topological groups.*

Proof. Let $\pi : G \rightarrow G/N$ be the mapping of projection. By the theory of groups the mapping $\bar{f}$ is an isomorphism of groups. Moreover, for any open subset V of G/N , one has $\bar{f}(V) = f(\pi^{-1}(V))$, which is an open subset of G' . Hence $\bar{f}$ is an open mapping. Finally, if V' is an open subset of G' , one has

$$\pi^{-1}(\bar{f}^{-1}(V')) = f^{-1}(V'),$$

which is an open subset of G . hence $\bar{f}^{-1}(V')$ is an open subset of G/N . $\square$

5.2.22 Proposition. *Let $(G, *, \mathcal{T})$ be a connected topological group, e be the neutral element of G and U be an open neighbourhood of e . Then G is generated by U , namely any element of G can be written as the composition of a family of elements of U .*

Proof. By replacing U by $U \cap \iota(U)$, we may assume that $U = \iota(U)$.

For any positive integer n , let

$$U_n = \underbrace{U * \cdots * U}_{n \text{ copies of } U}.$$

By Proposition 5.2.9, we obtain (by induction on n) that U_n is an open subset of X . Let

$$H = \bigcup_{n \in \mathbb{N}, n \geq 1} U_n.$$

Note that H is a sub-monoid of G . Moreover, since $\iota(U) = U$, we obtain that H is actually a subgroup of G . Hence H is an open subgroup of G , which is also closed (by Proposition 5.2.12). Hence $H = G$. $\square$

5.2.23 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group and H be a subgroup of G . Suppose that H and G/H are both connected, then G is also connected.*

Proof. Suppose by contradiction that U_1 and U_2 are disjoint non-empty open subsets of G such that $U_1 \cup U_2 = G$. Let $\pi : G \rightarrow G/H$ be the mapping of projection. It is an open mapping. Since $G/H = \pi(U_1) \cup \pi(U_2)$ and since G/H is connected, we obtain that $\pi(U_1) \cap \pi(U_2)$ is not empty. Let $a_1 \in U_1$ and $a_2 \in U_2$ such that $\pi(a_1) = \pi(a_2)$. Let $H' = a_1 * H = a_2 * H$. Since H is connected, so is H' . However, $H' \cap U_1$ and $H' \cap U_2$ are two non-empty open subsets of H' which are disjoint and of union H' . This contradicts the connectedness of H' . $\square$

5.2.24 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group and H be the connected component of the neutral element e of G . Then H is a closed and normal subgroup of G . Moreover, for any $x \in G$, the connected component of x in G is $x * H$.*

Proof. Since H is a connected component in G , by Proposition 5.1.13 we obtain that H is closed.

We check that H is a subgroup of G . Let $a \in H$. Note that $\iota(a) * H$ is connected since $L_{\iota(a)} : G \rightarrow G$ is a homeomorphism. Since $e \in \iota(a) * H$, we obtain that $\iota(a) * H \subseteq H$, which means that $\iota(a) * b \in H$ for any $(a, b) \in H \times H$. Hence H is a subgroup of G .

For any $a \in G$, $a * H * \iota(a)$ is a connected subset of G containing e . Hence $a * H * \iota(a) \subseteq H$. Therefore, H is a normal subgroup of G .

For any $x \in G$, the set $x * H$ is a connected subset of G containing x . Conversely, if A is a connected subset of G containing x , then $\iota(x) * A$ is a connected subset of G containing e . Hence $\iota(x) * A \subseteq H$ and $A \subseteq x * H$. $\square$

5.2.25 Proposition. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous homomorphism of additive group of real numbers. Then f is of the form*

$$f(x) = xf(1).$$

Proof. By the universal property of $(\mathbb{Z}, +)$, one has

$$\forall n \in \mathbb{Z}, \quad f(n) = nf(1).$$

Moreover, for any $m \in \mathbb{N}_{>0}$, one has

$$nf(1) = f(n) = f(nm/m) = mf(n/m),$$

so that

$$f(n/m) = \frac{n}{m}f(1).$$

Finally, since $\mathbb{Q}$ is dense in $\mathbb{R}$ and f is continuous, by expressing any real number as a limit of rational numbers, we obtain that

$$\forall x \in \mathbb{R}, \quad f(x) = xf(1).$$

$\square$

5.2.26 Corollary. *Let $g : \mathbb{R}^\times \rightarrow \mathbb{R}_{>0}$ be a continuous homomorphism of multiplicative groups. Then there exists $t \in \mathbb{R}$ such that $g(y) = |y|^t$ for any $y \in \mathbb{R}$.*

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping that sends $x \in \mathbb{R}$ to $\ln(g(e^x))$. One has

$$f(x_1 + x_2) = \ln(g(e^{x_1+x_2})) = \ln(g(e^{x_1})g(e^{x_2})) = f(x_1) + f(x_2).$$

Hence f is a continuous homomorphism of groups. By Proposition 5.2.25, one has

$$f(x) = xf(1) = x \ln(g(e)).$$

Therefore, for any $y \in \mathbb{R}_{>0}$, one has

$$g(y) = g(e^{\ln(y)}) = e^{f(\ln(y))} = e^{\ln(y) \ln(g(e))} = y^{\ln(g(e))}.$$

Finally $g(1) = g(-1)g(-1) = 1$. Since $g(-1) > 0$, we get $g(-1) = 1$. Hence

$$\forall y \in \mathbb{R}, \quad g(y) = |y|^{\ln(g(e))}.$$

□

5.3 Locally compact Hausdorff spaces

5.3.1 Definition. Let $(X, \mathcal{T})$ be a topological space. We say that $(X, \mathcal{T})$ is locally compact if any element x of X has a compact neighbourhood.

5.3.2 Example. The real line $\mathbb{R}$ is a locally compact Hausdorff topological space.

5.3.3 Definition. Let $(X, \mathcal{T})$ be a topological space and $f : X \rightarrow \mathbb{R}$ be a mapping. We call *support* of f the closure of $\{x \in X \mid f(x) \neq 0\}$, denoted as $\text{Supp}(f)$. If $\text{Supp}(f)$ is a compact subset of X , we say that f is of *compact support*. We denote by $C_c(X)$ the set of all continuous mappings of compact support from X to $\mathbb{R}$. It is a vector subspace of $\mathbb{R}^X$.

5.3.4 Proposition. Let $(X, \mathcal{T})$ be a locally compact Hausdorff space.

- (1) Let $x \in X$ and F be a closed subset of X such that $x \notin F$. There exists a compact neighbourhood C of x such that $C \cap F = \emptyset$.
- (2) Let x be an element of X and V_x be an open neighbourhood of x . There exists an open neighbourhood W_x of x such that $\overline{W_x} \subseteq V_x$ and that $\overline{W_x}$ is compact.
- (3) Let Y be a compact subset of X and V be an open subset of X such that $Y \subseteq V$. There exists an open subset W of X such that $\overline{W}$ is compact and that $Y \subseteq W \subseteq \overline{W} \subseteq V$.

Proof. (1) Let N be a compact neighbourhood of x . Since $N \cap F$ is the intersection of a compact subset and a closed subset of X , it is a compact subset of X . Since X is Hausdorff, there exists open subsets U and V of X such that $x \in U \subseteq N$, $N \cap F \subseteq V$ and $U \cap V = \emptyset$ (see Proposition 8.9.10 of FAAI). Note that $\overline{U} \subseteq N$, so $\overline{U}$ is compact. Moreover,

$$\overline{U} \cap F \subseteq N \cap F \subseteq V.$$

So $\overline{U} \cap F \subseteq \overline{U} \cap V = \emptyset$.

(2) Let F be $X \setminus V_x$. It is a closed subset of X . Since $x \notin F$, there exists a compact neighbourhood C of x such that $C \cap F = \emptyset$. Let W_x be the interior of C . It is an open neighbourhood of x . Moreover, $\overline{W}_x \subseteq C \subseteq V_x$.

(3) For any $y \in Y$, let W_y be an open neighbourhood of y such that $\overline{W}_y$ is compact and is contained in V . Since Y is compact, there is a finite subset $\{y_1, \dots, y_n\}$ of Y such that $Y \subseteq W := W_{y_1} \cup \dots \cup W_{y_n}$. Note that

$$\overline{W} = \overline{W}_{y_1} \cup \dots \cup \overline{W}_{y_n}.$$

Hence $\overline{W}$ is compact, and $Y \subseteq W \subseteq \overline{W} \subseteq V$. $\square$

5.3.5 Theorem (Uryson's Lemma). *Let X be a locally compact Hausdorff space. Let K be a compact subset of X and U be an open subset of X such that $K \subseteq U$. There exists a continuous mapping $f : X \rightarrow [0, 1]$ such that*

- (1) *for any $y \in K$, $f(y) = 1$,*
- (2) *f is of compact support and $\text{Supp}(f) \subseteq U$.*

Proof. By Proposition 5.3.4, there exists an open subset W of X such that $\overline{W}$ is compact and that $K \subseteq W \subseteq \overline{W} \subseteq U$.

We will construct a family $(U_r)_{r \in \mathbb{Q} \cap]0, 1]}$ of open subsets of X that satisfies the following conditions:

- (a) $K \subseteq U_1$,
- (b) for any $r \in \mathbb{Q} \cap]0, 1]$, $\overline{U}_r \subseteq W$ (in particular $\overline{U}_r$ is compact),
- (c) for any r and s in $\mathbb{Q} \cap]0, 1]$ such that $r < s$, one has $\overline{U}_s \subseteq U_r$.

For this purpose, we write the elements of $\mathbb{Q} \cap]0, 1]$ as a sequence $(r_n)_{n \in \mathbb{N}}$ such that $r_0 = 1$. By Proposition 5.3.4, there exists an open subset U_{r_0} of X such that $K \subseteq U_{r_0} \subseteq \overline{U}_{r_0} \subseteq W$. Suppose that $U_{r_0}, \dots, U_{r_n}$ have been chosen. We sort the elements of $\{r_0, \dots, r_n\}$ into a strictly increasing sequence

$$1 = s_0 > s_1 > \dots > s_n.$$

Let $i \in \{0, \dots, n\}$ be the largest index such that $s_i > r_{n+1}$. We pick an open subset $U_{r_{n+1}}$ of X such that $\overline{U}_{r_{n+1}}$ is compact, and that

$$\overline{U}_{s_i} \subseteq U_{r_{n+1}} \subseteq \overline{U}_{r_{n+1}} \subseteq U_{s_{i+1}}$$

when $i < n$, and

$$\overline{U}_{s_i} \subseteq U_{r_{n+1}} \subseteq \overline{U}_{r_{n+1}} \subseteq W$$

when $i = n$. By induction, we construct a family $(U_r)_{r \in \mathbb{Q} \cap]0, 1]}$ that satisfies the conditions (a)-(c) above.

For any $x \in X$, we let

$$f(x) = \sup\{r \in \mathbb{Q} \cap]0, 1] \mid x \in U_r\},$$

with the convention that $\sup \emptyset$ is equal to 0. Note that

$$\{x \in X \mid f(x) \neq 0\} \subseteq W.$$

Hence $\text{Supp}(f) \subseteq \overline{W}$. In particular, f is of compact support, and $\text{Supp}(f) \subseteq \overline{W} \subseteq U$. Moreover, since $K \subseteq U_1$, we obtain that the restriction of f to K is constant of value 1.

It remains to prove that f is continuous. Let $t \in [0, 1]$. By definition,

$$\{x \in X \mid f(x) > t\} = \{x \in X \mid \exists r \in \mathbb{Q} \cap]t, 1], x \in U_r\} = \bigcup_{r \in \mathbb{Q} \cap]t, 1]} U_r,$$

which is an open subset of X . Moreover, for any $t \in]0, 1]$,

$$\{x \in X \mid f(x) \geq t\} = \bigcap_{r \in \mathbb{Q} \cap]0, t[} U_r.$$

In fact, if $f(x) \geq t$, then, for any $r \in \mathbb{Q} \cap]0, t[$, one has $f(x) > r$ and hence $x \in U_r$. Conversely, if for any $r \in \mathbb{Q} \cap]0, t[$ one has $x \in U_r$ (and by definition $f(x) \geq r$), then

$$f(x) \geq \sup(\mathbb{Q} \cap]0, t[) = t.$$

Note that

$$\bigcap_{r \in \mathbb{Q} \cap]0, t[} U_r \subseteq \bigcap_{r \in \mathbb{Q} \cap]0, t[} \overline{U_r} \subseteq \bigcap_{r \in \mathbb{Q} \cap]0, t[} \bigcap_{s \in \mathbb{Q} \cap]r, t[} U_s = \bigcap_{r \in \mathbb{Q} \cap]0, t[} U_r.$$

Therefore, we obtain that $\{x \in X \mid f(x) \geq t\}$ is a closed subset of X and hence $\{x \in X \mid f(x) < t\}$ is an open subset of X .

Let $I =]a, b[$ be an open interval, where a and b are real numbers such that $a < b$. If $a \geq 0$ and $b > 1$, then

$$f^{-1}(I) = \{x \in X \mid f(x) > a\};$$

if $a < 0$, $b \leq 1$, then

$$f^{-1}(I) = \{x \in X \mid f(x) < b\};$$

if $\{a, b\} \subseteq [0, 1]$, then

$$f^{-1}(I) = \{x \in X \mid f(x) > a\} \cap \{x \in X \mid f(x) < b\}.$$

In all cases $f^{-1}(I)$ is an open subset of X . Since any open subset of $\mathbb{R}$ can be written as a union of open intervals, we obtain that f is a continuous mapping. $\square$

5.3.6 Corollary. *Let $(X, \mathcal{T})$ be a locally compact Hausdorff space. Then the topology $\mathcal{T}$ identifies with the coarsest topology that makes all mappings in $C_c(X)$ continuous.*

Proof. By definition any mapping in $C_c(X)$ is continuous when we consider the topology $\mathcal{T}$ on X .

Let $\mathcal{T}'$ be a topology such that all mappings in $C_c(X)$ are continuous with respect to $\mathcal{T}'$. Let U be a non-empty set that belongs to $\mathcal{T}$. For any $x \in U$, we pick a compact neighbourhood K_x of x that is contained in U . By Theorem 5.3.5, there exists an element $f_x \in C_c(X)$ such that $\text{Supp}(f_x) \subseteq U$ and that

$$K_x \subseteq V_x := \{y \in X \mid f_x(y) > 0\}.$$

Therefore,

$$U = \bigcup_{x \in U} K_x \subseteq \bigcup_{x \in U} V_x \subseteq U,$$

which shows that $U \in \mathcal{T}'$. □

5.3.7 Proposition. *Let $(X, \mathcal{T})$ be a locally compact Hausdorff space, which has a countable topological basis. There exists a sequence $(U_n)_{n \in \mathbb{N}}$ of open subsets of X , such that,*

- (1) *for any $n \in \mathbb{N}$, $\overline{U}_n$ is compact and is contained in U_{n+1} .*
- (2) $\bigcup_{n \in \mathbb{N}} U_n = X$.

Proof. Let $\mathcal{B}$ be a countable topological basis of $\mathcal{T}$ and let $\mathcal{B}_c$ be the set of $V \in \mathcal{B}$ such that $\overline{V}$ is compact. We claim that $\mathcal{B}_c$ is still a topological basis of $\mathcal{T}$. Let U be an element of $\mathcal{T}$. For any $x \in U$, by Proposition 5.3.4, there exists an open neighbourhood W_x of x such that $\overline{W}_x$ is compact and contained in U . Since $\mathcal{B}$ is a topological basis of $\mathcal{T}$, there exists an element V_x of $\mathcal{B}$ such that $x \in V_x \subseteq W_x$. Note that $\overline{V}_x \subseteq \overline{W}_x$. Hence $\overline{V}_x$ is compact. Therefore, any element of $\mathcal{B}$ can be written as a union of elements of $\mathcal{B}_c$, which proves that $\mathcal{B}_c$ is a topological basis of $\mathcal{T}$.

Since $\mathcal{B}$ is countable, so is $\mathcal{B}_c$. Let $(V_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{B}_c$ such that $\mathcal{B}_c = \{V_n \mid n \in \mathbb{N}\}$. We construct a sequence $(U_n)_{n \in \mathbb{N}}$ in a recursive way as follows. First, we let $U_0 = V_0$. Suppose that $U_0, \dots, U_n$ are constructed so that each $\overline{U}_i$ is compact, $V_i \subseteq U_i$ and

$$\forall i \in \{0, \dots, n-1\}, \quad \overline{U}_i \subseteq U_{i+1}.$$

Note that $\overline{U}_n \cup \overline{V}_{n+1}$ is compact, By Proposition 5.3.4, there exists an open subset U_{n+1} of X such that $\overline{U}_{n+1}$ is compact and

$$\overline{U}_n \cup \overline{V}_{n+1} \subseteq U_{n+1}.$$

Note that

$$X = \bigcup_{n \in \mathbb{N}} V_n \subseteq \bigcup_{n \in \mathbb{N}} U_n \subseteq X.$$

Hence the proposition is proved. $\square$

5.3.8 Lemma. *Let X be a topological space that has a countable topological basis. Then any open cover $(U_i)_{i \in I}$ has a countable subcover.*

Proof. Let $(V_n)_{n \in \mathbb{N}}$ be a family of open subsets of X that forms a topological basis. For any $x \in X$, we choose $i(x) \in I$ such that $x \in U_{i(x)}$ and let $n(x) \in \mathbb{N}$ such that $x \in V_{n(x)} \subseteq U_{i(x)}$. Let

$$A = \{n(x) \mid x \in X\} \subseteq \mathbb{N}.$$

This is a countable set. One has

$$X = \bigcup_{m \in A} V_m.$$

For any $m \in A$, we pick $x_m \in X$ such that $n(x_m) = m$. Note that

$$X = \bigcup_{m \in A} V_m \subseteq \bigcup_{m \in A} U_{i(x_m)}.$$

Hence $(U_i)_{i \in I}$ has a countable subcover. $\square$

5.3.9 Proposition. *Let X be a non-empty locally compact topological space. Let $(F_n)_{n \in \mathbb{N}}$ be a family of closed subsets of X . If*

$$X = \bigcup_{n \in \mathbb{N}} F_n,$$

then there exists $N \in \mathbb{N}$ such that F_N° is not empty.

Proof. Suppose by contradiction that all F_n° are empty. Let U_0 be an open subset of X such that $\overline{U_0}$ is compact. Suppose that non-empty open subsets $U_0, \dots, U_n$ of X have been chosen (so that $\overline{U_i}$ is compact and is contained in $U_{i-1} \setminus F_i$). We let U_{n+1} be a non-empty subset of X such that $\overline{U_{n+1}} \subseteq U_n \setminus F_{n+1}$. Note that $(\overline{U_n})_{n \in \mathbb{N}}$ is a decreasing sequence of non-empty compact subsets of X . Hence their intersection is not empty. This contradicts the assumption that

$$\emptyset \neq X = \bigcup_{n \in \mathbb{N}} F_n.$$

$\square$

5.3.10 Proposition. *Let $(G, *, \mathcal{T})$ be a topological group, X be a topological space, and $\varphi : G \times X \rightarrow X$ be a continuous left action which is transitive. Assume that both G and X are locally compact and have countable topological bases. Let x_0 be an element of X and H be the stabilizer of x_0 . Then the mapping $f : G/H \rightarrow X$ sending $a * H$ to $\varphi(a, x_0)$ is a homeomorphism.*

Proof. Let e be the neutral element of G . By Proposition 5.2.17, it suffices to prove that the mapping f is an open mapping.

Let V be a non-empty open subset of G . Let $x \in f(V)$ and let $s \in G$ such that $f(s) = x$. By Proposition 5.2.10, there exists an open neighbourhood U_0 of e such that

$$s * \iota(U_0) * U_0 \subseteq V.$$

Let U be an open neighbourhood of e such that $\overline{U}$ is compact and contained in U_0 .

Since G has a countable topological basis, by Lemma 5.3.8, we obtain that the open cover $(a * U)_{a \in G}$ has a countable sub-cover $(a_n * U)_{n \in \mathbb{N}}$. Note that

$$G = \bigcup_{n \in \mathbb{N}} a_n * U = \bigcup_{n \in \mathbb{N}} a_n * \overline{U},$$

so that

$$X = \bigcup_{n \in \mathbb{N}} L_{a_n}(f(\overline{U})).$$

Since $\overline{U}$ is compact, so is $L_{a_n}(f(\overline{U}))$. By Proposition 5.3.9, we obtain that there exists $n \in \mathbb{N}$ such that $L_{a_n}(f(\overline{U}))$ has a non-empty interior, which implies that $f(\overline{U})$ also has a non-empty interior W .

Let $y \in W$ and b be an element of $\overline{U}$ such that $f(b) = y$. Then

$$x = f(s) = L_{s*\iota(b)}f(b) = L_{s*\iota(b)}(y) \in L_{s*\iota(b)}(W) \subseteq f(s * \iota(\overline{U}) * \overline{U}) \subseteq f(V).$$

Thus $L_{s*\iota(b)}(W)$ is an open neighbourhood of x in $f(V)$. Hence $f(V)$ is an open subset of X . $\square$

5.4 Radon measure

5.4.1 Definition. Let $(X, \mathcal{T})$ be a locally compact Hausdorff space having a countable topological basis. We call *positive linear functional* on X any $\mathbb{R}$ -linear mapping $I : C_c(X) \rightarrow \mathbb{R}$ such that, for any $f \in C_c(X)$ satisfying

$$\forall x \in X, \quad f(x) \geq 0,$$

one has $I(f) \geq 0$.

5.4.2 Theorem. Let $(X, \mathcal{T})$ be a locally compact Hausdorff space having a countable topological basis, and $I : C_c(X) \rightarrow \mathbb{R}$ be a positive linear functional. Then I is an integral operator and determines a unique σ -finite Borel measure μ_I on X such that

$$\forall f \in C_c(X), \quad \int_X f(x) \mu_I(dx) = I(f).$$

The measure μ_I is called the Radon measure on the Boreal σ -algebra of X corresponding to I .

Proof. Let f and g be two elements of $C_c(X)$. Then $\inf\{f, g\}$ is still a continuous mapping, and

$$\text{Supp}(\inf\{f, g\}) \subseteq \text{Supp}(f) \cup \text{Supp}(g).$$

Hence $\inf\{f, g\}$ is compactly supported.

Let $(f_n)_{n \in \mathbb{N}}$ be a decreasing sequence in $C_c(X)$ that converges pointwise to the zero mapping. Let K be the support of f_0 . It is a compact subset of X . By Uryson's lemma, one can pick a continuous mapping of compact support $f : X \rightarrow [0, 1]$ such that $f(x) = 1$ for any $x \in K$. Then, for any $n \in \mathbb{N}$, one has

$$\forall x \in X, \quad f_n(x) \leq \|f_n\|_{L^\infty} f(x),$$

where

$$\|f_n\|_{L^\infty} := \sup_{y \in X} |f_n(y)| = \sup_{y \in K} f_n(y).$$

Since I is a positive linear functional, one has

$$I(f_n) \leq \|f_n\|_{L^\infty} I(f).$$

It remains to check that

$$\lim_{n \rightarrow +\infty} \|f_n\|_{L^\infty} = 0.$$

Since the sequence $(\|f_n\|_{L^\infty})_{n \in \mathbb{N}}$ is decreasing and bounded from below by 0, if it does not converge to 0, then there exists $\varepsilon > 0$ such that $\|f_n\|_{L^\infty} > \varepsilon$ for any $n \in \mathbb{N}$. For any $n \in \mathbb{N}$, $K_n = \{x \in X \mid f_n(x) \geq \varepsilon\}$. Then $(K_n)_{n \in \mathbb{N}}$ is a decreasing sequence of non-empty compact subsets of X . Thus

$$\bigcap_{n \in \mathbb{N}} K_n$$

is not empty. This contradicts the assumption that $(f_n)_{n \in \mathbb{N}}$ converges pointwisely to 0. □

5.4.3 Proposition. *Let $(X, \mathcal{T})$ be a locally compact Hausdorff space having a countable topological basis. Let $I : C_c(X) \rightarrow \mathbb{R}$ be a positive linear functional and μ_I be the corresponding Radon measure.*

- (1) *For any compact subset K of X , one has $\mu_I(K) < +\infty$.*
- (2) *For any open subset U of X , one has*

$$\mu_I(U) = \sup_{\substack{K \subseteq U \\ K \text{ compact}}} \mu_I(K) = \sup_{\substack{(f: X \rightarrow [0,1]) \in C_c(X) \\ \text{Supp}(f) \subseteq U}} I(f).$$

(3) For any Borel subset A of X , one has

$$\mu_I(A) = \inf_{U \in \mathcal{T}, A \subseteq U} \mu_I(U).$$

(4) For any Borel subset A of X , one has

$$\mu_I(A) = \sup_{\substack{K \subseteq A \\ K \text{ compact}}} \mu_I(K)$$

Proof. (1) By Uryson's lemma, there exists $(f : X \rightarrow [0, 1]) \in C_c(X)$ such that $f(x) = 1$ for any $x \in K$. Thus

$$\mu_I(K) \leq I(f) < +\infty.$$

(2) Note that U is also locally compact and Hausdorff, and has a countable topological basis. By Proposition 5.3.7, there is an increasing sequence $(K_n)_{n \in \mathbb{N}}$ of compact subsets of U such that $U = \bigcup_{n \in \mathbb{N}} K_n$. Hence

$$\sup_{\substack{K \subseteq U \\ K \text{ compact}}} \mu_I(K) \leq \mu_I(U) = \lim_{n \rightarrow +\infty} \mu_I(K_n) \leq \sup_{\substack{K \subseteq U \\ K \text{ compact}}} \mu_I(K).$$

Hence the equality

$$\mu_I(U) = \sup_{\substack{K \subseteq U \\ K \text{ compact}}} \mu_I(K)$$

holds.

If $f : X \rightarrow [0, 1]$ is a continuous mapping such that $\text{Supp}(f) \subseteq U$, then by definition one has $I(f) \leq \mu_I(U)$. Conversely, for any compact subset K contained in U , there exists a mapping $f : X \rightarrow [0, 1]$ in $C_c(X)$ such that $f(x) = 1$ for any $x \in K$ and that $\text{Supp}(f) \subseteq U$. Thus $\mu_I(U) \geq I(f) \geq \mu_I(K)$. Therefore,

$$\mu_I(U) \geq \sup_{\substack{(f: X \rightarrow [0, 1]) \in C_c(X) \\ \text{Supp}(f) \subseteq U}} I(f) \geq \sup_{\substack{K \subseteq U \\ K \text{ compact}}} \mu_I(K) = \mu_I(U).$$

Hence the equality

$$\mu_I(U) = \sup_{\substack{(f: X \rightarrow [0, 1]) \in C_c(X) \\ \text{Supp}(f) \subseteq U}} I(f).$$

holds.

(3) Without loss of generality, we may assume that $\mu_I(A) < +\infty$ (in the case where $\mu_I(A) = +\infty$, for any open subset U containing A , one has $\mu_I(U) = +\infty$). By definition,

$$\mu_I(A) = \inf_{\substack{f \in C_c(X)^\dagger \\ f \geq \mathbf{1}_A}} I(f).$$

We claim that, for any $f \in C_c(X)^\dagger$, and any $t \in \mathbb{R}$, the set

$$U_{f,t} := \{x \in X \mid f(x) > t\}$$

is open. In fact, there exists an increasing sequence $(f_n)_{n \in \mathbb{N}}$ in $C_c(X)$ so that f is the pointwise limit of $(f_n)_{n \in \mathbb{N}}$. Hence

$$U_{f,t} = \bigcup_{n \in \mathbb{N}} \{x \in X \mid f_n(x) > t\}.$$

Since each f_n is continuous, we obtain that $\{x \in X \mid f_n(x) > t\}$ is open, and hence $\{x \in X \mid f(x) > t\}$ is also open.

We now choose a decreasing sequence $(g_n)_{n \in \mathbb{N}}$ in $C_c(X)^\dagger$ such that $g_n \geq \mathbb{1}_A$ for any $n \in \mathbb{N}$ and that

$$\lim_{n \rightarrow +\infty} I(g_n) = \mu_I(A).$$

Let $\epsilon \in]0, 1[$. Note that, for any $n \in \mathbb{N}$,

$$U_{g_n, 1-\epsilon} = \{x \in X \mid g_n(x) > 1-\epsilon\}$$

is an open subset of X containing A . Moreover, one has

$$g_n \geq (1-\epsilon)\mathbb{1}_{U_{g_n, 1-\epsilon}},$$

so that

$$\mu_I(U_{g_n, 1-\epsilon}) \leq \frac{1}{1-\epsilon} I(g_n).$$

Taking the infimum with respect to n , we get

$$\inf_{U \in \mathcal{T}, A \subseteq U} \mu_I(U) \leq (1-\epsilon)\mu_I(A).$$

Since ϵ is arbitrary, we get

$$\inf_{U \in \mathcal{T}, A \subseteq U} \mu_I(U) = \mu_I(A).$$

(4) We first treat the case where A is of finite measure. Let $\epsilon > 0$. Let U be an open subset of X such that $A \subseteq U$ and that $\mu_I(U) \leq \mu_I(A) + \epsilon$. By (2), there is a compact subset F of U such that $\mu_I(U \setminus F) \leq \epsilon$. Since $\mu_I(U \setminus A) \leq \epsilon$, by (3) there exists an open subset V of X such that $U \setminus A \subseteq V$ and that $\mu(V) \leq 2\epsilon$. Let $K = F \setminus V$. Then K is a compact subset of X and

$$K \subseteq F \setminus (U \setminus A).$$

Since $F \subseteq U$, we obtain that $K \subseteq A$. Finally,

$$\mu_I(A \setminus K) \leq \mu_I(U \setminus K) = \mu_I(U \setminus F) + \mu_I(F \setminus K) \leq \mu_I(U \setminus F) + \mu_I(V) \leq 3\epsilon.$$

Hence the statement holds when $\mu_I(A) < +\infty$.

In the case where $\mu_I(A) = +\infty$, by Proposition 5.3.7 there exists an increasing sequence $(A_n)_{n \in \mathbb{N}}$ of Borel subsets of A such that

$$A = \bigcup_{n \in \mathbb{N}} A_n$$

and $\mu_I(A_n) < +\infty$ for any $n \in \mathbb{N}$. For any $n \in \mathbb{N}$ there exists a compact subset K_n of A_n such that

$$\mu_I(K_n) \geq \mu_I(A_n) - \frac{1}{2^n}.$$

Taking the limit when $n \rightarrow +\infty$, we get

$$\lim_{n \rightarrow +\infty} \mu_I(K_n) = +\infty.$$

□

5.5 Haar measure

In this section, we fix a locally compact Hausdorff topological group $(G, *, \mathcal{T})$, which is assumed to have a countable topological basis. We denote by $C_c^+(G)$ the set of continuous mappings $f : G \rightarrow \mathbb{R}_{\geq 0}$ of compact support, which does not vanish identically. Note that any element $f \in C_c(G)$ can be written as the difference of two elements in $C_c^+(G)$. Moreover, we denote by $\|\cdot\|_{\sup}$ the superior norm on $C_c(G)$, which is defined as

$$\|f\| := \sup_{x \in G} |f(x)|.$$

For any $a \in G$, we denote by $L_a : C_c(G) \rightarrow C_c(G)$ the mapping that sends $f \in C_c(G)$ to

$$(x \in G) \mapsto f(a * x).$$

Similarly, we denote by $R_a : C_c(G) \rightarrow C_c(G)$ the mapping that sends $f \in C_c(G)$ to

$$(x \in G) \mapsto f(x * a).$$

Let $\iota : C_c(G) \rightarrow C_c(G)$ the mapping that sends $f \in C_c(G)$ to

$$(x \in G) \mapsto f(\iota(x)).$$

Note that ι , L_a and R_a are isomorphisms of real vector spaces, which send non-negative mappings to non-negative mappings.

5.5.1 Definition. Let $I : C_c(G) \rightarrow \mathbb{R}$ be a non-zero positive linear function. We say that I defines a *left Haar measure* if $I(\tau_a(f)) = I(f)$ for any $a \in G$ and any $f \in C_c(G)$. The corresponding Boreal measure μ_I is called a *left Haar measure* on G .

Similarly, we say that I defines a *right Haar measure* if $I(\sigma_a(f)) = I(f)$ for any $a \in G$ and any $f \in C_c(G)$. The corresponding Borel measure μ_I is called a *right Haar measure* on G .

5.5.2 Definition. Let f and g be two elements of $C_c^+(G)$, we denote by $(f : g)$ the element

$$\inf \left\{ \sum_{i=1}^n c_i \mid n \in \mathbb{N}, (c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n, \exists (a_i)_{i=1}^n \in G^n, f \leq \sum_{i=1}^n c_i L_{a_i}(g) \right\}$$

5.5.3 Proposition. (1) Let f and g be elements of $C_c^+(G)$. Then $(f : g)$ is finite and bounded from below by $\|f\|_{\sup}/\|g\|_{\sup}$.

(2) Let f and g be elements of $C_c^+(G)$. For any $b \in G$, one has

$$(f : g) = (L_b(f) : g).$$

Moreover, for any $\lambda > 0$, one has $(\lambda f : g) = \lambda(f : g)$.

(3) Let f_1, f_2 and g be elements of $C_c^+(G)$. One has

$$(f_1 + f_2 : g) \leq (f_1 : g) + (f_2 : g).$$

(4) Let f_1, f_2 and g be elements of $C_c^+(G)$. Suppose that $f_1 \leq f_2$. Then

$$(f_1 : g) \leq (f_2 : g).$$

(5) Let f, g and h be elements of $C_c^+(G)$. Then

$$(f : h) \leq (f : g)(g : h).$$

Proof. (1) Let $n \in \mathbb{N}$, $(c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n$, $(a_i)_{i=1}^n \in G^n$, such that

$$f \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Then, for any $x \in G$ one has

$$f(x) \leq \sum_{i=1}^n c_i g(a_i * x) \leq \|g\|_{\sup} \sum_{i=1}^n c_i.$$

Taking the supremum with respect to x , we obtain that

$$\|f\|_{\sup} \leq \|g\|_{\sup} \sum_{i=1}^n c_i,$$

or equivalently

$$\sum_{i=1}^n c_i \geq \frac{\|f\|_{\sup}}{\|g\|_{\sup}}.$$

Taking the infimum, we get

$$(f : g) \geq \frac{\|f\|_{\sup}}{\|g\|_{\sup}}.$$

Let K be the support of f and

$$U = \{x \in G \mid g(x) > \|g\|_{\sup}/2\}.$$

By definition, K is a compact subset of G and U is an open subset of G . Let b be an arbitrary element of U . Note that

$$K \subseteq \bigcup_{a \in K} L_{a * \iota(b)}(U).$$

Hence there exists a subset $\{a_1, \dots, a_n\}$ of K such that

$$K \subseteq \bigcup_{i=1}^n L_{a_i * \iota(b)}(U).$$

Thus

$$f \leq \frac{2\|f\|_{\sup}}{\|g\|_{\sup}} \sum_{i=1}^n L_{a_i * \iota(b)}(g),$$

so that

$$(f : g) \leq \frac{2n\|f\|_{\sup}}{\|g\|_{\sup}}.$$

(2) Let $n \in \mathbb{N}$, $(c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n$, $(a_i)_{i=1}^n \in G^n$, such that

$$f \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Then we obtain

$$L_b(f) \leq \sum_{i=1}^n c_i L_{b * a_i}(g), \quad \lambda f \leq \sum_{i=1}^n \lambda c_i L_{a_i} g.$$

Therefore,

$$(L_b(f) : g) \leq \sum_{i=1}^n c_i, \quad (\lambda f : g) \leq \lambda \sum_{i=1}^n c_i.$$

Taking the infimum, we obtain

$$(L_b(f) : g) \leq (f : g), \quad (\lambda f : g) \leq \lambda(f : g).$$

Replacing b by $\iota(b)$ and f by $L_b(f)$ in the first inequality, we get

$$(f : g) \leq (L_b(f) : g).$$

Similarly, replacing f by λf and λ by λ^{-1} , we get

$$(f : g) \leq \lambda^{-1}(\lambda f : g).$$

Therefore, we obtain

$$(L_b(f) : g) = (f : g), \quad (\lambda f : g) = \lambda(f : g).$$

(3) Let $n \in \mathbb{N}$, $(c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n$, $(a_i)_{i=1}^n \in G^n$, such that

$$f_1 \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Let $m \in \mathbb{N}$, $(c'_j)_{j=1}^m \in \mathbb{R}_{\geq 0}^m$, $(b_j)_{j=1}^m \in G^m$ such that

$$f_2 \leq \sum_{j=1}^m c'_j L_{b_j}(g).$$

Then one has

$$f_1 + f_2 \leq \sum_{i=1}^n c_i L_{a_i}(g) + \sum_{j=1}^m c'_j L_{b_j}(g).$$

Hence

$$f_1 + f_2 \leq \sum_{i=1}^n c_i + \sum_{j=1}^m c'_j.$$

Taking the infimum, we obtain

$$(f_1 + f_2 : g) = (f_1 : g) + (f_2 : g).$$

(4) Let $n \in \mathbb{N}$, $(c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n$, $(a_i)_{i=1}^n \in G^n$, such that

$$f_2 \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Since $f_1 \leq f_2$, one also has

$$f_1 \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Hence

$$(f_1 : g) \leq \sum_{i=1}^n c_i.$$

Taking the infimum, we obtain

$$(f_1 : g) \leq (f_2 : g).$$

(5) Let $n \in \mathbb{N}$, $(c_i)_{i=1}^n \in \mathbb{R}_{\geq 0}^n$, $(a_i)_{i=1}^n \in G^n$, such that

$$f \leq \sum_{i=1}^n c_i L_{a_i}(g).$$

Let $m \in \mathbb{N}$, $(c'_j)_{j=1}^m \in \mathbb{R}_{\geq 0}^m$, $(b_j)_{j=1}^m \in G^m$ such that

$$g \leq \sum_{j=1}^m c'_j L_{b_j}(h).$$

Then one has

$$f \leq \sum_{i=1}^n \sum_{j=1}^m c_i c'_j L_{a_i * b_j}(h),$$

which implies that

$$(f : h) \leq \left(\sum_{i=1}^n c_i \right) \left(\sum_{j=1}^m c'_j \right).$$

Taking the infimum, we get

$$(f : h) \leq (f : g)(g : h).$$

□

5.5.4 Lemma. *Let $(G, *, \mathcal{T})$ be a locally compact Hausdorff topological group. Let f be an element of $C_c(G)$. For any $\varepsilon > 0$ there exists an open neighbourhood V of the neutral element of G such that $V = \iota(V)$, and that for any $a \in V$,*

$$\|L_a(f) - f\|_{\sup} \leq \varepsilon, \quad \|R_a(f) - f\|_{\sup} \leq \varepsilon.$$

Proof. Let e be the neutral element of G . Since f is continuous, for any $x \in X$, there is an open neighbourhood U_x of e such that

$$\forall a \in U_x, \quad \|f(a * x) - f(x)\| \leq \frac{\varepsilon}{2}, \quad \|f(x * a) - f(x)\| \leq \frac{\varepsilon}{2}.$$

By Proposition 5.2.10 (1), there exists an open neighbourhood V_x of e such that $\iota(V_x) = V_x$ and $V_x * V_x \subseteq U_x$.

Let K be the support of f . Since K is compact, there exists a finite subset $\{x_1, \dots, x_n\}$ of K such that

$$K \subseteq \bigcup_{i=1}^n R_{x_i}(V_{x_i}), \quad K \subseteq \bigcup_{i=1}^n L_{x_i}(V_{x_i}).$$

We let

$$V = \bigcap_{i=1}^n V_{x_i}.$$

Let x be an element of G and $a \in V$. If $x \in K$, then there exists $i \in \{1, \dots, n\}$ such that $x \in R_{x_i}(V_{x_i})$, or equivalently $x * \iota(x_i) \in V_{x_i}$. Thus one has

$$a * x = a * (x * \iota(x_i)) * x_i \in R_{x_i}(U_{x_i}), \quad x = (x * \iota(x_i)) * x_i \in R_{x_i}(V_{x_i}).$$

Hence

$$|f(a * x) - f(x)| \leq |f(a * x) - f(x_i)| + |f(x) - f(x_i)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

If $a * x \in K$, by the above result one has (since $\iota(a) \in V$)

$$|f(x) - f(a * x)| = |f(\iota(a) * (a * x)) - f(a * x)| \leq \varepsilon.$$

If $\{x, a * x\} \cap K = \emptyset$, then $f(x) = f(a * x) = 0$, and hence

$$|f(a * x) - f(x)| = 0 \leq \varepsilon.$$

Similarly, if $x \in K$, then there exists $j \in \{1, \dots, n\}$ such that $x \in L_{x_j}(V_{x_j})$, or equivalently $\iota(x_j) * x \in V_{x_j}$. Thus one has

$$x * a = x_j * (\iota(x_j) * x) * a \in L_{x_j}(U_{x_j}), \quad x = x_j * (\iota(x_j) * x) \in L_{x_j}(V_{x_j}).$$

Hence

$$|f(x * a) - f(x)| \leq |f(x * a) - f(x_j)| + |f(x_j) - f(x)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

If $x * a \in K$, then

$$|f(x) - f(x * a)| = |f((x * a) * \iota(a)) - f(x * a)| \leq \varepsilon.$$

If $\{x * a, x\} \cap K = \emptyset$, then

$$|f(x) - f(x * a)| = 0 \leq \varepsilon.$$

□

5.5.5 Theorem. *Let $(G, *, \mathcal{T})$ be a locally compact Hausdorff topological group, that has a countable topological basis. There exists a positive linear functional $I : C_c(G) \rightarrow \mathbb{R}$, that defines a left Haar measure.*

Proof. (1) Let g be an arbitrary element of $C_c^+(G)$. For $h \in C_c^+(G)$, we define a mapping $I_h : C_c^+(G) \rightarrow \mathbb{R}_{>0}$ as follows:

$$\forall f \in C_c^+(G), \quad I_h(f) := \frac{(f : h)}{(g : h)}.$$

By Proposition 5.5.3 (5), one has

$$(g : f)^{-1} \leq I_h(f) \leq (f : g).$$

Moreover, by Proposition 5.5.3 (1) and (2), for any $a \in G$, one has $I_h(L_a(f)) = I_h(f)$; for any $\lambda > 0$, one has $I_h(\lambda f) = \lambda I_h(f)$. By Proposition 5.5.3 (3), for any elements f_1 and f_2 of $C_c^+(G)$, one has

$$I_h(f_1 + f_2) \leq I_h(f_1) + I_h(f_2).$$

5.5.6 Lemma. *Let f_1 and f_2 be elements of $C_c^+(G)$. For any $\varepsilon > 0$, there exists an open neighbourhood V of the neutral element e of G , such that, for any $h \in C_c^+(G)$ with $\text{Supp}(h) \subseteq V$, one has*

$$I_h(f_1 + f_2) \geq I_h(f_1) + I_h(f_2) - \varepsilon.$$

Proof. Let f_0 be an element of $C_c^+(G)$ such that

$$\forall x \in \text{Supp}(f_1) \cup \text{Supp}(f_2), \quad f_0(x) = 1.$$

Let δ denote a positive parameter that we will precise later. Let u_1 and u_2 be elements of $C_c^+(G)$ defined as

$$u_i(x) = \begin{cases} \frac{f_i(x)}{f_1(x) + f_2(x) + \delta f_0(x)}, & \text{if } f_0(x) > 0, \\ 0 & \text{if } f_0(x) = 0. \end{cases}$$

By definition, $u_i(x) = 0$ if x does not belong to $\text{Supp}(f_i)$. Hence u_i is continuous on $G \setminus \text{Supp}(f_i)$. Moreover, the equality

$$u_i(x) = \frac{f_i(x)}{f_1(x) + f_2(x) + \delta f_0(x)}$$

holds on $\{x \in G \mid f_0(x) > 0\}$. Hence u_i is continuous on $\{x \in G \mid f_0(x) > 0\}$. Since

$$\{x \in G \mid f_0(x) > 0\} \subseteq \text{Supp}(f_i),$$

we obtain that u_i is continuous. By definition u_i takes values in $[0, 1]$ and $\text{Supp}(u_i) = \text{Supp}(f_i)$. Hence $u_i \in C_c^+(G)$.

By Lemma 5.5.4, there exists an open neighborhood V of the neutral element of G , such that, for any $i \in \{1, 2\}$

$$\forall (x, y) \in G \times G, \quad \iota(y) * x \in V \implies |u_i(x) - u_i(y)| \leq \delta.$$

Suppose that h is an element of $C_c^+(G)$ such that $\text{Supp}(h) \subseteq V$. Suppose that $n \in \mathbb{N}$, $(c_j)_{j=1}^n \in \mathbb{R}_{\geq 0}^n$ and $(a_j)_{j=1}^n \in G^n$ are such that

$$f_1 + f_2 + \delta f_0 \leq \sum_{j=1}^n c_j L_{\iota(x_j)}(h).$$

Then

$$\begin{aligned} f_i(x) &= (f_1(x) + f_2(x) + \delta f_0(x))u_i(x) \leq \sum_{j=1}^n c_j h(\iota(x_j) * x)u_i(x) \\ &\leq \sum_{j=1}^n c_j h(\iota(x_j) * x)(u_i(x_j) + \delta), \end{aligned}$$

where the last inequality comes from the assumption that $\iota(x_j) * x \in V$ implies $|u_i(x) - u_i(x_j)| \leq \delta$. Hence,

$$(f_1 : h) + (f_2 : h) \leq \sum_{j=1}^n c_j (u_1(x_j) + \delta + u_2(x_j) + \delta) \leq \sum_{j=1}^n c_j (1 + 2\delta),$$

where the second inequality comes from

$$\forall x \in G, \quad u_1(x) + u_2(x) \leq 1.$$

Taking the infimum, we get

$$(f_1 : h) + (f_2 : h) \leq (1 + 2\delta)(f_1 + f_2 + \delta f_0 : h) \leq (1 + 2\delta)((f_1 + f_2 : h) + \delta(f_0 : h)),$$

or equivalently

$$\begin{aligned} I_h(f_1) + I_h(f_2) &\leq (1 + 2\delta)(I_h(f_1 + f_2) + \delta I_h(f_0)) \\ &= I_h(f_1 + f_2) + \delta(2I_h(f_1 + f_2) + I_h(f_0) + 2\delta I_h(f_0)). \end{aligned}$$

Taking δ sufficiently small such that

$$\delta(2I_h(f_1 + f_2) + I_h(f_0) + 2\delta I_h(f_0)) \leq \varepsilon,$$

we get the desired result. □

We now continue with the proof of the theorem. Let B be the family of subsets of $C_c^+(G)$ of the form

$$\Theta(V) := \{h \in C_c^+(G) \mid h \subseteq V\},$$

where V is a neighborhood of the neutral element of G . Note that, if V_1 and V_2 are neighborhoods of the neutral elements of G , then

$$\Theta(V_1) \cap \Theta(V_2) = \Theta(V_1 \cap V_2).$$

Therefore, $\mathcal{B}$ forms a filter on $C_c^+(G)$. Let U be an ultrafilter containing B . For any $f \in C_c^+(G)$, since $I_h(f)$ takes value in a compact interval $[(g : f)^{-1}, (f : g)]$ when h varies, the family $(I_h(f))_{h \in C_c^+(G)}$ has a limit along the ultrafilter U . We denote by $I(f)$ the limit

$$\lim_U I_h(f).$$

Note that for any h and f in $C_c^+(G)$ and any $\lambda > 0$, one has

$$I_h(\lambda f) = \lambda I_h(f).$$

Taking the limit along U , we get

$$I(\lambda f) = \lambda I(f).$$

Similarly, for any $a \in G$ one has

$$I_h(L_a(f)) = I_h(f).$$

Taking the limit along U , we get

$$I(L_a(f)) = I(f).$$

Let f_1 and f_2 be elements of $C_c^+(G)$, by Lemma 5.5.6, for any $\varepsilon > 0$, there exists a neighborhood V of the neutral element of G such that

$$I_h(f_1) + I_h(f_2) - \varepsilon \leq I_h(f_1 + f_2) \leq I_h(f_1) + I_h(f_2)$$

for any $h \in \Theta(V)$. Therefore, by taking the limit along U , we get

$$I(f_1 + f_2) = I_h(f_1) + I_h(f_2).$$

Finally, since any element $f \in C_c(G)$ can be written as the difference of two elements in $C_c^+(G)$, we obtain that the mapping $I : C_c^+(G) \rightarrow \mathbb{R}_{>0}$ extends to a positive linear functional on $C_c(G)$. $\square$

5.5.7 Remark. Let $(G, *, \mathcal{T})$ be a locally compact Hausdorff topological group that has a countable topological basis. Let $I : C_c(G) \rightarrow \mathbb{R}$ be a positive linear functional that defines a left Haar measure, and μ_I be the Borel measure corresponding to I . Then the set H of bounded Borel measurable mappings $f : G \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\forall a \in G, \quad \int_G f(x) \mu_I(dx) = \int_G f(a * x) \mu_I(dx)$$

forms a λ -family containing $C_c^+(G)$. Since $C - c^+(G)$ is stable under multiplication, by the monotone class theorem, H is equal to the set of all bounded Borel measurable mappings from G to $\mathbb{R}_{\geq 0}$. By the monotone convergence theory of Beppo Levi, we deduce that, for any Borel measurable mapping $f : G \rightarrow [0, +\infty]$, one has

$$\forall a \in G, \quad \int_G f(x) \mu_I(dx) = \int_G f(a * x) \mu_I(dx).$$

The measure μ_I is called a *left Haar measure* of G .

5.5.8 Proposition. Let $(G, *, \mathcal{T})$ be a locally compact Hausdorff topological group, that has a countable topological basis. If I_1 and I_2 are positive linear functionals on $C_c(G)$, that define Haar measures. Then there exists $\lambda > 0$ such that $I_2 = \lambda I_1$.

Proof. Let μ_1 and μ_2 be Borel measure on G corresponding to I_1 and I_2 respectively. We fix a compact neighborhood C of the neutral element of G such that $\iota(C) = C$.

Let f and g be elements of $C_c^+(G)$. Let $\varepsilon > 0$. By Lemma 5.5.4, there exists an open neighborhood V of the neutral element of G such that $\iota(V) = V$ and that, for any $x \in G$ and any $a \in V$, one has

$$|f(x * a) - f(a * x)| \leq \varepsilon, \quad |g(x * a) - g(a * x)| \leq \varepsilon.$$

Let $h \in C_c^+(G)$ with $\text{Supp}(h) \subseteq V$. By replacing h by $h + h \circ \iota$, we may assume that $h \circ \iota = h$. Thus,

$$I_1(h)I_2(f) = \int_G \int_G h(y)f(x) \mu_2(dx) \mu_1(dy).$$

Since μ_1 is a left Haar measure, one has

$$I_1(h)I_2(f) = \int_G \int_G h(y)f(y * x) \mu_2(dx) \mu_1(dy).$$

Similarly,

$$\begin{aligned}
 I_1(f)I_2(h) &= \int_G \int_G h(x)f(y) \mu_1(dx)\mu_2(dy) \\
 &= \int_G \int_G h(\iota(y) * x)f(y) \mu_1(dx)\mu_2(dy) \\
 &= \int_G \int_G h(\iota(x) * y)f(y) \mu_1(dx)\mu_2(dy) \\
 &= \int_G \int_G h(\iota(x) * y)f(y) \mu_2(dy)\mu_1(dx) \\
 &= \int_G \int_G h(y)f(x * y) \mu_2(dy)\mu_1(dx),
 \end{aligned}$$

where the third equality comes from the assumption $h \circ \iota = h$, and the fourth equality comes from Fubini's theorem. Therefore, we get

$$\begin{aligned}
 |I_1(h)I_2(f) - I_1(f)I_2(h)| &\leq \int_G \int_G h(y)|f(x * y) - f(y * x)| \mu_2(dy)\mu_1(dx) \\
 &\leq \varepsilon \mu_1(A)I_2(h),
 \end{aligned}$$

where $A = (\text{Supp}(f) * C) \cup (C * \text{Supp}(f))$. Similarly,

$$|I_1(h)I_2(g) - I_1(g)I_2(h)| \leq \varepsilon \mu_1(B)I_2(h),$$

where $B = (\text{Supp}(g) * C) \cup (C * \text{Supp}(g))$. Therefore, we obtain

$$\left| \frac{I_1(h)}{I_2(h)} - \frac{I_1(f)}{I_2(f)} \right| \leq \frac{\varepsilon \mu_1(A)}{I_2(f)}, \quad \left| \frac{I_1(h)}{I_2(h)} - \frac{I_1(g)}{I_2(g)} \right| \leq \frac{\varepsilon \mu_1(B)}{I_2(g)},$$

which leads to

$$\left| \frac{I_1(f)}{I_2(f)} - \frac{I_1(g)}{I_2(g)} \right| \leq \varepsilon \left| \frac{\mu_1(A)}{I_2(f)} + \frac{\mu_1(B)}{I_2(g)} \right|.$$

Since $\varepsilon > 0$ is arbitrary, we obtain

$$\frac{I_1(f)}{I_2(f)} = \frac{I_1(g)}{I_2(g)},$$

which concludes the statement. $\square$

5.5.9 Remark. Let $(G, *, \mathcal{T})$ be a topological group. Then the opposite group $(G, \tilde{*})$ equipped with the same topology $\mathcal{T}$ also forms a topological group. Therefore, we deduce from Theorem 5.5.5 and Proposition 5.5.8 the existence and the uniqueness up to a positive factor of a positive linear functional I from $C_c(G)$ to $\mathbb{R}$ that defines a right Haar measure μ_I . Note that for any Borel measurable mapping $f : G \rightarrow [0, +\infty]$, one has

$$\forall a \in G, \quad \int_G f(x * a) \mu_I(dx) = \int_G f(x) \mu_I(dx).$$

5.6 Change of variables

5.6.1 Definition. Let $(G, *)$ be a group. If a and b are two elements of G , we call *commutator* of a and b the element

$$[a, b]_* := a * b * \iota(a) * \iota(b).$$

Note that $a * b = b * a$ if and only if $[a, b]_*$ is the neutral element of G .

We call the *commutator subgroup* of G the subgroup of G generated by the commutators, denoted as $[G, G]_*$.

5.6.2 Notation. Let $(G, *)$ be a group. If $g \in G$, we denote by $C_g : G \rightarrow G$ the mapping that sends $x \in G$ to $g * x * \iota(g)$. Note that

$$C_g(x * y) = g * (x * y) * \iota(g) = g * x * \iota(g) * g * y * \iota(g) = C_g(x) * C_g(y).$$

Hence C_g is a homomorphism of groups, called the *conjugacy* by g .

5.6.3 Proposition. Let $(G, *)$ be a group and a, b and G be elements of G . One has

$$\begin{aligned}\iota([a, b]_*) &= [b, a]_*, \\ C_g([a, b]_*) &= [C_g(a), C_g(b)]_*.\end{aligned}$$

In particular, $[G, G]_*$ is a normal subgroup of G .

Proof. One has

$$\iota([a, b]_*) = \iota(\iota(b)) * \iota(\iota(a)) * \iota(b) * \iota(a) = [b, a]_*.$$

Moreover, since C_g is a homomorphism of groups,

$$\begin{aligned}C_g([a, b]_*) &= C_g(a * b * \iota(a) * \iota(b)) \\ &= C_g(a) * C_g(b) * \iota(C_g(a)) * \iota(C_g(b)) \\ &= [C_g(a), C_g(b)]_*.\end{aligned}$$

The first equality shows that any element of $[G, G]_*$ is a composition of commutators. Hence, the second equality shows that, for any $x \in [G, G]_*$ and any $g \in G$, one has $C_g(x) \in [G, G]_*$. Hence $[G, G]_*$ is a normal subgroup of G . $\square$

5.6.4 Definition. Let K be a field and V be a finite-dimensional vector space over K . We denote by $\text{GL}(V)$ the group of all K -linear bijections from V to V . It is called the *general linear group* of V . Note that the mapping

$$\det : \text{GL}(V) \longrightarrow K^\times$$

is a group homomorphism. We denote by $\text{SL}(V)$ its kernel, called the *special linear group* of V .

5.6.5 Proposition. *Let $(G, \star)$ be a group. The functor from the category of all commutative groups to the category of sets, that sends any commutative group $(H, \star)$ to the set of homomorphism of groups from G to H , is representable by the quotient group $G/[G, G]_\star$.*

Proof. Let $\pi : G \rightarrow G/[G, G]_\star$ be the homomorphism of projection. For any elements a and b of G , one has

$$\pi([a, b]_\star) = [\pi(a), \pi(b)]_\star$$

identifies with the neutral element of $G/[G, G]_\star$. Hence $\pi(a) \star \pi(b) = \pi(b) \star \pi(a)$. Since π is surjective, we obtain that $(G/[G, G]_\star, \star)$ is a commutative group.

Let $f : G \rightarrow H$ be a homomorphism of groups from G to a commutative group $(H, \star)$. For any elements a and b of G , one has

$$f([a, b]_\star) = f(a \star b \star \iota(a) \star \iota(b)) = [f(a), f(b)]_\star.$$

Since H is a commutative group, $[f(a), f(b)]_\star$ is the neutral element of H . Hence f factorizes in a unique way as

$$G \xrightarrow{\pi} G/[G, G]_\star \xrightarrow{\tilde{f}} H.$$

□

5.6.6 Theorem. *Let K be a field and V be a finite-dimensional vector space over K . Assume that, either $\dim_K(V) \neq 2$, or K is not isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Then the following equality holds:*

$$[\mathrm{GL}(V), \mathrm{GL}(V)]_\circ = \mathrm{SL}(V). \quad (5.1)$$

Proof. Since $\det : \mathrm{GL}(V) \rightarrow K^\times$ is a homomorphism of groups from $\mathrm{GL}(V)$ to a commutative group, we obtain that $\det$ sends elements of $[\mathrm{GL}(V), \mathrm{GL}(V)]_\circ$ to 1. Hence $[\mathrm{GL}(V), \mathrm{GL}(V)]_\circ$ is contained in the special linear group of V .

If V is the zero vector space, then $\mathrm{GL}(V)$ and $\mathrm{SL}(V)$ are both the group of one element. Hence the equality (5.1) holds. If V is of dimension 1, then $\det : \mathrm{GL}(V) \rightarrow K^\times$ is an isomorphism, and $\mathrm{SL}(V)$ and $[\mathrm{GL}(V), \mathrm{GL}(V)]_\circ$ are both the subgroup formed by the neutral element. Hence the equality (5.1) also holds.

In the rest of the proof, we fix a basis $(e_i)_{i=1}^n$ of V over K . For distinct elements i and j in $\{1, \dots, n\}$ and $\lambda \in K$, we denote by $S_{i,j}(\lambda)$ the K -linear automorphism that sends $x := a_1e_1 + \dots + a_ne_n$ to

$$x + \lambda a_j e_i.$$

In other words, one has

$$S_{i,j}(\lambda)(e_j) = e_j + \lambda e_i,$$

and

$$\forall k \in \{1, \dots, n\} \setminus \{j\}, \quad S_{i,j}(\lambda)(e_k) = e_k.$$

Note that $S_{i,j}(\lambda)$ belongs to $\text{SL}(V)$ since

$$S_{i,j}(\lambda)(e_1) \wedge \dots \wedge S_{i,j}(\lambda)(e_n) = e_1 \wedge \dots \wedge e_{j-1} \wedge (e_j + \lambda e_i) \wedge e_{j+1} \wedge \dots \wedge e_n = e_1 \wedge \dots \wedge e_n.$$

Moreover, by definition, the inverse of $S_{i,j}(\lambda)$ is $S_{i,j}(-\lambda)$.

For any bijection $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, we let $P_\sigma : V \rightarrow V$ be the K -linear mapping that sends $a_1 e_1 + \dots + a_n e_n$ to

$$a_{\sigma(1)} e_1 + \dots + a_{\sigma(n)} e_n = a_1 e_{\sigma^{-1}(1)} + \dots + a_n e_{\sigma^{-1}(n)}.$$

Then one has $P_{\sigma^{-1}} = P_\sigma^{-1}$. Moreover, $P_{\sigma^{-1}} \circ S_{i,j}(\lambda) \circ P_\sigma$ sends $a_1 e_1 + \dots + a_n e_n$ to

$$P_{\sigma^{-1}}(a_{\sigma(1)} e_1 + \dots + a_{\sigma(n)} e_n + \lambda a_{\sigma(j)} e_i) = a_{\sigma(1)} e_{\sigma(1)} + \dots + a_{\sigma(n)} e_{\sigma(n)} + \lambda a_{\sigma(j)} e_{\sigma(i)}.$$

Hence

$$P_{\sigma^{-1}} \circ S_{i,j}(\lambda) \circ P_\sigma = S_{\sigma(i), \sigma(j)}(\lambda) \quad (5.2)$$

5.6.7 Lemma. *Any element of $\text{SL}(V)$ can be written as a composition of elements of the form $S_{i,j}(\lambda)$. Moreover, any element $f \in \text{GL}(V)$ can be decomposed as $f = g \circ h$, where $g \in \text{SL}(V)$ and h is of the form*

$$h(a_1 e_1 + a_2 e_2 + \dots + a_n e_n) = a_1 \det(f) e_1 + a_2 e_2 + \dots + a_n e_n$$

Proof. Let Θ be the family of $f \in \text{GL}(V)$ that can be written as $f = g \circ h$ with $g \in \text{SL}(V)$ and

$$h(a_1 e_1 + a_2 e_2 + \dots + a_n e_n) = a_1 \det(f) e_1 + a_2 e_2 + \dots + a_n e_n.$$

We prove by induction on n that $\Theta = \text{GL}(V)$, namely any element $f \in \text{GL}(V)$ belongs to Θ . The case where $n = 0$ or $n = 1$ is trivial. In what follows, we assume that $n \geq 2$ and that the induction hypothesis holds for the vector subspace $W = \text{Span}_K(\{e_1, \dots, e_{n-1}\})$ and the basis $(e_i)_{i=1}^{n-1}$.

Note that, for any $g \in \text{GL}(V)$, any distinct elements i and j in $\{1, \dots, n\}$, and any $\lambda \in K$, one has

$$g \in \Theta \iff S_{i,j}(\lambda) \circ g \in \Theta.$$

Therefore, in the reasoning, we may replace f by $S_{i,j}(\lambda) \circ f$ without loss of generality.

For any $f \in \text{GL}(V)$ and any $(i, j) \in \{1, \dots, n\}^2$, let $c_{i,j}(f)$ be the coefficient of e_j in the decomposition of $f(e_i)$ into a linear combination of $e_1, \dots, e_n$. Since f

is invertible, $c_{n,1}(f), \dots, c_{n,n}(f)$ cannot be 0 at the same time. First, consider the case when $c_{n,j}(f) \neq 0$ for some $j \in \{1, \dots, n-1\}$. Let

$$\lambda = \frac{1 - c_{n,n}(f)}{c_{n,j}(f)}.$$

One has

$$(S_{n,j}(\lambda) \circ f)(e_n) = f(e_n) + \lambda c_{n,j}(f)e_n = c_{n,1}(f)e_1 + \dots + c_{n,n-1}(f)e_{n-1} + e_n.$$

In the case where $c_{n,1}(f) = \dots = c_{n,n-1}(f) = 0$, by replacing f by $S_{1,n}(1) \circ f$ we can reduce the problem to the previous case. Hence, without loss of generality, we may assume that $c_{n,n}(f) = 1$. Furthermore, by replacing f by

$$S_{1,n}(-c_{n,1}(f)) \circ \dots \circ S_{n-1,n}(-c_{n,n-1}(f)) \circ f$$

we may assume that

$$c_{n,1}(f) = \dots = c_{n,n-1}(f) = 0.$$

Let $g: W \rightarrow W$ be the automorphism defined as

$$g(a_1e_1 + \dots + a_{n-1}e_{n-1}) = \sum_{i=1}^{n-1} a_i \sum_{j=1}^{n-1} c_{i,j}(f)e_j.$$

Applying the induction hypothesis to g , we reduce the problem to the case where $f(e_n) = e_n$, $f(e_1) = c_{1,1}(f)e_1 + c_{1,n}(f)e_n$, and

$$\forall i \in \{2, \dots, n-1\}, \quad f(e_i) = e_i + c_{i,n}(f)e_n.$$

Note that

$$\begin{aligned} & f(e_1) \wedge \dots \wedge f(e_n) \\ &= (c_{1,1}(f)e_1 + c_{1,n}(f)e_n) \wedge (e_2 + c_{2,n}(f)e_n) \wedge \dots \wedge (e_{n-1} + c_{n-1,n}(f)e_n) \wedge e_n \\ &= c_{1,1}(f)e_1 \wedge \dots \wedge e_n. \end{aligned}$$

Hence $\det(f) = c_{1,1}(f)$. Finally, the mapping

$$S_{n,1}(-c_{1,n}(f)/c_{1,1}(f)) \circ S_{n,2}(-c_{2,n}(f)) \circ \dots \circ S_{n,n-1}(-c_{n-1,n}(f)) \circ f$$

sends $a_1e_1 + \dots + a_ne_n$ to

$$a_1 \det(f)e_1 + a_2e_2 + \dots + a_ne_n,$$

as required. □

We now come back to the proof of the theorem. For any $b \in K^\times$, we denote by D_b the element of $\text{GL}(V)$ that sends $a_1e_1 + \cdots + a_ne_n$ to $a_1be_1 + a_2e_2 + \cdots + a_ne_n$.

If $n = 2$ and K is not isomorphic to $\mathbb{Z}/2\mathbb{Z}$, then there exists $b \in K \setminus \{0, 1\}$. In this case

$$[S_{1,2}(1), D_b]_\circ = S_{1,2}(1) \circ D_b \circ S_{1,2}(-1) \circ D_{b^{-1}}$$

sends $a_1e_1 + a_2e_2$ to

$$\begin{aligned} & (S_{1,2}(1) \circ D_b \circ S_{1,2}(-1))(a_1b^{-1}e_1 + a_2e_2) \\ &= (S_{1,2}(1) \circ D_b)(a_1b^{-1}e_1 + a_2e_2 - a_2e_1) \\ &= S_{1,2}(1)((a_1 - ba_2)e_1 + a_2e_2) \\ &= (a_1 + (1 - b)a_2)e_1 + a_2e_2 = S_{1,2}(1 - b)(a_1e_1 + a_2e_2). \end{aligned}$$

Hence

$$[S_{1,2}(1), D_b]_\circ = S_{1,2}(1 - b).$$

If $n > 2$, then $[S_{1,3}(1), S_{3,2}(1)]_\circ$ sends $x = a_1e_1 + \cdots + a_ne_n$ to

$$\begin{aligned} & (S_{1,3}(1) \circ S_{3,2}(1) \circ S_{1,3}(-1) \circ S_{3,2}(-1))(x) \\ &= (S_{1,3}(1) \circ S_{3,2}(1) \circ S_{1,3}(-1))(x - a_2e_3) \\ &= (S_{1,3}(1) \circ S_{3,2}(1))(x - a_2e_3 - (a_3 - a_2)e_1) \\ &= S_{1,3}(1)(x - a_2e_3 - (a_3 - a_2)e_1 + a_2e_3) \\ &= x - (a_3 - a_2)e_1 + a_3e_1 = x + a_2e_1 = S_{1,2}(1)(x). \end{aligned}$$

Hence, in both cases there exists $t \in K^\times$ such that

$$S_{1,2}(t) \in [\text{GL}(V), \text{GL}(V)]_\circ.$$

For general $\lambda \in K^\times$, one has

$$S_{1,2}(\lambda) = D_{\lambda t^{-1}} \circ S_{1,2}(t) \circ D_{\lambda^{-1}t}$$

since the latter sends $x = a_1e_1 + \cdots + a_ne_n$ to

$$\begin{aligned} & (D_{\lambda t^{-1}} \circ S_{1,2}(t))(\lambda^{-1}ta_1e_1 + a_2e_2 + \cdots + a_ne_n) \\ &= D_{\lambda t^{-1}}((\lambda^{-1}ta_1 + ta_2)e_1 + a_2e_2 + \cdots + a_ne_n) \\ &= (a_1 + \lambda a_2)e_1 + a_2e_2 + \cdots + a_ne_n \\ &= S_{1,2}(\lambda)(a_1e_1 + \cdots + a_ne_n). \end{aligned}$$

Finally, for a general pair (i, j) of distinct elements in $\{1, \dots, n\}$, there is a bijection $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ that sends 1 to i and 2 to j . By (5.2) we obtain that $S_{i,j}(\lambda) \in [\text{GL}(V), \text{GL}(V)]_\circ$. Therefore, by Lemma 5.6.7 we deduce

$$[\text{GL}(V), \text{GL}(V)]_\circ = \text{SL}(V).$$

□

5.6.8 Corollary. *We keep the notation and the assumptions of Theorem. Let $\varphi : \text{GL}(V) \rightarrow K^\times$ be a homomorphism of groups. There exists an endomorphism of groups $\psi : K^\times \rightarrow K^\times$ such that $\varphi = \psi \circ \det$.*

Proof. By Theorem 5.6.6, $\text{SL}(V)$ is the commutator subgroup of $\text{GL}(V)$. Hence φ sends elements of $\text{SL}(V)$ to 1.

We fix a basis $(e_i)_{i=1}^n$ of V over K . For any $b \in K^\times$, let $D_b : V \rightarrow V$ be the mapping that sends $a_1e_1 + \cdots + a_ne_n$ to

$$a_1be_1 + a_2be_2 + \cdots + a_nbe_n.$$

Let $\psi : K^\times \rightarrow K^\times$ the mapping that sends $b \in K^\times$ to $\varphi(D_b)$. It is a homomorphism of groups since

$$\psi(bb') = \varphi(D_{bb'}) = \varphi(D_b \circ D_{b'}) = \varphi(D_b)\varphi(D_{b'}) = \psi(b)\psi(b').$$

By Lemma 5.6.7, for any $f \in \text{GL}(V)$, there exists $g \in \text{SL}(V)$ such that

$$f = g \circ D_{\det(f)}.$$

Hence

$$\varphi(f) = \varphi(g)\varphi(D_{\det(f)}) = \psi(\det(f)).$$

□

5.6.9 Theorem. *Let V be a finite-dimensional vector space over $\mathbb{R}$, and ν be a Haar measure on V . For any $f \in \text{GL}(V)$ one has*

$$f_*(\nu) = |\det(f)|^{-1}\nu.$$

In other words, for any Borel subset A of V , one has

$$\nu(f(A)) = |\det(f)|\nu(A).$$

Proof. Let $(e_i)_{i=1}^n$ be a basis of V and $g : \mathbb{R}^n \rightarrow V$ be the mapping that sends $(a_1, \dots, a_n)$ to $a_1e_1 + \cdots + a_ne_n$. Note that g is an isomorphism of topological groups (namely an isomorphism of groups which is a homeomorphism). Let η be the Lebesgue measure on $\mathbb{R}$. Then $\eta^{\otimes n}$ is a Haar measure on $\mathbb{R}^n$, and $g_*(\eta^{\otimes n})$ is a Haar measure on V . Since Haar measures on V differ by a positive multiplicative factor, we may assume without loss of generality that $\nu = g_*(\eta^{\otimes n})$.

For any $u \in C_c(V)$ and any $y \in V$, one has

$$\begin{aligned} \int_V u(x+y) f_*(\nu)(dx) &= \int_V u(f(x)+y) \nu(dx) = \int_V u(f(x+f^{-1}(y))) \nu(dx) \\ &= \int_V u(f(x)) \nu(dx) = \int_V u(x) f_*(\nu)(dx). \end{aligned}$$

Therefore, $f_*(\nu)$ is a Haar measure on V . There then exists an element $\varphi(f) \in \mathbb{R}_{>0}$ such that $f_*(\nu) = \varphi(f)\nu$. Moreover, by definition, for any elements f_1 and f_2 in $\text{GL}(V)$, one has

$$(f_2 \circ f_1)_*(\nu) = f_{2,*}(f_{1,*}(\nu)) = \varphi(f_2)\varphi(f_1)\nu.$$

Hence φ is a morphism of groups from $\text{GL}(V)$ to $\mathbb{R}_{>0}$. By Corollary 5.6.8, there exists a homomorphism of groups $\psi : \mathbb{R}^\times \rightarrow \mathbb{R}_{>0}$ such that $\varphi = \psi \circ \det$. In particular, if $f \in \text{SL}(V)$ one has $\varphi(f) = 1$.

Let b be an element of V . For any $u \in C_c(V)$ one has

$$\begin{aligned} \int_V u(x) D_{b,*}(g_*(\eta^{\otimes n})) &= \int_{\mathbb{R}^n} u(D_b(g(t_1, \dots, t_n))) \eta^{\otimes n}(d(t_1, \dots, t_n)) \\ &= \int_{\mathbb{R}^n} u(bt_1 e_1 + t_2 e_2 + \dots + t_n e_n) \eta^{\otimes n}(d(t_1, \dots, t_n)) \\ &= \int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} u(bt_1 e_1 + t_2 e_2 + \dots + t_n e_n) \eta(dt_1) \right) \eta^{\otimes(n-1)}(d(t_2, \dots, t_n)). \end{aligned}$$

5.6.10 Lemma. *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a non-negative Borel mapping. For any $b \in \mathbb{R}^\times$ one has*

$$\int_{\mathbb{R}} h(bt) dt = |b|^{-1} \int_{\mathbb{R}} h(t) dt.$$

Proof. The above argument applied to the additive group $\mathbb{R}$ shows that there exists a homomorphism $\lambda : \mathbb{R}^\times \rightarrow \mathbb{R}_{>0}$ such that

$$\int_{\mathbb{R}} h(bt) dt = \lambda(b) \int_{\mathbb{R}} h(t) dt$$

for any $b \in \mathbb{R}_{>0}$ and any non-negative Borel mapping $h : \mathbb{R} \rightarrow \mathbb{R}$. Fixing one $h \in C_c^+(\mathbb{R})$, by dominated convergence theorem we obtain that λ is continuous. By Corollary 5.2.26, there exists $\theta \in \mathbb{R}$ such that $\lambda(b) = |b|^\theta$. Finally, we consider the case where $b = 2$ and $h = \mathbb{1}_{[0,2[}$ to get

$$\int_{\mathbb{R}} \mathbb{1}_{[0,1[}(t) dt = \lambda(2) \int_{\mathbb{R}} \mathbb{1}_{[0,2[}(t) dt,$$

so that $\lambda(2) = 1/2$. Hence

$$\forall b \in \mathbb{R}, \quad \lambda(b) = |b|^{-1}.$$

□

By the above lemma, we get

$$\int_V u(x) D_{b,*}(g_*(\eta^{\otimes n})) = |b|^{-1} \int_V u(x) g_*(\eta^{\otimes n})(dx).$$

Therefore, by Lemma 5.6.7, we obtain

$$\forall f \in \text{GL}(V), \quad \psi(f) = |\det(f)|^{-1},$$

as required. $\square$

5.6.11 Theorem (Change of variables). *Let E be a finite dimensional vector space over $\mathbb{R}$ and ν be a Haar measure on E . Let U and V be open subsets of E , and $\varphi : U \rightarrow V$ be a C^1 -diffeomorphism. Then, for any Borel measurable mapping $f : V \rightarrow \mathbb{R}_{\geq 0}$, one has*

$$\int_V f(y) \nu(dy) = \int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx).$$

Proof. By choosing a basis of E over $\mathbb{R}$, we may assume without loss of generality that E is of the form $\mathbb{R}^n$, where n is a natural number. The case where $n = 0$ is trivial. In the rest of the proof, we assume that $n \geq 1$. We equip $\mathbb{R}^n$ with the sup norm $\|\cdot\|$, which is defined as

$$\|(x_1, \dots, x_n)\| := \max\{|x_1|, \dots, |x_n|\}.$$

Let Θ be the set of hyperrectangles of the form

$$A = \prod_{i=1}^n]a_i, b_i]$$

such that $\{a_1, \dots, a_n, b_1, \dots, b_n\} \subseteq \mathbb{Q}$ and $\overline{A} \subseteq U$. If

$$b_1 - a_1 = \dots = b_n - a_n,$$

then we say that A is a hypercube, and the difference $b_i - a_i$ is called the *side length* of the hypercube.

5.6.12 Lemma. *For any hypercube A in Θ , one has*

$$\nu(\varphi(A)) \leq \sup_{x \in A} \|D\varphi(x)\|^n \nu(A). \quad (5.3)$$

Proof. Let ℓ be the side length of the hypercube A . By the mean value inequality, for any y and y' in A , one has

$$\|\varphi(y) - \varphi(y')\| \leq \left(\sup_{x \in A} \|D\varphi(x)\| \right) \|y - y'\|.$$

Therefore, $\varphi(A)$ is contained in a hypercube of sidelength

$$\left(\sup_{x \in A} \|D\varphi(x)\| \right) \ell.$$

Therefore, the inequality holds. $\square$

5.6.13 Lemma. *For any hyperrectangle A in Θ , one has*

$$\nu(\varphi(A)) \leq \int_A |\det(D\varphi(x))| \nu(dx).$$

Proof. Let ε be a positive real number. Since the vertices of A have rational coordinates, we can decompose A into a disjoint union of finitely many hypercubes $(A_i)_{i \in I}$ in Θ , such that, for any $i \in I$ and any $(x, y) \in A_i^2$, one has

$$\|D\varphi(y)^{-1} \circ D\varphi(x)\| \leq (1 + \varepsilon)^{1/n}.$$

This comes from the fact that $D\varphi$ is continuous and hence is uniformly continuous on the compact subset $\overline{A}$ of U . Indeed, one has

$$\begin{aligned} \|D\varphi(y)^{-1} \circ D\varphi(x)\| &\leq 1 + \|D\varphi(y)^{-1} \circ D\varphi(x) - \text{Id}_{\mathbb{R}^n}\| \\ &\leq 1 + \|D\varphi(y)^{-1}\| \cdot \|D\varphi(x) - D\varphi(y)\|. \end{aligned}$$

By the intermediate value theorem, for any $i \in I$ there exists $x_i \in A_i$ such that

$$|\det(D\varphi(x_i))| \nu(A_i) = \int_{A_i} |\det(D\varphi(x))| \nu(dx).$$

By Lemma 5.6.12, one has

$$\nu(D\varphi(x_i)^{-1}(\varphi(A_i))) \leq \sup_{x \in A_i} \|D\varphi(x_i)^{-1} \circ D\varphi(x)\|^n \nu(A_i) \leq (1 + \varepsilon) \nu(A_i).$$

Since

$$\varphi(A_i) = D\varphi(x_i)(D\varphi(x_i)^{-1}(\varphi(A_i))),$$

by Theorem 5.6.9 we get

$$\begin{aligned} \nu(\varphi(A_i)) &= |\det(D\varphi(x_i))| \nu(D\varphi(x_i)^{-1}(\varphi(A_i))) \leq |\det(D\varphi(x_i))| (1 + \varepsilon) \nu(A_i) \\ &= (1 + \varepsilon) \int_{A_i} |\det(D\varphi(x))| \nu(dx). \end{aligned}$$

Taking the sum with respect to $i \in I$, we obtain

$$\nu(\varphi(A)) \leq (1 + \varepsilon) \int_A |\det(D\varphi(x))| \nu(dx).$$

Since ε is arbitrary, we obtain the desired inequality. $\square$

Note that Θ is countable and forms a set semiring on U . Moreover, for any non-empty open subset U' of U , one has

$$U' = \bigcup_{\substack{A \in \Theta \\ A \subseteq U'}} A.$$

Therefore, U' can be written as the union of a countable family of pairwise disjoint elements of Θ . By the monotone convergence theorem of Beppo Levi, we obtain that, for any non-empty open subset U' of U , one has

$$\nu(\varphi(U')) \leq \int_{U'} |\det(D\varphi(x))| \nu(dx).$$

Since $\nu \circ \varphi$ and

$$A \mapsto \int_A |\det(D\varphi(x))| \nu(dx)$$

are Radon measures on U , by Proposition 5.4.3 (3), we obtain that, for any Borel subset A of U , one has

$$\nu(\varphi(A)) \leq \int_A |\det(D\varphi(x))| \nu(dx). \quad (5.4)$$

Therefore, for any non-negative Borel mapping $f : V \rightarrow \mathbb{R}$, one has

$$\int_V f(y) \nu(dy) \leq \int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx).$$

In fact, we can write f as the pointwise limit of the increasing sequence

$$f_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{f^{-1}([k/2^n, (k+1)/2^n])}.$$

Applying (5.4), we obtain

$$\begin{aligned} \int_V f_n(y) \nu(dy) &= \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \nu(f^{-1}([k/2^n, (k+1)/2^n])) \\ &\leq \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \int_{\varphi^{-1}(f^{-1}([k/2^n, (k+1)/2^n]))} |\det(D\varphi(x))| \nu(dx) \\ &= \int_U f_n(\varphi(x)) |\det(D\varphi(x))| \nu(dx). \end{aligned}$$

Taking the monotone limit when $n \rightarrow +\infty$, we get

$$\int_V f(y) \nu(dy) \leq \int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx).$$

Applying the inequality to φ^{-1} , we obtain that

$$\int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx) \leq \int_V f(y) |\det(D\varphi(\varphi^{-1}(y)))| |\det(D\varphi^{-1}(y))| \nu(dy).$$

Since

$$D\varphi^{-1}(y) = D\varphi(\varphi^{-1}(y))^{-1},$$

we obtain

$$\int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx) \leq \int_V f(y) \nu(dy).$$

Hence the equality

$$\int_V f(y) \nu(dy) = \int_U f(\varphi(x)) |\det(D\varphi(x))| \nu(dx)$$

holds. □

Exercises

Throughout the following exercises, n denotes a positive integer.

1. Let R be a commutative ring with unity, and $M(n, R)$ be the ring of $n \times n$ matrices over R . Define the general linear group of degree n over R as

$$\mathrm{GL}(n, R) := \{A \in M(n, R) \mid \det(A) \in R^\times\},$$

and the special linear group of degree n over R as

$$\mathrm{SL}(n, R) := \{A \in \mathrm{GL}(n, R) \mid \det(A) = 1_R\}.$$

Let $Z_n(R) := \{aI_n \mid a \in R^\times\} \subseteq \mathrm{GL}(n, R)$ be the subgroup of invertible scalar matrices, and define the projective general and special linear group of degree n over R as

$$\begin{aligned} \mathrm{PGL}(n, R) &:= \mathrm{GL}(n, R)/Z_n(R), \\ \mathrm{PSL}(n, R) &:= \mathrm{SL}(n, R)/(Z_n(R) \cap \mathrm{SL}(n, R)). \end{aligned}$$

- (a) Prove that the determinant map induces a group isomorphism

$$\mathrm{GL}(n, R)/\mathrm{SL}(n, R) \xrightarrow{\sim} R^\times.$$

- (b) Prove that $\mathrm{SL}(n, R)$ is a normal subgroup of $\mathrm{GL}(n, R)$.
- (c) Let p be a prime number, $\mathbb{F}_p$ be the finite field of p elements. Compute the cardinality of the following groups:

$$\mathrm{GL}(n, \mathbb{F}_p), \mathrm{SL}(n, \mathbb{F}_p), \mathrm{PGL}(n, \mathbb{F}_p), \mathrm{PSL}(n, \mathbb{F}_p).$$

2. Define the action of the group $(\mathbb{R}, +)$ on the space $\mathbb{R}$ by

$$\phi: (\mathbb{R}, +) \times \mathbb{R} \rightarrow \mathbb{R}, \quad \phi(a, x) = x + a.$$

- (a) Prove that ϕ defines a continuous group action, i.e., ϕ is a group action and is a continuous map.
 - (b) Prove that for $x \neq 0$, the volume form dx is invariant under the action of $\mathbb{R}$.
3. Define the action of the group $(\mathbb{R}^\times, \times)$ on the space $\mathbb{R} \setminus \{0\}$ by

$$\phi: (\mathbb{R}^\times, \times) \times \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}, \quad \phi(a, x) = ax.$$

- (a) Prove that ϕ defines a continuous group action, i.e., ϕ is continuous.
 - (b) Prove that for $x \in \mathbb{R} \setminus \{0\}$, the volume form $x^{-1}dx$ is invariant under the action of $\mathbb{R}^\times$.
4. Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, which is always endowed with the Euclidean topology. There is a natural group isomorphism $M(n, \mathbb{K}) \cong \mathbb{K}^{n^2}$, and we endow the topology on $M(n, \mathbb{K})$ by pulling back the topology from $\mathbb{K}^{n^2}$.

The Frobenius norm on $M(n, \mathbb{K})$ is defined by

$$\|A\| := \sqrt{\sum_{i,j} |A_{ij}|^2}.$$

For any matrix $A \in M(n, \mathbb{K})$, the sets

$$U(A, r) := \{B \in M(n, \mathbb{K}) \mid \|B - A\| \leq r\} \quad (r > 0)$$

forms a basis of open neighborhoods of A in $M(n, \mathbb{K})$.

- (a) Prove that the Frobenius norm is a norm on $M(n, \mathbb{K})$, i.e., for any $a \in \mathbb{K}$ and $A, B \in M(n, \mathbb{K})$, we have:

$$\begin{aligned} \|A + B\| &\leq \|A\| + \|B\|; \\ \|aA\| &= \|a\| \cdot \|A\|; \\ \|A\| = 0 &\iff A = 0. \end{aligned}$$
 - (b) Prove that for any $A, B \in M(n, \mathbb{K})$ we have $\|AB\| \leq \|A\| \cdot \|B\|$.
 - (c) Prove that for any matrix $A \in M(n, \mathbb{K})$, if $\|A\| < 1$, then the matrix $I + A$ is invertible.
5. Every matrix $A \in M(n, \mathbb{C})$ can be written uniquely as $A = X + Yi$, where $X, Y \in M(n, \mathbb{R})$. Define

$$\rho_n: M(n, \mathbb{C}) \rightarrow M(2n, \mathbb{R}), \quad \rho_n(X + Yi) = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}.$$

- (a) Prove that ρ_n is an injective homomorphism of $\mathbb{R}$ -algebras.
 - (b) Prove that for any $A \in M(n, \mathbb{C})$, we have $\|\rho_n(A)\| = \sqrt{2} \cdot \|A\|$.
 - (c) Prove that for any $A \in M(n, \mathbb{C})$, we have $\det(\rho_n(A)) = |\det(A)|^2$.
 - (d) Prove that ρ_n is continuous.
6. Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, endow $SL(n, \mathbb{K})$ and $GL(n, \mathbb{K})$ with the subspace topology of $M(n, \mathbb{K})$.

- (a) Prove that $GL(n, \mathbb{K})$ is a open subset of $M(n, \mathbb{K})$, $SL(n, \mathbb{K})$ is a closed subset of $M(n, \mathbb{K})$, and $GL(n, \mathbb{K})$ is a topological group with neutral element I_n .
- (b) The Heine-Borel theorem states that for any $m \geq 1$, a subspace of $\mathbb{R}^m$ is compact if and only if it is bounded and closed. Based on this theorem, prove that $GL(n, \mathbb{K})$ is Hausdorff and locally compact.
- (c) Prove that $\rho_n(GL(n, \mathbb{C})) \subseteq GL(2n, \mathbb{R})$, $\rho_n(SL(n, \mathbb{C})) \subseteq SL(2n, \mathbb{R})$, and the restriction $\rho_n|_{GL(n, \mathbb{C})}$ is a continuous group homomorphism.
- (d) Prove that $GL(n, \mathbb{C})$, $SL(n, \mathbb{C})$ and $SL(n, \mathbb{R})$ are path-connected, but $GL(n, \mathbb{R})$ is not connected.
- (e) Define

$$GL_n^+(\mathbb{R}) := \{A \in GL(n, \mathbb{R}) \mid \det(A) > 0\},$$

$$GL_n^-(\mathbb{R}) := \{A \in GL(n, \mathbb{R}) \mid \det(A) < 0\}$$

both endowed with the subspace topology. Prove that both of them are path-connected, and they are the two connected components of $GL(n, \mathbb{R})$.

7. We define the quotient space induced by a group action.

- (a) Let G be a group, and X be a topological space with [left] G -action. Define $x \sim y$ if and only if there exists some $g \in G$ such that $g \cdot x = y$, where $\cdot$ is the group action of G on X . Prove that $\sim$ is an equivalence relation on X . We define X/G as the quotient space of X by $\sim$.
- (b) Define the action of $\mathbb{Z}/2\mathbb{Z}$ on the cylinder $X = S^1 \times [0, 1]$ by

$$\phi: \mathbb{Z}/2\mathbb{Z} \times X \rightarrow X, \quad \phi(\bar{1}, (\theta, t)) = (\theta + \pi, t).$$

Describe the quotient space $X_1 := X/(\mathbb{Z}/2\mathbb{Z})$.

- (c) Define the action of $\mathbb{Z}/2\mathbb{Z}$ on the cylinder $X = S^1 \times [0, 1]$ by

$$\psi: \mathbb{Z}/2\mathbb{Z} \times X \rightarrow X, \quad \psi(\bar{1}, (\theta, t)) = (\theta + \pi, 1 - t).$$

Describe the quotient space $X_2 := X/(\mathbb{Z}/2\mathbb{Z})$.

8. A second-countable space is a topological space whose topology has a countable basis. This concept is crucial in the following proposition:

Proposition. Let G be a topological group, and X be a topological space with continuous left G -action which is transitive. Assume that both G and X are locally compact and second-countable. Let $x_0 \in X$ be an element of X and H be the stabilizer of x_0 . Then the mapping $f: G/H \rightarrow X$ sending gH to $g \cdot x_0$ is a homeomorphism.

- (a) Prove that any countable product of a second-countable space is second-countable.
 - (b) Prove that every subspace of a second-countable space is second-countable.
 - (c) Prove that subspaces of $\mathbb{R}^n$ and $\mathbb{C}^n$ are second-countable.
 - (d) Prove that a locally path-connected and connected topological space is path-connected.
9. Let $\tau \in \mathbb{C}$ such that $\text{Im}(\tau) > 0$. Define the action of $\Lambda := \mathbb{Z} + \tau\mathbb{Z}$ on $\mathbb{C}$ by

$$\phi: \Lambda \times \mathbb{C} \rightarrow \mathbb{C}, \quad \phi(a, z) = z + a.$$

Prove that ϕ defines a group action, and that $\mathbb{C}/\Lambda$ is homeomorphic to the torus $\mathbb{T} := S^1 \times S^1$, where $S^1 := \{z \in \mathbb{C} \mid |z| = 1\}$.

10. Let $\mathbb{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$ be the upper half-plane. For any matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{R})$ and $z \in \mathbb{H}$, define

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z := \frac{az + b}{cz + d}. \quad (*)$$

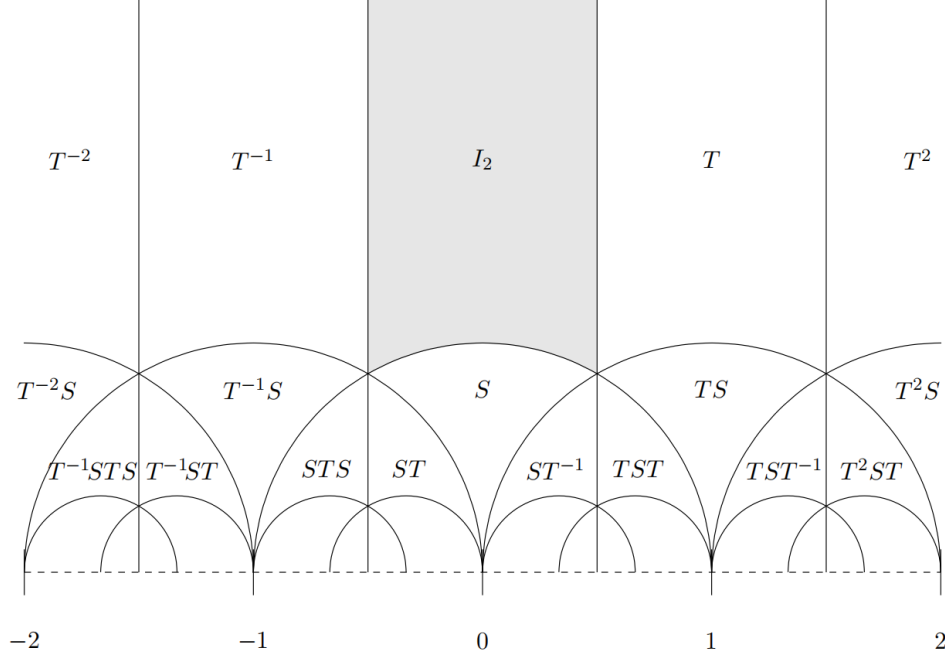
- (a) Prove that for any $A \in \text{SL}_2(\mathbb{R})$ and $z \in \mathbb{H}$, we have $A \cdot z \in \mathbb{H}$.
- (b) Prove that $(*)$ defines a group action of $\text{SL}_2(\mathbb{R})$ on $\mathbb{H}$.
- (c) For the coordinate $z = x + iy \in \mathbb{H}$ ($x, y \in \mathbb{R}$), define the volume form

$$\omega = \omega(z) = \frac{dx \wedge dy}{y^2}.$$

Prove that ω is invariant under the action of $\text{SL}_2(\mathbb{R})$ on $\mathbb{H}$, i.e., for any $A \in \text{SL}_2(\mathbb{R})$, we have $\omega(A \cdot z) = \omega(z)$.

- (d) Prove that $\mathbb{H} \cong \text{SL}_2(\mathbb{R})/\text{SO}(2)$, where $\text{SO}(2) := \{A \in \text{SL}_2(\mathbb{R}) \mid A^T A = 1\}$.

11. Continue to consider the action of $\mathrm{SL}_2(\mathbb{Z}) \subseteq \mathrm{SL}_2(\mathbb{R})$ on the upper half-plane $\mathbb{H}$ defined by $(*)$ in the preceding exercise. Define the fundamental region $F := \{z \in \mathbb{H} \mid |\mathrm{Re}(z)| \leq \frac{1}{2}, |z| \geq 1\}$.



- (a) Let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ be two elements of $\mathrm{SL}_2(\mathbb{Z})$. Prove that $S^2 = (ST)^3 = -I_2$.
- (b) Let $G = \langle S, T \rangle$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by S and T . Prove that for any $z \in \mathbb{H}$, there exists some $A \in G$ such that $A \cdot z \in F$.
- (c) Prove that for any z in the interior of F and $A \in \mathrm{SL}_2(\mathbb{Z})$, if $A \cdot z \in F$, then $A \in \{\pm I_2\}$.
- (d) Prove that $\mathrm{SL}_2(\mathbb{Z})$ is generated by S and T . (In fact, there are group presentations $\mathrm{SL}_2(\mathbb{Z}) \cong \langle x, y \mid x^4 = 1, x^2 = y^3 \rangle$ and $\mathrm{PSL}_2(\mathbb{Z}) \cong \langle x, y \mid x^2 = y^3 = 1 \rangle$.)
12. Let V be a n -dimensional real vector space equipped with a non-degenerate symmetric bilinear form $\beta: V \times V \rightarrow \mathbb{R}$. The isometry group of (V, β) is the set of linear transformations of V which preserve β , i.e.,

$$\mathrm{Isom}(V, \beta) := \{A \in \mathrm{GL}(V) \mid \beta(Ax, Ay) = \beta(x, y), \forall x, y \in V\}.$$

Let B be the matrix of β with respect to a basis $B = \{e_1, \dots, e_n\}$, then

$$\mathrm{Isom}(V, \beta) \xrightarrow{\sim} \mathrm{Isom}(n, B) := \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A^T B A = B\}.$$

- (a) If B_1, B_2 are two matrices of β with respect to two different bases B_1 and B_2 , then there is a continuous group isomorphism $\text{Isom}(n, B_1) \cong \text{Isom}(n, B_2)$. Describe this isomorphism explicitly.
- (b) Recall that there exists integers p, q ($0 \leq p, q \leq n$, $p + q = n$) determined by (V, β) , such that

$$B = I_{p,q} := \begin{pmatrix} I_p & 0 \\ 0 & -I_q \end{pmatrix}$$

for some choice of basis. In this case, β is said to be of signature (p, q) . Prove that the bilinear form determined by the quadratic form $q(X) := \det \begin{pmatrix} x_1 & x_2 \\ x_2 & x_3 \end{pmatrix} = x_1 x_3 - x_2^2$ is of signature $(1, 2)$.

- (c) Define the generalized orthogonal group as

$$\text{O}(p, q) := \text{Isom}(n, I_{p,q}), \quad \text{O}(n) := \text{O}(n, 0),$$

and the special orthogonal group (or rotation group) of degree n as

$$\text{SO}(n) := \text{O}(n) \cap \text{SL}(n, \mathbb{R}),$$

both endowed with the subspace topology of $\text{GL}_n(\mathbb{R})$. Prove that $\text{O}(n)$ is compact, and the index $[\text{O}(n) : \text{SO}(n)] = 2$.

- (d) Recall that any matrix $A \in \text{SO}(n)$ can be decomposed into

$$A = S^{-1} \cdot \text{diag}(1, \dots, 1, R(\theta_1), \dots, R(\theta_k)) \cdot S,$$

where $S \in \text{SO}(n)$, $R(\theta) := \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, and $\theta_i \in \mathbb{R}$ for all $1 \leq i \leq k$.

Based on this fact, prove that $\text{SO}(n)$ is path-connected.

13. We introduce the symplectic group and the unitary group.

- (a) The symplectic group is defined as the set of linear transformations of a $2n$ -dimensional vector space V over a field $\mathbb{K}$, which preserve a non-degenerate skew-symmetric (i.e., anti-symmetric) bilinear form. V is called a symplectic vector space, and the symplectic group of V is denoted by $\text{Sp}(V)$.

When $\text{char}(\mathbb{K}) \neq 2$, there exists a basis for V , such that the symplectic form is represented by

$$J_{2n} := \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Prove that for this basis, $\text{Sp}(V)$ can be identified with

$$\text{Sp}(2n, \mathbb{K}) := \{A \in \text{GL}(2n, \mathbb{K}) \mid A^T J_{2n} A = J_{2n}\}.$$

- (b) Define the unitary group of degree n as

$$U(n) := \{A \in M(n, \mathbb{C}) \mid A^* A = 1\},$$

where $A^* := \overline{A}^T$ is the conjugate transpose of A . The special unitary group of degree n is defined as

$$SU(n) := U(n) \cap SL(n, \mathbb{C}).$$

Prove that $\rho_n(U(n)) = O(2n) \cap Sp(2n, \mathbb{R})$, where the intersection is taken in $GL(2n, \mathbb{R})$.

- (c) Recall that any unitary matrix $A \in U(n)$ can be diagonalized by another unitary matrix $S \in U(n)$, i.e.,

$$A = S^* \cdot \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) \cdot S,$$

where $\theta_i \in \mathbb{R}$ for all $1 \leq i \leq n$. Based on this fact, prove that $U(n)$ is path-connected.

- 14.** The sphere of dimension n is defined as $S^n := \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$.

- (a) Show that the space $S^0 = \{\pm 1\}$, which has a group structure.
 (b) The circle S^1 can be identified with the group of unit complex numbers:

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}, \quad z \cdot w = zw, \quad z^{-1} = \overline{z},$$

which is the compact abelian group $U(1)$. Prove that there are group isomorphisms between $\mathbb{R}/\mathbb{Z}$, S^1 , $U(1)$ and $SO(2)$.

- (c) The quaternion algebra $\mathbb{H}$ is a 4-dimensional real vector space with basis $\{1, i, j, k\}$ satisfying

$$i^2 = j^2 = k^2 = ijk = -1.$$

Any quaternion is written as $q = a + bi + cj + dk$ for $a, b, c, d \in \mathbb{R}$. Define the conjugate $\overline{q} := a - bi - cj - dk$ and the norm $|q| := |q\overline{q}|^{1/2}$. Prove that $|q| = \sqrt{a^2 + b^2 + c^2 + d^2}$.

- (d) The 3-sphere S^3 is identified with the group of unit quaternions:

$$S^3 = \{q = a + bi + cj + dk \in \mathbb{H} \mid |q| = \sqrt{a^2 + b^2 + c^2 + d^2} = 1\},$$

where the multiplication is given by quaternion multiplication. Prove that there is a group isomorphism between S^3 and $SU(2)$.

- 15.** (a) Prove that there is a group isomorphism between $U(n)/SU(n)$ and $U(1)$.

- (b) Define the action of $O(n+1)$ on S^n as

$$O(n+1) \times S^n \rightarrow S^n, \quad (A, x = (x_1, \dots, x_{n+1})^T) \mapsto Ax,$$

where $x = (x_1, \dots, x_{n+1})^T \in \mathbb{R}^{n+1}$ is a column vector with $\|x\| = 1$. Prove that this action is transitive and induces $O(n+1)/O(n) \cong S^n$.

- (c) Define the action of $U(n+1)$ on S^{2n+1} as

$$U(n+1) \times S^{2n+1} \rightarrow S^{2n+1}, \quad (A, z = (z_1, \dots, z_{n+1})^T) \mapsto Az,$$

where $z = (z_1, \dots, z_{n+1})^T \in \mathbb{C}^{n+1}$ is a column vector with $\|z\| = 1$, and S^{2n+1} is identified with $\{z \in \mathbb{C}^{n+1} \mid \|z\| = 1\}$. Prove that this action is transitive and induces $U(n+1)/U(n) \cong S^{2n+1}$.

- (d) Prove by induction that $SO(n)$ and $U(n)$ are path-connected.

- 16.** Define the action of $SL(2, \mathbb{R})$ on $\mathbb{R}^2$ by $A \cdot x = Ax$ for $A \in SL(2, \mathbb{R})$ and $x = (x_1, x_2)^T \in \mathbb{R}^2$. This action induces an action on the 3-dimensional space $\text{Sym}_2(\mathbb{R})$ of real symmetric 2×2 matrices by

$$A \cdot X = AXA^T, \quad \text{where } X = \begin{pmatrix} x_1 & x_2 \\ x_2 & x_3 \end{pmatrix} \in \text{Sym}_2(\mathbb{R}).$$

Consider the quadratic form of signature $(1, 2)$ on $\text{Sym}_2(\mathbb{R})$:

$$q(X) = \det(X) = x_1x_3 - x_2^2.$$

Prove that the action of $SL(2, \mathbb{R})$ preserves q , i.e., $q(A \cdot X) = q(X)$ for all $A \in SL(2, \mathbb{R})$ and $X \in \text{Sym}_2(\mathbb{R})$. Conclude that there exists an injective group homomorphism from $SL(2, \mathbb{R})$ to the generalized orthogonal group $O(1, 2)$.

- 17.** Let G be a topological group, H be a closed subgroup of G of finite index.
- (a) Prove that H is also an open subgroup of G , hence is clopen (i.e., is both open and closed).
 - (b) Prove that every left coset gH and right coset Hg for $g \in G$ is clopen.
 - (c) If G is connected, show that $H = G$.
- 18.** Let G be a topological group, X be a topological space with a continuous G -action $\phi: G \times X \rightarrow X$. For $x \in X$, let

$$G_x = \text{Stab}_G(x) := \{g \in G \mid g \cdot x = x\}, \quad O_x := \{g \cdot x \mid g \in G\}$$

be the stabilizer and orbit of x , respectively.

- (a) Prove that the orbit map $G \rightarrow O_x$, $g \mapsto g \cdot x$ is continuous.
 - (b) Prove that if G is compact and X is Hausdorff, then both G_x and O_x are compact, and O_x is closed.
- 19.** Let G_1 and G_2 be topological groups. Let $G = G_1 \times G_2$ be the group of direct product, equipped with the product topology and componentwise multiplication

$$(g_1, g_2) \cdot (h_1, h_2) = (g_1 h_1, g_2 h_2).$$

- (a) Prove that G is a topological group.
 - (b) Suppose that for $i \in \{1, 2\}$, G_i acts continuously on a topological space X_i . Prove that G acts continuously on the product space $X := X_1 \times X_2$.
- 20.** Let G be the group defined by the presentation $\langle a, b \mid aba^{-1} = b^{-1} \rangle$. The Klein bottle can be defined as the quotient space of $\mathbb{R}^2$ by the following action:

$$\phi: G \times \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad a \cdot (x, y) = (x + 1, 1 - y), \quad b \cdot (x, y) = (x, y + 1).$$

Prove that ϕ is a well-defined group action.

- 21.** Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. The real projective space $\mathbb{RP}^n$ (when $\mathbb{K} = \mathbb{R}$) and complex projective space $\mathbb{CP}^n$ (when $\mathbb{K} = \mathbb{C}$) are the quotient spaces of $\mathbb{K}^{n+1} \setminus \{0\}$ by the multiplicative action of $\mathbb{K}^\times$ via scalar multiplication:

$$x \mapsto ax, \quad a \in \mathbb{K}^\times.$$

Elements are represented by homogeneous coordinates

$$[x_1 : x_2 : \cdots : x_{n+1}],$$

where $(x_1, \dots, x_{n+1}) \in \mathbb{K}^{n+1} \setminus \{0\}$, and the equivalence relation is given by

$$[x_1 : \cdots : x_{n+1}] = [ax_1 : \cdots : ax_{n+1}] \quad \text{for all } a \in \mathbb{K}^\times.$$

- (a) Prove the disjoint unions $\mathbb{RP}^n = \mathbb{RP}^{n-1} \sqcup \mathbb{R}^n$, $\mathbb{CP}^n = \mathbb{CP}^{n-1} \sqcup \mathbb{C}^n$.
 - (b) Describe the action of $\text{PGL}(n+1, \mathbb{R})$ on $\mathbb{RP}^n$ and the action of $\text{PGL}(n+1, \mathbb{C})$ on $\mathbb{CP}^n$.
- 22.** Let $E \subseteq \mathbb{CP}^2$ be the solution set defined in homogeneous coordinates $[x : y : z]$ by the equation $y^2 z = x^3 - x z^2$, i.e.,

$$E := \{[x : y : z] \in \mathbb{CP}^2 \mid y^2 z = x^3 - x z^2\}.$$

As a topological space, E is homeomorphic to the torus $\mathbb{T}^2 = S^1 \times S^1$. Prove that $\mathbb{Z}/4\mathbb{Z}$ has a group action on E via the homeomorphism $\sigma: E \rightarrow E$ defined by

$$\sigma([x : y : z]) = [-x : iy : z].$$

23. Prove the following results:

- (a) $\mathbb{RP}^n \cong S^n/(\mathbb{Z}/2\mathbb{Z})$, $\mathbb{CP}^n \cong S^{2n+1}/S^1$. The $\mathbb{Z}/2\mathbb{Z}$ -action is the antipodal map, and the S^1 -action is the scalar multiplication by unit complex numbers.
- (b) $\mathbb{RP}^1 \cong S^1$, $\mathbb{CP}^1 \cong S^2$, $\mathbb{RP}^3 \cong \mathrm{SO}(3)$.
- (c) $\mathbb{RP}^n \cong \mathrm{O}(n+1)/(\mathrm{O}(n) \times \mathrm{O}(1))$, $\mathbb{CP}^n \cong \mathrm{U}(n+1)/(\mathrm{U}(n) \times \mathrm{U}(1))$.

Chapter 6

Reduction of endomorphisms

6.1 Principal ideal domain

6.1.1 Definition. Let R be a commutative unitary ring. Recall that an ideal of R is by definition a sub- R -module of R . If an ideal I of R is generated by one element, we say that I is a *principal ideal*. If all ideals of R are principal ideals, we say that R is a *principal ideal ring*.

Let a and b be elements of R . Note that $aR \subseteq bR$ if and only if there exists $c \in R$ such that $a = bc$. We denote by $b \mid a$ this condition and say that b is a *divisor* of a and a is a *multiple* of b .

6.1.2 Definition. Let R be a commutative unitary ring. If the neutral elements for the addition and for the multiplication are different, and if

$$\forall (s, t) \in (R \setminus \{0\}) \times (R \setminus \{0\}), \quad st \in (R \setminus \{0\}),$$

then we say that R is an integral domain. If R is a principal ideal ring and an integral domain in the same time, we say that R is a *principal ideal domain*.

6.1.3 Proposition. Let R be an integral domain, and a and b be two elements of R . Then $aR = bR$ if and only if there exists $c \in R^\times$ such that $a = bc$.

Proof. Assume that there exists $c \in R^\times$ such that $a = cb$. By definition one has $aR \subseteq bR$. Since c is invertible, one has $b = c^{-1}a$ and hence $bR \subseteq aR$. Therefore, $aR = bR$.

Conversely, assume that $aR = bR$, then there exists x and y in R such that $a = bx$ and $b = ay$. If $a = 0$, then $b = 0$. In this case one has $a = b \cdot 1$. If $a \neq 0$, then, from $a = bx = ayx$ we deduce that $yx = 1$, namely $x \in R^\times$. $\square$

6.1.4 Definition. Let K be a commutative unitary ring. Recall that a non-zero polynomial P in $K[T]$ can be written in the form

$$a_0 + a_1T + \cdots + a_nT^n,$$

where $a_0, \dots, a_n$ are elements of K , and $a_n \neq 0$. The natural number n is called the *degree* of P and is denoted as $\deg(P)$. The coefficient a_n is called the *leading coefficient* of P and is denoted as $c(P)$. If $c(P) = 1$, then we say that P is *monic*. By convention, the degree of the zero polynomial is defined to be $-\infty$, and the zero polynomial is considered as a monic polynomial.

6.1.5 Proposition. *Let K be an integral domain. Then the ring $K[T]$ of polynomials of coefficients in K is an integral domain. It is a principal ideal domain if K is a field.*

Proof. Since K is an integral domain, if P and Q are non-zero polynomials in $K[T]$, then PQ has as degree $\deg(P) + \deg(Q)$ and $c(PQ) = c(P)c(Q)$. Hence $K[T]$ is also an integral domain.

We will then check that $K[T]$ is an integral domain under the hypothesis that K is a field. Let I be an ideal of $K[T]$. If I is the zero ideal, then it is generated by one element 0. If I is non-zero, then there exists a non-zero polynomial $P \in I$ such that $\deg(P)$ is minimal. By the Euclidean division of polynomials, for any $F \in K[T]$, there exists a unique pair of polynomial Q and R in $K[T]$ such that $F = PQ + R$ and $\deg(R) < \deg(P)$. Since P is of minimal degree among all non-zero polynomials in I , we obtain that $R = 0$. Hence I is generated by P . $\square$

6.1.6 Remark. Let K be a field and I be an ideal of $K[T]$. There then exists a unique monic polynomial $P \in I$ such that I is generated by P .

6.1.7 Definition. Let A be an integral domain. We say that an element $a \in A \setminus A^\times$ is *irreducible* if a can not be decomposed into the product of two elements in $A \setminus A^\times$. In the case where A is of the form $K[T]$ with K being a field, an irreducible element of $K[T]$ is called an *irreducible polynomial*.

6.1.8 Definition. Let A be a commutative unitary ring and I be an ideal of A . We say that I is a *prime ideal* if the quotient ring A/I is an integral domain. This is equivalent to require that $I \neq A$ and

$$\forall (a, b) \in (A \setminus I) \times (A \setminus I), \quad ab \in A \setminus I.$$

We say that I is a *maximal ideal* if the quotient ring A/I is a field. By definition, a maximal ideal is always a prime ideal.

6.1.9 Proposition. *Let A be a commutative unitary ring. The maximal ideals of A are precisely the maximal element in the set of all ideals of A that are not equal to A , equipped with the order of inclusion $\subseteq$.*

Proof. Let I be an ideal of A .

Suppose that A/I is a field. If J is another ideal of A such that $J \not\supseteq I$, then there exists $a \in J \setminus I$. The equivalence class of a in A/I being invertible, there exists $b \in A$ such that $1 - ab \in I$. In particular, $1 = (1 - ab) + ab \in J$, which means that $J = A$.

Conversely, suppose that I is a maximal element among ideals that are not equal to A . If a is an element of $A \setminus I$ such that the class of a in A/I is not invertible, then the set

$$J = \{ab + c \mid b \in A, c \in I\}$$

is an ideal of A containing I , which is different from A since 1 does not belong to J . This contradicts the assumption that I is maximal. Therefore, any non-zero element of A/I is invertible, which means that A/I is a field. $\square$

6.1.10 Proposition. *Let A be an integral domain and a be an element of A . If the ideal I of A generated by a is a prime ideal, then a is an irreducible element of A .*

Proof. Suppose that a can be decomposed into the product of two elements b and c in $A \setminus A^\times$. Since I is a prime ideal, b and c cannot be simultaneously in $A \setminus I$. Suppose that $b \in I$, then there exists $d \in A$ such that $b = ad = bcd$. Hence $cd = 1$, which contradicts the assumption that c is in $A \setminus A^\times$. $\square$

6.1.11 Proposition. *Let A be a principal ideal domain. If a is an irreducible element of A , then the ideal generated by a is a maximal ideal.*

Proof. Let I be the ideal generated by a . By Proposition 6.1.9, it suffices to show that I is not strictly contained in an ideal that is different from A . Suppose by contradiction that there exists an ideal J of A such that $I \subsetneq J \subsetneq A$. Since A is a principal ideal domain, there exists $b \in J$ such that J is generated by b . Since I is contained in J , there exists $c \in A$ such that $a = bc$. Note that b is not a unit since $J \neq A$. Moreover c is not unit since $I \neq J$. This contradicts the assumption that a is irreducible. $\square$

6.1.12 Definition. Let R be a commutative unitary ring. We say that R is *Noetherian* if, for any increasing sequence

$$I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots \subseteq I_n \subseteq I_{n+1} \subseteq \dots$$

of ideals of R , there exists $n \in \mathbb{N}$ such that

$$I_n = I_{n+1} = \dots$$

6.1.13 Proposition. *A commutative unitary ring R is Noetherian if and only if all ideals of R are finitely generated. In particular, any principal ideal ring is Noetherian.*

Proof. Assume that R is Noetherian. Let I be an ideal of R . Assume that I is not finitely generated. Then one can construct recursively a strictly increasing sequence

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n \subsetneq I_{n+1} \subsetneq \dots$$

This contradicts the assumption that R is Noetherian.

Conversely, assume that any ideal of R is finitely generated. Let

$$I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots \subseteq I_n \subseteq I_{n+1} \subseteq \dots$$

be an increasing sequence of ideals of R . Then $I = \bigcup_{n \in \mathbb{N}} I_n$ is an ideal of R . By the assumption, I is generated by finitely many elements $x_1, \dots, x_p$. Let $n \in \mathbb{N}$ such that $\{x_1, \dots, x_p\} \subseteq I_n$. Then, for any $m \in \mathbb{N}$ such that $m \geq n$, one has $\{x_1, \dots, x_p\} \subseteq I_m$ and hence $I_m = I$. $\square$

6.1.14 Corollary. *Let R be a Noetherian commutative unitary ring. If I is an ideal of R such that $I \neq R$, then I is contained in a maximal ideal of R .*

Proof. Suppose that I is not contained in any maximal ideal of R , then one can construct recursively a strictly increasing sequence

$$I = I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n \subsetneq I_{n+1} \subsetneq \dots$$

of ideals of R such that $I_n \neq R$ for any $n \in \mathbb{N}$. This contradicts the assumption that R is Noetherian. $\square$

6.1.15 Remark. The statement of Corollary 6.1.14 is true for any commutative unitary ring under the axiom of choice. Let R be a commutative unitary ring and I be an ideal of R such that $I \neq R$. Let Θ be the set of all ideals J of R such that $I \subseteq J \subsetneq R$. The set Θ is non-empty and ordered by the relation of inclusion $\subseteq$. Moreover, if Θ_0 is a non-empty totally ordered subset of Θ , then $\bigcup_{J \in \Theta_0} J$ is an upper bound of Θ_0 in Θ . Hence by Zorn's lemma we obtain that Θ admits a maximal element, which is necessarily a maximal ideal.

6.1.16 Theorem. *Let A be a principal ideal domain. Any non-zero element $a \in A \setminus A^\times$ can be decomposed as*

$$a = p_1 \cdots p_n,$$

where $p_1, \dots, p_n$ are irreducible elements of A . Moreover, if

$$a = p_1 \cdots p_n = q_1 \cdots q_m$$

are two decompositions of a into a product of irreducible elements, then one has $n = m$, and there exists a permutation $\sigma \in S_n$ such that

$$\forall i \in \{1, \dots, n\}, \quad p_i A = q_{\sigma(i)} A.$$

Proof. We first prove the existence of the decomposition.

If a is irreducible, then $a = a$ is a decomposition. Otherwise, by Corollary 6.1.14, aA is strictly contained in a maximal ideal of A , which is, by Proposition 6.1.10, of the form p_1A , where p_1 is an irreducible element of A . Thus there exists $a_1 \in A \setminus A^\times$ such that $a = p_1a_1$. Iterating this procedure, we obtain a sequence of decompositions

$$a = p_1 \cdots p_i a_i,$$

where $p_1, \dots, p_i$ are irreducible elements of A and $a_i \in A \setminus A^\times$. This procedure should terminate at certain step $i \in \mathbb{N}$ (so that a_i is irreducible) since otherwise we obtain a strictly increasing sequence

$$aA \subsetneq a_1A \subsetneq a_2A \subsetneq \dots$$

of ideals of A , which contradicts the fact that A is a Noetherian ring. Therefore, we obtain the existence of the decomposition $a = p_1 \cdots p_n$.

To prove the uniqueness of the decomposition, we reason by induction on the minimal number of irreducible factors in all possible decompositions. In the case where a is irreducible, by definition a cannot be written as a product of more than one element in $A \setminus A^\times$. We now suppose that a can be decomposed as the product of n irreducible elements $p_1, \dots, p_n$ of A (with $n \geq 2$) and that a cannot be written as the product of less than n irreducible elements. Moreover, we suppose as the induction hypothesis the uniqueness up to permutation and unit factors of the decomposition for elements that can be written as the product of less than n irreducible elements. If a can be decomposed as $q_1 \cdots q_m$, where $q_1, \dots, q_m$ are irreducible elements, then by

$$a = q_1 \cdots q_m \in p_n A$$

we deduce that there exists $i \in \{1, \dots, m\}$ such that $q_i \in p_n A$ (because $p_n A$ is a maximal ideal and hence a prime ideal). By permutation we may assume that $q_m \in p_n A$ and hence $q_m A \subseteq p_n A$. Since $q_m A$ is a maximal ideal, we obtain $q_m A = p_n A$. Moreover, by Proposition 6.1.3, there exists $\lambda \in A^\times$ such that $q_m = \lambda p_n$. Therefore, we obtain

$$p_1 \cdots p_{n-1} = (\lambda q_1) \cdots q_{m-1}.$$

Note that $\lambda q_1 A = q_1 A$. By the induction hypothesis, we obtain that n and m , and there exists a permutation $\tau \in S_{n-1}$ such that $p_i A = q_{\tau(i)} A$ for any $i \in \{1, \dots, n-1\}$. The theorem is thus proved. $\square$

6.1.17 Remark. Let K be a field and M be the set of all monic irreducible polynomials in $K[T]$.

By Proposition 6.1.5 and Theorem 6.1.16 any non-zero polynomial F in $K[T]$ can be decomposed as

$$F(T) = c(F)P_1(T)\cdots P_n(T),$$

where $c(F)$ is the leading coefficient of F and $P_1(T), \dots, P_n(T)$ are monic irreducible polynomials in $K[T]$. Moreover, the decomposition is unique up to permutation of $P_1, \dots, P_n$. For any $P \in M$, we denote by $\text{ord}_P(F)$ the number of occurrence of P among $P_1, \dots, P_n$. Then we can rewrite the irreducible decomposition of F as

$$F(T) = c(F) \prod_{P \in M} P(T)^{\text{ord}_P(F)} := c(F) \prod_{\substack{P \in M \\ \text{ord}_P(F) \neq 0}} P(T)^{\text{ord}_P(F)}.$$

Note that, if F and G are two non-zero polynomials in $K[T]$, then, for any $P \in M$, one has

$$\text{ord}_P(FG) = \text{ord}_P(F) + \text{ord}_P(G), \quad \text{ord}_P(F + G) \geq \min\{\text{ord}_P(F), \text{ord}_P(G)\},$$

where by convention $\text{ord}_P(0) = +\infty$. Moreover, $F \mid G$ if and only if

$$\forall P \in M, \quad \text{ord}_P(F) \leq \text{ord}_P(G).$$

6.1.18 Proposition. *Let A be a commutative unitary ring. The set of all ideals of A , equipped with the relation of inclusion is order complete*

Proof. Let Θ be a family of ideals of A . If Θ is not empty, then $\bigcap_{I \in \Theta} I$ is an ideal of A . It is certainly the infimum of Θ . In the case where Θ is empty, the ideal A is the infimum of Θ . Moreover, the ideal generated by $\bigcup_{I \in \Theta} I$ is the supremum of Θ , we denote it as $\sum_{I \in \Theta} I$. $\square$

6.1.19 Definition. Let K be a field, n be a positive integer, and $F_1, \dots, F_n$ be polynomials in $K[T]$. We call *greatest common divisor* of $F_1, \dots, F_n$ the unique monic polynomial that generates the ideal

$$\sum_{i=1}^n F_i(T)K[T],$$

denoted as $\gcd(F_1, \dots, F_n)$. By definition, there exists polynomials $G_1, \dots, G_n$ in $K[T]$ such that

$$\gcd(F_1, \dots, F_n)(T) = F_1(T)G_1(T) + \cdots + F_n(T)G_n(T).$$

We call *least common multiple* of $F_1, \dots, F_n$ the unique monic polynomial that generates the ideal

$$\bigcap_{i=1}^n F_i(T)K[T],$$

denoted as $\text{lcm}(F_1, \dots, F_n)$.

6.1.20 Proposition. *Let K be a field, n be a positive integer and $F_1, \dots, F_n$ be polynomials in $K[T]$. For any irreducible monic polynomial P , one has*

$$\begin{aligned} \text{ord}_P(\gcd(F_1, \dots, F_n)) &= \min_{i \in \{1, \dots, n\}} \text{ord}_P(F_i), \\ \text{ord}_P(\text{lcm}(F_1, \dots, F_n)) &= \max_{i \in \{1, \dots, n\}} \text{ord}_P(F_i) \end{aligned}$$

Proof. By definition, for any $i \in \{1, \dots, n\}$, one has

$$\gcd(F_1, \dots, F_n) \mid F_i, \quad F_i \mid \text{lcm}(F_1, \dots, F_n).$$

Hence

$$\begin{aligned} \text{ord}_P(\gcd(F_1, \dots, F_n)) &\leq \text{ord}_P(F_i), \\ \text{ord}_P(\text{lcm}(F_1, \dots, F_n)) &\geq \text{ord}_P(F_i) \end{aligned}$$

Let M be the set of monic irreducible polynomials in $K[T]$. Let

$$F = \prod_{P \in M} P^{\min\{\text{ord}_P(F_1), \dots, \text{ord}_P(F_n)\}}.$$

Then, for any $i \in \{1, \dots, n\}$, F is a divisor of F_i . Hence $F \mid \gcd(F_1, \dots, F_n)$. Therefore,

$$\min\{\text{ord}_P(F_1), \dots, \text{ord}_P(F_n)\} \leq \text{ord}_P(\gcd(F_1, \dots, F_n)).$$

Similarly, let

$$G = \prod_{P \in M} P^{\max\{\text{ord}_P(F_1), \dots, \text{ord}_P(F_n)\}}.$$

Then, for any $i \in \{1, \dots, n\}$, G is a multiple of F_i . Hence $\text{lcm}(F_1, \dots, F_n) \mid G$. Therefore,

$$\max\{\text{ord}_P(F_1), \dots, \text{ord}_P(F_n)\} \geq \text{ord}_P(\text{lcm}(F_1, \dots, F_n)).$$

□

6.2 Cayley-Hamilton theorem

6.2.1 Definition. Let K be a commutative unitary ring. Recall that a K -algebra is by definition a K -module A equipped with a composition law $A \times A \rightarrow A$, $(a, b) \mapsto ab$ such that A equipped with the addition and with this composition law forms a unitary ring, and that

$$\forall (\lambda, a, b) \in K \times A \times A, \quad \lambda(ab) = (\lambda a)b = a(\lambda b).$$

Let A be a K -algebra and V be a K -module. We equip $A \otimes V$ with the following structure of left A -module

$$A \times (A \otimes V) \longrightarrow A \otimes V, \quad (a, b \otimes x) \longmapsto ab \otimes x.$$

6.2.2 Proposition. *Let K be a commutative unitary ring, A be a K -algebra and V be a free K -module of finite rank n . Let $(e_i)_{i=1}^n$ be a basis of V . Then $(1 \otimes e_i)_{i=1}^n$ is a basis of $A \otimes V$.*

Proof. Let M be a left A -module. One has following bijections which are functorial on M :

$$\begin{aligned} \text{Hom}_A(A \otimes V, M) &\cong \{f : A \times V \rightarrow M \mid f \text{ is } K\text{-bilinear and } f(a, x) = af(1, x)\} \\ &\cong \text{Hom}_K(V, M) \cong M^{\{e_1, \dots, e_n\}} \cong M^{\{1 \otimes e_1, \dots, 1 \otimes e_n\}}. \end{aligned}$$

Therefore, $A \otimes V$ is a free A -module generated by $(1 \otimes e_i)_{i=1}^n$. $\square$

6.2.3 Definition. Let K be a commutative unitary ring, V be a K -module and $\varphi \in \text{End}_K(V)$, where $\text{End}_K(V)$ denotes the set of all K -linear endomorphisms of V . For any polynomial

$$F(T) = a_0 + a_1T + \dots + a_dT^d \in K[T],$$

we denote by $F(\varphi) \in \text{End}_K(V)$ the K -linear endomorphism

$$a_0 \text{Id}_V + a_1\varphi + \dots + a_d\varphi^d.$$

Note that $F \mapsto F(\varphi)$ defines a homomorphism of K -algebras from $K[T]$ to $\text{End}_K(V)$. Since $K[T]$ is commutative, we obtain that, for any polynomials F and G in $K[T]$, one has

$$F(\varphi) \circ G(\varphi) = G(\varphi) \circ F(\varphi).$$

We equip with V with a structure of $K[T]$ -module as follows:

$$K[T] \times V \longrightarrow V, \quad (F(T), x) \longmapsto F(\varphi)(x).$$

This mapping is K -bilinear and hence defines a K -linear mapping $K[T] \otimes V \rightarrow V$ that sends $F \otimes x$ to $F(\varphi)(x)$. Note that, if we consider $K[T] \otimes V$ as a $K[T]$ -module and consider the above structure of $K[T]$ -module on V , then this K -linear map is actually $K[T]$ -linear. In fact, for any polynomials F and G in $K[T]$ and any $x \in V$ one has

$$(FG)(\varphi)(x) = (F(\varphi) \circ G(\varphi))(x) = F(\varphi)(G(\varphi)(x)).$$

6.2.4 Lemma. *Let K be a commutative unitary ring and V be a free K -module of finite rank n . Let $\varphi : V \rightarrow V$ be a K -linear endomorphism of V . Then one has*

$$(\varphi^a)^\vee = (\varphi^\vee)^a.$$

Proof. Let $(e_i)_{i=1}^n$ be an arbitrary basis of V and let $(e_i^\vee)_{i=1}^n$ be its dual basis. One has

$$e_1^\vee \wedge \varphi^\vee(e_2^\vee) \wedge \cdots \wedge \varphi^\vee(e_n^\vee) = e_1^\vee \wedge (e_2^\vee \circ \varphi) \wedge \cdots \wedge (e_n^\vee \circ \varphi).$$

Therefore,

$$\begin{aligned} & (e_1^\vee \wedge \varphi^\vee(e_2^\vee) \wedge \cdots \wedge \varphi^\vee(e_n^\vee))(e_1 \wedge \cdots \wedge e_n) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) e_1^\vee(e_{\sigma(1)}) e_2^\vee(\varphi(e_{\sigma(2)})) \cdots e_n^\vee(\varphi(e_{\sigma(n)})) \end{aligned}$$

Since $e_1^\vee(e_i) = 0$ for any $i \in \{2, \dots, n\}$, we obtain that

$$\begin{aligned} & (e_1^\vee \wedge \varphi^\vee(e_2^\vee) \wedge \cdots \wedge \varphi^\vee(e_n^\vee))(e_1 \wedge \cdots \wedge e_n) \\ &= (e_1^\vee \wedge \cdots \wedge e_n^\vee)(e_1 \wedge \varphi(e_2) \wedge \cdots \wedge \varphi(e_n)) \\ &= (e_1^\vee \cdots \wedge e_n^\vee)(\varphi^a(e_1) \wedge e_2 \wedge \cdots \wedge e_n) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) e_{\sigma(1)}^\vee(\varphi^a(e_1)) e_{\sigma(2)}^\vee(e_2) \cdots e_{\sigma(n)}^\vee(e_n) \\ &= ((e_1^\vee \circ \varphi^a) \wedge e_2^\vee \cdots \wedge e_n^\vee)(e_1 \wedge \cdots \wedge e_n). \end{aligned}$$

Hence we obtain

$$(\varphi^a)^\vee(e_1^\vee) = (\varphi^\vee)^a(e_1^\vee).$$

When the basis $(e_i)_{i=1}^n$, $e_1^\vee$ runs over the set $V^\vee \setminus \{0\}$. Hence we get the required equality. $\square$

6.2.5 Lemma. *Let K be a commutative unitary ring and V be a free K -module of finite rank n . Let $\varphi : V \rightarrow V$ and $\psi : V \rightarrow V$ be K -linear endomorphisms of V . If $\varphi^\vee = \psi^\vee$, then $\varphi = \psi$.*

Proof. Let $(e_i)_{i=1}^n$ be a basis of V over K and let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. By definition, one has $e_i^\vee \circ \varphi = e_i^\vee \circ \psi$ for any $i \in \{1, \dots, n\}$. Note that, for any $x \in V$, one has

$$x = \sum_{i=1}^n e_i^\vee(x) e_i.$$

Therefore, for any $y \in V$, one has

$$\varphi(y) = \sum_{i=1}^n (e_i^\vee \circ \varphi)(y) e_i = \sum_{i=1}^n (e_i^\vee \circ \psi)(y) e_i = \psi(y).$$

$\square$

6.2.6 Proposition. *Let K be a commutative unitary ring and V be a free K -module of finite rank n . Let $\varphi : V \rightarrow V$ be a K -linear endomorphism of V . Then one has*

$$\varphi \circ \varphi^a = \det(\varphi) \text{Id}_V.$$

Proof. Applying Theorem 4.7.1 to $\varphi^\vee$, by Lemma 6.2.4 and Proposition 4.6.7 we get

$$(\varphi \circ \varphi^a)^\vee = (\varphi^a)^\vee \circ \varphi^\vee = \det(\varphi^\vee) \text{Id}_{V^\vee} = \det(\varphi) \text{Id}_{V^\vee} = \det(\varphi) \text{Id}_V^\vee.$$

Hence, by Lemma 6.2.5, we obtain the desired equality. $\square$

6.2.7 Theorem (Cayley-Hamilton). *Let K be a commutative unitary ring, V be a free K -module of finite rank n , and $\varphi \in \text{End}_K(V)$. Let $T \text{Id}_V - \varphi$ denote the $K[T]$ -linear endomorphism $T \otimes \text{Id}_V - 1 \otimes \varphi$ of $K[T] \otimes V$ that sends $F(T) \otimes x$ to*

$$TF(T) \otimes x - F(T) \otimes \varphi(x),$$

and let $P_\varphi(T) = \det(T \text{Id}_V - \varphi)$. Then one has

$$P_\varphi(\varphi) = 0$$

in $\text{End}_K(V)$.

Proof. Note that, for any element of the form $F(T) \otimes x$ in $K[T] \otimes V$, the image of

$$(T \text{Id}_V - \varphi)(F(T) \otimes x)$$

in V by the $K[T]$ -linear mapping

$$K[T] \otimes V \longrightarrow V$$

described in Definition 6.2.3 vanishes. This is because

$$\varphi \circ F(\varphi) = F(\varphi) \circ \varphi.$$

We then deduce that, for any $\xi \in K[T] \otimes V$, the image of $(T \text{Id}_V - \varphi)(\xi)$ by the $K[T]$ -linear mapping

$$K[T] \otimes V \longrightarrow V$$

also vanishes.

By Proposition 6.2.6, one has

$$P_\varphi(T) \text{Id}_{K[T] \otimes V} = (T - \varphi) \circ (T - \varphi)^a.$$

Let $x \in V$. Applying this equality to $1 \otimes x$, we get

$$P_\varphi(T) \otimes x = P_\varphi(T)(1 \otimes x) = (T \text{Id}_V - \varphi)(\xi),$$

with $\xi = (T \text{Id}_V - \varphi)^a(1 \otimes x)$. Taking the image in V by the $K[T]$ -linear mapping

$$K[T] \otimes V \longrightarrow V$$

described in Definition 6.2.3, we get

$$P_\varphi(\varphi)(x) = 0.$$

$\square$

6.2.8 Definition. Let K be a commutative unitary ring, V be a free K -module of finite rank n . For any $\varphi \in \text{End}_K(V)$, the polynomial

$$P_\varphi(T) = \det(T \text{Id}_V - \varphi) \in K[T]$$

is called the *characteristic polynomial* of the endomorphism φ .

Let $(e_i)_{i=1}^n$ be a basis of V over K . Then $(1 \otimes e_i)_{i=1}^n$ is a basis of the $K[T]$ -module $K[T] \otimes V$. By abuse of notation, for any $x \in V$, we still use the notation x to denote the element $1 \otimes x$ in $K[T] \otimes V$ and thus view V as a sub- K -module of $K[T] \otimes V$.

With this notation, one has

$$(T \text{Id}_V - \varphi)(e_i) = Te_i - \varphi(e_i).$$

Therefore,

$$(T \text{Id}_V - \varphi)(e_1) \wedge \cdots \wedge (T \text{Id}_V - \varphi)(e_n) = (Te_1 - \varphi(e_1)) \wedge \cdots \wedge (Te_n - \varphi(e_n))$$

can be rewritten as

$$\begin{aligned} T^n(e_1 \wedge \cdots \wedge e_n) - T^{n-1} \sum_{i=1}^n (e_1 \wedge \cdots \wedge e_{i-1} \wedge \varphi(e_i) \wedge e_{i+1} \wedge \cdots \wedge e_n) + \\ \cdots + (-1)^n \varphi(e_1) \wedge \cdots \wedge \varphi(e_n) \end{aligned}$$

Thus P_φ is a monic polynomial of degree n , and the coefficient of T^0 of the polynomial P_φ is $(-1)^n \det(\varphi)$. We denote by $\text{tr}(\varphi)$ the opposite of the coefficient of T^{n-1} in the characteristic polynomial P_φ , namely

$$\text{tr}(\varphi)(e_1 \wedge \cdots \wedge e_n) = \sum_{i=1}^n (e_1 \wedge \cdots \wedge e_{i-1} \wedge \varphi(e_i) \wedge e_{i+1} \wedge \cdots \wedge e_n).$$

We write the characteristic polynomial of φ into the form

$$P_\varphi(T) = T^n - s_1(\varphi)T^{n-1} + \cdots + (-1)^d s_d(\varphi).$$

Note that $s_d(\varphi) = \det(\varphi)$ and $s_1(\varphi) = \text{tr}(\varphi)$.

6.2.9 Remark. Assume that P_φ split into $P_\varphi(T) = (T - \lambda_1) \cdots (T - \lambda_d)$. Then one has

$$s_\ell(\varphi) = \sum_{1 \leq i_1 < i_2 < \cdots < i_\ell \leq n} \lambda_{i_1} \cdots \lambda_{i_\ell}.$$

By definition, for any $(x_1, \dots, x_n) \in V^n$, one has

$$(Tx_1 + \varphi(x_1)) \wedge \cdots \wedge (Tx_n + \varphi(x_n)) = (T^n + s_1(\varphi)T^{n-1} + \cdots + s_d(\varphi))x_1 \wedge \cdots \wedge x_n.$$

Therefore,

$$s_\ell(\varphi)(x_1 \wedge \cdots \wedge x_n) = \sum_{\substack{I=(i_1, \dots, i_\ell) \\ 1 \leq i_1 < \dots < i_\ell \leq n}} z_1^I \wedge \cdots \wedge z_n^I,$$

where

$$z_j^I = \begin{cases} x_j, & j \notin I, \\ \varphi(x_j), & j \in I. \end{cases}$$

Let $F_\ell : \text{End}_K(V)^\ell \rightarrow K$, be the multilinear mapping defined as

$$F_\ell(\varphi_1, \dots, \varphi_\ell)x_1 \wedge \cdots \wedge x_d = \sum_{\substack{I=(i_1, \dots, i_\ell) \\ 1 \leq i_1 < \dots < i_\ell \leq n}} u_1^I \wedge \cdots \wedge u_d^I,$$

with

$$u_j^I = \begin{cases} x_j, & j \notin I, \\ \varphi_{i_k}(x_j), & j = i_k. \end{cases}$$

Note that F_ℓ is a multilinear form, and $s_\ell(\varphi) = F_\ell(\varphi, \dots, \varphi)$. Hence s_ℓ is a homogeneous polynomial of degree ℓ on $\text{End}_K(V)$.

6.3 Reduction of linear endomorphisms

In this section, we fix a field K .

6.3.1 Definition. Let V be a finite-dimensional vector space over K . If φ is a K -linear endomorphism of V , we let λ_φ be the homomorphism of K -algebras from $K[T]$ to $\text{End}_K(V)$ that sends $F \in K[T]$ to $F(\varphi)$. By Cayley-Hamilton Theorem, the characteristic polynomial P_φ belongs to the kernel of λ_φ . Hence the kernel of λ_φ is a non-zero ideal of $K[T]$. We denote by $Q_\varphi(T)$ the unique monic polynomial that generates $\text{Ker}(\lambda_\varphi)$. Note that a polynomial $F \in K[T]$ satisfies $F(\varphi) = 0$ if and only if $Q_\varphi \mid F$. We call Q_φ the *minimal polynomial* of φ .

6.3.2 Proposition. Let V be a finite-dimensional vector space over K , $\varphi \in \text{End}_K(V)$, and W be a vector subspace of V such that $\varphi(W) \subseteq W$. Let $\psi : W \rightarrow W$ be the restriction of φ to W . Then $Q_\psi \mid Q_\varphi$.

Proof. Note that $Q_\varphi(\varphi) = 0$. Hence, for any $x \in W$,

$$Q_\varphi(\psi)(x) = Q_\varphi(\varphi)(x) = 0.$$

Hence $Q_\varphi(\psi) = 0$, which leads to $Q_\psi \mid Q_\varphi$. □

6.3.3 Definition. Let V be a vector space over K , I be a set and $(V_i)_{i \in I}$ be a family of vector subspace of V . Recall that, in the case where the K -linear mapping

$$\bigoplus_{i \in I} V_i \longrightarrow V, \quad (x_i)_{i \in I} \longmapsto \sum_{i \in I} x_i$$

is an isomorphism of K -vector spaces, we say that V is a *direct sum* of $(V_i)_{i \in I}$. Note that, if for any $i \in I$, $(e_{i,j})_{j \in J_i}$ is a basis of V_i , then

$$e_{i,j}, \quad i \in I, j \in J_i$$

forms a basis of V over K .

6.3.4 Proposition. Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. Let $V_1, \dots, V_n$ be vector subspaces of V . We assume that, for any $i \in \{1, \dots, n\}$, one has $\varphi(V_i) \subseteq V_i$. Let $\varphi_i : V_i \rightarrow V_i$ be the restriction of φ to V_i .

(1) If $V = V_1 + \dots + V_n$, then $Q_\varphi(T) = \text{lcm}(Q_{\varphi_1}(T), \dots, Q_{\varphi_n}(T))$.

(2) If V is the direct sum of $V_1, \dots, V_n$, then $P_\varphi(T) = P_{\varphi_1}(T) \cdots P_{\varphi_n}(T)$

Proof. (1) Let $G = \text{lcm}(Q_{\varphi_1}, \dots, Q_{\varphi_n})$. For any $i \in \{1, \dots, n\}$, let $F_i \in K[T]$ such that $G = F_i Q_{\varphi_i}$. Let $x \in V$, we write x as $x_1 + \dots + x_n$ with $x_i \in V_i$. One has

$$G(\varphi)(x) = \sum_{i=1}^n G(\varphi)(x_i) = \sum_{i=1}^n G(\varphi_i)(x_i) = \sum_{i=1}^n F_i(\varphi_i) Q_{\varphi_i}(\varphi_i)(x_i).$$

Hence G is a vanishing polynomial of φ and thus $Q_\varphi \mid G$.

Conversely, by Proposition 6.3.2, one has $Q_{\varphi_i} \mid Q_\varphi$ for any $i \in \{1, \dots, n\}$. Hence $G \mid Q_\varphi$.

(2) For any $i \in \{1, \dots, n\}$, let $(e_{i,j})_{j=1}^{d_i}$ be a basis of V_i . Moreover,

$$(Te_{i,1} - \varphi(e_{i,1})) \wedge \dots \wedge (Te_{i,d_i} - \varphi(e_{i,d_i})) = P_{\varphi_i}(T) e_{i,1} \wedge \dots \wedge e_{i,d_i}.$$

Hence $P_\varphi(T) = P_{\varphi_1}(T) \cdots P_{\varphi_n}(T)$. □

6.3.5 Theorem. Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. Let $F_1, \dots, F_n$ be non-zero polynomials in $K[T]$ and $F = F_1 \cdots F_n$. Assume that $F_1, \dots, F_n$ are pairwise coprime, that is, for any indices i and j in $\{1, \dots, n\}$ such that $i \neq j$, F_i and F_j do not have any common irreducible factor. Then $\text{Ker}(P(\varphi))$ is the direct sum of $\text{Ker}(F_1(\varphi)), \dots, \text{Ker}(F_n(\varphi))$.

Proof. Without loss of generality, we may assume that all polynomials F_i are monic. For any $i \in \{1, \dots, n\}$, let

$$G_i = F_1 \cdots F_{i-1} F_{i+1} \cdots F_n.$$

Since $F_1, \dots, F_n$ are pairwise coprime, one has $\gcd(G_1, \dots, G_n) = 1$. In fact, for any irreducible monic polynomial P , either P does not divide any polynomial among $F_1, \dots, F_n$ and hence $P \nmid \gcd(G_1, \dots, G_n)$, or there exists $i \in \{1, \dots, n\}$ such that $P \mid F_i$. In this case, for any $j \in \{1, \dots, n\} \setminus \{i\}$ one has $P \nmid F_j$ and hence $P \nmid G_i$. Thus there exist polynomials $A_1, \dots, A_n$ in $K[T]$ such that

$$A_1(T)G_1(T) + \dots + A_n(T)G_n(T) = 1.$$

In particular, for any $x \in V$, one has

$$x = A_1(\varphi)G_1(\varphi)(x) + \dots + A_n(\varphi)G_n(\varphi)(x).$$

Note that, if $x \in \text{Ker}(F(\varphi))$, then, for any $i \in \{1, \dots, n\}$, one has

$$F_i(\varphi)A_i(\varphi)G_i(\varphi)(x) = A_i(\varphi)F(\varphi)(x) = 0,$$

namely $A_i(\varphi)G_i(\varphi)(x)$ belongs to $\text{Ker}(F_i(\varphi))$. Hence

$$\text{Ker}(F(\varphi)) = \text{Ker}(F_1(\varphi)) + \dots + \text{Ker}(F_n(\varphi)).$$

If $(x_1, \dots, x_n) \in \text{Ker}(F_1(\varphi)) \times \dots \times \text{Ker}(F_n(\varphi))$ such that $x_1 + \dots + x_n = x$, then, for any $i \in \{1, \dots, n\}$, one has

$$\begin{aligned} A_i(\varphi)G_i(\varphi)(x) &= \sum_{j=1}^n A_i(\varphi)G_i(\varphi)(x_j) = A_i(\varphi)G_i(\varphi)(x_i) \\ &= \sum_{j=1}^n A_j(\varphi)F_j(\varphi)(x_i) = x_i. \end{aligned}$$

Therefore, $\text{Ker}(F(\varphi))$ is the direct sum of $\text{Ker}(F_1(\varphi)), \dots, \text{Ker}(F_n(\varphi))$. $\square$

6.3.6 Remark. Let V be a finite-dimensional vector space over K , and $\varphi \in \text{End}_K(V)$. Let F be a non-zero polynomial in $K[T]$. For any $x \in \text{Ker}(F(\varphi))$, one has

$$F(\varphi)(\varphi(x)) = \varphi(F(\varphi)(x)) = 0.$$

Hence $\varphi(\text{Ker}(F(\varphi))) \subseteq \text{Ker}(F(\varphi))$.

Suppose that $F(\varphi) = 0$, and that F is decomposed as

$$F(T) = F_1(T) \cdots F_n(T),$$

where $F_1, \dots, F_n$ are pairwise coprime. Then, by Theorem 6.3.5, $V = \text{Ker}(F(\varphi))$ is the direct sum of $\text{Ker}(F_1(\varphi)), \dots, \text{Ker}(F_n(\varphi))$.

6.3.7 Proposition. *Let V be a finite-dimensional vector space over K and d be its dimension. Let $\varphi \in \text{End}_K(V)$. Let P_φ and Q_φ be respectively the characteristic polynomial and the minimal polynomial of φ . Then one has*

$$Q_\varphi \mid P_\varphi, \quad P_\varphi \mid Q_\varphi^d.$$

In particular, P_φ and Q_φ have the same irreducible factors.

Proof. By Theorem 6.2.7, one has $P_\varphi(\varphi) = 0$. Hence $Q_\varphi \mid P_\varphi$. Suppose that Q_φ is of the form

$$T^n + a_1 T^{n-1} + \cdots + a_n.$$

Then, in $\text{End}_{K[T]}(K[T] \otimes V)$ one has

$$Q_\varphi(T) \text{Id}_{K[T] \otimes V} = Q_\varphi(T) \otimes \text{Id}_V - 1 \otimes Q_\varphi(\varphi) = (T \text{Id}_V - \varphi) \sum_{i=0}^{n-1} a_i \sum_{j=1}^{i-1} T^j \otimes \varphi^{i-1-j}.$$

Taking the determinant, we get

$$Q_\varphi(T)^d = P_\varphi(T) \det \left(\sum_{i=0}^{n-1} a_i \sum_{j=1}^{i-1} T^j \otimes \varphi^{i-1-j} \right).$$

Therefore, $P_\varphi \mid Q_\varphi^d$. □

6.3.8 Definition. Let V be a finite-dimensional vector spaces over K and $\varphi \in \text{End}_K(V)$. We say that φ is *diagonalizable* (or *semi-simple*) if there exists a basis of V which is composed of eigenvectors of φ .

6.3.9 Proposition. *Let V be a finite-dimensional vector spaces over K and $\varphi \in \text{End}_K(V)$. Let λ be an element of K . The following statements are equivalent:*

- (1) λ is an eigenvalue of V ,
- (2) λ is a root of the minimal polynomial Q_φ of φ ,
- (3) λ is a root of the characteristic polynomial P_φ of φ .

Proof. The equivalence of (2) and (3) comes from Proposition 6.3.7. In fact, λ is a root of a polynomial if and only if $T - \lambda$ is an irreducible component of the polynomial.

If λ is an eigenvalue of V , then $W := \text{Ker}(\lambda \text{Id}_V - \varphi)$ is non-zero, and the restriction of φ to W is given by λId_W . Hence $Q_{\varphi|_W}(T) = (T - \lambda)$. By Proposition 6.3.2, we obtain that $(T - \lambda)$ divide Q_φ .

Conversely, if Q_φ is of the form

$$Q_\varphi(T) = (T - \lambda)F(T),$$

then $F(\varphi)$ is non-zero. There then exists $y \in V$ such that $x = F(\varphi)(y)$ is non-zero. Moreover,

$$0 = (\varphi - \lambda \text{Id}_V)(x) = \varphi(x) - \lambda x.$$

Hence λ is an eigenvalue of φ . □

6.3.10 Theorem. *Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. Then φ is diagonalizable if and only if its minimal polynomial Q_φ can be decomposed as*

$$Q_\varphi(T) = (T - \lambda_1) \cdots (T - \lambda_n),$$

where $\lambda_1, \dots, \lambda_n$ are distinct elements of K . Moreover, when φ is diagonalizable, one has

$$Q_\varphi(T) = \prod_{\lambda \in \text{Sp}(\varphi)} (T - \lambda),$$

where $\text{Sp}(\varphi)$ denotes the set of all eigenvalues of φ .

Proof. For any $\lambda \in K$, we let $E_\lambda(\varphi)$ the kernel of $\lambda \text{Id}_V - \varphi$.

Assume that

$$Q_\varphi(T) = (T - \lambda_1) \cdots (T - \lambda_n).$$

Since $\lambda_1, \dots, \lambda_n$ are distinct, for any indices i and j in $\{1, \dots, n\}$ such that $i \neq j$, the polynomials $T - \lambda_i$ and $T - \lambda_j$ are coprime. Hence, by Theorem 6.3.5, V is the direct sum of $E_{\lambda_1}(\varphi), \dots, E_{\lambda_n}(\varphi)$. Therefore, V is diagonalizable.

Conversely, assume that $(e_i)_{i=1}^d$ is a basis of V which is composed of eigenvectors of φ . For any $i \in \{1, \dots, d\}$, let μ_i be the eigenvalue of e_i . Then, each Ke_i is stable under the action of φ . Moreover, if we denote by $\varphi_i : Ke_i \rightarrow Ke_i$ the restriction of φ , then one has $Q_{\varphi_i} = T - \mu_i$. By Proposition 6.3.4,

$$Q_\varphi = \text{lcm}(Q_{\varphi_1}, \dots, Q_{\varphi_d}) = \prod_{\lambda \in \{\mu_1, \dots, \mu_d\}} (T - \lambda).$$

In particular, Q_φ divides

$$\prod_{\lambda \in \text{Sp}(\varphi)} (T - \lambda).$$

Finally, if λ is an eigenvalue of φ , then $\varphi(E_\lambda(\varphi)) \subseteq E_\lambda(\varphi)$. If we denote by $\psi_\lambda : E_\lambda(\varphi) \rightarrow E_\lambda(\varphi)$ the restriction of φ , then, by Proposition 6.3.2,

$$Q_{\psi_\lambda} = T - \lambda$$

divides Q_φ . Therefore,

$$Q_\varphi = \prod_{\lambda \in \text{Sp}(\varphi)} (T - \lambda).$$

□

6.4 Dunford decomposition

In this section, K denotes a field.

6.4.1 Definition. Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. We say that φ is *nilpotent* if there exists $k \in \mathbb{N}_{\geq 1}$ such that $\varphi^k = 0$.

6.4.2 Proposition. Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$ be a nilpotent endomorphism of V . Let d be the dimension of V . Then $\varphi^d = 0$.

Proof. Since φ is nilpotent, its minimal polynomial has only one irreducible factor T . Hence so is its characteristic polynomial. Thus we obtain $P_\varphi(T) = T^d$, which leads to $\varphi^d = 0$ by Cayley-Hamilton theorem. $\square$

6.4.3 Theorem (Dunford decomposition). Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. Suppose that the characteristic polynomial of φ is of the form

$$P_\varphi(T) = (T - \lambda_1)^{\alpha_1} \cdots (T - \lambda_r)^{\alpha_r},$$

where $\lambda_1, \dots, \lambda_r$ are distinct elements of K , and $\alpha_1, \dots, \alpha_r$ are positive integers. Then there exists a unique pair (s, n) in $\text{End}_K(V)^2$ which satisfies the following conditions:

- (1) s is diagonalizable and n is nilpotent,
- (2) $s \circ n = n \circ s$,
- (3) $\varphi = s + n$.

Moreover, both s and n belong to the image of

$$K[T] \rightarrow \text{End}_K(V), \quad F \mapsto F(\varphi).$$

Proof. For any $i \in \{1, \dots, r\}$, let

$$G_i = (T - \lambda_1)^{\alpha_1} \cdots (T - \lambda_{i-1})^{\alpha_{i-1}} (T - \lambda_{i+1})^{\alpha_{i+1}} \cdots (T - \lambda_r)^{\alpha_r}.$$

Since $\gcd(G_1, \dots, G_r) = 1$, there exists polynomials $A_1, \dots, A_r$ such that that

$$1 = A_1 G_1 + \cdots + A_r G_r.$$

For any $i \in \{1, \dots, r\}$, let $V_i = \text{Ker}((\varphi - \lambda_i \text{Id}_V)^{\alpha_i})$, and let $\pi_i = A_i(\varphi) \circ G_i(\varphi)$. Note that

$$(\varphi - \lambda_i \text{Id}_V)^{\alpha_i} \circ \pi_i = 0,$$

which implies that $\pi_i(V) \subseteq V_i$ and $\pi_i \circ \pi_j = 0$ once $i \neq j$. Moreover, for any $x \in V$, one has

$$x = \pi_1(x) + \cdots + \pi_r(x).$$

By Theorem 6.3.5, V is the direct sum of $V_1, \dots, V_r$. Therefore, if $x \in V_i$, then $\pi_i(x) = x$. In particular, one has $\pi_i(V) = V_i$ and $\pi_i^2 = \pi_i$.

Let $s : V \rightarrow V$,

$$s(x) := \lambda_1 \pi_1 + \cdots + \lambda_r \pi_r.$$

For $x \in V_i$, one has $s(x) = \lambda_i x$. Hence V has a basis that is composed of eigenvectors of s , namely s is diagonalizable. Since each π_i can be written as a polynomial of φ , so does s .

Let

$$n = \varphi - s = (\varphi \circ \pi_1 + \cdots + \varphi \circ \pi_r) - (\lambda_1 \pi_1 + \cdots + \lambda_r \pi_r) = \sum_{i=1}^r (\varphi - \lambda_i \text{Id}_V) \circ \pi_i.$$

Therefore,

$$n^d = (\varphi - \lambda_1 \text{Id}_V)^d \circ \pi_1 + \cdots + (\varphi - \lambda_r \text{Id}_V)^d \circ \pi_r = 0.$$

Moreover, by definition, n is a polynomial of φ . Hence $s \circ n = n \circ s$.

It remains to prove the uniqueness of the decomposition. Let $\varphi = s' + n'$ be another decomposition with s' diagonalizable, n' nilpotent and $s' \circ n' = n' \circ s'$. We deduce that

$$s' \circ \varphi = \varphi \circ s', \quad n' \circ \varphi = \varphi \circ n'.$$

Let x be an eigenvector of s' and λ be the corresponding eigenvalue. Then one has

$$n'(x) = \varphi(x) - \lambda x,$$

and hence, by $n' \circ \varphi = \varphi \circ n'$, we get $x \in \text{Ker}(\varphi - \lambda \text{Id}_V)^d$. Therefore, one has $\det(\lambda \text{Id}_V - \varphi) = 0$ and $\lambda \in \{\lambda_1, \dots, \lambda_r\}$. Note that, for any $i \in \{1, \dots, r\}$, one has

$$V_i = \text{Ker}((\varphi - \lambda_i \text{Id}_V)^{\alpha_i}) \subseteq \text{Ker}((\varphi - \lambda_i \text{Id}_V)^d),$$

and V is both direct sum of $(V_i)_{i=1}^r$ and $(\text{Ker}(\varphi - \lambda_i \text{Id}_V)^d)_{i=1}^r$. Hence one has

$$V_i = \text{Ker}(\varphi - \lambda_i \text{Id}_V)^d.$$

Since s' is diagonalizable, V is the direct sum of $(E_{\lambda_i}(s'))_{i=1}^r$ and $E_{\lambda_i}(s') \subseteq V_i$. Hence we get $E_{\lambda_i}(s') = V_i$ and

$$s' = \sum_{i=1}^r \lambda_i \pi_i = s.$$

□

6.5 Jordan-Chevalley decomposition

In this section, we fix a field K .

6.5.1 Definition. Let V be a finite-dimensional vector space over K and $\varphi : V \rightarrow V$ be a K -linear endomorphism. We say that φ is a *Jordan endomorphism* if there exists $\lambda \in K$ such that both the characteristic polynomial and the minimal polynomial of φ are of the form $(T - \lambda)^d$, with $d = \dim_K(V)$.

6.5.2 Lemma. Let $\varphi \in \text{End}_K(V)$, $n \in \mathbb{N}_{\geq 1}$ and $x \in V$. Assume that $\varphi^n(x) = 0$ and $\varphi^{n-1}(x) \neq 0$. Then the family

$$x, \varphi(x), \dots, \varphi^{n-1}(x)$$

is linearly independent.

Proof. Let $(a_1, \dots, a_n) \in K^n$ such that

$$a_1x + a_2\varphi(x) + \dots + a_n\varphi^{n-1}(x) = 0.$$

We prove by induction that, for any $k \in \{1, \dots, n\}$, $a_k = 0$. First, one has

$$0 = \varphi^{n-1}(a_1x + a_2\varphi(x) + \dots + a_n\varphi^{n-1}(x)) = a_1\varphi^{n-1}(x).$$

Hence $a_1 = 0$. Suppose that $a_1 = \dots = a_k = 0$. Then

$$0 = \varphi^{n-1-k}(a_{k+1}\varphi^k(x) + \dots + a_n\varphi^{n-1}(x)) = a_{k+1}\varphi^{n-1}(x).$$

Hence $a_{k+1} = 0$. □

6.5.3 Remark. Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$ be a Jordan endomorphism and λ be the root of its characteristic polynomial. Let $\psi = \varphi - \lambda \text{Id}_V$. Pick $x \in V \setminus \text{Ker}(\psi^{d-1})$, $d = \dim_K(V)$. Then the family

$$x, \psi(x), \dots, \psi^d(x)$$

forms a basis of V . Note that the matrix of φ with respect to the above basis is of the form

$$\begin{pmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{pmatrix}$$

6.5.4 Theorem. *Let V be a finite-dimensional vector space over K and $\varphi \in \text{End}_K(V)$. Suppose that the minimal polynomial of φ splits into a product of monic polynomials of degree 1. Then there exist vector subspaces $V_1, \dots, V_\ell$ of V such that*

- (1) *for any $i \in \{1, \dots, \ell\}$, $\varphi(V_i) \subseteq V_i$,*
- (2) *V is a direct sum of $V_1, \dots, V_\ell$,*
- (3) *the restriction $\varphi_i : V_i \rightarrow V_i$ of φ to V_i is a Jordan endomorphism.*

Proof. By Theorem 6.3.5, we may assume without loss of generality that the minimal polynomial Q_φ is of the form $(T - \lambda)^\alpha$, where $\lambda \in K$ and α is a positive integer.

Let $\psi := \varphi - \lambda \text{Id}_V$. We reason by induction on the dimension d of V . In the case where $d = 1$, the minimal polynomial of φ is already $(T - \lambda)^d$ and hence φ is a Jordan endomorphism.

In the remaining part of the proof, we suppose that φ is *not* a Jordan endomorphism. Let $x \in V \setminus \text{Ker}(\psi^{\alpha-1})$ and let W be the vector subspace generated by $\{x, \psi(x), \dots, \psi^{\alpha-1}(x)\}$. By the same argument as in the proof of the previous proposition, we obtain that

$$x, \psi(x), \dots, \psi^{\alpha-1}(x)$$

forms a basis of W . Moreover, one has $\psi(W) \subseteq W$ and hence $\varphi(W) \subseteq W$. Let $\varphi_1 : W \rightarrow W$ be the restriction of φ . Note that the minimal polynomial of φ_1 is $(T - \lambda)^\alpha$. Hence φ_1 is a Jordan endomorphism.

Let $f : V \rightarrow K$ be a K -linear form such that $f(\psi^{\alpha-1}(x)) \neq 0$. Let H be the vector subspace of $V^\vee$ generated by

$$\{f, f \circ \psi, f \circ \psi^2, \dots, f \circ \psi^{\alpha-1}\} = \{f, \psi^\vee(f), (\psi^\vee)^2(f), \dots, (\psi^\vee)^{\alpha-1}(f)\}.$$

Since

$$(\psi^\vee)^\alpha = (\psi^\alpha)^\vee = 0 \text{ and } (\psi^\vee)^{\alpha-1}(f)(x) = f(\psi^{\alpha-1}(x)),$$

we obtain that the family

$$f, f \circ \psi, \dots, f \circ \psi^{\alpha-1}$$

forms a basis of H . Therefore, the vector subspace

$$U := \{y \in V \mid \forall h \in H, h(y) = 0\}$$

of V has dimension $d - \alpha$. Moreover, if $(b_1, \dots, b_\alpha)$ is an element of K^α such that $(b_1, \dots, b_\alpha) \neq (0, \dots, 0)$ and $i \in \{1, \dots, \alpha\}$ is the smallest index such that $b_i \neq 0$, then one has

$$f \circ \psi^{\alpha-1-i}(b_1x + b_2\psi(x) + \dots + b_\alpha\psi^{\alpha-1}(x)) = b_i f(\psi^{\alpha-1}(x)) \neq 0.$$

Hence

$$b_1x + b_2\psi(x) + \cdots + b_\alpha\psi^\alpha(x) \notin U.$$

This shows that $U \cap W = \{0\}$ since $\dim(U) = d - \alpha$ and $\dim(W) = \alpha$. We obtain that V is the direct sum of U and W . By definition, U is stable by the action of ψ and hence $\varphi(U) \subseteq U$. Hence, by induction we get the desired result. $\square$

6.5.5 Definition. The vector subspaces $V_1, \dots, V_\ell$ are called Jordan blocs of φ .

6.5.6 Remark. Let $\varphi = s + \psi$ be the Dunford decomposition of φ . Then

- (1) $\dim(\text{Ker}(\psi))$ is equal to the number of Jordan blocs.
- (2) $\dim(\text{Ker}(\psi^i)) - \dim(\text{Ker}(\psi^{i-1}))$ is equal to the number of Jordan blocs of dimension $\geq i$.

6.6 Behaviour under the conjugation

In this section, let K be a field.

6.6.1 Definition. Let V be a finite-dimensional vector space over K . Note that $\text{GL}(V)$ acts from left on $\text{End}_K(V)$ as following

$$\text{GL}(V) \times \text{End}_K(V) \longrightarrow \text{End}_K(V), \quad (g, \varphi) \longmapsto g \circ \varphi \circ g^{-1}.$$

6.6.2 Theorem. Let V be a finite-dimensional vector space over K . Let $(g, \varphi) \in \text{GL}(V) \times \text{End}_K(V)$. Then $g \circ \varphi \circ g^{-1}$ and φ have the same characteristic polynomial. They also have the same minimal polynomial.

Proof. By abuse of notation, for any $\psi \in \text{End}_K(V)$, we still use the notation ψ to denote the endomorphism $1 \otimes \psi$ of $K[T] \otimes V$. With this notation, one has

$$\begin{aligned} \det(T \text{Id}_V - g \circ \varphi \circ g^{-1}) &= \det(g \circ (T \text{Id}_V - \varphi) \circ g^{-1}) \\ &= \det(g) \det(T \text{Id}_V - \varphi) \det(g^{-1}) = \det(T \text{Id}_V - \varphi). \end{aligned}$$

Moreover, for any positive integer n , one has

$$(g \circ \varphi \circ g^{-1})^n = g \circ \varphi^n \circ g^{-1}.$$

Therefore, for any $F \in K[T]$,

$$g \circ F(\varphi) \circ g^{-1} = F(g \circ \varphi \circ g^{-1}).$$

Therefore,

$$\{F \in K[T] \mid F(\varphi) = 0\} = \{F \in K[T] \mid F(g \circ \varphi \circ g^{-1}) = 0\}.$$

Hence $Q_\varphi = Q_{g \circ \varphi \circ g^{-1}}$. $\square$

6.6.3 Theorem. *Let V and W be finite-dimensional vector spaces over K , $\varphi : V \rightarrow W$ and $\psi : W \rightarrow V$ be K -linear mappings. Then one has*

$$P_{\psi \circ \varphi}(T)T^{\dim(W)} = P_{\varphi \circ \psi}(T)T^{\dim(V)}.$$

In particular,

$$\text{tr}(\varphi \circ \psi) = \text{tr}(\psi \circ \varphi).$$

Proof. We consider the K -linear mapping

$$\Phi : V \oplus W \longrightarrow V \oplus W$$

that sends (x, y) to $(\psi(\varphi(x)), \varphi(x))$. Note that

$$P_{\Phi}(T) = P_{\psi \circ \varphi}(T)T^{\dim(W)}.$$

In fact, if $(x_i)_{i=1}^n$ and $(y_m)_{i=1}^m$ are respectively bases of V and W , then (here we view V and W as vector subspaces of $V \oplus W$)

$$\begin{aligned} & (T \text{Id}_{V \oplus W} - \Phi)(x_1) \wedge \cdots \wedge (T \text{Id}_{V \oplus W} - \Phi)(x_n) \\ & \quad \wedge (T \text{Id}_{V \oplus W} - \Phi)(y_1) \wedge \cdots \wedge (T \text{Id}_{V \oplus W} - \Phi)(y_m) \\ &= (Tx_1 - \psi(\varphi(x_1)) - \varphi(x_1)) \wedge \cdots \wedge (Tx_n - \psi(\varphi(x_n)) - \varphi(x_n)) \\ & \quad \wedge Ty_1 \wedge \cdots \wedge Ty_m \\ &= (Tx_1 - \psi(\varphi(x_1))) \wedge \cdots \wedge (Tx_n - \psi(\varphi(x_n))) \wedge Ty_1 \wedge \cdots \wedge Ty_m. \end{aligned}$$

Let $f : V \oplus W \rightarrow V \oplus W$ that sends (x, y) to $(x - \psi(y), y)$. Then f is invertible and $f^{-1}(x, y) = (x + \psi(y), y)$. One has

$$\begin{aligned} & (f \circ \Phi \circ f^{-1})(x, y) = (f \circ \Phi)(x + \psi(y), y) \\ &= f(\psi(\varphi(x)) + \psi(\varphi(\psi(y))), \varphi(x) + \varphi(\psi(y))) \\ &= (0, \varphi(x) + \varphi(\psi(y))). \end{aligned}$$

Thus $P_{f \circ \Phi \circ f^{-1}}(T) = P_{\varphi \circ \psi}(T)T^{\dim(V)}$. Therefore, we get

$$P_{\psi \circ \varphi}(T)T^{\dim(W)} = P_{\varphi \circ \psi}(T)T^{\dim(V)}.$$

□

Exercises

Throughout the following exercises, n denotes a positive integer. Unless otherwise stated, all matrices are over the complex number field $\mathbb{C}$.

1. Let A be a 3×3 matrix with three eigenvalues $1, -2, 3$.
 - (a) Find $|A|$.
 - (b) Find the eigenvalues of A^{-1} and the adjugate matrix A^* .
 - (c) Find the eigenvalues of $A^2 + 2A + I_3$.
2. Let A be an $n \times n$ matrix satisfying $(A + I_n)^k = O_n$ for some positive integer k . Find $|A|$.
3. Let A be an $n \times n$ matrix satisfying $2A^2 - 3A - 5I_n = O_n$. Prove that $2A + I_n$ is invertible.
4. Let A be an $n \times n$ matrix, and the sum of elements in each row of A is a . Prove that:
 - (a) $\lambda = a$ is an eigenvalue of A , and $(1, 1, \dots, 1)^T$ is the corresponding eigenvector.
 - (b) When A is invertible, find the sum of elements in each row of A^{-1} and $2A^{-1} - 3A$, respectively.
5. Find the eigenvalues of the following real matrices (with $a, b \neq 0$):
 - (a) $A = (a_{ij})_{n \times n}$, where $a_{ij} = \begin{cases} 1 & i = j, \\ a & i \neq j. \end{cases}$
 - (b) $B = (b_{ij})_{n \times n}$, where $b_{ij} = \begin{cases} 1 & i = j, \\ b & i - j = \pm 1, \\ 0 & \text{otherwise.} \end{cases}$
6. Let A be an $n \times n$ matrix satisfying $A^2 = I_n$. Prove that:
 - (a) The eigenvalues of A can only be 1 or -1 ;
 - (b) If all eigenvalues of A are 1 , then $A = I_n$.
7. (a) Let A be an $m \times n$ matrix and B be an $n \times m$ matrix with $m \geq n$. Prove that

$$|\lambda I_m - AB| = \lambda^{m-n} |\lambda I_n - BA|.$$

- (b) Let $a_i (i = 1, 2, \dots, n, n \geq 3)$ be real numbers satisfying $a_1 + a_2 + \dots + a_n = 0$. Find the eigenvalues of the matrix

$$A = \begin{pmatrix} a_1^2 + 1 & a_1 a_2 + 1 & \cdots & a_1 a_n + 1 \\ a_2 a_1 + 1 & a_2^2 + 1 & \cdots & a_2 a_n + 1 \\ \vdots & \vdots & \ddots & \vdots \\ a_n a_1 + 1 & a_n a_2 + 1 & \cdots & a_n^2 + 1 \end{pmatrix}.$$

8. Find the following integrals:

- (a) $\int_{-1}^1 x^3 \cos(x^4 + 1) e^{-x^2} dx$, $\int_0^4 \sqrt{2x + 1} dx$, $\int_0^{+\infty} x e^{-x^2} dx$.
 (b) $\Gamma(n) := \int_0^{+\infty} x^{n-1} e^{-x} dx$ for $n \geq 1$.
 (c) $\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}} dx$, $\int_0^\pi \cos \theta d\theta$, $\int_0^\pi \cos^2 \theta \sin \theta d\theta$.

9. Let $\Delta_n := \{\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n \mid x_i \geq 0, \sum_{i=1}^n x_i \leq 1\}$ be the standard n -simplex. Compute the volume $\text{Vol}(\Delta_n) = \int_{\Delta_n} d\mathbf{x}$.

10. Let $\Phi : U \rightarrow V$ be the diffeomorphism with:

- domain $U = \{(u, v) \mid u > 0, |v| < 1\}$
- codomain $V = \{(x, y) \mid x > 0, y > 0\}$
- rule: $x = \frac{u(1+v)}{2}$, $y = \frac{u(1-v)}{2}$ (forward: $u = x + y$, $v = \frac{x-y}{x+y}$).

- (a) Compute $\det(D\Phi) = |\partial(x, y)/\partial(u, v)|$.
 (b) Evaluate $I = \iint_D \cos\left(\frac{x-y}{x+y}\right) dx dy$, where $D = \{(x, y) \mid x > 0, y > 0, x + y \leq 1\}$.

11. Let $\Phi : U \rightarrow V$ be the diffeomorphism with:

- domain $U = \{(u, v) \mid u > 0, v \in \mathbb{R}\}$.
- codomain $V = \{(x, y) \mid x > |y|\}$.
- rule: $x = u \cosh v = \frac{1}{2}u(e^v + e^{-v})$, $y = u \sinh v = \frac{1}{2}u(e^v - e^{-v})$.

- (a) Compute $\det(D\Phi) = |\partial(x, y)/\partial(u, v)|$.
 (b) Evaluate $I = \iint_D \sqrt{x^2 - y^2} dx dy$, where $D = \{(x, y) \mid |x - y| \leq e, |x + y| \leq e, |x| \geq |y|\}$.

12. Let $\Phi : U \rightarrow V$ be the diffeomorphism with:

- domain $U = \{(r, \theta) \mid r > 0, 0 < \theta < 2\pi\}$.

- codomain $V = \mathbb{R}^2 \setminus \{(x, 0) \mid x \geq 0\}$.
- rule: $(x, y) = \Phi(r, \theta) := (r \cos \theta, r \sin \theta)$.

- (a) Compute $\det(D\Phi) = |\partial(x, y)/\partial(r, \theta)|$.
- (b) Evaluate $I = \int_{\mathbb{R}} e^{-x^2} dx$. (Hint: Compute the value of I^2 .)
- (c) Let $A \in M_n(\mathbb{R})$ be a real symmetric positive-definite matrix and $\mathbf{b} \in \mathbb{R}^n$. Evaluate $I(A, \mathbf{b}) = \int_{\mathbb{R}^n} e^{-\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}} d\mathbf{x}$.

- 13.** Recall that there is an action of $\mathrm{SL}_2(\mathbb{Z})$ on the upper half-plane $\mathbb{H}$ (see the Exercises for Chapter 5). Let $Y(1)$ be the quotient space of $\mathbb{H}$ under this group action, then the $\mathrm{SL}_2(\mathbb{Z})$ -invariant measure $\frac{dx dy}{y^2}$ on $\mathbb{H}$ induces a measure on $Y(1)$.

Let $F := \{z \in \mathbb{H} \mid |\operatorname{Re}(z)| \leq \frac{1}{2}, |z| \geq 1\}$ be the fundamental region, then there is a diffeomorphism between the interior of F and the complement of a line in $Y(1)$. Hence we have

$$\operatorname{Vol}(Y(1)) = \operatorname{Vol}(F) = \int_F \frac{dx dy}{y^2}.$$

Compute the volume of $Y(1)$ based on the above equality. For $N \geq 2$, what is the volume of the quotient space $Y(N)$ of $\mathbb{H}$ under the action of $\Gamma(N) := \{A \in \mathrm{SL}_2(\mathbb{Z}) \mid A \equiv I_2 \pmod{N}\} \subseteq \mathrm{SL}_2(\mathbb{Z})$?

- 14.** Let $\Phi: U \rightarrow V$ be the diffeomorphism with:

- domain $U = \{(r, \theta, \varphi) \mid r > 0, 0 < \theta < \pi, 0 < \varphi < 2\pi\}$.
- codomain $V = \mathbb{R}^3 \setminus \{(x, 0, z) \mid x \geq 0, z \in \mathbb{R}\}$.
- rule: $\Phi(r, \theta, \varphi) := (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$.

- (a) Compute $\det(D\Phi) = |\partial(x, y, z)/\partial(r, \theta, \varphi)|$.
- (b) Evaluate the volume $\operatorname{Vol}(B)$ of the unit ball $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.
- (c) Compute the surface area of the unit sphere $S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$.

- 15.** Let S^2 be the unit sphere equipped with the surface measure

$$dS = \sin \theta d\theta d\varphi, \quad 0 \leq \theta \leq \pi, 0 \leq \varphi < 2\pi,$$

where the standard spherical and Cartesian coordinates satisfy

$$x = \sin \theta \cos \varphi, \quad y = \sin \theta \sin \varphi, \quad z = \cos \theta.$$

For complex-valued continuous functions on S^2 , define the complex inner product

$$\langle f, g \rangle = \iint_{S^2} f \bar{g} \, dS,$$

and the induced norm $\|f\| = \sqrt{\langle f, f \rangle}$.

We consider the following family of polynomial functions restricted on S^2 , which are the first few spherical harmonics up to degree 2:

$$\begin{aligned} f_{0,0} &= 1, & f_{1,\pm 1} &= x \pm iy, \\ f_{1,0} &= z, & f_{2,\pm 1} &= z(x \pm iy), \\ f_{2,0} &= z^2 - \frac{1}{3}, & f_{2,\pm 2} &= (x^2 - y^2) \pm 2ixy. \end{aligned}$$

- (a) Express each function above in spherical coordinates (θ, φ) .
- (b) Prove that these functions are mutually orthogonal with respect to the given inner product.
- (c) Compute the norm of each function.

16. Let α, β, γ be the three interior angles of a triangle. Prove that

$$f(x, y, z) = x^2 + y^2 + z^2 - 2xy \cos \alpha - 2yz \cos \beta - 2zx \cos \gamma$$

is non-negative for any $x, y, z \in \mathbb{R}$. (Hint: Show that this quadratic form is positive semi-definite.)

17. Let A be an $n \times n$ matrix. Is A always diagonalizable in each of the following cases?

- (a) $A^k = I_n$, where $k \geq 2$.
- (b) $A^k = A$, where $k \geq 2$.
- (c) $A^k = O_n$, where $k \geq 2$.
- (d) $A^3 - 2A^2 + A = O_n$.

18. Let A, B be $n \times n$ matrices such that $AB = BA$.

- (a) Show that there exists a nonzero vector $v \in \mathbb{C}^n$ which is both an eigenvector of A and B .
- (b) Show that if all the eigenvalues of A are distinct, then B is diagonalizable.
- (c) Show that if both A and B are diagonalizable, then there exists an invertible matrix P such that both $P^{-1}AP$ and $P^{-1}BP$ are diagonal.

19. The rank $\text{rank}(A)$ of a matrix A is the maximum number of linearly independent rows (or columns) of A , which equals the dimension of its row space (or column space). The rank of a matrix is invariant under elementary transformations.

- (a) Let A be an $m \times n$ real matrix. Show that for any $x \in \mathbb{R}^n$, the vector $Ax = 0$ if and only if $A^T Ax = 0$. Deduce that $\text{rank}(A^T A) = \text{rank}(A)$. (Hint: We have $\text{rank}(A) + \dim N(A) = n$, where $N(A) := \{x \in \mathbb{R}^n \mid Ax = 0\}$ is the null space of A .)
- (b) Do we always have $\text{rank}(AB) = \text{rank}(BA)$?
- (c) Suppose that A is an $n \times n$ matrix such that $A^2 = O_n$. What is the maximum possible rank of A ?
- (d) Let $A \in \mathbb{M}_{m \times n}$, $B \in \mathbb{M}_{n \times p}$, $C \in \mathbb{M}_{p \times q}$ be matrices. Prove that the following two block matrices are equivalent via elementary transformations:

$$\begin{pmatrix} ABC & O \\ O & B \end{pmatrix} \sim \begin{pmatrix} O & AB \\ -BC & B \end{pmatrix}$$

Deduce the Frobenius rank inequality:

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC).$$

What conclusion can be drawn if $p = n$ and $B = I_n$?

20. Let

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

- (a) Show that A_1, A_2 and A_3 have the same eigenvalues.
- (b) Determine the similarity relations among A_1, A_2 and A_3 .

21. Let the square matrices

$$A = \begin{pmatrix} -2 & 0 & 0 \\ 2 & x & 2 \\ 3 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & y \end{pmatrix}$$

be similar. Find the values of x and y . (Hint: Similar matrices have the same trace, determinant and eigenvalues.)

22. Let A and B be nonzero $n \times n$ matrices satisfying $A^2 + A = O_n$, $B^2 + B = O_n$, and $AB = O_n$. Prove that:

- (a) -1 is a common eigenvalue of A and B ;
 (b) A and B admit no common eigenvectors corresponding to the eigenvalue -1 .
- 23.** Let A be an $m \times m$ matrix, B be an $n \times n$ matrix. Prove that the matrices A, B have a common eigenvalue if and only if there exists a nonzero matrix $X \in \mathbb{M}_{m \times n}$ such that $AX = XB$.
- 24.** Let A be an $n \times n$ matrix with n distinct eigenvalues $\lambda_1, \dots, \lambda_n$. For each $1 \leq i \leq n$, define the Lagrange polynomial $L_i(\lambda)$ as

$$L_i(\lambda) = \prod_{j: j \neq i} (\lambda - \lambda_j) / (\lambda_i - \lambda_j).$$

- (a) For each $1 \leq i \leq n$, let $A_i := L_i(A)$. Prove the following four properties: $A_i^2 = A_i$, $A_i A_j = O_n$ for all $j \neq i$, $AA_i = \lambda_i A_i$, and $\sum_{i=1}^n A_i = I_n$.
- (b) Show that for any polynomial function $f(\lambda)$, we have the Sylvester's formula $f(A) = \sum_{i=1}^n f(\lambda_i) A_i$. In fact, this formula can be extended to any function f that is analytic on a simply connected domain containing all eigenvalues of A .
- (c) Compute A^k for $k \geq 2$, where

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

- 25.** Let A be a linear transformation on an n -dimensional vector space V over a field F . Suppose that with respect to a basis $\alpha_1, \alpha_2, \dots, \alpha_n$ of V , the matrix of A is given by

$$A = \begin{pmatrix} & & & & a_1 \\ 0 & & & a_2 & \\ & & \ddots & & \\ & a_{n-1} & & & 0 \\ a_n & & & & \end{pmatrix}.$$

- (a) Can V be decomposed into a direct sum of 2-dimensional or 1-dimensional invariant subspaces of A ?
- (b) Determine the sufficient and necessary conditions of A is diagonalizable.
- 26.** Let A be a linear transformation on a vector space V over a field F . Let $f(x), g(x), h(x) \in F[x]$ such that $(f(x), h(x)) = 1$ and let

$$(f(x), g(x)) = d(x), \quad [f(x), g(x)] = m(x).$$

Prove that

(a)

$$\ker d(A) = \ker f(A) \cap \ker g(A),$$

$$\ker m(A) = \ker f(A) + \ker g(A).$$

(b)

$$\text{rank}[f(A)g(A)h(A)] = \text{rank}[f(A)g(A)] + \text{rank}[g(A)h(A)] - \text{rank}(g(A)).$$

27. Find the minimal polynomial of the following matrices over a field K , and determine whether they are diagonalizable.

(a)

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)

$$B = \begin{pmatrix} 7 & -1 & -7 & 1 \\ -1 & 7 & 1 & -7 \\ 7 & -1 & -7 & 1 \\ -1 & 7 & 1 & -7 \end{pmatrix}$$

28. Suppose A, B, C are arbitrary $n \times n$ matrices over complex number field .

(a) A is similar to a upper triangular matrix.

(b) If $\text{tr}(A^k) = 0, k = 1, 2, \dots, n$, then A is nilpotent.

(c) If A diagonalizable and $\text{tr}(A^k) = 0, k = 1, 2, \dots, n$, then $A = 0$.

(d) Assume $AB - BA = C$. If $CA = AC$, then C is nilpotent.

29. Let $A = \text{diag}(A_1, A_2, \dots, A_s)$ be an $n \times n$ matrix over a field K , where each A_i is an $n_i \times n_i$ upper triangular matrix with all diagonal entries equal to a_i , for $i = 1, 2, \dots, s$. Prove that A is diagonalizable if and only if each A_i ($i = 1, 2, \dots, s$) is a scalar matrix.

30. Let A be a linear transformation on an n -dimensional vector space V over a field F , with $n > 1$. Prove that if $\text{rank}(A) = 1$, then A is either nilpotent or diagonalizable; when A is nilpotent, it is not diagonalizable.

- 31.** Find the Jordan canonical form of the following matrices A over a field K and find an invertible matrix P such that $P^{-1}AP = J$. :

(a)

$$\begin{pmatrix} 4 & -5 & 2 \\ 5 & -7 & 3 \\ 6 & -9 & 4 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 1 & -3 & 4 \\ 4 & -7 & 8 \\ 6 & -7 & 7 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 13 & 16 & 16 \\ -5 & -7 & -6 \\ 6 & -8 & -7 \end{pmatrix}$$

(d)

$$\begin{pmatrix} 3 & 0 & 8 \\ 3 & -1 & 6 \\ -2 & 0 & -5 \end{pmatrix}$$

(e)

$$\begin{pmatrix} 3 & -4 & 0 & 2 \\ 4 & -5 & -2 & 4 \\ 0 & 0 & 3 & -2 \\ 0 & 0 & 2 & -1 \end{pmatrix}$$

- 32.** Let a be a nonzero element in a field F . Find the Jordan canonical form of $J_r(a)^2$.

- 33.** Let A be an $n \times n$ block upper triangular matrix over a field F :

$$A = \begin{pmatrix} A_1 & A_3 \\ 0 & A_2 \end{pmatrix},$$

where A_1 and A_2 are $n_1 \times n_1$ and $n_2 \times n_2$ matrices respectively. Prove that if A_1 and A_2 have Jordan canonical forms J_1 and J_2 respectively, and the characteristic polynomial $f_1(\lambda)$ of A_1 is coprime to the characteristic polynomial $f_2(\lambda)$ of A_2 , then the Jordan canonical form of A is $\text{diag}(J_1, J_2)$.

34. Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Find $A^{100} + 2A^{90} + 3A^{60}$.

35. Let $A = D + N$ be the Dunford decomposition of $A \in M_n(\mathbb{C})$. Prove that for any polynomial $f(x) \in \mathbb{C}[x]$, the semisimple part of the Dunford decomposition of $f(A)$ is $f(D)$.

36. Let V be a 3-dimensional linear space over a field K , and let $\alpha_1, \alpha_2, \alpha_3$ be a basis of V . The corresponding dual basis in V^* is f_1, f_2, f_3 . Let

$$\beta_1 = 2\alpha_1 + \alpha_2 + 2\alpha_3, \quad \beta_2 = \alpha_1 + 2\alpha_2 - 2\alpha_3, \quad \beta_3 = -2\alpha_1 + 2\alpha_2 + \alpha_3.$$

Prove that $\beta_1, \beta_2, \beta_3$ form a basis of V , and find the corresponding dual basis g_1, g_2, g_3 in V^* (expressed in terms of f_1, f_2, f_3).

37. Let A, P be $n \times n$ complex matrices, and let P be invertible. Prove that

$$e^{P^{-1}AP} = P^{-1}e^AP.$$

38. Let A be a symmetric transformation on an n -dimensional real inner product space V . For any $a \in V$ with $a \neq 0$, define

$$F(a) = \frac{(a, Aa)}{(a, a)}.$$

Prove:

- (a) $F(ka) = F(a)$, $\forall k \in \mathbb{R}^*$;
- (b) Let A be an $n \times n$ real symmetric matrix with all eigenvalues arranged in nonincreasing order:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Prove that for any nonzero column vector $\alpha \in \mathbb{R}^n$,

$$\lambda_n \leq \frac{\alpha^\top A \alpha}{\|\alpha\|^2} \leq \lambda_1.$$

- (c) $F(a)$ attains its minimum at ξ , where ξ is a unit eigenvector of A corresponding to the smallest eigenvalue.

39. A unitary matrix is a complex square matrix U that satisfies

$$U^*U = UU^* = I,$$

where $U^* = \overline{U}^T$ is the conjugate transpose (Hermitian adjoint) of U , and I is the identity matrix.

Let H be an $n \times n$ Hermite matrix. Prove that:

- (a) $I - iH$ and $I + iH$ are both invertible;
 - (b) $A = (I - iH)(I + iH)^{-1}$ is unitary, and -1 is not an eigenvalue of A .
- 40.** Find all 2×2 unitary matrices.
- 41.** Denote by V the set of 2×2 Hermite matrices with trace zero.
- (a) Write down the general form of elements in V ;
 - (b) Prove that V is a linear space over the real numbers;
 - (c) Find a basis and the dimension of V ;
 - (d) Prove that for every nonzero $H \in V$, there exists a unitary matrix A with determinant 1 such that

$$AHA^{-1} = \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix},$$

where $c > 0$.

- 42.** Is a symmetric matrix over the complex field a Hermite matrix? Is it a normal matrix?
- 43.** Let A be a linear transformation on an n -dimensional linear space V over a field F , and suppose that the characteristic polynomial $f(\lambda)$ of A factors into linear factors in $F[\lambda]$. Prove that if λ_1 is an eigenvalue of A with multiplicity r_1 , then

$$\text{rank}(A - \lambda_1 I)^{r_1} = n - r_1.$$

- 44.** Let A, B be $n \times n$ complex matrices such that $AB - BA = A$.
- (a) Prove that A is nilpotent.
 - (b) If B is nilpotent with nilpotency index $l > 1$, then $A = 0$.
- 45.** Determine whether the following complex matrices A have square roots. If so, find a square root of A .

$$(a) \begin{pmatrix} 4 & -5 & 2 \\ 5 & -7 & 3 \\ 6 & -9 & 4 \end{pmatrix}$$

$$(b) \begin{pmatrix} 1 & -3 & 4 \\ 4 & -7 & 8 \\ 6 & -7 & 7 \end{pmatrix}$$

46. Let A be an $n \times n$ matrix over a number field K , and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be all the roots (possibly with repetitions) of the characteristic polynomial $|\lambda I - A|$ in the complex numbers. Prove:

(a) For any polynomial $g(x)$ over the complex numbers,

$$|g(A)| = g(\lambda_1)g(\lambda_2)\cdots g(\lambda_n).$$

(b) For any polynomial $f(x)$ over K , $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)$ are all the roots of the characteristic polynomial $|\lambda I - f(A)|$ in the complex numbers. Consequently, if λ_1 is an eigenvalue of A with multiplicity l_1 , then $f(\lambda_1)$ is an eigenvalue of $f(A)$ with multiplicity at least l_1 .

47. Let A be a 2×2 matrix over the real numbers. Prove that if $|A| < 0$, then A is diagonalizable.

Chapter 7

Differential geometry

7.1 Differential forms

In this section, we fix a finite-dimensional vector space E over $\mathbb{R}$. We denote by E^* the dual vector space of E .

7.1.1 Definition. Let U be an open subset of E , and p be a natural number. We call p -form on U any mapping from U to $\Lambda^p(E^*)$. For any $\ell \in \mathbb{N} \cup \{\infty\}$, we denote by $\Omega_\ell^p(U)$ the vector space of all p -forms on U which are of class C^ℓ . Note that a 0-form on U is just a mapping from U to $\mathbb{R}$. The set $\Omega_\infty^p(U)$ is often denoted as $\Omega^p(U)$ for simplicity.

If $\alpha : U \rightarrow \Lambda^p(E^*)$ is a p -form on U . We define the *support* of α to be the closure of the set

$$\{x \in U \mid \alpha(x) \neq 0\}$$

in U , denoted as $\text{Supp}(\alpha)$. If $\text{Supp}(\alpha)$ is compact, then we say that α is of compact support. We denote by $\Omega_c^p(U)$ the set of all continuous p -forms on U which are of compact support.

Let p and q be natural numbers, α be a p -form on U and β be a q -form on U , we denote by $\alpha \wedge \beta$ be the $(p+q)$ -form on U which is given by the pointwise wedge product of α and β , namely,

$$(\alpha \wedge \beta)(x) := \alpha(x) \wedge \beta(x).$$

Note that $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$.

7.1.2 Remark. Let r be a positive integer and $p_1, \dots, p_r$ be natural numbers. The mapping

$$\bigwedge^{p_1}(E^*) \times \dots \times \bigwedge^{p_r}(E^*) \longrightarrow \bigwedge^{p_1 + \dots + p_r}(E^*)$$

is multilinear and hence is of class C^∞ .

Let U be an open subset of E . For any $i \in \{1, \dots, r\}$, let α_i be a p_i -form on U . Let $\omega = \alpha_1 \wedge \dots \wedge \alpha_r$. If $\alpha_1, \dots, \alpha_r$ are continuous (resp. differentiable, of class C^ℓ), so is ω .

7.1.3 Proposition. *Let K be a commutative unitary ring, V and W be K -modules. We assume that V is a free K -module of finite rank. The mapping*

$$\Phi : V^\vee \otimes W \longrightarrow \text{Hom}_K(V, W), \quad \varphi \otimes w \longmapsto (x \mapsto \varphi(x)w)$$

is an isomorphism.

Proof. We first show that the above mapping is well defined. In fact, it is induced by the bilinear mapping $V^\vee \times W \rightarrow \text{Hom}_K(W)$ that sends $(\varphi, w) \in V^\vee \otimes W$ to

$$(x \in V) \longmapsto \varphi(x)w.$$

Let $(e_i)_{i=1}^n$ be a basis of V and $(e_i^\vee)_{i=1}^n$ be its dual basis. We construct a K -linear mapping $\Psi : \text{Hom}_K(V, W) \rightarrow V^\vee \otimes W$ that sends $f \in \text{Hom}_K(V, W)$ to

$$\sum_{i=1}^n e_i^\vee \otimes f(e_i).$$

We claim that Ψ is the inverse of Φ . In fact, for $(\varphi, w) \in V^\vee \times W$, one has

$$\Psi(\Phi(\varphi \otimes w)) = \sum_{i=1}^n e_i^\vee \otimes \varphi(e_i)w = \sum_{i=1}^n \varphi(e_i)e_i^\vee \otimes w = \varphi \otimes w;$$

for any $f \in \text{Hom}_K(V, W)$

$$\Phi(\Psi(f)) = \sum_{i=1}^n \Phi(e_i^\vee \otimes f(e_i)).$$

It sends every $x = a_1e_1 + \dots + a_ne_n \in V$ to

$$\sum_{i=1}^n e_i^\vee(x)f(e_i) = \sum_{i=1}^n a_i f(e_i) = f(x),$$

so that $\Phi(\Psi(f)) = f$. □

7.1.4 Definition. Let U be an open subset of E and p be a natural number. Let $\alpha : U \rightarrow \wedge^p(E^*)$ be a p -form that is differentiable. Recall that the differential of α defines a mapping from U to $\text{Hom}_{\mathbb{R}}(E, \wedge^p(E^*))$. We identify $\text{Hom}_{\mathbb{R}}(E, \wedge^p(E^*))$ with $E^* \otimes \wedge^p(E^*)$ via the isomorphism established in Proposition 7.1.3 and consider $D\alpha$ as a mapping from U to $E^* \otimes \wedge^p(E^*)$. We denote by $d\alpha$ the composite of $D\alpha$ with the $\mathbb{R}$ -linear mapping

$$E^* \otimes \wedge^p(E^*) \longrightarrow \wedge^{p+1}(E^*), \quad \varphi \otimes \alpha \longmapsto \varphi \wedge \alpha.$$

7.1.5 Example. Let U be an open subset of E and $f : U \rightarrow \mathbb{R}$ be a differentiable mapping. Then by definition, one has $df = Df$. If p is a natural number and $\alpha : U \rightarrow \Lambda^p(E^*)$ is a p -form. Taking a basis $(x_i)_{i=1}^n$ of E^* , we could write α in a unique way into a sum

$$(x \in U) \mapsto \sum_{1 \leq i_1 < \dots < i_p \leq n} f_{i_1, \dots, i_p}(x) dx_{i_1} \wedge \dots \wedge dx_{i_p},$$

where each $f_{i_1, \dots, i_p}$ is a mapping from U to $\mathbb{R}$. Moreover, f is continuous (resp. differentiable, of class C^ℓ) if and only if each $f_{i_1, \dots, i_p}$ is continuous (resp. differentiable, of class C^ℓ). Thus, in the study of p -forms, we may often assume without loss of generality that the p -form is of the form $f\eta$, where f is a mapping from U to $\mathbb{R}$ and η is an element of $\Lambda^p(E^*)$. In the case where f is differentiable, by definition, one has

$$d(f\eta) = df \wedge \eta.$$

7.1.6 Proposition. Let p and q be natural numbers, U be an open subset of E , $\alpha : U \rightarrow \Lambda^p(E^*)$ and $\beta : U \rightarrow \Lambda^q(E^*)$ be differentiable mappings. Then the following equality holds:

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta.$$

Proof. By (multi)linearity we may assume that α and β are of the form $f\alpha_0$ and $g\beta_0$, where f and g are differentiable mappings from U to $\mathbb{R}$, and $\alpha_0 \in \Lambda^p(E^*)$, $\beta_0 \in \Lambda^q(E^*)$. Thus,

$$\alpha \wedge \beta = fg \alpha_0 \wedge \beta_0.$$

Therefore, by Leibniz's rule, one has

$$d\alpha \wedge \beta = f dg \wedge \alpha_0 \wedge \beta_0 + g df \alpha_0 \wedge \beta_0.$$

Moreover, one has

$$\begin{aligned} d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta &= df \wedge \alpha_0 \wedge (g\beta_0) + (-1)^p f \alpha_0 \wedge dg \wedge \beta_0 \\ &= g df \wedge \alpha_0 \wedge \beta_0 + f dg \wedge \alpha_0 \wedge \beta_0 = d(\alpha \wedge \beta). \end{aligned}$$

□

7.1.7 Definition. Let K be a commutative unitary ring, n be a natural number, and V be a K -module. Let $\pi : V^n \rightarrow V^{\otimes n}$ be the universal n -linear mapping. Note that S_n acts from the left on the set V^n by permuting the coordinates:

$$(\sigma, (x_1, \dots, x_n)) \mapsto (x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)})$$

The composite of π with the mapping

$$V^n \longrightarrow V^{\otimes n}, \quad (x_1, \dots, x_n) \mapsto (x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)})$$

is n -linear, hence induces a K -linear endomorphism of $V^{\otimes n}$ that sends $x_1 \otimes \cdots \otimes x_n$ to $x_{\sigma^{-1}(1)} \otimes \cdots \otimes x_{\sigma^{-1}(n)}$. Thus we obtain a right action of S_n on $V^{\otimes n}$. If an element T of $V^{\otimes n}$ is invariant by the action of any element σ in S_n , we say that T is *symmetric*. Note that T is *not* an element in the symmetric algebra of V , but an element in the tensor algebra of V .

7.1.8 Proposition. *Let K be a commutative unitary ring, n be a natural number, and $(V_i)_{i=1}^n$ be a family of K -modules. Then there is a homomorphism of K -modules*

$$\Phi : V_1^\vee \otimes \cdots \otimes V_n^\vee \longrightarrow (V_1 \otimes \cdots \otimes V_n)^\vee$$

that sends $\varphi_1 \otimes \cdots \otimes \varphi_n \in V_1^\vee \otimes \cdots \otimes V_n^\vee$ to

$$(x_1 \otimes \cdots \otimes x_n) \longmapsto \varphi_1(x_1) \cdots \varphi_n(x_n).$$

It is an isomorphism when $V_1, \dots, V_n$ are free K -module of finite type.

Proof. We first show that Φ is well defined. Let $(\varphi_1, \dots, \varphi_n)$ be an element of $V_1^\vee \times \cdots \times V_n^\vee$. Note that the mapping

$$V_1 \times \cdots \times V_n \longrightarrow K, \quad (x_1, \dots, x_n) \longmapsto \varphi_1(x_1) \cdots \varphi_n(x_n)$$

is n -linear, it induced by the universal property of tensor product a K -linear mapping

$$\lambda_{(\varphi_1, \dots, \varphi_n)} : V_1 \otimes \cdots \otimes V_n \longrightarrow K.$$

Moreover, the uniqueness part of the universal property shows that the mapping

$$V_1^\vee \times \cdots \times V_n^\vee \longrightarrow (V_1 \otimes \cdots \otimes V_n)^\vee, \quad (\varphi_1, \dots, \varphi_n) \longmapsto \lambda_{(\varphi_1, \dots, \varphi_n)}$$

is n -linear, and hence induces a K -linear mapping from

$$V_1^\vee \otimes \cdots \otimes V_n^\vee \longrightarrow (V_1 \otimes \cdots \otimes V_n)^\vee$$

that sends $\varphi_1 \otimes \cdots \otimes \varphi_n$ to $\lambda_{(\varphi_1, \dots, \varphi_n)}$.

Assume that $V_1, \dots, V_n$ are free K -module of finite type. For any $i \in \{1, \dots, n\}$, let $(e_{i,j})_{j=1}^{r_i}$ be a basis of V_i and let $(e_{i,j}^\vee)_{j=1}^{r_i}$ be its dual basis. We construct a K -linear mapping

$$\Psi : (V_1 \otimes \cdots \otimes V_n)^\vee \longrightarrow V_1^\vee \otimes \cdots \otimes V_n^\vee$$

that sends $\psi \in (V_1 \otimes \cdots \otimes V_n)^\vee$ to

$$\sum_{(j_1, \dots, j_n) \in \prod_{i=1}^n \{1, \dots, r_i\}} \psi(e_{1,j_1} \otimes \cdots \otimes e_{n,j_n}) e_{1,j_1}^\vee \otimes \cdots \otimes e_{n,j_n}^\vee.$$

We check that Ψ is the inverse of Φ . By definition, for $\psi \in (V_1 \otimes \cdots \otimes V_n)^\vee$,

$$\Phi\left(\sum_{(j_1, \dots, j_n) \in \prod_{i=1}^n \{1, \dots, r_i\}} \psi(e_{1,j_1} \otimes \cdots \otimes e_{n,j_n}) e_{1,j_1}^\vee \otimes \cdots \otimes e_{n,j_n}^\vee\right)$$

sends $e_{1,j_1} \otimes \cdots \otimes e_{n,j_n}$ to

$$\psi(e_{1,j_1} \otimes \cdots \otimes e_{n,j_n}),$$

so that $\Phi(\Psi(\psi)) = \psi$. Conversely, for any $(\varphi_1, \dots, \varphi_n) \in V_1^\vee \times \cdots \times V_n^\vee$, Ψ sends $\Phi(\varphi_1 \otimes \cdots \otimes \varphi_n)$ to

$$\sum_{(j_1, \dots, j_n) \in \prod_{i=1}^n \{1, \dots, r_i\}} \varphi_1(e_{1,j_1}) \cdots \varphi_n(e_{n,j_n}) e_{1,j_1}^\vee \otimes \cdots \otimes e_{n,j_n}^\vee.$$

By multilinearity of the tensor product, this is equal to

$$\left(\sum_{j_1=1}^{r_1} \varphi_1(e_{1,j_1}) e_{1,j_1}^\vee\right) \otimes \cdots \otimes \left(\sum_{j_n=1}^{r_n} \varphi_n(e_{n,j_n}) e_{n,j_n}^\vee\right) = \varphi_1 \otimes \cdots \otimes \varphi_n.$$

□

7.1.9 Proposition. *Let K be a commutative unitary ring, n be a natural number and V be K -module. Let φ be an element of $(V^\vee)^{\otimes n}$ and let $\lambda_\varphi : V^n \rightarrow K$ be the n -linear form on V^n corresponding to $\Phi(\varphi)$, where*

$$\Phi : (V^\vee)^{\otimes n} \longrightarrow (V^{\otimes n})^\vee$$

is the K -linear mapping defined in Proposition 7.1.8. If φ is symmetric, then, for any $\sigma \in S_n$, one has

$$\forall (x_1, \dots, x_n) \in V^n, \quad \lambda_\varphi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = \lambda_\varphi(x_1, \dots, x_n).$$

The converse is true when V is a free K -module of finite type.

Proof. Note that φ can be written in the form

$$\sum_{i=1}^m \varphi_{i,1} \otimes \cdots \otimes \varphi_{i,n},$$

where $\varphi_{i,j} \in V^\vee$. In this case, one has

$$\lambda_\varphi(x_1, \dots, x_n) = \sum_{i=1}^m \varphi_{i,1}(x_1) \cdots \varphi_{i,n}(x_n).$$

If φ is symmetric, for any $\sigma \in S_n$, one has

$$\varphi = \sum_{i=1}^m \varphi_{i,\sigma^{-1}(1)} \otimes \cdots \otimes \varphi_{i,\sigma^{-1}(n)},$$

so that

$$\begin{aligned}\lambda_\varphi(x_1, \dots, x_n) &= \sum_{i=1}^m \varphi_{i, \sigma^{-1}(1)}(x_1) \cdots \varphi_{i, \sigma^{-1}(n)}(x_n) \\ &= \sum_{i=1}^m \varphi_{i, 1}(x_{\sigma(1)}) \cdots \varphi_{i, n}(x_{\sigma(n)}) = \lambda_\varphi(x_{\sigma(1)}, \dots, x_{\sigma(n)}).\end{aligned}$$

Assume now that V is a free K -module of finite type. Let $(e_j)_{j=1}^r$ be a basis of V , and $(e_j^\vee)_{j=1}^r$ be its dual basis. Then one has

$$\varphi = \sum_{(j_1, \dots, j_n) \in \{1, \dots, r\}^n} \lambda_\varphi(e_{j_1}, \dots, e_{j_n}) e_{j_1}^\vee \otimes \cdots \otimes e_{j_n}^\vee.$$

For any $\sigma \in S_n$, one has

$$\begin{aligned}& \sum_{(j_1, \dots, j_n) \in \{1, \dots, r\}^n} \lambda_\varphi(e_{j_1}, \dots, e_{j_n}) e_{j_{\sigma^{-1}(1)}}^\vee \otimes \cdots \otimes e_{j_{\sigma^{-1}(n)}}^\vee \\ &= \sum_{(j_1, \dots, j_n) \in \{1, \dots, r\}^n} \lambda_\varphi(e_{j_{\sigma(1)}}, \dots, e_{j_{\sigma(n)}}) e_{j_1}^\vee \otimes \cdots \otimes e_{j_n}^\vee.\end{aligned}$$

If, for any $(x_1, \dots, x_n) \in V^n$, one has

$$\lambda_\varphi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = \lambda_\varphi(x_1, \dots, x_n),$$

then the equality

$$\sum_{(j_1, \dots, j_n) \in \{1, \dots, r\}^n} \lambda_\varphi(e_{j_1}, \dots, e_{j_n}) e_{j_{\sigma^{-1}(1)}}^\vee \otimes \cdots \otimes e_{j_{\sigma^{-1}(n)}}^\vee = \varphi$$

holds. □

7.1.10 Proposition. *Let K be a commutative unitary ring, n be a positive integer such that $n \geq 1$, and V be a K -module. Let φ be a symmetric tensor in $V^{\otimes n}$. Suppose that 2 is invertible in K . Then the image of φ by the mapping*

$$\pi : V^{\otimes n} \longrightarrow \wedge^n(V), \quad x_1 \otimes \cdots \otimes x_n \longmapsto x_1 \wedge \cdots \wedge x_n$$

is zero.

Proof. Let σ be a transposition, that is a permutation which exchanges two elements of $\{1, \dots, n\}$ while fixing other elements. By definition, one has $\sigma(\varphi) = \varphi$, and $\pi(\sigma(\varphi)) = -\pi(\varphi)$. Hence we obtain that $\pi(\varphi) = -\pi(\varphi)$, which means that $\pi(\varphi) = 0$ since 2 is invertible in K . □

7.1.11 Proposition. *Let p be a natural number, U be an open subset of E and $\alpha : U \rightarrow \wedge^p(E^*)$ be a p -form on U . We assume that α is of class C^2 . Then $d^2(\alpha) = 0$.*

Proof. By the linearity of d , we may assume that α is of the form $f\beta$, where β is an element of $\Lambda^p(E^*)$, and f is a mapping of class C^2 from U to $\mathbb{R}$. In this case, one has $d^2(\alpha) = d^2f \wedge \beta$. Note that $d^2f = d(Df)$. Therefore, for any $x \in U$, $d^2f(x)$ is equal to the image of $D^2f(x)$ in $\Lambda^2(E^*)$. By what we have seen in the first semester, $D^2f(x)$ is a symmetric bilinear form on E . By Propositions 7.1.9 and 7.1.10, we obtain that $d^2f = 0$. $\square$

7.1.12 Definition. We also fix a non-zero element ξ of $\det(E)$. Note that ξ is of the form $e_1 \wedge \cdots \wedge e_n$, where $(e_i)_{i=1}^n$ is a basis of E over $\mathbb{R}$. Let μ_ξ be a Haar measure on E such that the measure of the parallelotope

$$\{a_1e_1 + \cdots + a_ne_n \mid (a_1, \dots, a_n) \in]0, 1]^n\}$$

is equal to 1. By Theorem 5.6.9, the choice of the Haar measure μ_ξ only depends on the non-zero element ξ but not on the choice of the basis $(e_i)_{i=1}^n$ of E such that $\xi = e_1 \wedge \cdots \wedge e_n$. Moreover, for any non-zero real number b , one has

$$\mu_{b\xi} = |b|^{-1} \mu_\xi.$$

Let U be an open subset of E and $\alpha : U \rightarrow \Lambda^n(E^*)$ be an n -form. If the mapping

$$(x \in U) \longmapsto \alpha(x)(\xi) \in \mathbb{R}$$

is integrable with respect to the Haar measure μ_ξ , then we say that the differential form α is integrable, and we denote by

$$\int_U \alpha$$

the integral

$$\int_U \alpha(x)(\xi) \mu_\xi(dx).$$

Note that the integral only depends on the orbit of ξ in $\det(E)$ by the action of $\mathbb{R}_{>0}$, which is called an *orientation* of E .

Let p be an element of $\{0, \dots, n\}$ and let $\alpha : U \rightarrow \Lambda^p(E^*)$ be a p -form. If, for any $\beta \in \Omega_c^{n-p}(U)$, $\alpha \wedge \beta$ is an integrable n -form, then we say that α is *locally integrable*. We denote by $\Omega_{\text{loc}}^p(U)$ the set of all locally integrable p -forms on U .

7.2 Localization and mollification

In this section, we fix a finite-dimensional vector space E over $\mathbb{R}$.

7.2.1 Lemma. *The mapping*

$$b : \mathbb{R} \longrightarrow [0, 1], \quad b(t) := \begin{cases} \exp\left(\frac{1}{t^2-1}\right), & |t| < 1, \\ 0, & |t| \geq 1. \end{cases}$$

is smooth (namely, of class C^∞)

Proof. We first prove that, for any natural number n ,

$$\lim_{x \rightarrow +\infty} x^n e^{-x} = 0.$$

In fact, by the equality

$$\forall x \in \mathbb{R}, \quad e^x = \sum_{k \in \mathbb{N}} \frac{x^k}{k!},$$

we obtain that $e^x \geq x^{n+1}/(n+1)!$, so that

$$\forall x \in \mathbb{R}_{>0}, \quad 0 \leq x^n e^{-x} \leq \frac{(n+1)!}{x}.$$

Taking the limit when $x \rightarrow +\infty$, we get

$$\lim_{x \rightarrow +\infty} x^n e^{-x} = 0.$$

We now prove by induction that, the mapping b is n times differentiable and $b^{(n)}$ is of the form

$$b^{(n)}(t) = \begin{cases} \frac{F_n(t)}{(t^2-1)^{2n}} b(t), & |t| < 1 \\ 0, & |t| \geq 1, \end{cases}$$

where F_n is a polynomial. By definition, the statement is true when $n = 0$. Suppose that the statement is true for n , then the function $b^{(n)}$ is differentiable on $] -1, 1[$ and, on this interval,

$$\begin{aligned} b^{(n+1)}(t) &= \frac{F_n(t)}{(t^2-1)^{2n}} b'(t) - \frac{4ntF_n(t)}{(t^2-1)^{2n+1}} b(t) + \frac{F'_n(t)}{(t^2-1)^{2n}} b(t) \\ &= \frac{-2tF_n(t) - 4nt(t^2-1)F_n(t) + (t^2-1)^2 F'_n(t)}{(t^2-1)^{2n+2}} b(t). \end{aligned}$$

Moreover, $b^{(n)}$ is also differentiable on $\mathbb{R} \setminus [-1, 1]$, and $b^{(n+1)}(t) = 0$ for any $t \in \mathbb{R} \setminus [-1, 1]$.

Note that, by the change of variable $x = -1/(t^2-1)$, we obtain

$$\lim_{|t| \rightarrow 1-} \frac{b(t)}{(t^2-1)^{2n+1}} = \lim_{x \rightarrow +\infty} x^{2n+1} e^{-x} = 0.$$

As a consequence, we obtain that

$$\lim_{t \rightarrow 1} \frac{b^{(n)}(t)}{t-1} = \lim_{t \rightarrow -1} \frac{b^{(n)}(t)}{t+1} = 0.$$

In particular, $b^{(n)}$ is differentiable at 1 and -1 , and $b^{(n+1)}(1) = b^{(n+1)}(-1) = 0$. $\square$

7.2.2 Proposition. *Let U be an open subset of E and K be a compact subset of E such that $K \subseteq U$. There exists a smooth mapping $\rho : E \rightarrow [0, 1]$ such $\rho(x) = 1$ for any $x \in K$ and $\text{Supp}(\rho)$ is compact and is contained in U .*

Proof. Without loss of generality, we may assume that $E = \mathbb{R}^n$, where n is a positive integer. We equip $\mathbb{R}^n$ with the norm $\|\cdot\|$ defined as

$$\|(x_1, \dots, x_n)\| = \max_{i \in \{1, \dots, n\}} |x_i|.$$

Let $\chi : \mathbb{R}^n \rightarrow \mathbb{R}$ which is defined as

$$\chi(x_1, \dots, x_n) = Cb(x_1) \cdots b(x_n),$$

where C is a positive constant such that

$$C \left(\int_{\mathbb{R}} b(t) dt \right)^n = 1.$$

Note that Fubini's theorem shows that

$$\int_{\mathbb{R}^n} \chi(x_1, \dots, x_n) d(x_1, \dots, x_n) = 1,$$

where $d(x_1, \dots, x_n)$ denotes Lebesgue measure on $\mathbb{R}^n$. For any $\varepsilon \in \mathbb{R}_{>0}$, we denote by χ_ε the mapping from E to $\mathbb{R}$ defined as

$$\chi_\varepsilon(x) := \frac{1}{\varepsilon^n} \chi(x/\varepsilon).$$

By the formula of change of variables, one has

$$\int_{\mathbb{R}^n} \chi_\varepsilon(x_1, \dots, x_n) d(x_1, \dots, x_n) = 1.$$

For any $\varepsilon > 0$, we let

$$K_\varepsilon = \bigcup_{y \in K} \overline{B}(y, \varepsilon).$$

Clearly, for any $x \in K_\varepsilon$, one has

$$d(x, K) := \inf_{y \in K} d(x, y) \leq \varepsilon.$$

Conversely, if x is an element of E such that $d(x, K) \leq \varepsilon$, then there exists a sequence $(y_n)_{n \in \mathbb{N}}$ of elements in K such that

$$\lim_{n \rightarrow +\infty} \|x - y_n\| \leq \varepsilon.$$

By extracting a subsequence, we may assume without loss of generality $(y_n)_{n \in \mathbb{N}}$ converges to some $y \in K$. Hence $d(x, y) \leq \varepsilon$. Therefore, we obtain that

$$K_\varepsilon = \{x \in E \mid d(x, K) \leq \varepsilon\}$$

is a compact subset of E .

We claim that there exists $\varepsilon > 0$ such that $K_{2\varepsilon} \subseteq U$. Let $(\varepsilon_n)_{n \in \mathbb{N}}$ be a decreasing sequence that converges to 0. Suppose by contradiction that, for any $n \in \mathbb{N}$ there exists $x_n \in K_{\varepsilon_n} \setminus U$. Since K_{ε_0} is compact, by extracting a subsequence, we may assume that $(x_n)_{n \in \mathbb{N}}$ converges to some element x . Since U is open, we obtain that $x \notin U$. However, one has $d(x, K) = 0$ and hence $x \in K \subseteq U$. This leads to a contradiction.

We now let ρ be the convolution of $\mathbb{1}_{K_\varepsilon}$ and χ_ε , which is defined as

$$\forall x \in \mathbb{R}^n, \quad \rho(x) := \int_{\mathbb{R}^n} \mathbb{1}_{K_\varepsilon}(y) \chi_\varepsilon(x - y) \, dy.$$

Since χ_ε is smooth of compact support, by the theorem of differentiability under integral, we obtain that ρ is also smooth. Moreover, by the formula of change of variables and by the fact that $\chi_\varepsilon(y)$ is an even function, one has

$$\rho(x) = \int_{\mathbb{R}^n} \mathbb{1}_{K_\varepsilon}(x - y) \chi_\varepsilon(y) \, dy = \int_{\mathbb{R}^n} \mathbb{1}_{K_\varepsilon}(x + y) \chi_\varepsilon(y) \, dy.$$

For any $x \in K$, one has

$$\rho(x) = \int_{\mathbb{R}^n} \chi_\varepsilon(y) \, dy = 1.$$

For any $x \notin K_{2\varepsilon}$, one has

$$\rho(x) = \int_{K_{2\varepsilon}} \mathbb{1}_{K_\varepsilon}(x + y) \chi_\varepsilon(y) \, dy = 0.$$

Hence the support of ρ is contained in $K_{2\varepsilon}$. In particular ρ is of compact support and $\text{Supp}(\rho) \subseteq U$. $\square$

7.2.3 Remark. Let U be an open subset of E , K be a compact subset of U and $f : U \rightarrow \mathbb{R}$ be a mapping. We pick a smooth mapping $\rho : E \rightarrow \mathbb{R}$ of compact support such that $\rho(x) = 1$ for any $x \in K$ and that the support of ρ is contained in U . The mapping

$$(x \in E) \longmapsto \begin{cases} f(x)\rho(x), & x \in U, \\ 0, & x \in E \setminus U. \end{cases}$$

coincides with f on K , of compact support contained in U , and is of class C^ℓ if f is of class C^ℓ .

7.2.4 Lemma. *Let Ω be a set, $S \subseteq \mathbb{R}^\Omega$ be a Riesz space and $I : S \rightarrow \mathbb{R}$ be an integral operator. Then S is dense in $L^1(I)$, namely, for any $f \in L^1(I)$ and any $\varepsilon > 0$ there exists $g \in S$ such that $I(|f - g|) \leq \varepsilon$.*

Proof. Without loss of generality, we may assume that f takes non-negative values. For any $\varepsilon > 0$, there exists $u \in S^\dagger$ such that $f \leq u$ and that

$$I(u) - I(f) \leq \varepsilon/2.$$

Let $(g_n)_{n \in \mathbb{N}} \in S^\mathbb{N}$ be an increasing sequence that converges pointwise to u . For any $n \in \mathbb{N}$, let $f_n = \inf\{f, g_n\}$. One has $f_n \leq g_n$ and $g_n - f_n \leq g_n - f \leq u - f$. Hence one has

$$I(|g_n - f_n|) = I(g_n - f_n) \leq I(u - f) \leq \varepsilon/2.$$

Moreover, the sequence $(f - f_n)_{n \in \mathbb{N}}$ is decreasing and converges pointwise to 0. Since $I : L^1(I) \rightarrow \mathbb{R}$ is an integral operator, one has

$$\lim_{n \rightarrow +\infty} I(f - f_n) = 0.$$

Hence, there exists $n \in \mathbb{N}$ such that

$$I(|f - f_n|) = I(f - f_n) \leq \varepsilon/2.$$

Hence

$$I(|g_n - f|) \leq \varepsilon.$$

□

7.2.5 Proposition. *We equip E with a norm $\|\cdot\|$ and with a Haar measure μ . Let $\chi : E \rightarrow [0, 1]$ be a smooth mapping with support contained in*

$$B := \{x \in E \mid \|x\| \leq 1\},$$

such that

$$\int_E \chi(x) \mu(dx) = 1.$$

Let $f : E \rightarrow \mathbb{R}$ be a mapping that is locally integrable with respect to the measure μ . For any $\varepsilon > 0$, we let $\chi_\varepsilon : E \rightarrow \mathbb{R}$ be the mapping defined as

$$\chi_\varepsilon(x) := \frac{1}{\varepsilon} \chi(x/\varepsilon),$$

where n is the dimension of E . Let f_ε be the convolution of f and χ_ε , which is defined as

$$f_\varepsilon(x) := \int_E f(x - y) \chi_\varepsilon(y) dy = \int_E f(y) \chi_\varepsilon(x - y) dy.$$

(1) Suppose that f is continuous at some $x_0 \in E$, then

$$\lim_{\varepsilon \rightarrow 0} f_\varepsilon(x_0) = f(x_0).$$

(2) Let U be an open subset of E , on which the mapping f is uniformly continuous. For any compact subset K of E such that $K \subseteq U$, the family $(f_\varepsilon)_{\varepsilon > 0}$ converges uniformly to f on K .

(3) Suppose that f is integrable on E . Then

$$\lim_{\varepsilon \rightarrow 0} \int_E |f_\varepsilon(x) - f(x)| \mu(dx) = 0.$$

Proof. For any $\varepsilon > 0$, let B_ε be the ball $\{x \in E \mid \|x\| \leq \varepsilon\}$.

(1) Since

$$\int_E \chi_\varepsilon(y) \mu(dy) = 1,$$

we obtain that

$$\begin{aligned} |f_\varepsilon(x_0) - f(x_0)| &= \left| \int_E (f(x_0 - y) - f(x_0)) \chi_\varepsilon(y) dy \right| \\ &\leq \int_E |f(x_0 - y) - f(x_0)| \chi_\varepsilon(y) dy \\ &\leq \sup_{y \in B_\varepsilon} |f(x_0 - y) - f(x_0)|. \end{aligned}$$

Since f is continuous at x_0 , one has

$$\lim_{\varepsilon \rightarrow 0} \sup_{y \in B_\varepsilon} |f(x_0 - y) - f(x_0)| = 0,$$

which leads to

$$\lim_{\varepsilon \rightarrow 0} f_\varepsilon(x_0) = f(x_0).$$

(2) Let ε_0 be a positive number such that

$$K_\varepsilon := \{x \in E \mid d(x, K) \leq \varepsilon\} \subseteq U.$$

By (1), for any $x \in K$ one has

$$\sup_{x \in K} |f_\varepsilon(x) - f(x)| \leq \sup_{x \in K} \sup_{y \in B_\varepsilon} |f(x - y) - f(x)| \leq \sup_{\substack{(a,b) \in U \times U \\ \|a-b\| \leq \varepsilon}} |f(a) - f(b)|.$$

Hence

$$\lim_{\varepsilon \rightarrow 0} \sup_{x \in K} |f_\varepsilon(x) - f(x)| = 0.$$

(3) We first consider the case where f belongs to $C_c(E)$. Let K be the support of f . Note that the mapping f is uniformly continuous on $\mathbb{R}$. Let

$$K_1 = \{y \in E \mid d(y, K) \leq 1\}.$$

For $\varepsilon < 1$, the support of the function f_ε is contained in K_1 . Moreover, by (2), one has

$$\limsup_{\varepsilon \rightarrow 0} \sup_{x \in E} |f_\varepsilon(x) - f(x)| = 0.$$

Since, for any $\varepsilon \in]0, 1[$

$$0 \leq \int_E |f_\varepsilon(x) - f(x)| \mu(dx) \leq \mu(K_1) \sup_{x \in E} |f_\varepsilon(x) - f(x)|,$$

we obtain that

$$\lim_{\varepsilon \rightarrow 0} \int_E |f_\varepsilon(x) - f(x)| \mu(dx) = 0.$$

We now consider the general case. Let $\delta > 0$. By Lemma 7.2.4, there exists $g \in C_c(E)$ such that

$$\int_E |f(x) - g(x)| \mu(dx) \leq \delta.$$

Note that

$$\begin{aligned} \int_E |f_\varepsilon(x) - g_\varepsilon(x)| \mu(dx) &= \int_E \left| \int_E (f(y) - g(y)) \chi_\varepsilon(x - y) \mu(dy) \right| \mu(dx) \\ &\leq \int_E \int_E |f(y) - g(y)| \chi_\varepsilon(x - y) \mu(dy) \mu(dx) \\ &= \int_E \int_E |f(y) - g(y)| \chi_\varepsilon(x - y) \mu(dx) \mu(dy) \\ &= \int_E |f(y) - g(y)| \mu(dy). \end{aligned}$$

Therefore, by the triangle inequality, we get

$$\int_E |f_\varepsilon(x) - f(x)| \mu(dx) \leq \int_E |g_\varepsilon(x) - g(x)| \mu(dx) + 2\delta,$$

which leads to

$$\limsup_{\varepsilon \rightarrow 0} \int_E |f_\varepsilon(x) - f(x)| \mu(dx) \leq 2\delta.$$

Since $\delta > 0$ is arbitrary, we get

$$\lim_{\varepsilon \rightarrow 0} \int_E |f_\varepsilon(x) - f(x)| \mu(dx) = 0.$$

□

7.2.6 Theorem (Taylor's formula). *Let $(E, \|\cdot\|)$ be a Banach space, U be an open subset of E , $a \in U$ and $h \in E$. Suppose that, for any $t \in [0, 1]$, one has $a + th \in U$. Let $f : U \rightarrow \mathbb{R}$ be a mapping of class C^{n+1} . Then the following equality holds:*

$$f(a + h) = \sum_{k=0}^n \frac{1}{k!} D^k f(a)(h, \dots, h) + \int_0^1 \frac{(1-t)^n}{n!} D^{n+1} f(a + th)(h, \dots, h) dt.$$

Proof. We reason by induction on n . In the case where $n = 0$, one has

$$\frac{d}{dt} f(a + th) = Df(a + th)(h),$$

so that

$$\int_0^t Df(a + th)(h) dt = f(a + h) - f(a).$$

Suppose that the formula holds for n . By the relation

$$\frac{d}{dt} D^n f(a + th)(h, \dots, h) = D^{n+1} f(a + th)(h, \dots, h),$$

by the integration by parts, we get

$$\begin{aligned} & \int_0^1 \frac{(1-t)^n}{n!} D^{n+1} f(a + th)(h, \dots, h) dt \\ &= -\frac{1}{n!} D^n f(a)(h, \dots, h) + \int_0^1 \frac{(1-t)^{n-1}}{n!} D^n f(a + th)(h, \dots, h) dt. \end{aligned}$$

Thus, by induction we obtain the desired result. $\square$

7.2.7 Remark. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be a convex open subset of E , namely U is an open subset of E such that, for any $(x, y) \in U \times U$ and any $t \in [0, 1]$, one has $tx + (1-t)y \in U$. Let $a \in U$ and $f : U \rightarrow \mathbb{R}$ be a mapping of class C^ℓ , where $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. By Theorem 7.2.6, one has

$$f(a + h) = f(a) + \int_0^1 Df(a + th)(h) dt.$$

Let $(e_i)_{i=1}^n$ be a basis and $(e_i^\vee)_{i=1}^n$ be its dual basis. Then one can write $f(a + h)$ as

$$f(a) + \sum_{i=1}^n e_i^\vee(h) \int_0^1 Df(a + th)(e_i) dt.$$

In other word, there exists continuous mappings $g_1, \dots, g_n$ from U to $\mathbb{R}$ such that

$$\forall x \in U, \quad f(x) = f(a) + \sum_{i=1}^n g_i(x) e_i^\vee(x - a),$$

where

$$g_i(x) = \int_0^1 Df(tx + (1-t)a)(e_i) dt.$$

In particular,

$$g_i(a) = Df(a)(e_i),$$

and g_i is of class $C^{\ell-1}$.

7.2.8 Definition. Let k be a commutative unitary ring and A be a commutative k -algebra. Let $P : A \rightarrow k$ be a homomorphism of k -algebras. For any $f \in A$, we denote by $f(P)$ the image of f in k by the homomorphism P . We call *point derivation* at P any homomorphism of k -modules $\partial : A \rightarrow k$ which satisfies the following Leibniz rule:

$$\forall (f, g) \in A \times A, \quad \partial(fg) = f(P)\partial(g) + g(P)\partial(f).$$

We denote by $\text{Der}_P(A)$ the set of all point derivations at P of the k -algebra A . It forms a sub- k -module of k^A .

7.2.9 Example. Let $(E, \|\cdot\|)$ be a Banach space over $\mathbb{R}$, U be a non-empty open subset of E , and $P \in U$. Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. By abuse of notation, we use P to denote the evaluation mapping

$$(f \in C^\ell(U)) \mapsto f(P).$$

This is a homomorphism of $\mathbb{R}$ -algebras. For any $v \in E$, the mapping

$$(f \in C^\ell(U)) \mapsto Df(P)(v)$$

is a point derivation at P . For any $c \in k$ we define $\partial(c)$ via identification $c \mapsto c \cdot 1_A$.

7.2.10 Proposition. Let k be a commutative unitary ring, A be a commutative k -algebra, $P : A \rightarrow k$ be a homomorphism of k -algebras and I be the kernel of P .

- (1) Let $\partial : A \rightarrow k$ be a point derivation at P . For any $f \in I^2$, one has $\partial(f) = 0$. Hence ∂ induces a homomorphism of k -modules from I/I^2 to k .
- (2) Suppose that 2 is not a zero divisor in k (namely there does not exist $b \in k \setminus \{0\}$ such that $2b = 0$). For any $c \in k$ one has $\partial(c) = 0$.
- (3) Suppose that 2 is not a zero divisor in k . The mapping

$$\Phi : \text{Der}_P(A) \longrightarrow \text{Hom}_k(I/I^2, k), \quad \partial \mapsto ([f] \mapsto \partial(f))$$

is an isomorphism of k -modules.

Proof. (1) By definition, any element u of I^2 can be written as a sum

$$\sum_{i=1}^n f_i g_i,$$

where $f_1, \dots, f_n, g_1, \dots, g_n$ are elements of I . In particular,

$$f_1(P) = \dots = f_n(P) = g_1(P) = \dots = g_n(P) = 0.$$

By Leibniz law, one has

$$\partial(u) = \sum_{i=1}^n \partial(f_i g_i) = \sum_{i=1}^n (f_i(P) \partial(g_i) + g_i(P) \partial(f_i)) = 0.$$

(2) By definition, one has

$$\partial(1) = \partial(1^2) = \partial(1) + \partial(1) = 2\partial(1).$$

Since 2 is not a zero divisor, we obtain that $\partial(1) = 0$. Since ∂ is k -linear, for any $c \in k$ one has $\partial(c) = 0$.

(3) For any k -linear mapping $\lambda : I/I^2 \rightarrow k$, we let $\Psi(\lambda)$ be the mapping

$$A \longrightarrow k, \quad f \longmapsto \lambda([f - f(P)]).$$

We claim that $\Psi(\lambda)$ is a derivation and that

$$\Psi : \text{Hom}_k(I/I^2, k) \longrightarrow \text{Der}_P(A)$$

is the inverse of Φ .

Let f and g be two elements of A . One has $(f - f(P))(g - g(P)) \in I^2$, and hence

$$\begin{aligned} [fg - f(P)g(P)] &= [f(P)(g - g(P)) + g(P)(f - f(P))] \\ &= f(P)[g - g(P)] + g(P)[f - f(P)] \end{aligned}$$

in I/I^2 . Hence

$$\Psi(\lambda)(fg) = f(P)\Psi(\lambda)(g) + g(P)\Psi(\lambda)(f).$$

Finally, for any $\partial \in \text{Der}_k(A)$, by (2) one has

$$\Psi(\Phi(\partial))(f) = \Phi(\partial)([f - f(P)]) = \partial(f - f(P)) = \partial(f).$$

For any $\lambda \in \text{Hom}_k(I/I^2, k)$, one has

$$\Phi(\Psi(\lambda))([f]) = \Psi(\lambda)(f) = \lambda(f - f(P)) = \lambda(f)$$

since $f(P) = 0$. □

7.2.11 Proposition. *Let E be a finite-dimensional vector space over $\mathbb{R}$, $\ell \in \mathbb{N} \cup \{\infty\}$, U be an open subset of E and $P \in U$. Let $\partial : C^\ell(U) \rightarrow \mathbb{R}$ be a point derivation at P . Then ∂ is local at P , namely, for any elements f and g of $C^\ell(U)$ that coincide on a neighbourhood of P , one has $\partial(f) = \partial(g)$.*

Proof. By linearity it suffices to prove that, if $f \in C^\ell(U)$ vanishes on an open neighbourhood W of a contained in U , then one has $\partial f = 0$. Let K be a compact neighbourhood of P contained in W , and let $\chi : U \rightarrow [0, 1]$ be a smooth mapping such that $\chi(x) = 1$ for any $x \in K$ and that $\chi(y) = 0$ for any $y \in E \setminus W$. By definition, the function $\chi(f)$ vanishes identically. Hence

$$0 = \partial(\chi f) = \chi(P)\partial f + f(P)\partial(\chi) = \partial(f)$$

since $\chi(P) = 1$ and $f(P) = 0$. □

7.2.12 Proposition. *Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E and $P \in U$. Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. Let I be the set of $f \in C^\ell(U)$ such that $f(P) = 0$. Then the $\mathbb{R}$ -linear mapping*

$$\Phi : E^* \longrightarrow I/I^2, \quad \varphi \longmapsto [(\varphi - \varphi(P))|_U]$$

is injective. It is an isomorphism of vector spaces over $\mathbb{R}$ when $\ell = \infty$ and U is convex.

Proof. Note that

$$C^\ell(U) \longrightarrow E^*, \quad f \longmapsto Df(P)$$

is an $\mathbb{R}$ -linear mapping. Moreover, for any $v \in E$, the mapping

$$C^\ell(U) \longrightarrow E^*, \quad f \longmapsto Df(P)(v)$$

is a point derivation at P . Hence, for any $f \in I^2$, one has $Df(P) = 0$. Thus the above $\mathbb{R}$ -linear mapping induces by passing to quotient an $\mathbb{R}$ -linear mapping

$$\Psi : I/I^2 \longrightarrow E^*, \quad [f] \longmapsto Df(P).$$

Clearly, for any $\varphi \in E^*$, one has $\Psi(\Phi(\varphi)) = \varphi$. Hence Φ is injective.

In the rest of the proof, we suppose that I is the set of all $f \in C^\infty(U)$ such that $f(P) = 0$. Let $(e_i)_{i=1}^n$ be a basis of E and $(\varphi_i)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. By Remark 7.2.7, for any $f \in I$, one can write f as

$$f = \sum_{i=1}^n g_i(\varphi_i - \varphi_i(P)),$$

where $g_i \in C^\infty(U)$ and $g_i(P) = Df(P)(e_i)$. In particular, $f \in I^2$ if and only if $Df(P) = 0$. Therefore, for any $f \in I$, one has

$$f - Df(P) + Df(P)(P) \in I^2,$$

namely

$$[f] = [Df(P) - Df(P)(P)].$$

It hence belongs to the image of Φ . □

7.3 Vector fields

Throughout this section we fix a finite-dimensional vector space E over $\mathbb{R}$ and an open set $U \subseteq E$.

7.3.1 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$ and $U \subseteq E$ be an open subset of E . We call *tangent vector field* on U any mapping $X : U \rightarrow E$.

Given a tangent vector field X and a differentiable mapping $f : U \rightarrow \mathbb{R}$, we let denote by $X(f)$ the mapping from U to $\mathbb{R}$ that sends $x \in U$ to $Df(x)(X(x))$. In particular, if f is a constant mapping, then $X(f) = 0$.

7.3.2 Proposition. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , and X be a tangent vector field on U . For any differentiable mappings f and g from U to $\mathbb{R}$, one has

$$X(fg) = fX(g) + gX(f).$$

In particular X is $\mathbb{R}$ -linear.

Proof. We have

$$D(fg) = fD(g) + gD(f).$$

Applying this equality to X , we obtain $X(fg) = fX(g) + gX(f)$. □

7.3.3 Remark. One can easily see that:

- (1) If $(e_i)_{i=1}^n$ is a basis of E and $(e_i^\vee)_{i=1}^n$ be its dual basis, then any tangent vector field X can be written as

$$\forall x \in U \quad X(x) = \sum_{i=1}^n e_i^\vee(X(x))e_i = \sum_{i=1}^n X(e_i^\vee)(x)e_i.$$

In particular, X is of class C^m if and only if, for any $\varphi \in E^*$, $X(\varphi)$ is of class C^m . If X and Y are tangent vector fields such that $X(\varphi) = Y(\varphi)$ for any $\varphi \in E^*$, then $X = Y$.

- (2) If X is of class C^m , for any $f : U \rightarrow \mathbb{R}$ of class C^{m+1} , $X(f)$ is of class C^m .

7.3.4 Proposition. *Let X and Y be tangent vector fields of class C^1 on U . There exists a unique tangent vector field $[X, Y]$ of class C^0 on U such that, for any $h \in C^2(U)$,*

$$[X, Y](h) = X(Y(h)) - Y(X(h))$$

Proof. By linearity, we can assume that $X = fv$ and $Y = gw$, where $(f, g) \in C^1(U)^2$, $(v, w) \in E^2$. So,

$$\begin{aligned} X(Y(h)) &= f\partial_v(Y(h)) = f\partial_v(g\partial_w(h)) = f\partial_v(g)\partial_w(h) + fg\partial_v\partial_w(h), \\ Y(X(h)) &= g\partial_w(X(h)) = g\partial_w(f\partial_v(h)) = g\partial_w(f)\partial_v(h) + fg\partial_w\partial_v(h). \end{aligned}$$

Since h is of class C^2 , one has $\partial_v\partial_w(h) = \partial_w\partial_v(h)$. Therefore, We obtain

$$X(Y(h)) - Y(X(h)) = f\partial_v(g)\partial_w(h) - g\partial_w(f)\partial_v(h).$$

If we take

$$[X, Y] = f\partial_v(g)w - g\partial_w(f)v$$

then

$$[X, Y](h) = X(Y(h)) - Y(X(h)).$$

The uniqueness of $[X, Y]$ comes from the fact that a tangent vector field is uniquely determined by its value at linear forms. $\square$

7.3.5 Remark. The vector field $[X, Y]$ is called Lie bracket of X and Y . It has the following properties:

- (1) $(X, Y) \mapsto [X, Y]$ is $\mathbb{R}$ linear.
- (2) If X and Y are of class C^m , $m \geq 1$, then $[X, Y]$ is of class C^{m-1} .

7.3.6 Definition. Let E and E' be finite-dimensional vector spaces over $\mathbb{R}$, U and U' be open subsets of E and E' respectively, and $\psi : U \rightarrow U'$ be a C^1 diffeomorphism. If X is a tangent vector field of U , we denote by $\psi_*(X) : U' \rightarrow E'$ the mapping that sends $y \in U'$ to

$$D\psi(\psi^{-1}(y))(X(\psi^{-1}(y))).$$

It is a tangent vector field on U' . Note that, if X is of class C^m and ψ is of class C^{m+1} , then $\psi_*(X)$ is of class C^m .

7.3.7 Proposition. *Let E and E' be finite-dimensional vector spaces over $\mathbb{R}$, U and U' be open subsets of E and E' respectively, and $\psi : U \rightarrow U'$ be a C^2 diffeomorphism. If X and Y are tangent vector field of class C^1 on U , then one has*

$$\psi_*([X, Y]) = [\psi_*(X), \psi_*(Y)].$$

Proof. Let $h \in C^2(U')$ and let $f = h \circ \psi$. By definition, for any tangent vector field Z on U , for any $x \in U$ one has

$$\psi_*(X)(h)(\psi(x)) = Dh(\psi(x))(D\psi(x)(X(x))) = D(h \circ \psi)(x)(X(x)) = X(h \circ \psi)(x).$$

Therefore, $\psi_*(X)(h) = X(h \circ \psi) \circ \psi^{-1}$. In particular,

$$\begin{aligned} [\psi_*(X), \psi_*(Y)](h) &= \psi_*(X)(Y(h \circ \psi) \circ \psi^{-1}) - \psi_*(Y)(X(h \circ \psi) \circ \psi^{-1}) \\ &= X(Y(h \circ \psi)) \circ \psi^{-1} - Y(X(h \circ \psi)) \circ \psi^{-1} \\ &= [X, Y](h \circ \psi) \circ \psi^{-1} = \psi_*([X, Y])(h). \end{aligned}$$

□

7.3.8 Remark. Let K be a commutative unitary ring and V be a K -module. Let $p \in \mathbb{N}$.

For any $(\varphi_1, \dots, \varphi_p) \in (V^\vee)^p$, the mapping

$$V^p \longrightarrow K, \quad (e_1, \dots, e_p) \longmapsto \det(\varphi_i(e_j))_{(i,j) \in \{1, \dots, p\}^2}$$

is multilinear and alternating. Hence induces a K -linear form

$$\tilde{\Phi}(\varphi_1, \dots, \varphi_p) : \Lambda^p V \longrightarrow K.$$

Moreover, the mapping $\tilde{\Phi} : (V^\vee)^p \rightarrow (\Lambda^p V)^\vee$ is multilinear and alternating. Hence it induces a K -linear mapping

$$\Phi : \Lambda^p(V^\vee) \longrightarrow (\Lambda^p V)^\vee.$$

By abuse of notation, if α is an element of $\Lambda^p(V^\vee)$, for any $(x_1, \dots, x_p) \in V^p$, we denote by

$$\alpha(x_1, \dots, x_p)$$

the value $\Phi(\alpha)(x_1 \wedge \dots \wedge x_p)$. In particular, if $\theta \in V^\vee$ and $\beta \in \Lambda^p(V^\vee)$, then

$$(\theta \wedge \beta)(x_0, \dots, x_p) = \sum_{i=0}^p (-1)^i \theta(x_i) \beta(x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_p).$$

This comes from the expansion of a determinant by a row.

Suppose that V is a free K -module of finite type and $(e_i)_{i=1}^n$ is a basis of V over K . Let $(e_i^\vee)_{i=1}^n$ be the dual basis of $(e_i)_{i=1}^n$. Then

$$(e_{i_1}^\vee \wedge \dots \wedge e_{i_p}^\vee)_{1 \leq i_1 < \dots < i_p \leq n}$$

forms a basis of $\Lambda^p(V^\vee)$. Note that Φ sends this basis to the dual basis of

$$(e_{i_1} \wedge \dots \wedge e_{i_p})_{1 \leq i_1 < \dots < i_p \leq n}.$$

Therefore, in the case where V is a free K -module of finite type, the above K -linear mapping Φ is actually an isomorphism of K -modules.

7.3.9 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , and ω be a p -form on U . If $X_1, \dots, X_p$ are tangent vector fields on U , we denote by $\omega(X_1, \dots, X_p) : U \rightarrow \mathbb{R}$ the mapping sending $x \in U$ to

$$\omega(x)(X_1(x), \dots, X_p(x)) \in \mathbb{R}.$$

7.3.10 Proposition. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , $m \in \mathbb{N}_{\geq 1}$, and X be a tangent vector field of class C^m on U . For any p -form ω of class C^m on U , there exists a unique p -form $L_X(\omega)$ of class C^{m-1} on U such that

$$L_X(\omega)(X_1, \dots, X_p) = X(\omega(X_1, \dots, X_p)) - \sum_{i=1}^p \omega(X_1, \dots, [X, X_i], \dots, X_p). \quad (7.1)$$

Proof. We first check that the term on the right hand side of the equality, viewed as a function on $(X_1, \dots, X_p)$, is $C^1(U)$ -linear. Let $f \in C^1(U)$ and $j \in \{1, \dots, p\}$. One has

$$\begin{aligned} X(\omega(X_1, \dots, fX_j, \dots, X_p)) &= X(f\omega(X_1, \dots, X_j, \dots, X_p)) \\ &= X(f)\omega(X_1, \dots, X_j, \dots, X_p) + fX(\omega(X_1, \dots, X_j, \dots, X_p)) \end{aligned}$$

Moreover, if we replace X_j by fX_j , then the sum in (7.1) becomes

$$\begin{aligned} &\omega(X_1, \dots, [X, fX_j], \dots, X_p) + f \sum_{i \in \{1, \dots, p\} \setminus \{j\}} \omega(X_1, \dots, [X, X_i], \dots, X_p) \\ &= X(f)\omega(X_1, \dots, X_p) + f \sum_{i=1}^p \omega(X_1, \dots, [X, X_i], \dots, X_p) \end{aligned}$$

since

$$\begin{aligned} [X, fX_j](h) &= X(fX_j(h)) - fX_j(X(h)) \\ &= X(f)X_j(h) + fX(X_j(h)) - fX_j(X(h)) \\ &= (X(f)X_j + [X, X_j])(h). \end{aligned}$$

Taking the difference, we obtain that the right hand side of (7.1) is $C^1(U)$ -linear in its j -th coordinate.

Let i and j be indices in $\{1, \dots, n\}$ such that $i < j$. If $X_i = X_j$, then $\omega(X_1, \dots, X_p) = 0$ since ω is alternating. Similarly, the sum in (7.1) becomes

$$\omega(X_1, \dots, [X, X_i], \dots, X_j, \dots, X_p) + \omega(X_1, \dots, X_i, \dots, [X, X_j], \dots, X_p) = 0.$$

By Remark 7.3.8, there exists a unique p -form $L_X(\omega)$ such that (7.1) holds. Finally, in the case where $X_1, \dots, X_p$ are constant vector fields, the right hand side of (7.1) is of class C^{m-1} . Hence $L_X(\omega)$ is of class C^{m-1} . $\square$

7.3.11 Proposition. *Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , and X be a tangent vector field of class C^1 on U . Let p and q be two natural numbers. Let α and β be p -form and q -form on class C^1 on U , respectively. Then the following equality holds:*

$$L_X(\alpha \wedge \beta) = L_X(\alpha) \wedge \beta + \alpha \wedge L_X(\beta).$$

Proof. We reason by induction on p . We first consider the case where $p = 0$. In this case α is given by a mapping $f \in C^1(U)$. By definition,

$$\begin{aligned} L_X(f\beta)(X_1, \dots, X_q) &= X(f\beta(X_1, \dots, X_q)) - \sum_{i=1}^q f\beta(X_1, \dots, [X, X_i], \dots, X_q) \\ &= X(f)\beta(X_1, \dots, X_q) + fX(\beta(X_1, \dots, X_q)) - \sum_{i=1}^q f\beta(X_1, \dots, [X, X_i], \dots, X_q) \end{aligned}$$

So $L_X(f\beta) = X(f)\beta + fL_X(\beta)$, as required.

We then consider the case where α is a 1-form. By definition (in developing the determinant), one has

$$(\alpha \wedge \beta)(X_0, \dots, X_q) = \sum_{i=0}^q (-1)^i \alpha(X_i) \beta(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_q).$$

Therefore,

$$\begin{aligned} &L_X(\alpha \wedge \beta)(X_0, \dots, X_q) \\ &= X((\alpha \wedge \beta)(X_0, \dots, X_q)) - \sum_{i=0}^q (\alpha \wedge \beta)(X_0, \dots, [X, X_i], \dots, X_q) \\ &= \sum_{i=0}^q (-1)^i X(\alpha(X_i)) \beta(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_q) \\ &\quad - \sum_{i=0}^q (-1)^i \alpha([X, X_i]) \beta(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_q) \\ &\quad + \sum_{j=0}^q (-1)^j \alpha(X_j) X(\beta(X_0, \dots, X_{j-1}, X_{j+1}, \dots, X_q)) \\ &\quad - \sum_{i=0}^q \sum_{j=0}^{i-1} (-1)^j \alpha(X_j) \beta(X_0, \dots, X_{j-1}, X_{j+1}, \dots, [X, X_i], \dots, X_q) \\ &\quad - \sum_{i=0}^q \sum_{j=i+1}^q (-1)^j \alpha(X_j) \beta(X_0, \dots, [X, X_i], \dots, X_{j-1}, X_{j+1}, \dots, X_q) \\ &= (L_X(\alpha) \wedge \beta + \alpha \wedge L_X(\beta))(X_0, \dots, X_q). \end{aligned}$$

We now treat the case where $p \geq 2$ in assuming the equality is true when α is a $(p-1)$ -form. By multilinearity we may assume that α is of the form $\theta \wedge \omega$, where θ and ω are respectively 1-form and $(p-1)$ -form of class C^1 . By the induction hypothesis, one has

$$\begin{aligned} L_X(\theta \wedge \omega \wedge \beta) &= L_X(\theta) \wedge (\omega \wedge \beta) + \theta \wedge L_X(\omega \wedge \beta) \\ &= L_X(\theta) \wedge (\omega \wedge \beta) + \theta \wedge L_X(\omega) \wedge \beta + \theta \wedge \omega \wedge L_X(\beta) \\ &= (L_X(\theta) \wedge \omega + \theta \wedge L_X(\omega)) \wedge \beta + \theta \wedge \omega \wedge L_X(\beta) \\ &= L_X(\theta \wedge \omega) \wedge \beta + \theta \wedge \omega \wedge L_X(\beta), \end{aligned}$$

as required. $\square$

7.3.12 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , and X be a tangent vector field on U . If $p \in \mathbb{N}_{\geq 1}$ and ω is a p -form on U , we denote by $\iota_X(\omega)$ the $(p-1)$ -form on U such that

$$\iota_X(\omega)(X_1, \dots, X_{p-1}) = \omega(X, X_1, \dots, X_{p-1}).$$

7.3.13 Proposition. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , and X be a tangent vector field on U . Let p and q be natural numbers, and α and β be respectively p -form and q -form of class C^1 on U . Then one has

$$\iota_X(\alpha \wedge \beta) = \iota_X(\alpha) \wedge \beta + (-1)^p \alpha \wedge \iota_X(\beta).$$

Proof. We reason by induction on p . The case where $p = 0$ is trivial. We now consider the case where $p = 1$. For simplifying the notation, we let $X_0 = X$. One has

$$\begin{aligned} \iota_X(\alpha \wedge \beta)(X_1, \dots, X_q) &= (\alpha \wedge \beta)(X_0, \dots, X_q) \\ &= (\iota_X(\alpha) \wedge \beta)(X_1, \dots, X_q) + \sum_{i=1}^q (-1)^i \alpha(X_i) \iota_X(\beta)(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_q) \\ &= (\iota_X(\alpha) \wedge \beta - \alpha \wedge \iota_X(\beta))(X_1, \dots, X_q). \end{aligned}$$

In the following, we assume that $p \geq 2$ and that the equality holds when α is a $(p-1)$ -form. By multilinearity it suffices to treat the case where $\alpha = \theta \wedge \omega$, where θ is a 1-form and ω is a $(p-1)$ -form. One has

$$\begin{aligned} \iota_X(\alpha \wedge \beta) &= \iota_X(\theta \wedge \omega \wedge \beta) = \iota_X(\theta) \wedge \omega \wedge \beta - \theta \wedge \iota_X(\omega \wedge \beta) \\ &= \iota_X(\theta) \wedge \omega \wedge \beta - \theta \wedge \iota_X(\omega) \wedge \beta + (-1)^p \theta \wedge \omega \wedge \iota_X(\beta) \\ &= \iota_X(\theta \wedge \omega) \wedge \beta + (-1)^p \theta \wedge \omega \wedge \iota_X(\beta), \end{aligned}$$

as required. $\square$

7.3.14 Theorem (Cartan's formula). *Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E and X be a tangent vector field of class C^1 on U . For any $p \in \mathbb{N}$ and any p -form ω of class C^1 , one has*

$$L_X(\omega) = \iota_X(d\omega) + d\iota_X(\omega),$$

where, in the case where $p = 0$, $d\iota_X(\omega) = 0$ by convention.

Proof. We denote by $\delta_X(\omega)$ the term on the right hand side of the equality. By definition, for any $f \in C^1(U)$, one has

$$L_X(f) = \delta_X(f) = X(f).$$

Moreover, for any 1-form ω of class C^1 , by definition

$$L_X(\omega)(Y) = X(\omega(Y)) - \omega([X, Y]).$$

Moreover,

$$\iota_X(d\omega)(Y) + d\iota_X(\omega)(Y) = d\omega(X, Y) + Y(\omega(X)).$$

We claim that

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]).$$

To check this equality, we may assume without loss of generality that ω is of the form $f dg$, where $f \in C^1(U)$ and $g \in E^*$. In this case $d\omega = df \wedge dg$. Hence

$$d\omega(X, Y) = X(f)Y(g) - Y(f)X(g).$$

Moreover,

$$\begin{aligned} X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]) &= X(fY(g)) - Y(fX(g)) - f[X, Y](g) \\ &= X(f)Y(g) + fX(Y(g)) - Y(f)X(g) - fY(X(g)) - f[X, Y](g) \\ &= X(f)Y(g) - Y(f)X(g), \end{aligned}$$

as required.

Since any p -form of class C^1 is a $C^1(U)$ -linear combination of p -forms of the form $d\theta_1 \wedge \cdots \wedge d\theta_p$, where $(\theta_1, \dots, \theta_p) \in E^*$, to prove that $L_X = \delta_X$, by induction it suffices to check that, for any 1-form α and any p -form β , one has

$$\delta_X(\alpha \wedge \beta) = \delta_X(\alpha) \wedge \beta + \alpha \wedge \delta_X(\beta).$$

By definition,

$$\begin{aligned} \delta_X(\alpha \wedge \beta) &= \iota_X(d(\alpha \wedge \beta)) + d(\iota_X(\alpha \wedge \beta)) \\ &= \iota_X(d\alpha \wedge \beta - \alpha \wedge d\beta) + d(\iota_X(\alpha) \wedge \beta - \alpha \wedge \iota_X(\beta)) \\ &= \iota_X(d\alpha) \wedge \beta + d\alpha \wedge \iota_X(\beta) - \iota_X(\alpha) \wedge d\beta + \alpha \wedge \iota_X(d\beta) \\ &\quad + d(\iota_X(\alpha)) \wedge \beta + \iota_X(\alpha) \wedge d\beta - d\alpha \wedge \iota_X(\beta) + \alpha \wedge d(\iota_X(\beta)) \\ &= \delta_X(\alpha) \wedge \beta + \alpha \wedge \delta_X(\beta). \end{aligned}$$

□

7.3.15 Theorem. *Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E , $p \in \mathbb{N}$, and ω be a p -form of class C^1 on U . For any tangent vector fields $X_0, \dots, X_p$ of class C^1 then*

$$\begin{aligned} d\omega(X_0, \dots, X_p) &= \sum_{i=0}^p (-1)^i X_i(\omega(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_p)) + \\ &\quad \sum_{0 \leq i < j \leq p} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_{j-1}, X_{j+1}, \dots, X_p) \end{aligned}$$

Proof. We reason by induction on p . In the case where $p = 0$, ω is a mapping of class C^1 from U to $\mathbb{R}$. Hence, by definition

$$d\omega(X_0) = X_0(\omega).$$

In the case where $p = 1$, the formula reads

$$d\omega(X_0, X_1) = X_0(\omega(X_1)) - X_1(\omega(X_0)) - \omega([X_0, X_1]).$$

This has already been proved in the proof of Cartan's formula.

Suppose that the equality holds for $(p-1)$ -forms. By definition,

$$d\omega(X_0, \dots, X_p) = \iota_{X_0}(d\omega)(X_1, \dots, X_p) = L_{X_0}(\omega)(X_1, \dots, X_p) - d(\iota_{X_0}(\omega))(X_1, \dots, X_p).$$

By definition,

$$\begin{aligned} &L_{X_0}(\omega)(X_1, \dots, X_p) \\ &= X_0(\omega(X_1, \dots, X_p)) - \sum_{i=1}^p \omega(X_1, \dots, [X_0, X_i], \dots, X_p) \\ &= X_0(\omega(X_1, \dots, X_p)) + \sum_{i=1}^p (-1)^i \omega([X_0, X_i], X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_p) \end{aligned}$$

Moreover, by the induction hypothesis,

$$\begin{aligned} d(\iota_{X_0}\omega)(X_1, \dots, X_p) &= \sum_{i=1}^p (-1)^{i-1} X_i(\omega(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_p)) - \\ &\quad \sum_{1 \leq i < j \leq p} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_{j-1}, X_{j+1}, \dots, X_p). \end{aligned}$$

Taking the difference, we obtain the desired result. $\square$

7.4 Submanifolds

7.4.1 Theorem (Implicit Function Theorem). *Let E, F, G be Banach spaces over $\mathbb{R}$, U be an open subset of E , V be an open subset of F , and $(x_0, y_0) \in U \times V$. Let $f : U \times V \rightarrow G$ be a mapping that satisfies:*

- (1) $f(x_0, y_0) = 0$,
- (2) f is of class C^n on a neighbourhood of (x_0, y_0) , where $n \in \mathbb{N}_{\geq 1} \cup \{\infty\}$,
- (3) the partial derivative $D_y f(x_0, y_0) : F \rightarrow G$ is a bounded bijection.

Then there exist open subsets U_0 and V_0 of U and V respectively such that $x_0 \in U_0$ and $y_0 \in V_0$, and a unique mapping $\varphi : U_0 \rightarrow V_0$ of class C^n , such that $\varphi(x_0) = y_0$ and

$$\forall x \in U_0, \quad f(x, \varphi(x)) = 0.$$

Moreover,

$$D\varphi(x) = -(D_y f(x, \varphi(x)))^{-1} \circ D_x f(x, \varphi(x)).$$

Proof. Let $g : U \times V \rightarrow E \times G$ be the mapping defined as

$$g(x, y) = (x, f(x, y)).$$

Note that g is differentiable and

$$Dg(x_0, y_0)(u, v) = (u, D_x f(x_0, y_0)(u) + D_y f(x_0, y_0)(v)).$$

This mapping is bounded. Moreover, its inverse is given by

$$(a, b) \mapsto (a, D_y f(x_0, y_0)^{-1}(b - D_x f(x_0, y_0)(a))).$$

By the local inversion theorem, we obtain that there exists an open subset W of $E \times F$ such that $(x_0, y_0) \in W \subseteq U \times V$ the restriction of g to W defines a C^n diffeomorphism from W to $g(W)$. Note that $g(x_0, y_0) = (x_0, 0)$. We choose an open subset U_0 of E such that $x_0 \in U_0 \subseteq U$ and that, for any $x \in U_0$, $(x, 0) \in g(W)$. Thus, for any $x \in U_0$, there exists a unique element $\varphi(x) \in V$ such that $(x, \varphi(x)) \in W$ and $f(x, \varphi(x)) = 0$. Actually φ is given by the composite of the following mappings

$$U_0 \xrightarrow{x \mapsto (x, 0)} g(W) \xrightarrow{g^{-1}} W \xrightarrow{(x, y) \mapsto y} V.$$

Hence φ is of class C^n . Moreover, differentiating the constant function

$$(x \in U_0) \mapsto f(x, \varphi(x)),$$

we get

$$D_x f(x, \varphi(x)) + D_y f(x, \varphi(x)) D\varphi(x) = 0,$$

which leads to

$$D\varphi(x) = -(D_y f(x, \varphi(x)))^{-1} \circ D_x f(x, \varphi(x)).$$

□

7.4.2 Example. Consider

$$f : \mathbb{R}^2 \ni (x, y) \mapsto x^2 + y^2 - 1 \in \mathbb{R} \quad \text{and} \quad (x_0, y_0) = (0, 1).$$

Then $f(x_0, y_0) = 0$ and one easily checks that

$$\frac{\partial f}{\partial y}(x, y) = 2y \quad \text{and} \quad \frac{\partial f}{\partial y}(x_0, y_0) = 2.$$

Hence

$$\varphi(0) = \frac{\frac{\partial f}{\partial x}(0, 1)}{\frac{\partial f}{\partial y}(0, 1)} = 0.$$

7.4.3 Definition. Let E and F be Banach spaces over $\mathbb{R}$, U be an open subset of E and $f : U \rightarrow F$ be a differentiable mapping. If, for any $x \in U$, $Df(x) : E \rightarrow F$ is injective (resp. surjective), then we say that f is an *immersion* (resp. *submersion*).

7.4.4 Theorem. Let n be a natural number and E be a finite-dimensional vector space of dimension n over $\mathbb{R}$ and M be a subset of E . Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$ and $d \in \{0, \dots, n\}$. The following conditions are equivalent.

- (1) For any $a \in M$, there exists an open neighbourhood U of a , an open neighbourhood V of 0 , a vector subspace F of dimension d of E and a C^ℓ diffeomorphism $f : U \rightarrow V$ such that $f(a) = 0$ and $f(U \cap M) = V \cap F$.
- (2) For any $a \in M$, there exist an open neighbourhood U of a , a vector space G of dimension $n - d$ over $\mathbb{R}$ together with a submersion $g : U \rightarrow G$ of class C^ℓ , such that $g(a) = 0$ and $U \cap M = g^{-1}(\{0\})$.
- (3) For any $a \in M$, there exist an open neighbourhood U of a , a vector space H of dimension d over $\mathbb{R}$, an open neighbourhood W of 0 in H , and an immersion $\varphi : W \rightarrow E$ of class C^ℓ such that $\varphi(0) = a$ and that φ defines a homeomorphism between W and $U \cap M$.

Proof. “(1) $\implies$ (2)”: Take $G = E/F$ and let $g : U \rightarrow G$ be the composite of f with the projection mapping $\pi : E \rightarrow G$. For any $x \in U$, $Df(x)$ is an isomorphism. Hence $Dg(x) = \pi \circ Df(x)$ is surjective. Thus we obtain that g is a submersion. Note that

$$g^{-1}(\{0\}) = f^{-1}(\pi^{-1}(\{0\})) = f^{-1}(F \cap V) = U \cap M.$$

“(2) $\implies$ (3)”: By definition, $Dg(a) : E \rightarrow G$ is surjective. We let H be the kernel of $Dg(a)$ and let $u : G \rightarrow E$ be an $\mathbb{R}$ -linear mapping such that $Dg(a) \circ u = \text{Id}_G$. Note that E is a direct sum of H and $u(G)$. Consider the mapping

$$h : H \times u(G) \longrightarrow G, \quad (x, y) \longmapsto g(x + y + a)$$

and observe that

$$D_y h(0, 0) = Dg(a)|_{u(G)}$$

is an isomorphism. By the implicit function theorem, there exists an open neighbourhood W of 0 in H and a mapping $\psi : W \rightarrow u(G)$ of class C^ℓ such that $\psi(0) = 0$ and

$$\forall x \in W, \quad h(x, \psi(x)) = g(x + \psi(x) + a) = 0,$$

namely $x + \psi(x) + a \in U \cap M$. Let $\varphi : W \rightarrow E$ be defined as

$$\forall x \in W, \quad \varphi(x) = x + \psi(x) + a.$$

Then $\varphi(0) = a$, and

$$D\varphi(x) = i_H + D\psi(x),$$

where $i_H : H \rightarrow E$ is the mapping of inclusion. Since $D\psi(x)$ takes value in $u(G)$ and E is a direct sum of H and $u(G)$, we obtain that $D\varphi(x)$ is injective. Therefore, φ is an immersion. Similarly, since E is a direct sum of H and $u(G)$, we obtain that φ is injective. Its inverse is the restriction of the (translated) projection to H along $u(G)$ to the image of φ , which is also continuous. Hence φ is a homeomorphism to its image.

“(3) $\implies$ (1)”: Without loss of generality, we may assume that $a = 0$. Let F be the image of $D\varphi(0)$. It is a vector subspace of dimension d of E . We choose a vector subspace F' of E such that E is a direct sum of F and F' . Let $\pi : E \rightarrow F$ be the linear projection along F' . In other words, if x is an element of E that is written in a unique way as $u + u'$ where $(u, u') \in F \times F'$, then π sends x to u . Similarly, we let $\pi' : E \rightarrow F'$ be the linear projection along F .

Note that there exists an open subset W_0 of H such that $0 \in W_0 \subseteq W$ and that the restriction of $\pi \circ \varphi$ to W_0 is a C^ℓ -homeomorphism. Therefore, by possibly shrinking W , we may assume that $\pi \circ \varphi$ is already a C^ℓ -homeomorphism.

We now construct a mapping $f : U \rightarrow E$ such that

$$f(x) = x - (\pi' \circ \varphi)((\pi \circ \varphi)^{-1}(\pi(x))).$$

Let V be the image of f . Note that the inverse of f sends $y \in V$ to

$$y + (\pi' \circ \varphi)((\pi \circ \varphi)^{-1}(\pi(y))).$$

Moreover, both f and f^{-1} are of class C^ℓ . We obtain that f is a C^ℓ diffeomorphism. Finally, $x \in U \cap M$ if and only if there exists $b \in W$ such that $\varphi(b) = x$. If this happens, then

$$f(x) = x - (\pi' \circ \varphi)(b) = x - \pi'(x) = \pi(x) \in V \cap F.$$

Conversely, if $f(x) \in V \cap F$, then

$$(\pi' \circ \varphi)((\pi \circ \varphi)^{-1}(\pi(x))) = \pi'(x).$$

Since

$$\pi(\varphi((\pi \circ \varphi)^{-1}(\pi(x)))) = \pi(x),$$

we obtain that

$$x = \pi'(x) + \pi(x) = \varphi((\pi \circ \varphi)^{-1}(\pi(x))).$$

□

7.4.5 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$ and M be a subset of E . Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$, and $d \in \{0, \dots, n\}$. If M satisfies the equivalent conditions in Theorem 7.4.4, then we say that M is a C^ℓ submanifold of dimension d of E .

7.4.6 Example. Let E be a finite-dimensional vector space over $\mathbb{R}$. Any non-empty open subset U of E is a submanifold of E .

7.4.7 Example. Let E be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of E and $f : U \rightarrow \mathbb{R}$ be a mapping. Let $M = \{x \in U \mid f(x) = 0\}$. We assume that M is not empty and that $Df(x)$ is non-zero for any $x \in M$. Then M is a submanifold of E . Such a submanifold is called a *hypersurface* in E .

7.5 Tangent space

7.5.1 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$ and M be a submanifold of E . Let $a \in M$. We call *tangent vector* of M at a any element of E of the form $\gamma'(0)$, where γ is a differentiable mapping from an open neighbourhood of 0 in $\mathbb{R}$ to E , whose image is contained in M . We denote by $T_a M$ the set of all tangent vectors of M at a , called the *tangent space* of M at a .

7.5.2 Theorem. Let E be a finite-dimensional vector space over $\mathbb{R}$ and n be the dimension of E . Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$ and M be a submanifold of class C^ℓ of E and $a \in M$. Let d be the dimension of M . Then the tangent space $T_a M$ is a vector subspace of dimension d of E . Moreover, the following statements hold.

- (1) Suppose that U is an open neighbourhood of a , V is an open neighbourhood of 0, F is a vector subspace of dimension d of E and $f : U \rightarrow V$ is a C^ℓ -diffeomorphism such that $f(a) = 0$ and $f(U \cap M) = V \cap F$. Then $T_a M = (Df(a))^{-1}(F)$.
- (2) Suppose that U is an open neighbourhood of a , G is a vector space of dimension $n-d$ over $\mathbb{R}$ and $g : U \rightarrow G$ is a submersion of class C^ℓ such that $U \cap M = g^{-1}(\{0\})$, then $T_a M = \text{Ker}(Dg(a))$.

- (3) Suppose that U is an open neighbourhood of a , H is a vector space of dimension d over $\mathbb{R}$, W is an open subset of H and $\varphi : W \rightarrow E$ is an immersion of class C^ℓ such that $a \in \varphi(W)$ and that φ defines a homeomorphism between W and $U \cap M$, then one has $T_a M = D\varphi(\varphi^{-1}(a))(H)$.

Proof. Let d be the dimension of M .

We first use the notation of (3). Without loss of generality, we may assume that $\varphi^{-1}(a) = 0$. Let v be an element of H . There exists $\varepsilon > 0$ such that $tv \in W$ for any $t \in]-\varepsilon, \varepsilon[$. Let $\gamma :]-\varepsilon, \varepsilon[\rightarrow E$ that sends t to $\varphi(tv)$. By definition the image of γ is contained in M . Moreover, one has $\gamma'(0) = D\varphi(0)(v)$. This shows that $D\varphi(0)(H) \subseteq T_a(M)$. Thus $T_a(M)$ contains a vector subspace of dimension d .

We now switch to the notation of (2). Let γ be a mapping of class C^1 from an open neighbourhood I of 0 in $\mathbb{R}$ to E such that $\gamma(0) = a$ and $\gamma(I) \subseteq M$. By possibly shrinking I , we may assume that $\gamma(I) \subseteq U$. Note that $g \circ \gamma$ is a constant mapping taking 0 as its value. Thus

$$D(g \circ \gamma(0)) = Dg(a)(\gamma'(0)) = 0.$$

This shows that $\gamma'(0)$ belongs to the kernel of $Dg(a)$. Thus $T_a M \subseteq \text{Ker}(Dg(a))$, namely $T_a M$ is contained in a vector subspace of dimension d of E . Combining the above arguments, we obtain that $T_a M$ is a vector subspace of dimension d of E , which is equal to $D\varphi(0)(H)$ and to $\text{Ker}(Dg(a))$.

Finally, under the notation of (1), for any mapping γ of class C^1 from an open neighbourhood I of 0 in $\mathbb{R}$ to E such that $\gamma(I) \subseteq M$ and $\gamma(0) = a$, the composite mapping $f \circ \gamma$ is of class C^1 and takes value in F . Therefore,

$$(f \circ \gamma)'(0) = Df(a)(\gamma'(0)) \in F,$$

which means that $\gamma'(0) \in (Df(a))^{-1}(F)$. Hence one has $T_a M \subseteq (Df(a))^{-1}(F)$. Since both $T_a M$ and $(Df(a))^{-1}(F)$ have dimension d over K , we obtain that they are actually equal. $\square$

7.5.3 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$, M be a C^ℓ submanifold of E , $a \in M$ and U be an open neighbourhood of a . Let H be a vector space over $\mathbb{R}$, W be an open subset of H and $\varphi : W \rightarrow E$ be an immersion of class C^ℓ such that $a \in \varphi(W)$, that defines a homeomorphism between W and $U \cap M$, then we say that φ is a *local chart* of M around a . By Theorem 7.5.2, we obtain that, for any $x \in W$, one has

$$T_{\varphi(x)} M = D\varphi(x)(H).$$

Let V be an open subset of M , F be a finite-dimensional vector space over $\mathbb{R}$ and $f : V \rightarrow F$ be a mapping. Let m be an element of $\mathbb{N} \cup \{\infty\}$ such that $m \leq \ell$. If, for any local chart $\varphi : W \rightarrow M$ such that $\varphi(W) \subseteq V$, the mapping

$$f \circ \varphi : \varphi^{-1}(V) \longrightarrow F$$

is differentiable (resp. of class C^m), then we say that the mapping f is *differentiable* (resp. *of class C^m*). We denote by $C^m(V, F)$ the set of all mappings of class C^m from V to F . In the case where $F = \mathbb{R}$, $C^m(V, \mathbb{R})$ is also denoted as $C^m(V)$.

7.5.4 Example. Let $(E, \langle \cdot, \cdot \rangle)$ be a Euclidean space (i.e. finite-dimensional inner product space over $\mathbb{R}$) and M be a C^ℓ submanifold of E , where $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. Let a be an element of M and $\pi : E \rightarrow T_a M$ be the orthogonal projection. Let $\psi : W \rightarrow M$ be a local chart of M around a . The composite mapping

$$\eta : W \xrightarrow{\psi} M \xrightarrow{x \mapsto \pi(x-a)} T_a M$$

sends $\psi^{-1}(a)$ to 0, and

$$D\eta(\psi^{-1}(a)) = \pi \circ D\psi(\psi^{-1}(a))$$

is an isomorphism. Therefore, by the local inversion theorem, there is an open subset W' of W containing $\psi^{-1}(a)$ such that η defines a C^ℓ homeomorphism from W' to $\eta(W')$. In particular, the mapping

$$\tilde{\psi} : \eta(W') \rightarrow M, \quad w \mapsto \psi(\eta^{-1}(w))$$

is also a local chart. We call it a *canonical local chart* around a . Note that, such a local chart is the inverse of the orthogonal projection to the tangent space, hence locally it does not depend on the choice of the auxiliary local chart ψ (in other words, two such local charts only differ by their domains of definition).

7.5.5 Remark. Let E be a finite-dimensional vector space over $\mathbb{R}$, M be a C^ℓ submanifold of E , and $a \in M$. Let $\psi_1 : W_1 \rightarrow M$ and $\psi_2 : W_2 \rightarrow M$ be two local charts of M around a . Then $\psi_2^{-1} \circ \psi_1$ locally defines a C^ℓ homeomorphism around $\psi_1^{-1}(a)$. This can be seen via the canonical local chart around a after choosing an inner product on E . In particular, if F is a finite-dimensional vector space over $\mathbb{R}$, V is an open subset of M and $f : V \rightarrow F$ is a mapping, then f is of class C^m ($m \in \mathbb{N} \cup \{\infty\}$, $m \leq \ell$) if and only if, for any $a \in V$ there is a local chart $\psi : W \rightarrow M$ around a such that $\psi(W) \subseteq V$ and $f \circ \psi$ is of class C^m .

7.5.6 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$, M be a C^ℓ submanifold of dimension d of E . Let V be an open subset of M . We call *tangent vector field* of M on V any mapping X from V to E such that $X(a) \in T_a M$ for any $a \in V$.

For any $a \in M$, we denote by $T_a^* M$ the dual vector space of $T_a M$. Let V be an open subset of M . We call *p-form* on V any family

$$\omega = (\omega_a)_{a \in V} \in \prod_{a \in V} \bigwedge^p T_a^* M.$$

Note that each ω_a determines an alternating n -linear mapping from $(T_p M)^n$ to $\mathbb{R}$. We say that ω is of class C^m if, for any family $(X_1, \dots, X_p)$ of tangent vector fields on V which are of class C^m , the mapping

$$(a \in V) \mapsto \omega_a(X_1(a), \dots, X_p(a))$$

is of class C^m .

It turns out that the set of p -forms on V forms a $\mathbb{R}^V$ -module. We denote by $\Omega_m^p(V)$ the set of p -forms of class C^m on V . It forms a $C^m(V)$ -module.

7.5.7 Example. Let E be a finite-dimensional vector space over $\mathbb{R}$, M be a submanifold of class C^ℓ of E , where $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. Let V be an open subset of M and $f : V \rightarrow \mathbb{R}$ be a differentiable mapping. For any $a \in V$, we denote by $df(a)$ the following element

$$D(f \circ \psi)(\psi^{-1}(a)) \circ D\psi(\psi^{-1}(a))^{-1},$$

where ψ is a local chart of M at a . Note that $df(a)$ does not depend on the choice of the local chart ψ . In fact, if ψ_1 and ψ_2 are local charts around a , and $\eta = \psi_2^{-1} \circ \psi_1$. Let $x = \psi_1^{-1}(a)$ and $y = \eta(x) = \psi_2^{-1}(a)$. One has

$$\begin{aligned} D(f \circ \psi_1)(x) \circ D\psi_1(x)^{-1} &= D(f \circ \psi_2 \circ \eta)(x) \circ D(\psi_2 \circ \eta)(x)^{-1} \\ &= D(f \circ \psi_2)(y) \circ D\eta(x) \circ (D\psi_2(y) \circ D\eta(x))^{-1} = D(f \circ \psi_2)(y) \circ D\psi_2(y). \end{aligned}$$

The family $(df(a))_{a \in V}$ is a 1-form on V . The mapping from the vector space of all differentiable mappings on V to that of 1-forms on V , which sends f to df , is $\mathbb{R}$ -linear. Note that one has

$$d(fg) = f dg + g df \tag{7.2}$$

for any differentiable mappings f and g from V to $\mathbb{R}$.

Let X be a tangent vector field on V . If $f : V \rightarrow \mathbb{R}$ is a differentiable mapping, we denote by $X(f)$ the mapping from V to $\mathbb{R}$ that sends $a \in V$ to

$$df(a)(X(a)).$$

By (7.2) we obtain that, if f and g are differentiable mappings from V to $\mathbb{R}$, then

$$X(fg) = fX(g) + gX(f).$$

If $\psi : W \rightarrow M$ is a local chart such that $\psi(W) \subseteq V$, then one has

$$X(f) \circ \psi = D(f \circ \psi)((D\psi)^{-1}(X \circ \psi))$$

In particular, for any $m \in \mathbb{N} \cup \{\infty\}$ such that $m \leq \ell - 1$, if f is of class C^{m+1} and X is of class C^m , then $X(f)$ is of class C^m . This can be seen by the case of a canonical local chart, where $(D\psi)^{-1}$ is given by the orthogonal projection.

7.5.8 Example. Let $\ell \in \mathbb{N} \cup \{\infty\}$. Let M be a C^ℓ submanifold of E and U be an open subset of E containing M . Let p be a natural number and ω be a p -form on U . For each $a \in M \cap U$, the inclusion mapping $T_a M \rightarrow E$ induces by passing to dual a surjective mapping $\pi_a : E^* \rightarrow T_a^* M$. The family $(\pi_a(\omega_a))_{a \in M \cap U}$ then forms a p -form on $M \cap U$ which we denote as $\omega|_{M \cap U}$. If $X_1, \dots, X_p$ are tangent vector fields on $M \cap U$, by abuse of notation we denote by $\omega(X_1, \dots, X_p)$ the mapping from $M \cap U$ to $\mathbb{R}$ that sends $a \in M \cap U$ to $\pi_a(\omega_a)(X_1(a), \dots, X_p(a))$.

7.5.9 Definition. Let $\ell \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. Let M be a C^ℓ submanifold of E . If $\psi : W \rightarrow M$ is a local chart and Y is a tangent vector field on W , we denote by $\psi_*(Y)$ the tangent vector field on $\psi(W)$ defined as

$$\forall x \in W, \quad \psi_*(Y)(\psi(x)) = D\psi(x)(Y(x)).$$

Note that, for any tangent vector field X on $\psi(W)$, there exists a unique tangent vector field Y on W such that $X = \psi_*(Y)$. Actually this tangent vector field Y is given by

$$Y(x) = D\psi(x)^{-1}(X(\psi(x))).$$

Note that, for any differentiable mapping $f : \psi(W) \rightarrow \mathbb{R}$, one has

$$\psi_*(Y)(f)(\psi(x)) = D(f \circ \psi)(x)(D\psi(x)^{-1}(D\psi(x)(Y(x)))) = D(f \circ \psi)(x)(Y(x)).$$

In other words

$$\psi_*(Y)(f) \circ \psi = Y(f \circ \psi). \quad (7.3)$$

Therefore, for $m \in \mathbb{N} \cup \{\infty\}$ such that $m \leq \ell$, $\psi_*(Y)$ is of class C^m if and only if Y is of class C^m .

7.5.10 Proposition. Let $\ell \in \mathbb{N}_{\geq 2} \cup \{\infty\}$ and M be a C^ℓ submanifold of E . Let V be an open subset of M and X_1 and X_2 be two tangent vector field of class C^1 on V . There exists a unique tangent vector field $[X_1, X_2]$ on V such that, for any mapping $f : V \rightarrow \mathbb{R}$ of class C^2 , one has

$$[X_1, X_2](f) = X_1(X_2(f)) - X_2(X_1(f)).$$

Proof. Let $\psi : W \rightarrow M$ be a local chart of M such that $\psi(W) \subseteq V$. Let Y_1 and Y_2 be tangent vector fields on W such that

$$\psi_*(Y_1) = X_1|_V, \quad \psi_*(Y_2) = X_2|_V.$$

For any $f : V \rightarrow \mathbb{R}$ of class C^2 , by (7.3) one has

$$\begin{aligned} (X_1(X_2(f)) - X_2(X_1(f))) \circ \psi &= Y_1(X_2(f) \circ \psi) - Y_2(X_1(f) \circ \psi) \\ &= Y_1(Y_2(f \circ \psi)) - Y_2(Y_1(f \circ \psi)) = [Y_1, Y_2](f \circ \psi) = \psi_*[Y_1, Y_2](f) \circ \psi. \end{aligned}$$

Therefore, we obtain that there is a unique tangent vector field $[X_1, X_2]$ on V such that

$$[X_1, X_2](f) = X_1(X_2(f)) - X_2(X_1(f)),$$

which identifies with $\psi_*([Y_1, Y_2])$ on $\psi(W)$. $\square$

7.5.11 Definition. Let $\ell \in \mathbb{N}_{\geq 2} \cup \{\infty\}$ and M be a C^ℓ submanifold of E . Let V be an open subset of M , $p \in \mathbb{N}$ and ω be a p -form of class C^1 on V . We define $d\omega$ to be the $(p+1)$ -form on V such that, for any tangent vector fields $X_0, \dots, X_p$ of class C^1 , one has

$$\begin{aligned} d\omega(X_0, \dots, X_p) &= \sum_{i=0}^p (-1)^i X_i(\omega(X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_p)) + \\ &\quad \sum_{0 \leq i < j \leq p} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_{j-1}, X_{j+1}, \dots, X_p) \end{aligned}$$

Note that, if ω is of class C^m , where $m \in \mathbb{N}_{\geq 1} \cup \{\infty\}$, $m \leq \ell$, then $d\omega$ is of class C^{m-1} .

7.5.12 Remark. Let $\psi : W \rightarrow M$ be a local chart of M such that $\psi(W)$ is contained in V . If ω is a p -form on V , we denote by $\psi^*(\omega)$ the p -form on W such that, for any tangent vector field $Y_1, \dots, Y_p$ on W , one has

$$\psi^*(\omega)(Y_1, \dots, Y_p) = \omega(\psi_*(Y_1), \dots, \psi_*(Y_p)) \circ \psi.$$

By Theorem 7.3.15, we obtain that

$$d\psi^*(\omega) = \psi^*(d\omega).$$

7.6 Partition of unity

7.6.1 Definition. Let X be a set, $(U_i)_{i \in I}$ and $(V_j)_{j \in J}$ be covers of X . We say that $(V_j)_{j \in J}$ is *finer* than $(U_i)_{i \in I}$ if, for any $j \in J$, there exists $i \in I$ such that $V_j \subseteq U_i$.

Let $(X, \mathcal{T})$ be a topological space and $(U_i)_{i \in I}$ be a cover of X . We say that $(U_i)_{i \in I}$ is *locally finite* if, for any $x \in X$, there exists a neighbourhood V of x such that

$$\{i \in I \mid V \cap U_i \neq \emptyset\}$$

is finite.

We say that a topological space $(X, \mathcal{T})$ is *paracompact* if, for any any open cover $(U_i)_{i \in I}$ of X , there exists an open cover $(V_j)_{j \in J}$ which is locally finite and finer than $(U_i)_{i \in I}$.

7.6.2 Theorem. *Let $(X, \mathcal{T})$ be a locally compact Hausdorff space, which has a countable topological basis. Then $(X, \mathcal{T})$ is paracompact.*

Proof. Let $U = (U_i)_{i \in I}$ be an open cover of X . Let $\mathcal{B}$ be a topological basis of X . Then

$$\mathcal{B}' = \{V \in \mathcal{B} \mid \exists i \in I, V \subseteq U_i\}.$$

is also a topological basis of X . Hence, by replacing $(U_i)_{i \in I}$ by $\mathcal{B}'$, we may assume that $\{U_i \mid i \in I\}$ forms a topological basis of X .

By Proposition 5.3.7, there exists a sequence of open subsets $(V_n)_{n \in \mathbb{N}}$ such that, for any $n \in \mathbb{N}$, $\overline{V_n}$ is compact and is contained in V_{n+1} . Moreover, X is the union of all V_n . For any $n \in \mathbb{N}$, let $K_n = \overline{V_n} \setminus V_{n-1}$ and $W_n = V_{n+1} \setminus \overline{V_{n-2}}$ (with the convention that $\overline{V_{-2}} = V_{-1} = \emptyset$). Then K_n is compact, W_n is open and $K_n \subseteq W_n$. We choose a finite subset I_n of I such that $U_i \subseteq W_n$ for any $i \in I_n$ and that $K_n = \bigcup_{i \in I_n} U_i$. Let $J = \bigcup_{n \in \mathbb{N}} I_n$. Then $(U_i)_{i \in J}$ is finer than $(U_i)_{i \in I}$. Note that $W_n \cap W_m = \emptyset$ once $|n - m| \geq 2$. Hence the cover $(U_i)_{i \in J}$ is locally finite. $\square$

7.6.3 Theorem. *Let E be a finite-dimensional vector space over $\mathbb{R}$, $\ell \in \mathbb{N} \cup \{\infty\}$, and M be a C^ℓ manifold. Let $(U_i)_{i \in I}$ be an open cover of M . There exists a family $(\chi_i)_{i \in I}$ of mappings from M to $[0, 1]$ that verifies the following conditions:*

- (1) *for any $i \in I$, the support of χ_i is compact and contained in U_i ,*
- (2) *$(\text{Supp}(\chi_i))_{i \in I}$ forms a cover of M which is locally finite,*
- (3) *for any $x \in M$, $\sum_{i \in I} \chi_i(x) = 1$.*

The family $(\chi_i(x))_{i \in I}$ is called a partition of unity with respect to the open cover $(U_i)_{i \in I}$.

Proof. For each $P \in M$, we let

$$(\psi_j : W_j \rightarrow M)_{j \in J_P}$$

be a family of local charts around P , such that each $\psi_j(W_j)$ is contained in certain U_i and that $\{\psi_j(W_j) \mid j \in J_P\}$ forms a neighbourhood basis at P . For simplicity, we assume that $(J_P)_{P \in M}$ are pairwise disjoint and we denote by J the union of $(J_P)_{P \in M}$. For each $j \in J$, we let $f_j : W_j \rightarrow [0, 1]$ be a mapping of class C^∞ with compact support, such that $\text{Supp}(f_j) \subseteq W_j$. Let $W'_j = \{x \in W_j \mid f_j(x) > 0\}$. Then $\{\psi_j(W'_j) \mid j \in J\}$ forms a topological basis of M .

By the proof of the previous theorem, we observe that there exists a subset J' of J such that $(\psi_j(W'_j))_{j \in J'}$ forms a locally finite open cover of M . For any $j \in J'$, let $g_j = f_j \circ \psi_j^{-1}$. We extend the domain of definition of g_j to M by letting $g_j|_{M \setminus W_j} \equiv 0$.

Let

$$G = \sum_{j \in J'} g_j : M \rightarrow \mathbb{R}_{>0}.$$

This mapping is well defined since $(W'_j)_{j \in J'}$ is a locally finite cover. For any $j \in J'$, we pick $\lambda(j) \in I$ such that $\psi_j(W_j) \subset U_{\lambda(j)}$. We choose

$$\chi_i = \sum_{j \in J', \lambda(j)=i} \frac{g_j}{G}.$$

Then the family $(\chi_i)_{i \in I}$ satisfies the required conditions. $\square$

7.6.4 Definition. Let E be a finite-dimensional vector space over $\mathbb{R}$, and M be a C^ℓ submanifold of dimension d of E . We call *orientation field* of M any continuous mapping $\theta : M \rightarrow \Lambda^d(E)$ such that $\theta(P) \in \Lambda^d(T_P M) \setminus \{0\}$ for any $P \in M$. We say that a submanifold M is *orientable* if there exists an orientation field on M . We say that M is *oriented* if M is orientable and an orientation field θ on M has been chosen. In this case, by abuse of notation, we call θ an *orientation* on M .

7.6.5 Remark. Let E be a finite-dimensional vector space over $\mathbb{R}$, and M be a C^ℓ submanifold of dimension d of E . Equivalently, we can define orientation field of M as

$$\theta : M \rightarrow \Lambda^d(E)$$

satisfying the following conditions

- (1) for all $x \in M$ $\theta(x) = v_1 \wedge \cdots \wedge v_n$ and $T_x M = \{v \in E \mid v \wedge \theta(x) = 0\}$,
- (2) for all $x \in M$ $\theta(x) \neq 0$,
- (3) θ is continuous.

Notice, that the requirement that $\theta(x) = v_1 \wedge \cdots \wedge v_n$ is redundant as it is implied by the requirement $T_x M = \{v \in E \mid v \wedge \theta(x) = 0\}$. Indeed, let $e_1, \dots, e_d$ be a basis of $T_x M, x \in M$. We can extend it to the basis of E , by adding $e_{d+1}, \dots, e_n$. Then

$$e_i \wedge \theta(x) = 0, \quad i \in \{1, \dots, d\}$$

if and only if e_i appears in every summand of

$$\theta(x) = \sum a_i e_{\lambda(1)} \wedge \cdots \wedge e_{\lambda(d)}.$$

7.6.6 Proposition. Let M be a C^ℓ submanifold of E and θ be an orientation on E . Let F be a vector space of dimension d over $\mathbb{R}$, W be a connected open subset of F and $\psi : W \rightarrow E$ be a local chart. Let $\xi : W \rightarrow \Lambda^p(F)$ be a mapping such that the homomorphism $\Lambda^p(F) \rightarrow \Lambda^p(E)$ induced by $D\psi(x)$ (i.e. $v_1 \wedge \cdots \wedge v_p \mapsto D\psi(x)(v_1) \wedge \cdots \wedge D\psi(x)(v_p)$) sends $\xi(x)$ to $\theta(x)$. Then the orbit of $\xi(x)$ by the action of $\mathbb{R}_{>0}$ does not depend on the choice of $x \in W$. We call it the *orientation* of ψ .

Proof. Since θ is continuous, we obtain that ξ is also continuous (one can see it on a canonical local chart, where ξ is given by the composite of θ with the projection $\Lambda^p(E) \rightarrow \Lambda^p(F)$). Since W is connected, the image of W by ξ is also connected. Note that $\Lambda^p(F)$ is a real vector space of dimension 1, hence it is isomorphic to $\mathbb{R}$. $\Lambda^p(F) \setminus \{0\}$ has two connected components, which coincide with the two orbits by the action of $\mathbb{R}_{>0}$. Therefore, we obtain the desired result. $\square$

7.6.7 Definition. Let M be a C^ℓ submanifold of dimension d of E . We fix an orientation θ on M . Let $\psi : W \rightarrow M$ is a local chart such that W is connected, and ω be a d -form of compact support, whose support is contained in $\psi(W)$. If $\psi^*(\omega)$ is integrable on W , we say that ω is *integrable* and we denote by $\int_M \omega$ the integral of $\psi^*(\omega)$ on W with respect to the orientation of ψ .

7.6.8 Remark. Let M be a C^ℓ submanifold of dimension d of E , equipped with an orientation θ . Let $\psi_1 : W_1 \rightarrow M$ and $\psi_2 : W_2 \rightarrow M$ be two local charts of M such that W_1 and W_2 are connected. Let ω be a d -form of compact support, whose support is contained in a connected component of $\psi_1(W_1) \cap \psi_2(W_2)$. Then $\det(D\psi_2^{-1} \circ D\psi_1)$ only takes positive values. By the formula of change of variables in integrals, we obtain that

$$\int_{W_1} \psi_1^*(\omega) = \int_{W_2} \psi_2^*(\omega).$$

7.6.9 Definition. Let M be a C^ℓ submanifold of dimension d of E , equipped with an orientation θ . Let $(\psi_i : W_i \rightarrow M)_{i \in I}$ be a family of local charts of M such that each W_i is connected and that the family $(\psi_i(W_i))_{i \in I}$ forms an open cover of M . Let $(\chi_i)_{i \in I}$ be a partition of unity with respect to the open cover $(\psi_i(W_i))_{i \in I}$. Let ω be a d -form on M . In the case where ω is continuous with compact support, we define $\int_M \omega$ to be

$$\sum_{i \in I} \int_M \chi_i \omega.$$

Note that this sum is actually finite since ω is assumed to have a compact support. Moreover, the value of the sum does not depend on the choice of the covering local charts. In fact, if $(\varphi_j : V_j \rightarrow M)_{j \in J}$ is another family of local charts of M such that each V_j is connected and that $(\varphi_j(V_j))_{j \in J}$ covers M . Let $(\mu_j)_{j \in J}$ be a partition of unity with respect to the cover $(\varphi_j(V_j))_{j \in J}$. Then one has

$$\sum_{i \in I} \int_M \chi_i \omega = \sum_{i \in I} \int_M \chi_i \sum_{j \in J} \mu_j \omega = \sum_{j \in J} \int_M \sum_{i \in I} \chi_i \mu_j \omega = \sum_{j \in J} \int_M \mu_j \omega.$$

7.7 Stokes formula

In this section, we fix a finite-dimensional vector space E over $\mathbb{R}$. We also fix an element $\ell \in \mathbb{N}_{\geq 1} \cup \{+\infty\}$

7.7.1 Definition. Let F be a finite-dimensional vector space over $\mathbb{R}$. If U is an open subset of $\mathbb{R}_{\leq 0} \times F$, the intersection of U with $\{0\} \times F$ is called the *boundary part* of U , and is denoted as ∂U .

7.7.2 Definition. Let F and G be a finite-dimensional vector space over $\mathbb{R}$. Let U be an open subset of $\mathbb{R}_{\leq 0} \times F$. We say a mapping $f : U \rightarrow G$ is *differentiable* (resp. *of class C^ℓ*) if there exists an open subset $\tilde{U}$ of $\mathbb{R} \times F$, together with a mapping $\tilde{f} : \tilde{U} \rightarrow G$ which is differentiable (resp. of class C^ℓ), such that f identifies with the restriction of $\tilde{f}$ to U . If furthermore $\tilde{f}$ could be selected to be an open immersion, we say that f is an *open immersion*.

7.7.3 Remark. Let F be a finite-dimensional vector space over $\mathbb{R}$, U be an open subset of $\mathbb{R}_{\leq 0} \times F$, and $f : U \rightarrow E$ be a differentiable mapping. Let $\tilde{U}$ be an open subset of $\mathbb{R} \times F$ containing U and $\tilde{f} : \tilde{U} \rightarrow E$ be a differentiable mapping that extends f . Note that the value of $D\tilde{f}(P)(v)$ for $v \in \mathbb{R}_{\leq 0} \times F$ only depends on f (not on the choice of $\tilde{f}$). Since $\mathbb{R} \times F$ is generated by $\mathbb{R}_{\leq 0} \times F$, we obtain that $D\tilde{f}(P)$ is determined by f . We denote by $Df(P)$ this value.

7.7.4 Definition. Let F and G be finite-dimensional vector spaces over $\mathbb{R}$, U and V be open subsets of $\mathbb{R}_{\leq 0} \times F$ and $\mathbb{R}_{\leq 0} \times G$, respectively. We say that a bijection $f : U \rightarrow V$ is a *C^ℓ diffeomorphism* if there exists open subsets $\tilde{U}$ and $\tilde{V}$ of $\mathbb{R} \times F$ and $\mathbb{R} \times G$, respectively, together with a C^ℓ diffeomorphism $\tilde{f} : \tilde{U} \rightarrow \tilde{V}$, such that $\tilde{f}|_U = f$.

7.7.5 Proposition. Let F be a finite-dimensional vector space over $\mathbb{R}$ and U be an open subset of $\mathbb{R}_{\leq 0} \times F$. An element $x \in \mathbb{R} \times F$ belongs $U \setminus \partial U$ if and only if, for any $v \in \mathbb{R} \times F$, there exists $\varepsilon > 0$ and a differentiable mapping $\gamma : [-\varepsilon, \varepsilon] \rightarrow U$ such that $\gamma(0) = x$ and $\gamma'(0) = v$.

Proof. Note that $U \setminus \partial U$ is an open subset of $\mathbb{R}_{< 0} \times F$. If $x \in U \setminus \partial U$, for any $v \in \mathbb{R} \times F$, there exists $\varepsilon > 0$ such that $x + tv \in U \setminus \partial U$ for any $t \in [-\varepsilon, \varepsilon]$. Hence the mapping

$$\gamma : [-\varepsilon, \varepsilon] \rightarrow U, \quad \gamma(t) = x + tv$$

satisfies the required condition.

Assume now $x \in \partial U$. Let $\varepsilon > 0$ and $\gamma : [-\varepsilon, \varepsilon] \rightarrow U$ be a differentiable mapping such that $\gamma(0) = x$ and $\gamma'(0) = v$. We write $\gamma(t)$ into coordinates as $(\lambda(t), \mu(t))$. By definition, $\lambda(t) \leq 0$ for any $t \in [-\varepsilon, \varepsilon]$, and $\lambda(0) = 0$. Hence 0 is a point of local maximum of λ , which leads to $\lambda'(0) = 0$. Hence $\gamma'(0)$ cannot take arbitrary value in $\mathbb{R} \times F$. $\square$

7.7.6 Corollary. Let F and G be finite-dimensional vector spaces over $\mathbb{R}$ and U and V be open subsets of $\mathbb{R}_{\leq 0} \times F$ and $\mathbb{R}_{\leq 0} \times G$ respectively. Suppose that $f : U \rightarrow V$ is a C^ℓ -diffeomorphism. Then $f(\partial U) = \partial V$.

Proof. We first prove that $f(U \setminus \partial U) \subseteq V \setminus \partial V$. Let $x \in U \setminus \partial U$. For any $v \in \mathbb{R} \times F$, there exists $\varepsilon > 0$ and $\gamma : [-\varepsilon, \varepsilon] \rightarrow U$ such that $\gamma(0) = x$ and $\gamma'(0) = Df(x)^{-1}(v)$. Let $\mu = f \circ \gamma$. Then $\mu(0) = f(x)$ and $\mu'(0) = Df(x)(\gamma'(0)) = v$. Hence $f(x) \in V \setminus \partial V$.

Note that $f^{-1} : V \rightarrow U$ is also a C^ℓ -diffeomorphism. Hence $f^{-1}(V \setminus \partial V) \subseteq U \setminus \partial U$. Hence the statement holds. $\square$

7.7.7 Definition. Let M be a subset of E , d be a natural number such that $d \leq \dim(E)$. We say that M is a submanifold with boundary of dimension d of class C^ℓ of E if, for any $P \in M$, there exists an open neighbourhood U of P in E , a vector space F of dimension $d - 1$ over $\mathbb{R}$, an open subset $\widetilde{W}$ of $\mathbb{R} \times F$, together with an immersion

$$\widetilde{\psi} : \widetilde{W} \rightarrow E$$

of class C^ℓ , which defines a homeomorphism between $W = \widetilde{W} \cap (\mathbb{R}_{\leq 0} \times F)$ and $U \cap M$. By abuse of notation, we say that $\psi : W \rightarrow M$ is a *local chart* of M around P (note that ψ could be extended to an immersion on an open neighbourhood of W in $\mathbb{R} \times F$). We call *boundary* of M the union of all $\psi(\partial W)$, where $\psi : W \rightarrow M$ runs over the set of all local charts of M .

For any $P \in M$ and an arbitrary local chart $\psi : W \rightarrow M$ around P , where W is a subset of $\mathbb{R}_{\leq 0} \times F$, F being a vector space of dimension $d - 1$, we denote by $T_P M$ the image of $\mathbb{R} \times F$ by the $\mathbb{R}$ -linear mapping $D\psi(\psi^{-1}(P))$. Note that the chain rule of differential shows that the vector subspace $D\psi(\psi^{-1}(P))$ does not depend on the choice of the local chart ψ . In fact, if $\psi_1 : W_1 \rightarrow M$ and $\psi_2 : W_2 \rightarrow M$ are two local charts around P , then

$$\begin{aligned} D\psi_2(\psi_2^{-1}(P)) &= D(\psi_1 \circ \psi_1^{-1} \circ \psi_2)(\psi_2^{-1}(P)) \\ &= D\psi_1(\psi_1^{-1}(P)) \circ D(\psi_1^{-1} \circ \psi_2)(\psi_2^{-1}(P)). \end{aligned}$$

Since $\psi_1^{-1} \circ \psi_2$ is a diffeomorphism around $\psi_2^{-1}(P)$, we obtain that

$$D(\psi_1^{-1} \circ \psi_2)(\psi_2^{-1}(P))$$

is an isomorphism.

7.7.8 Remark. Let M be a submanifold of dimension d with boundary, P be an element of ∂M , and $\psi : W \rightarrow M$ be a local chart around P , where W is an open subset of $\mathbb{R}_{\leq 0} \times F$. By Corollary 7.7.6, we obtain that P belongs to the image of ∂W . Moreover, the restriction of ψ to ∂W defines a local chart of ∂W . In particular, ∂W is a C^ℓ submanifold (without boundary) of dimension $d - 1$ of E .

7.7.9 Definition. Let F be a finite-dimensional vector space over $\mathbb{R}$. Let ξ be a non-zero element of $\det(\mathbb{R} \times F)$ and $\eta \in \det(F)$ such that $\xi = (1, 0) \wedge \eta$ (here we

consider F as a vector subspace of $\mathbb{R} \times F$ in identifying $x \in F$ with $(0, x) \in \mathbb{R} \times F$. Then the orbit of η by the action of $\mathbb{R}_{>0}$ only depends on the orbit of ξ by the action of $\mathbb{R}_{>0}$. Thus, if we view the orbit of ξ as an orientation on $\mathbb{R} \times F$, then the orbit of η is called the orientation on F induced by the orientation given by ξ .

Let M be a submanifold with boundary of dimension d of E . We call *orientation* on M any continuous mapping θ from M to $\Lambda^d E$ such that $\theta(P) \in \Lambda^d(T_P M)$ and $\theta(P)$ is non-zero for any $P \in M$. Note that, if $\psi : W \rightarrow M$ is a local chart, where W is an open subset of $\mathbb{R}_{\leq 0} \times F$, F being a vector space of dimension d over $\mathbb{R}$, then θ determines an orientation on $\mathbb{R} \times F$, as indicated in Proposition 7.6.6, which furthermore determines an orientation on F , as described in Definition 7.7.9.

Let M be a submanifold with boundary of dimension d of E . We call *tangent vector field* on M any mapping X from M to E such that $X(P) \in T_P M$ for any $P \in M$. Let $\psi : W \rightarrow M$ be a local chart of M , there exists a unique tangent vector field Y on W such that the mapping $\psi_*(Y) : \psi(W) \rightarrow E$ sending $\psi(x)$ to $D\psi(x)(Y(x))$ is equal to $X|_{\psi(W)}$. If, for any local chart $\psi : W \rightarrow M$, the tangent vector field Y on W such that $\psi_*(Y) = X|_{\psi(W)}$ is of class C^m (where $m \in \mathbb{N} \cup \{\infty\}$, $m \leq \ell$), we day that the tangent vector field X is of class C^m .

Let M be a submanifold with boundary of dimension d of E . We call *p-form* on M any family $\omega = (\omega_a)_{a \in M}$, where ω_a belongs to $\Lambda^p(T_a^* M)$. If $X_1, \dots, X_p$ are tangent vector fields on M , we denote by $\omega(X_1, \dots, X_p)$ the mapping that sends $a \in M$ to

$$\omega_a(X_1(a), \dots, X_p(a)).$$

If for any tangent vector field $X_1, \dots, X_p$ of class C^m , the mapping $\omega(X_1, \dots, X_p)$ is of class C^m , we say that ω is of class C^m . If ω is of class C^1 , the differential of ω is constructed in a similar way as in Definition 7.5.11. For any $a \in \partial M$, $T_a \partial M$ is a vector subspace of $T_a M$ and hence $T_a^* \partial M$ is a quotient vector space of $T_a^* M$. We denote by $\omega|_{\partial M, a}$ the image of ω_a by the quotient mapping

$$\bigwedge^p T_a^* M \longrightarrow \bigwedge^p T_a^* \partial M.$$

7.7.10 Theorem (Stokes). *Let E be a finite-dimensional vector space over $\mathbb{R}$, M a C^1 -submanifold of dimension d with boundary in E . We fix an orientation on M . Let ω be a $d-1$ -form of class C^1 with compact support on M . Then the following equality holds:*

$$\int_M d\omega = \int_{\partial M} \omega|_{\partial M}.$$

Proof. By partition of unity, we may assume that the support of ω is contained in the image of a local chart. Hence we reduce the problem to the case where M is an open subset of $\mathbb{R}_{\leq 0} \times F$, where F is a vector space of dimension $d-1$. Furthermore, by extending the domain of definition of ω , we may reduce the problem to the cases

where $M = \mathbb{R}_{<0} \times F$ or $M = \mathbb{R}_{\leq 0} \times F$. By choosing a basis of F , we may assume that F is of the form $\mathbb{R}^{d-1}$. Let $x_1, \dots, x_d$ be the mappings from $\mathbb{R} \times \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ given by the projections to the d coordinates. We denote by μ the usual Haar measure on $\mathbb{R} \times \mathbb{R}^{d-1} = \mathbb{R}^d$ and assume that the orientation is chosen so that

$$\int_M f dx_1 \wedge \dots \wedge dx_d = \int_M f d\mu.$$

for any non-negative Borel measurable function f on $\mathbb{R}^d$.

We first treat the case where $M = \mathbb{R}_{<0} \times \mathbb{R}$. In this case, ∂M is the empty set. We write ω into the form

$$\sum_{i=1}^d f_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_d.$$

Then, one has

$$d\omega = \sum_{i=1}^d (-1)^{i+1} \frac{\partial f_i}{\partial x_i} dx_1 \wedge \dots \wedge dx_d.$$

Therefore,

$$\int_M d\omega = \sum_{i=1}^d \int_{\mathbb{R}^d} \frac{\partial f_i}{\partial x_i} d\mu.$$

Since f_i is of compact support,

$$\int_{\mathbb{R}} \frac{\partial f_i}{\partial x_i} dx_i = 0,$$

by Fubini's theorem, we obtain

$$\int_M d\omega = 0.$$

We now consider the case where $M = \mathbb{R}_{\leq 0} \times \mathbb{R}^{n-1}$. We still write ω into the form

$$\sum_{i=1}^d f_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_d.$$

The above argument still shows that

$$\int_M \frac{\partial f_i}{\partial x_i} dx_1 \wedge \dots \wedge dx_d = 0$$

for $i \in \{2, \dots, d\}$, and

$$\int_M \frac{\partial f_1}{\partial x_1} dx_1 \wedge \dots \wedge dx_d = \int_{\mathbb{R}^{d-1}} f_1(0, x_2, \dots, x_d) dx_2 \wedge \dots \wedge dx_d,$$

which equals

$$\int_{\partial M} \omega|_{\partial M}.$$

□

7.7.11 Example. Let M is a curve in $E = \mathbb{R}^3$ given by the parametrisation $\psi(t) = (\psi_1(t), \psi_2(t), \psi_3(t))$, $t \in [a, b]$. Our form ω is a 0-form, i.e. a function $\omega : M \rightarrow \mathbb{R}$, and $d\omega = \frac{\partial \omega}{\partial x_1} dx_1 + \frac{\partial \omega}{\partial x_2} dx_2 + \frac{\partial \omega}{\partial x_3} dx_3$ is a 1-form. We have

$$\int_M d\omega = \int_a^b d\omega(\psi(t))(\psi'(t))dt$$

It comes from the following:

$$\psi^*(d\omega) = \psi^*\left(\frac{\partial \omega}{\partial x_1} dx_1 + \frac{\partial \omega}{\partial x_2} dx_2 + \frac{\partial \omega}{\partial x_3} dx_3\right) = d\omega_{\psi(t)}(\psi'(t))dt$$

By the formula for the change of variable, and the fundamental theorem of analysis

$$\int_a^b d\omega(\psi(t))(\psi'(t))dt = \int_a^b \frac{d}{dt} \omega(\psi(t))dt = \omega(\psi(b)) - \omega(\psi(a)).$$

Exercises

When we talk about vector fields, paramtrisations, submanifolds, etc, they are of class C^∞ , or at least C^l for l large enough.

1. a) Let $a, b : \mathbb{R} \rightarrow \mathbb{R}^n$ be differentiable vector valued functions. Use coordinates to verify that

$$(a \cdot b)' = a' \cdot b + a \cdot b'.$$

- b) Let $a, b : \mathbb{R} \rightarrow \mathbb{R}^3$ be vector valued functions. Verify that

$$(a \times b)' = a' \times b + a \times b'.$$

2. Find the image of the following maps F and determine whether they are immersions/ embeddings (immersions that are homeomorphisms onto their images)/ submersions/ diffeomorphisms.
 - a) $F : S^1 \rightarrow \mathbb{R}^2$, $F(\theta) = (\sin \theta, \cos \theta \sin \theta)$,
 - b) $F : \mathbb{R} \rightarrow \mathbb{R}^2$, $F(x) = (x^2, 0)$,
 - c) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (x, y^3)$,
 - d) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (\exp x \cos y, \exp x \sin y)$,
 - e) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (x, y + x^3)$.

3. Let η be the C_0^∞ function on $\mathbb{R}$ defined by

$$\eta(x) = \begin{cases} \lambda \exp(\frac{1}{x^2-1}), & x \in [-1, 1], \\ 0, & |x| \geq 1. \end{cases}$$

where $\lambda > 0$ is chosen such that $\int_{\mathbb{R}} \eta = 1$. Set $\eta_\epsilon(x) = \epsilon^{-1} \eta(\epsilon^{-1}x)$. The family $\{\eta_\epsilon\}_{\epsilon \in (0,1]}$ is a standard mollifier.

Let χ be the characteristic function of $(0, 1)$ and write

$$\chi_\epsilon = \chi * \eta_\epsilon.$$

Prove that

- a) χ_ϵ is also C_0^∞ .
- b) As $\epsilon \rightarrow 0$, $\chi_\epsilon \rightarrow \chi$ in both L^1 and L^2 topologies.
- c) On each closed interval I contained in $(0, 1)$, $\chi_\epsilon \rightarrow 1$ uniformly as $\epsilon \rightarrow 0$.
- d) On each closed interval I contained in $(0, 1)$ and any $k \geq 1$, the k -th derivatives

$$\frac{d^k \chi_\epsilon}{dx^k} \rightarrow 0$$

uniformly as $\epsilon \rightarrow 0$.

4. Consider the parametrized curve

$$H = \{\gamma(t) = (\cos t, \sin t, t); t \in \mathbb{R}\} \subset \mathbb{R}^3.$$

- a) Prove that H is a smooth submanifold of $\mathbb{R}^3$.
- b) Calculate the tangent space of H at a given point $\gamma(t)$.
- c) The *curvature* of a parametrized curve $\gamma : \mathbb{R} \rightarrow \mathbb{R}^3$ is by definition the function

$$\kappa = \frac{|\gamma' \times \gamma''|}{|\gamma'|^3}.$$

What does it mean when a parametrized curve has zero curvature everywhere?

- d) Calculate the curvature of H .
- e) The *torsion* of a parametrized curve $\gamma : \mathbb{R} \rightarrow \mathbb{R}^3$ is by definition the function

$$\tau = \frac{(\gamma' \times \gamma'') \cdot \gamma'''}{|\gamma' \times \gamma''|^2}$$

Calculate the curvature of the curves $H_r, r \geq 0$, where

$$H_r = \{\gamma_r(t) = (\cos t, \sin t, rt); t \in \mathbb{R}\} \subset \mathbb{R}^3,$$

and conclude that the torsion of H_r is zero if and only if H_r is contained in a plane. This is in fact true for general parametrized curves in $\mathbb{R}^3$.

5. Consider the ellipse E parametrized by

$$E = \{\gamma(t) = (a \cos t, b \sin t), t \in [0, 2\pi]\} \subset \mathbb{R}^2,$$

where a, b are positive constants.

Calculate the curvature κ of E , and calculate the integral

$$\int_E \kappa \, ds := \int_0^{2\pi} \kappa |\gamma'| \, dt = 2\pi.$$

This identity in fact holds for arbitrary closed Jordan curves on the plane with a clockwise parametrization. It is called the *Hopf Umlausatz*, and is a special case of the *Gauss–Bonnet theorem*.

6. Consider the following quadratic surfaces in $\mathbb{R}^3$:

a) Consider the unit sphere

$$S^2 = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 + x_3^2 = 1\}.$$

Prove that S^2 is a smooth submanifold of $\mathbb{R}^3$ and calculate its tangent space at a given point $x \in S^2$.

b) Prove that

$$H = \{x_1^2 + x_2^2 - x_3^2 = 1\}$$

is a smooth submanifold of $\mathbb{R}^3$.

c) Let

$$C = \{x_1^2 + x_2^2 - x_3^2 = 0\}.$$

Determine whether C is a submanifold of $\mathbb{R}^3$.

7. Recall that a k -form ω is called *closed* if $d\omega = 0$, and called *exact* if $\omega = d\eta$ for a $(k-1)$ -form η . Let $U \subset \mathbb{R}^2$ be an open set and let ω be a 1-form on U . Prove that ω is exact if and only

$$\int_\gamma \omega = 0$$

for any closed piecewise smooth loop $\gamma \subset U$.

8. a) Consider the differential forms on $\mathbb{R}^2$

$$\omega = x^2 y \, dx + (x + y) \, dy,$$

and

$$\omega = x \, dx + y \, dy.$$

Compute $d\omega$, and determine whether ω is closed/exact.

- b) Consider the differential forms

$$\omega = \frac{-y \, dx + x \, dy}{x^2 + y^2}$$

defined on $\mathbb{R}^2 \setminus \{0\}$. Compute $d\omega$ and determine whether ω is closed/exact.

We conclude that exact forms are always closed, but the converse in general does not hold.

9. Let M be a manifold. For $k = 0, 1, \dots$, the k -th *de Rham cohomology* (with $\mathbb{R}$ coefficients) of M is by definition the real vector space

$$H^k(M, \mathbb{R}) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}.$$

In calculation we often write the following *de Rham cocomplex*

$$\Omega^{-1}(M) := 0 \rightarrow \Omega^0(M) \rightarrow \Omega^1(M) \rightarrow \Omega^2(M) \rightarrow \dots$$

where Ω^k stands for the space of k -forms, and the arrows are the operator d . If we name them by $d_k : \Omega^k \rightarrow \Omega^{k+1}$, then we can write

$$H^k(M, \mathbb{R}) = \ker(d_k) / \text{im}(d_{k-1}).$$

Let α, α' be closed p -forms and β, β' be closed q -forms on some smooth manifold. Prove that

- a) $\alpha \wedge \beta$ is closed.

- b) Suppose both $\alpha - \alpha'$ and $\beta - \beta'$ are exact, then

$$\alpha \wedge \beta - \alpha' \wedge \beta'$$

is exact.

Therefore, the wedge product of forms defines a product structure on cohomology:

$$H^p \times H^q \rightarrow H^{p+q}, ([\alpha], [\beta]) \mapsto [\alpha \wedge \beta].$$

- 10.** We prove *Poincaré's lemma* for $\mathbb{R}^2$, which asserts that closed forms on $\mathbb{R}^2$ are in fact exact, that is, the de Rham cohomology of the plane is zero. There are two cases to build:

a) Let ω be a closed 1-form defined on $\mathbb{R}^2$. Suppose $\omega = p dx + q dy$ where p, q are smooth functions, prove that there exists a smooth function f with

$$\partial_x f = p, \quad \partial_y f = q,$$

and the zero-form f satisfies

$$df = \omega.$$

b) Let $\omega = r dx \wedge dy$ be a closed 2-form. Find functions a, b such that $\partial_x b = \partial_y a + r$, and conclude that

$$d(a dx + b dy) = \omega.$$

- 11.** We calculate the de Rham cohomology groups of the unit circle $S^1 \subset \mathbb{R}^2$.

a) Prove first the easy cases: $H^0(S^1, \mathbb{R}) = \mathbb{R}$ and $H^k(S^1, \mathbb{R}) = 0$ for $k \geq 2$.

Consider the one form $\omega = -y dx + x dy$ defined on S^1 . Recall that

$$\int_{S^1} \omega = 2\pi$$

and ω is closed but not exact, by Exercise 8 b).

b) Prove that for any closed 1-form η defined on S^1 , there exists $c \in \mathbb{R}$ such that $\eta - c\omega$ is an exact form. Therefore we conclude that $H^1(S^1; \mathbb{R}) \cong \mathbb{R}$.

c) Explain that the calculation above also yields the k -th de Rham cohomology of $\mathbb{R}^2 \setminus \{0\}$ for $k = 0, 1$.

d) Try to determine $H^2(\mathbb{R}^2 \setminus \{0\}; \mathbb{R})$.

- 12.** Let $p_1 = (-1, 0), p_2 = (1, 0)$. We calculate the de Rham cohomology of $M = \mathbb{R}^2 \setminus \{p_1, p_2\}$.

a) Prove first the easy cases

$$H^k(M; \mathbb{R}) = \begin{cases} \mathbb{R}, & k = 0, \\ 0, & k \geq 3. \end{cases}$$

b) Try translations of the generator ω of $H^1(S^1; \mathbb{R})$ found in Exercise 11 to find closed forms ω_1, ω_2 that generate $H^1(M; \mathbb{R})$. That is, $H^1(M; \mathbb{R}) = \mathbb{R}^2$.

c) Prove that every 2-form ω on M is exact.

In conclusion,

$$H^k(M; \mathbb{R}) = \begin{cases} \mathbb{R}, & k = 0, \\ \mathbb{R}^2, & k = 1, \\ 0, & k \geq 2. \end{cases}$$

- 13.** A generalized version of Poincaré's lemma states that any closed forms are exact on a *star-shaped* region $S \subset \mathbb{R}^n$ for all $n \geq 0$, where S is called star-shaped if there exists $s_0 \in S$ such that for any $s \in S$, the segment

$$\{ts_0 + (1-t)s, t \in [0, 1]\}$$

is contained in S .

In this exercise we give a proof of Poincaré's lemma for an open ball $B = B(0, R)$ in $\mathbb{R}^n$, $n \geq 1$, although the proof can be easily generalized to star-shaped regions that are smooth submanifolds of $\mathbb{R}^n$.

Let ω be a closed smooth k -form with $k \geq 1$. We write the radial vector field

$$X = \sum_{i=1}^n x_i \partial_{x_i},$$

and let L_X be the associated Lie derivative.

For $t \in (-\infty, 0]$ we define the functions $\phi_t : B \rightarrow B$ by

$$\phi_t(x) = \exp(t)x.$$

It is called the *flow* of the vector field X .

- a) Use the definition of Lie derivatives to verify that

$$\frac{d\phi_t^* \omega}{dt} = \phi_t^*(L_X \omega).$$

- b) Apply Cartan's magic formula and prove that

$$\frac{d\phi_t^* \omega}{dt} = d(\phi_t^*(\iota_X \omega)).$$

- c) Prove that $\phi_0^* \omega = \omega$ and

$$\lim_{t \rightarrow -\infty} \phi_t^* \omega = 0.$$

- d) Integrate $\phi_t^* \omega$ on $(-\infty, 0]$ and conclude that $\omega = d\eta$, where

$$\eta = \int_{-\infty}^0 \phi_t^*(\iota_X \omega) dt.$$

14. Let G be a group equipped with the binary operation $\cdot$ that has also the structure of a smooth manifold. We say G is a *Lie group* if the binary operation

$$\cdot : G \times G \rightarrow G, (g, h) \mapsto gh$$

and the map of taking inverse

$$\iota : G \rightarrow G, g \mapsto g^{-1}$$

are both smooth. We identify $M_{n \times n}(\mathbb{R})$ with the Euclidean space $\mathbb{R}^{n^2}$. Clearly $M_{n \times n}(\mathbb{R})$ is a Lie group with respect to matrix addition. In this exercise we prove that the general linear group $\text{GL}(n, \mathbb{R})$ and the special linear group $\text{SL}(n, \mathbb{R})$ are Lie groups with respect to matrix multiplication.

- a) Prove that $\text{GL}(n, \mathbb{R})$ is a (disconnected) open subset of $M_{n \times n}(\mathbb{R})$ and thus has the structure of a smooth submanifold of $\mathbb{R}^{n^2}$.
 b) Prove that matrix multiplication and taking inverse are both smooth maps on $\text{GL}(n, \mathbb{R})$ using the explicit formulae. Conclude that $\text{GL}(n, \mathbb{R})$ is a Lie group.
 c) Let $A \in \text{SL}(n, \mathbb{R})$. Prove that for any $M \in M_{n \times n}(\mathbb{R})$,

$$\det(A + tM) = \det(A) \det(I + tA^{-1}M)$$

and for any $N \in M_{n \times n}(\mathbb{R})$,

$$\det(I + tN) = 1 + \text{tr } N \cdot t + o(t).$$

- d) Calculate the differential of $\det$ at $A \in \text{SL}(n, \mathbb{R})$ and prove that $\text{SL}(n, \mathbb{R})$ is a smooth submanifold of $\mathbb{R}^{n^2}$.
 15. Let $\text{SO}(2, \mathbb{R}) \subset M_{2 \times 2}(\mathbb{R})$ be the set of 2×2 orthogonal matrices with determinant 1.

- a) Prove that we can parametrize $\text{SO}(2, \mathbb{R})$ by

$$\left\{ \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}; t \in \mathbb{R} \right\},$$

- b) Prove that $\text{SO}(2, \mathbb{R})$ is a smooth submanifold in $M_{2 \times 2}(\mathbb{R})$
 c) Spell out the formulae for multiplication and taking inverse of $\text{SO}(2, \mathbb{R})$ matrices by the parametrization in a), and conclude that $\text{SO}(2, \mathbb{R})$ is a Lie group.

- 16.** Let V be a real vector space. We say V is a *Lie algebra* if there exists a binary operation

$$[\cdot, \cdot] : V \times V \rightarrow V,$$

(called the *Lie bracket*), such that $[\cdot, \cdot]$ is bilinear and antisymmetric, and the *Jacobi identity*

$$[[x, y], z] + [[y, z], x] + [[z, x], y] = 0$$

holds for all $x, y, z \in V$.

- 17.** Prove that the followings are Lie algebras:

- a) $\mathbb{R}^3$, with the Lie bracket defined by cross product of vectors.
- b) Let $X(M)$ be the space of all vector fields on a smooth manifold M , with the Lie bracket defined as in Proposition 7.3.4 of the lecture.
- c) $gl(n, \mathbb{R}) = M_{n \times n}(\mathbb{R})$, with the Lie bracket defined by the commutator

$$[A, B] = AB - BA, \quad A, B \in M_{n \times n}(\mathbb{R}).$$

- d)

$$sl(n, \mathbb{R}) = \{M \in M_{n \times n}(\mathbb{R}); \operatorname{tr} M = 0\},$$

with the same Lie bracket defined above.

- e)

$$so(n, \mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}); A + A^T = 0\},$$

with the same Lie bracket defined above.

- 18.** Let G be a Lie group and denote by $1 \in G$ the neutral element. We denote the tangent space $T_1 G$ by g . In this exercise we prove that g has the structure of a Lie algebra. It is usually referred to as *the* Lie algebra of the Lie group G .

For any $g \in G$ we define the left translation map

$$\ell_g : G \rightarrow G, \quad h \mapsto gh,$$

which is a diffeomorphism by the definition of Lie groups.

A vector field X on G is called *left-invariant* if

$$(d\ell_g)_h(X_h) = X_{gh}, \quad g, h \in G.$$

- a) Let $X_1 \in g$. Prove that X_1 extends to a left-invariant vector field X defined pointwise by

$$X_g = (d\ell_g)_1 X_1, \quad g \in G.$$

- b) Let X, Y be left-invariant vector fields. Suppose $X_1 = Y_1$, prove that $X = Y$. Therefore we can conclude that g is isomorphic to the vector space of left-invariant vector fields on G .
- c) Let X, Y be left-invariant vector fields on G . Prove that the Lie bracket $[X, Y]$ is left-invariant as well.
- 19.** Calculate the tangent spaces at the identity and prove that the Lie algebras of the Lie groups $GL(n, \mathbb{R})$, $SL(n, \mathbb{R})$, and $SO(2, \mathbb{R})$ are respectively $gl(n, \mathbb{R})$, $sl(n, \mathbb{R})$, and $so(2, \mathbb{R})$.
- 20.** In this exercise we prove some classical formulae in multivariable calculus, as low-dimensional cases of Stokes' theorem.
- a) Let P, Q be smooth functions on a compact, 2-dimensional submanifold $M \subset \mathbb{R}^2$ with boundary ∂M . Prove *Green's formula*:

$$\int_{\partial M} P dx + Q dy = \int_M (\partial_x Q - \partial_y P) dx dy.$$

- b) Let P, Q, R be smooth functions on a compact, 3-dimensional submanifold $M \subset \mathbb{R}^3$ with boundary ∂M . Prove the *divergence theorem*

$$\int_{\partial M} P dy dz + Q dz dx + R dx dy = \int_M (\partial_x P + \partial_y Q + \partial_z R) dx dy dz,$$

- c) Let P, Q, R be smooth functions on a compact, 2-dimensional submanifold $M \subset \mathbb{R}^3$ with dimensional boundary ∂M . Prove the *Kelvin–Stokes theorem*

$$\begin{aligned} \int_M (\partial_y R - \partial_z Q) dy dz + (\partial_z P - \partial_x R) dz dx + (\partial_x Q - \partial_y P) dx dy \\ = \int_{\partial M} P dx + Q dy + R dz. \end{aligned}$$

- 21.** a) Find the area of the annulus

$$A = \{(x, y) \in \mathbb{R}^2; 1 \leq x^2 + y^2 \leq 4\}.$$

- b) Find the volume of the closed unit ball

$$B = B(0, 1) = \{(x, y, z) \in \mathbb{R}^3; x^2 + y^2 + z^2 \leq 1\}.$$

- c) Consider the cylindrical surface

$$M = \{(x, y, z) \in \mathbb{R}^3; x^2 + y^2 = 1, -1 \leq z \leq 1\}$$

and let $\omega = y dz dx$. Calculate $\int_M \omega$.

22. Let $D \subset \mathbb{R}^2$ be a bounded domain with smooth boundary.

a) Show that the area of D can be written as

$$\text{Area}(D) = \frac{1}{2} \int_{\partial D} x \, dy - y \, dx.$$

b) Use this formula to compute the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

c) Calculate the volume of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

23. Let $\phi \in C_0^\infty(D)$, where $D \subset \mathbb{R}^n$ is an open domain. Prove that for a differentiable function f on $\mathbb{R}^2$ and any index $\alpha \in \mathbb{N}^n$ we have

$$\int_{\mathbb{R}^n} \phi \partial^\alpha f = (-1)^{|\alpha|} \int_{\mathbb{R}^n} f \partial^\alpha \phi.$$

24. a) Find a nowhere vanishing vector field on the unit circle S^1 and the 3-sphere S^3 .

Brouwer's *hairy ball theorem* asserts that the n -dimensional unit sphere S^n admits no nowhere-vanishing vector fields when n is even. The $n = 2$ case was first proved by Poincaré, which we will establish by contradiction:

Suppose X is a nowhere vanishing vector field on S^2 , then $e_1 = \|X\|^{-1}X$ is a nowhere-vanishing unit vector field.

b) Prove that there exists a unit vector field e_2 such that $\{e_1, e_2\}$ forms an orthonormal basis for $T_p S^2$ at any $p \in S^2$.

We extend $\{e_1, e_2\}$ to an orthonormal basis $\{e_1, e_2, N\}$ of $\mathbb{R}^3$.

c) Calculate the differential of the inner products of e_1, e_2, N with one another and prove that there exists 1-forms ω , θ_1 , and θ_2 such that the following *Weingarten's frame equations* hold:

$$\begin{cases} de_1 = \omega e_2 - \theta_1 N, \\ de_2 = -\omega e_1 - \theta_2 N, \\ dN = \theta_1 e_1 + \theta_2 e_2. \end{cases}$$

d) Prove *Cartan's structure equation*

$$d\omega = -\theta_1 \wedge \theta_2.$$

e) Prove that $\theta_1 \wedge \theta_2(e_1, e_2) = 1$ and conclude that $\theta_1 \wedge \theta_2$ is an area form on S^2 . It follows that

$$\int_{S^2} \theta_1 \wedge \theta_2 = 4\pi.$$

f) Reach the contradiction.

Chapter 8

Fourier series

8.1 Jensen inequality

8.1.1 Definition. Let E be a vector space over $\mathbb{R}$. Let S be a subset of E . We call a *convex combination* of elements of S any element $x \in E$ which can be written as

$$\lambda_1 x_1 + \cdots + \lambda_n x_n,$$

where n is a positive integer, $\lambda_1, \dots, \lambda_n$ are non-negative real numbers such that $\lambda_1 + \cdots + \lambda_n = 1$, and $(x_1, \dots, x_n) \in S^n$. We denote by $\text{conv}(S)$ the set of all convex combinations of elements of S and call it the *convex hull* of S . If $\text{conv}(S) = S$, then we say that S is a *convex* subset of E .

Let U be a convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. If the *epigraph* of f , which is defined as

$$\Gamma_+(f) := \{(x, t) \in U \times \mathbb{R} \mid f(x) \leq t\},$$

is a convex subset of $E \times \mathbb{R}$, then we say that f is *convex*. If $-f$ is convex, or equivalently, the *hypograph* of f , which is defined as

$$\Gamma_-(f) := \{(x, t) \in U \times \mathbb{R} \mid f(x) \geq t\},$$

is a convex subset of $E \times \mathbb{R}$, then we say that f is *concave*.

8.1.2 Proposition. Let E be a vector space over $\mathbb{R}$ and U be a subset of E . Then U is convex if and only if, for any $t \in [0, 1]$ and any $(x, y) \in U^2$, one has

$$tx + (1 - t)y \in U.$$

Proof. The necessity is clear. For the sufficiency, we reason by induction that any convex combination of n elements of U still belongs to U . The case where $n = 1$

and $n = 2$ are clear. In the following, we prove the passage from the case of n points to that of $n+1$ points. Let $x_1, \dots, x_{n+1}$ be elements of U , and $(\lambda_1, \dots, \lambda_{n+1}) \in \mathbb{R}_{\geq 0}^{n+1}$. Without loss of generality, we assume that $\lambda_1 > 0$. For $i \in \{1, \dots, n\}$, let

$$\mu_i = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n}.$$

Then

$$\mu_1 + \dots + \mu_n = 1.$$

by the induction hypothesis, we obtain that

$$y := \mu_1 x_1 + \dots + \mu_n x_n \in U.$$

Let

$$t = \lambda_1 + \dots + \lambda_n.$$

One has $1 - t = \lambda_{n+1}$. By the case of the convex combination of two elements, one has

$$ty + (1 - t)x_{n+1} = \lambda_1 x_1 + \dots + \lambda_n x_n + \lambda_{n+1} x_{n+1},$$

as required. $\square$

8.1.3 Theorem (Jensen inequality). *Let E be a vector space over $\mathbb{R}$, U be a convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. The following statements are equivalent.*

- (1) f is convex.
- (2) For any positive integer n , any $(x_1, \dots, x_n) \in U^n$ and $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}_{\geq 0}^n$ such that $\lambda_1 + \dots + \lambda_n = 1$, one has

$$f(\lambda_1 x_1 + \dots + \lambda_n x_n) \leq \lambda_1 f(x_1) + \dots + \lambda_n f(x_n).$$

- (3) For any $(x, y) \in U^2$ and any $t \in [0, 1]$, one has

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

Proof. Let $(x_1, \dots, x_n) \in U^n$ and $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}_{\geq 0}^n$ such that $\lambda_1 + \dots + \lambda_n = 1$. Then the inequality

$$f(\lambda_1 x_1 + \dots + \lambda_n x_n) \leq \lambda_1 f(x_1) + \dots + \lambda_n f(x_n)$$

can be interpreted as

$$(\lambda_1 x_1 + \dots + \lambda_n x_n, \lambda_1 f(x_1) + \dots + \lambda_n f(x_n)) = \lambda_1(x_1, f(x_1)) + \dots + \lambda_n(x_n, f(x_n))$$

belongs to the epigraph of f . Therefore, the first two conditions are equivalent.

The proof for the equivalence between (1) and (3) is similar. It suffices to apply 8.1.2. $\square$

8.1.4 Definition. Let E be a vector space over $\mathbb{R}$. We call *affine function* on E any mapping $f : E \rightarrow \mathbb{R}$ of the form

$$f(x) = \varphi(x) + C,$$

where $\varphi : E \rightarrow \mathbb{R}$ is an $\mathbb{R}$ -linear mapping and C is a constant. For any $(x, y) \in E \times E$ and any $t \in [0, 1]$, one has

$$\begin{aligned} f(tx + (1-t)y) &= \varphi(tx + (1-t)y) + C \\ &= t\varphi(x) + (1-t)\varphi(y) + C(t + (1-t)) \\ &= tf(x) + (1-t)f(y). \end{aligned}$$

Therefore, f is both convex and concave.

8.1.5 Proposition. Let E be a vector space over $\mathbb{R}$, U be a convex subset of E . Let $(f_i)_{i \in I}$ be a family of convex mappings from U to $\mathbb{R}$. Assume that, for any $x \in U$, one has $\sup_{i \in I} f_i(x) \in \mathbb{R}$. Then the mapping

$$f : U \longrightarrow \mathbb{R}, \quad f(x) := \sup_{i \in I} f_i(x)$$

is convex.

Proof. Let x and y be elements of U , and $t \in [0, 1]$. One has

$$\begin{aligned} f(tx + (1-t)y) &= \sup_{i \in I} f_i(tx + (1-t)y) \leq \sup_{i \in I} (tf_i(x) + (1-t)f_i(y)) \\ &\leq \left(\sup_{i \in I} tf_i(x) \right) + \left(\sup_{i \in I} (1-t)f_i(y) \right) = t \left(\sup_{i \in I} f_i(x) \right) + (1-t) \left(\sup_{i \in I} f_i(y) \right) \\ &= tf(x) + (1-t)f(y). \end{aligned}$$

Hence f is convex. □

8.1.6 Proposition. Let H be a Hilbert space, U be a convex open subset of H , and $f : U \rightarrow \mathbb{R}$ be a mapping. Assume that f is second order differentiable, and $Df(x)$ is semipositive (resp. seminegative) for any $x \in U$, then f is convex (resp. concave). Moreover, for any $a \in U$, one has

$$\begin{aligned} \forall x \in U, \quad f(x) &\geq f(a) + Df(a)(x - a) \\ (\text{resp. } \forall x \in U, \quad f(x) &\leq f(a) + Df(a)(x - a)). \end{aligned}$$

Proof. It suffices to treat the semipositive case.

We claim that, for any $a \in U$, one has

$$\forall x \in U, \quad f(x) \geq f(a) + Df(a)(x - a).$$

Let us consider the mapping

$$g : t \longmapsto f(a + t(x - a)).$$

This mapping is second order differentiable. Moreover, one has

$$g'(t) = Df(a + t(x - a))(x - a),$$

and

$$g''(t) = D^2f(a + t(x - a))(x - a, x - a).$$

By the Taylor's formula, we obtain that there exists $\theta \in [0, 1]$ such that

$$\begin{aligned} f(x) &= g(1) = g(0) + g'(0) + \frac{1}{2}g''(\theta) \\ &= f(a) + Df(a)(x - a) + \frac{1}{2}D^2f(a + \theta(x - a))(x - a, x - a) \\ &\geq f(a) + Df(a)(x - a). \end{aligned}$$

Hence, if we denote by f_a the affine function on H defined as

$$f_a(x) := f(a) + Df(a)(x - a),$$

then f is equal to $\sup_{a \in U} f_a$. Hence f is convex. $\square$

8.1.7 Example. Consider the exponential mapping $\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}$, $x \mapsto e^x$. One has $\exp'(x) = \exp(x)$ and hence $\exp''(x) = \exp(x) > 0$. Therefore, the exponential mapping is convex.

8.2 Hölder inequality

In this section, we fix a measure space (Ω, A, ν) . Moreover, in this section $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

8.2.1 Definition. Let p be a real number such that $p \geq 1$. If $f : \Omega \rightarrow \mathbb{K}$ is an A -measurable mapping, we denote by $\|f\|_{L^p}$ the element

$$\left(\int_{\Omega} |f(\omega)|^p \nu(d\omega) \right)^{1/p}.$$

We denote by $\mathcal{L}^p(\Omega, A, \nu)$ the set of all A -measurable mappings $f : \Omega \rightarrow \mathbb{K}$ such that $\|f\|_{L^p} < +\infty$. By definition, if λ is a real number, then

$$\|\lambda f\|_{L^p} = |\lambda| \cdot \|f\|_{L^p},$$

provided that the term on the right hand side is well defined.

8.2.2 Definition. For any measurable mapping f from Ω to $\mathbb{K}$, we denote by $\|f\|_{L^\infty}$ the element

$$\inf_{\substack{A \in \mathcal{A} \\ \nu(A)=0}} \sup_{\omega \in \Omega \setminus A} |f(\omega)|.$$

We denote by $\mathcal{L}^\infty(\Omega, A, \nu)$ the set of measurable mappings $f : \Omega \rightarrow \mathbb{K}$ such that $\|f\|_{L^\infty} < +\infty$. Note that the set

$$\{\omega \in \Omega \mid |f(\omega)| > \|f\|_{L^\infty}\} = \bigcup_{\varepsilon \in \mathbb{Q}_{>0}} \{\omega \in \Omega \mid |f(\omega)| > \|f\|_{L^\infty} + \varepsilon\}$$

is negligible.

It turns out that $\mathcal{L}^\infty(\Omega, A, \nu)$ forms a vector space over $\mathbb{K}$ and $\|\cdot\|_{L^\infty}$ is a seminorm on it. In fact, if f and g are elements of $\mathcal{L}^\infty(\Omega, A, \nu)$, for any $A \in \mathcal{A}$ such that $\nu(A) = 0$, one has

$$\sup_{\omega \in \Omega \setminus A} |(f+g)(\omega)| \leq \left(\sup_{\omega \in \Omega \setminus A} |f(\omega)| \right) + \left(\sup_{\omega \in \Omega \setminus A} |g(\omega)| \right).$$

Taking the infimum with respect to A , we obtain

$$\|f+g\|_{L^\infty} \leq \|f\|_{L^\infty} + \|g\|_{L^\infty}.$$

The null space of the seminorm $\|\cdot\|_{L^\infty}$ is the set of all negligible functions. We denote by $L^\infty(\Omega, A, \nu)$ the quotient vector space of $\mathcal{L}^\infty(\Omega, A, \nu)$ by the null space. The seminorm $\|\cdot\|_{L^\infty}$ induces by passing to quotient a norm on $L^\infty(\Omega, A, \nu)$ which we still denote as $\|\cdot\|_{L^\infty}$.

8.2.3 Theorem (Hölder inequality). *Let n be a positive integer, $p_1, \dots, p_n$ be elements of $\mathbb{R}_{\geq 1} \cup \{\infty\}$ such that*

$$\frac{1}{p_1} + \dots + \frac{1}{p_n} = 1,$$

where by convention $\frac{1}{\infty}$ is considered as 0. Let $f_1, \dots, f_n$ be A -measurable mappings from Ω to $\mathbb{K}$. One has

$$\|f_1 \cdots f_n\|_{L^1} \leq \prod_{j=1}^n \|f_j\|_{L^{p_j}},$$

provided that the product on the right hand side is well defined.

Proof. We may assume that $\|f_j\|_{L^{p_j}} > 0$ for any $j \in \{1, \dots, n\}$ since otherwise

$$\{\omega \in \Omega \mid f_j(\omega) \neq 0\}$$

is of measure 0 and hence $f_1 \cdots f_n$ is also negligible, which leads to

$$\|f_1 \cdots f_n\|_{L^1} = 0.$$

We can assume furthermore that $\|f_j\|_{L^p} < +\infty$ for any $j \in \{1, \dots, n\}$ since otherwise the inequality is trivial.

We reason by induction on n . The case where $n = 1$ is trivial. Assume that the statement is true for the case of $n - 1$. If at least one of $p_1, \dots, p_n$ is ∞ (without loss of generality, we suppose that $p_n = \infty$), then one has

$$\frac{1}{p_1} + \dots + \frac{1}{p_{n-1}} = 1,$$

and

$$\|f_1 \cdots f_n\|_{L^1} \leq \|f_1 \cdots f_{n-1}\|_{L^1} \cdot \|f_n\|_{L^\infty}.$$

By the induction hypothesis, we obtain the desired result.

In the rest of proof, we assume that $p_1, \dots, p_n$ are real numbers. For positive real numbers $x_1, \dots, x_n$, one has

$$x_1 \cdots x_n = \exp \left(\sum_{j=1}^n \frac{\ln(x_j^{p_j})}{p_j} \right).$$

Since $\exp(\cdot)$ is convex and

$$\sum_{j=1}^n \frac{1}{p_j} = 1,$$

by Jensen inequality one has

$$x_1 \cdots x_n \leq \sum_{j=1}^n \frac{1}{p_j} \exp(\ln(x_j^{p_j})) = \sum_{j=1}^n \frac{x_j^{p_j}}{p_j}.$$

By the above inequality, one has

$$\frac{|f_1 \cdots f_n|}{\|f_1\|_{L^{p_1}} \cdots \|f_n\|_{L^{p_n}}} \leq \sum_{j=1}^n \frac{|f_j|^{p_j}}{p_j \|f_j\|_{L^{p_j}}^{p_j}}.$$

Taking the integral with respect to ν , we get

$$\frac{1}{\|f_1\|_{L^{p_1}} \cdots \|f_n\|_{L^{p_n}}} \int_{\Omega} |f_1 \cdots f_n|(\omega) \nu(d\omega) \leq \frac{1}{p_1} + \dots + \frac{1}{p_n} = 1,$$

as required. $\square$

8.2.4 Corollary (Minkowski inequality). *Let f and g be elements of $\mathcal{L}^p(\Omega, A, \nu)$. Then $f + g \in \mathcal{L}^p(\Omega, A, \nu)$ and*

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

In particular, $\mathcal{L}^p(\Omega, A, \nu)$ is a vector space over $\mathbb{K}$ and $\|\cdot\|_{L^p}$ is a seminorm on $\mathcal{L}^p(\Omega, A, \nu)$.

Proof. If $p = 1$, the statement comes from the triangle inequality $|f + g| \leq |f| + |g|$.

In the following, we suppose that $p > 1$ and we let

$$q := (1 - p^{-1})^{-1} = \frac{p}{p-1},$$

so that $p^{-1} + q^{-1} = 1$. Without loss of generality, we may assume that f and g take non-negative values. We apply Hölder's inequality to f and $(f + g)^{p-1}$ to obtain

$$\int_{\Omega} f(f + g)^{p-1} d\nu \leq \|f\|_{L^p} \left(\int_{\Omega} (f + g)^p \right)^{(p-1)/p}.$$

Similarly, one has

$$\int_{\Omega} g(f + g)^{p-1} d\nu \leq \|g\|_{L^p} \left(\int_{\Omega} (f + g)^p \right)^{(p-1)/p}.$$

Taking the sum of these two inequalities, we get

$$\|f + g\|_{L^p}^p = \int_{\Omega} (f + g)^p d\nu \leq (\|f\|_{L^p} + \|g\|_{L^p}) \|f + g\|_{L^p}^{p-1},$$

so that

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

□

8.2.5 Remark. Note that the null subspace of $\mathcal{L}^p(\Omega, A, \nu)$ with respect to the seminorm $\|\cdot\|_{L^p}$ coincides with the vector space of negligible functions. We denote by $L^p(\Omega, A, \nu)$ the quotient of $\mathcal{L}^p(\Omega, A, \nu)$ by the vector subspace of negligible functions. The seminorm $\|\cdot\|_{\mathcal{L}^p}$ induces by passing to quotient a norm on $L^p(\Omega, A, \nu)$, which we still denote by $\|\cdot\|_{L^p}$.

8.2.6 Lemma. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence of elements of $\mathcal{L}^p(\Omega, A, \nu)$. Suppose that

$$C := \sum_{n \in \mathbb{N}} \|u_n\|_{L^p} < +\infty.$$

Then there exists subset $A \in \mathcal{A}$ with $\mu(A) = 0$ such that the sequence $(u_n(\omega))_{n \in \mathbb{N}}$ converges absolutely for any $\omega \in \Omega \setminus A$. Moreover, if we denote by $u : \Omega \rightarrow \mathbb{R}$ the mapping that sends $\omega \in \Omega \setminus A$ to $\sum_{n \in \mathbb{N}} u_n(\omega)$ and sends $\omega \in A$ to 0, then $u \in \mathcal{L}^p(\Omega, A, \nu)$, and

$$\lim_{n \rightarrow +\infty} \left\| u - \sum_{k=0}^n u_k \right\|_{L^p} = 0.$$

Proof. For any $n \in \mathbb{N}$, let $v_n = \sum_{k=0}^n |v_k|$ and

$$v = \sum_{n \in \mathbb{N}} |u_n| = \lim_{n \rightarrow +\infty} v_n.$$

Then the sequence $(v_n)_{n \in \mathbb{N}}$ is increasing. Moreover, in the case where $p < \infty$, by Corollary 8.2.4, one has

$$\|v_n\|_{L^p} \leq \sum_{k=0}^n \|u_k\|_{L^p} \leq C.$$

In the case where $p = \infty$, one has

$$\{\omega \in \Omega \mid v(\omega) = +\infty\} \subseteq \bigcup_{n \in \mathbb{N}} \{\omega \in \Omega \mid v_n(\omega) > C\}.$$

Hence

$$\mu(\{\omega \in \Omega \mid v(\omega) > C\}) = 0,$$

which leads to

$$\mu(\{\omega \in \Omega \mid v(\omega) = +\infty\}) = 0$$

and $\|v\|_{L^\infty} \leq C$. In the case where $p \neq \infty$, by the monotone convergence theorem of Beppo Levi, we obtain that

$$\int_{\Omega} v(\omega)^p \nu(d\omega) = \lim_{n \rightarrow +\infty} \int_{\Omega} v_n(\omega)^p \nu(d\omega) \leq C^p.$$

Therefore, the set

$$A := \{\omega \in \Omega \mid v(\omega) = +\infty\}$$

always has measure 0. Note that for any $\omega \in \Omega \setminus A$, the series

$$\sum_{n \in \mathbb{N}} u_n(\omega)$$

converges absolutely. We denote by $u(\omega)$ its limit. If $\omega \in A$, we let $u(\omega) = 0$. Since $|u| \leq v$ for any $\omega \in \Omega$, we obtain that $u \in \mathcal{L}^p(\Omega, A, \nu)$, and, in the case where $p \neq \infty$,

$$\|u\|_{L^p} \leq \left(\int_{\Omega} v(\omega)^p \nu(d\omega) \right)^{1/p} \leq \sum_{n \in \mathbb{N}} \|u_n\|_{L^p};$$

in the case where $p = \infty$, one has

$$\|u\|_{L^\infty} \leq \|v\|_{L^\infty} \leq \sum_{n \in \mathbb{N}} \|u_n\|_{L^\infty}.$$

The same argument also shows that

$$\left\| u - \sum_{k=0}^n u_k \right\|_{L^p} \leq \sum_{k \in \mathbb{N}, k \geq n} \|u_k\|_{L^p}.$$

Therefore,

$$\lim_{n \rightarrow +\infty} \left\| u - \sum_{k=0}^n u_k \right\|_{L^p} = 0.$$

□

8.2.7 Theorem. *Let p be a real number such that $p \geq 1$. The seminormed vector space $\mathcal{L}^p(\Omega, A, \nu)$ is complete, namely any Cauchy sequence in $\mathcal{L}^p(\Omega, A, \nu)$ converges.*

Proof. Let $(f_n)_{n \in \mathbb{N}}$ be a Cauchy sequence. We extract a subsequence $(f_{n_k})_{k \in \mathbb{N}}$ in a recursive way as follows: if $n_0, \dots, n_k$ are chosen, we let n_{k+1} be a natural number $> n_k$ such that

$$\forall (\ell, m) \in \mathbb{N}_{\geq n_{k+1}}, \quad \|f_\ell - f_m\|_{L^p} \leq 2^{-(k+1)}.$$

Thus

$$\|f_{n_0}\|_{L^p} + \sum_{k \in \mathbb{N}} \|f_{n_{k+1}} - f_{n_k}\|_{L^p} \leq \|f_{n_0}\|_{L^p} + \sum_{k \in \mathbb{N}} 2^{-k} = \|f_{n_0}\|_{L^p} + 2 < +\infty.$$

Hence the sequence $(f_{n_k})_{k \in \mathbb{N}}$ converges in $\mathcal{L}^p(\Omega, A, \nu)$. Since $(f_n)_{n \in \mathbb{N}}$ is a Cauchy sequence which admits a convergent subsequence, it also converges. □

8.2.8 Corollary. *The $\mathbb{K}$ -vector space $\mathcal{L}^2(\Omega, A, \nu)$ equipped with the inner product $\langle \cdot, \cdot \rangle_{L^2}$ defined as*

$$\langle f, g \rangle_{L^2} := \int_{\Omega} \overline{f(\omega)} g(\omega) \nu(d\omega),$$

is a Hilbert space.

8.2.9 Remark. From the proof of Lemma 8.2.6 and Theorem 8.2.7, we observe that, if $(f_n)_{n \in \mathbb{N}}$ is a convergent sequence in $\mathcal{L}^p(\Omega, A, \nu)$, then it has a subsequence $(f_{n_k})_{k \in \mathbb{N}}$ such that the series (with the convention that $f_{n_{-1}} = 0$)

$$\sum_{k \in \mathbb{N}} (f_{n_k} - f_{n_{k-1}})$$

converges absolutely ν -almost everywhere to the limit of $(f_n)_{n \in \mathbb{N}}$ (in $\mathcal{L}^p(\Omega, A, \nu)$).

8.2.10 Remark. Assume that $\nu(\Omega) < +\infty$. For any p and q in $\mathbb{R}_{\geq 0} \cup \{\infty\}$ such that $p < q$, one has

$$\mathcal{L}^q(\Omega, A, \nu) \subseteq \mathcal{L}^p(\Omega, A, \nu).$$

In fact, in the case where $q \in \mathbb{R}_{\geq 1}$, by the fact that $\mathbb{1}_{\Omega} \in \mathcal{L}^{q/(q-p)}(\Omega, A, \nu)$, we deduce from Hölder's inequality that, for any $f \in \mathcal{L}^q(\Omega, A, \nu)$, one has

$$\int_{\Omega} |f|^p d\nu \leq \left(\int_{\Omega} |f|^q d\nu \right)^{p/q} \cdot \left(\int_{\Omega} \mathbb{1}_{\Omega}^{q/(q-p)} d\nu \right)^{(q-p)/q}.$$

Hence

$$\|f\|_{L^p} \leq \|f\|_{L^q} \cdot \nu(\Omega)^{(q-p)/q} < +\infty.$$

In the case $q = \infty$, for any $f \in \mathcal{L}^p(\Omega, A, \nu)$, one has

$$\int_{\Omega} |f|^p d\nu \leq \|f\|_{L^\infty}^p \nu(\Omega) < +\infty.$$

8.3 Haar measure on a torus

In this section, $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

8.3.1 Definition. Recall that a Hilbert space over $\mathbb{K}$ is by definition a vector space V over $\mathbb{K}$ equipped with an inner product $\langle \cdot, \cdot \rangle$, such that V equipped with the norm $\|\cdot\|$ defined as

$$\|x\| := \langle x, x \rangle^{1/2}$$

is complete.

We say that an orthogonal family (resp. orthonormal family) $(e_i)_{i \in I}$ is an orthogonal basis (resp. orthonormal basis) if

$$\text{Vect}_{\mathbb{K}}(\{e_i \mid i \in I\})$$

is dense in V .

8.3.2 Proposition. Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $(e_i)_{i \in I}$ be an orthogonal family in V . Then it is an orthogonal basis if and only if

$$\{e_i \mid i \in I\}^\perp = \{0\}.$$

Proof. Without loss of generality, we may assume that $(e_i)_{i \in I}$ is an orthonormal basis. Let W be the closure of

$$\text{Vect}_{\mathbb{K}}(\{e_i \mid i \in I\}).$$

“ $\implies$ ” Suppose that $W = V$. If x is an element of V such that $\langle e_i, x \rangle = 0$ for any $i \in I$, by Proposition 3.2.7, we obtain that $x = p_W(x) = 0$.

“ $\impliedby$ ”: Suppose that $\{e_i \mid i \in I\}^\perp = \{0\}$. By Proposition 3.2.7, for any $x \in V$, one has

$$x - p_W(x) \in W^\perp \subseteq \{e_i \mid i \in I\}^\perp.$$

Hence $x - p_W(x) = 0$, which means that $x = p_W(x)$. □

In the rest of the section, we fix a finite-dimensional vector space over $\mathbb{R}$, equipped with a Haar measure vol . Let $(e_i)_{i=1}^n$ be a basis and $(e_i^\vee)_{i=1}^n$ be its dual basis. We let D be the parallelotope

$$\{a_1 e_1 + \dots + a_n e_n \mid (a_1, \dots, a_n) \in [0, 1]^n\},$$

and let $\text{vol}(D)$ be its measure.

Let M be the sub- $\mathbb{Z}$ -module E generated by $(e_j)_{j=1}^n$. Let N be the sub- $\mathbb{Z}$ -module of $E^\vee$ generated by $(e_j^\vee)_{j=1}^n$. For $\alpha \in N$, we let φ_α be the mapping from E to $\mathbb{C}$ that sends $x \in E$ to

$$\exp(2\pi i \alpha(x)).$$

This is a morphism of groups from $(E, +)$ to the unit circle in $\mathbb{C}$. Note that, for any $m \in M$, one has $\alpha(m) \in \mathbb{Z}$ and hence $\varphi_\alpha(m) = 1$. In particular, φ_α is M -periodic, namely,

$$\forall x \in E, \forall m \in M, \quad \varphi_\alpha(x + m) = \varphi_\alpha(x).$$

In particular, φ_α induces by passing to quotient a morphism of groups from E/M to the unit circle in $\mathbb{C}$, which we still denote as φ_α by abuse of notation. Moreover, if α and β are two elements of N , then

$$\varphi_{\alpha+\beta} = \varphi_\alpha \varphi_\beta.$$

We denote by $\mathcal{L}^2(D)$ the $\mathbb{C}$ -vector space of square-integrable Borel measurable functions on D . We equip $\mathcal{L}^2(D)$ with the following semi-inner product $\langle \cdot, \cdot \rangle$:

$$\forall (f, g) \in \mathcal{L}^2(D) \times \mathcal{L}^2(D), \quad \langle f, g \rangle := \frac{1}{\text{vol}(D)} \int_D \overline{f(x)} g(x) \text{vol}(dx).$$

Note that this inner product does not depend on the choice of the Haar measure vol . Let $L^2(D)$ be the quotient of $\mathcal{L}^2(D)$ by the vector subspace of negligible functions. Note that $\langle \cdot, \cdot \rangle$ induces by passing to quotient an inner product on $L^2(D)$ so that $L^2(D)$ becomes a Hilbert space. By abuse of notation, for any $f \in \mathcal{L}^2(D)$, we still use f to denote its equivalence class in $L^2(D)$.

8.3.3 Proposition. *The family $(\varphi_\alpha)_{\alpha \in N}$ forms an orthonormal family in the Hilbert space $L^2(D)$.*

Proof. We identify E with the product of $\mathbb{R}e_i$ and suppose that vol is the product of the usual Lebesgue measure on $\mathbb{R}e_j$ (namely the measure of $[0, 1]e_i$ is 1). Note that, for any $x \in E$, one has

$$\langle \varphi_\alpha, \varphi_\alpha \rangle = \frac{1}{\text{vol}(D)} \int_D 1 \text{vol}(dx) = 1.$$

Let α and β be two different element of N . Then

$$\langle \varphi_\alpha, \varphi_\beta \rangle = \frac{1}{\text{vol}(D)} \int_D \exp(2\pi i(\beta - \alpha)(x)) \text{vol}(dx).$$

Suppose that $\beta - \alpha$ is of the form

$$a_1 e_1^\vee + \cdots + a_n e_n^\vee,$$

where $(a_1, \dots, a_n) \in \mathbb{Z}^n$. Then

$$\int_D \exp(2\pi i(\beta - \alpha)(x)) \text{vol}(dx) = \prod_{j=1}^n \int_0^1 \exp(2\pi i a_j t) dt.$$

If $a_j \neq 0$, then

$$\int_0^1 \exp(2\pi i a_j t) dt = \left[\frac{\exp(2\pi i a_j t)}{2\pi i a_j} \right]_{t=0}^{t=1} = 0.$$

□

8.3.4 Proposition. *Let $\mathbb{T}_M$ be the quotient of E by the subgroup M . We denote by $\pi : E \rightarrow \mathbb{T}_M$ the quotient mapping. Then the restriction of π to D is an isomorphism of measurable spaces. In particular, if we equip $\mathbb{T}_M$ with the Haar measure η such that $\eta(\mathbb{T}_M) = 1$, then, for any integrable mapping $h : \mathbb{T}_M \rightarrow \mathbb{C}$, one has*

$$\int_{\mathbb{T}_M} h(t) \eta(dt) = \frac{1}{\text{vol}(D)} \int_D h(\pi(x)) \text{vol}(dx).$$

Proof. The restriction of π to D is a measurable bijection.

Moreover, the image of an open subset U of D by π is a Borel measurable subset. To see this, it suffices to treat the case where U is of the form $U_1 e_1 + \cdots + U_n e_n$, where each U_i is an open subset of $[0, 1[$. Let $p : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$ be the projection mapping. Then

$$p(U_i \cap]0, 1[)$$

is an open subset of $\mathbb{R}/\mathbb{Z}$ and $p(U_i \cap \{0\})$ is a closed subset of $\mathbb{R}/\mathbb{Z}$. Therefore, $\pi(U_i)$ is a Borel subset of $\mathbb{R}/\mathbb{Z}$. We then deduce that

$$\pi(U) = \prod_{i=1}^n p(U_i)[e_i]$$

is Borel measurable.

Clearly the mapping sending a Borel measurable set A in $\mathbb{T}_M$ to

$$\frac{\text{vol}(\pi^{-1}(A) \cap D)}{\text{vol}(D)}$$

is a measure. It suffices to check that it is a Haar measure. Let A be a Borel measurable subset of $\mathbb{T}_M$ and $t \in \mathbb{T}_M$. We pick an arbitrary $x \in D$ such that $\pi(x) = t$. Then

$$\begin{aligned}\pi^{-1}(A+t) \cap D &= (\pi^{-1}(A) + x) \cap D = \left(\bigcup_{m \in M} (\pi^{-1}(A) \cap D) + (x - m) \right) \cap D \\ &= \bigcup_{m \in M} \left(((\pi^{-1}(A) \cap D) + x) \cap (m + D) \right) - m.\end{aligned}$$

Therefore, by the fact that vol is a Haar measure, we obtain

$$\begin{aligned}\text{vol}(\pi^{-1}(A+t) \cap D) &= \sum_{m \in M} \text{vol}((\pi^{-1}(A \cap D) + x) \cap (m + D)) \\ &= \text{vol}((\pi^{-1}(A \cap D) + x)) = \text{vol}(\pi^{-1}(A \cap D))\end{aligned}$$

□

8.4 Convolution

In this section, we fix a locally compact Hausdorff topological group $(G, +)$. We assume in addition that $(G, +)$ is σ -compact (namely G is a union of countably many compact subsets) and that there exists a σ -finite measure η on G which is both a left and right Haar measure. In particular, for any non-negative measurable real-valued function h defined on G , one has

$$\int_G h(x) \eta(dx) = \int_G h(-x) \eta(dx).$$

8.4.1 Definition. Let f and g be two measurable mappings from G to $\mathbb{C}$. For $x \in G$, if the mapping

$$(y \in G) \longmapsto f(y)g(-y+x)$$

is integrable, we denote by $(f * g)(x)$ the integral

$$\int_G f(y)g(-y+x) \eta(dy).$$

Note that

$$(f * g)(x) = \int_G f(-y)g(y+x) \eta(dy) = \int_G f(x+(-y))g(y) \eta(dy).$$

In particular, in the case where $(G, +)$ is a commutative group, then $(f * g)(x)$ is well defined if and only if $(g * f)(x)$ is well defined, and one has

$$(f * g)(x) = (g * f)(x).$$

We say that f and g are convolvable if the set of $x \in G$ such that $(f * g)(x)$ is not well defined forms a negligible set. In the case where f and g are convolvable, for $x \in G$ such that $(f * g)(x)$ is not well defined, by convention we let $(f * g)(x) = 0$.

8.4.2 Theorem (Young's inequality). *Let p, q and r be elements of $\mathbb{R}_{\geq 1} \cup \{\infty\}$. Suppose that*

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r},$$

*with the convention that $1/\infty = 0$. Let $f \in L^p(G)$ and $g \in L^q(G)$. Then f and g are convolvable. Moreover, $f * g$ belongs to $\mathcal{L}^r(G)$ and satisfies*

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \cdot \|g\|_{L^q}.$$

Proof. Case 1. We assume that $r = \infty$. In the case where $q \neq \infty$, by Hölder's inequality, one has

$$\int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \leq \|f\|_{L^p} \cdot \left(\int_G |g(-y+x)|^q \eta(dy) \right)^{1/q} = \|f\|_{L^p} \cdot \|g\|_{L^q};$$

in the case where $q = \infty$, we still have

$$\int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \leq \|f\|_{L^1} \cdot \|g\|_{L^\infty}$$

Therefore, f and g are convolvable, and

$$|(f * g)(x)| \leq \|f\|_{L^p} \cdot \|g\|_{L^q}.$$

This leads to $f * g \in L^\infty(G)$ and $\|f * g\|_{L^\infty} \leq \|f\|_{L^p} \cdot \|g\|_{L^q}$.

Case 2. We now treat the case where $p = q = r = 1$. By Fubini's theorem, one has

$$\int_G \int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \eta(dx) = \int_G |f(y)| \int_G |g(-y+x)| \eta(dx) \eta(dy).$$

Since η is a Haar measure, it is equal to $\|f\|_{L^1} \cdot \|g\|_{L^1}$. In particular,

$$(y \in G) \longmapsto f(y)g(-y+x)$$

is integrable outside of a negligible subset of Ω , which shows that f and g are convolvable. Moreover,

$$\begin{aligned} \|f * g\|_{L^1} &= \int_G \left| \int_G f(y)g(-y+x) \eta(dy) \right| \eta(dx) \\ &\leq \int_G \int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \eta(dx) = \|f\|_{L^1} \cdot \|g\|_{L^1}. \end{aligned}$$

Case 3. We now suppose that $r \notin \{1, \infty\}$. In this case neither p nor q could be ∞ . Let a and b be elements of $\mathbb{R}_{\geq 1} \cup \{\infty\}$ such that

$$\frac{1}{p} + \frac{1}{a} = 1, \quad \frac{1}{q} + \frac{1}{b} = 1.$$

Then one has

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{r} = 1.$$

In the case where p, q, a and b are real numbers, by Hölder inequality, one has

$$\begin{aligned} & \int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \\ &= \int_G |f(y)|^{p/b} \cdot |g(-y+x)|^{q/a} \cdot (|f(y)|^{p/r} \cdot |g(-y+x)|^{q/r}) \eta(dy) \\ &\leq \left(\int_G |f(y)|^p \eta(dy) \right)^{1/b} \left(\int_G |g(-y+x)|^q \eta(dy) \right)^{1/a} \\ &\quad \left(\int_G |f(y)|^p \cdot |g(-y+x)|^q \eta(dy) \right)^{1/r} \\ &\leq \|f\|_{L^p}^{p/b} \cdot \|g\|_{L^q}^{q/a} \left(\int_G |f(y)|^p \cdot |g(-y+x)|^q \eta(dy) \right)^{1/r}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \int_G \left(\int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \right)^r \eta(dx) \\ &\leq \|f\|_{L^p}^{pr/b} \cdot \|g\|_{L^q}^{qr/a} \cdot \|f\|_{L^p}^p \cdot \|g\|_{L^q}^q = \|f\|_{L^p}^r \cdot \|g\|_{L^q}^r, \end{aligned}$$

Thus f and g are convolvable, and

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \cdot \|g\|_{L^q}.$$

Suppose that $p = 1$ and $q = r$, then

$$\begin{aligned} & \int_G |f(y)| \cdot |g(-y+x)| \eta(dy) = \int_G |f(y)|^{1/b} \cdot (|f(y)|^{1/r} \cdot |g(-y+x)|) \eta(dy) \\ &\leq \left(\int_G |f(y)| \eta(dy) \right)^{1/b} \left(\int_G |f(y)| \cdot |g(-y+x)|^r \eta(dy) \right)^{1/r} \\ &= \|f\|_{L^1}^{1/b} \left(\int_G |f(y)| \cdot |g(-y+x)|^r \eta(dy) \right)^{1/r}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \int_G \left(\int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \right)^r \eta(dx) \\ & \leq \|f\|_{L^1}^{r/b} \cdot \|f\|_{L^1} \cdot \|g\|_{L^q}^q = \|f\|_{L^1}^r \cdot \|g\|_{L^q}^r. \end{aligned}$$

Thus f and g are convolvable, and

$$\|f * g\|_{L^r} \leq \|f\|_{L^1} \cdot \|g\|_{L^q}.$$

Suppose that $q = 1$ and $p = r$, then

$$\begin{aligned} & \int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \\ & = \int_G |g(-y+x)|^{1/a} \cdot (|f(y)| \cdot |g(-y+x)|^{1/r}) \eta(dy) \\ & \leq \left(\int_G |g(-y+x)| \eta(dy) \right)^{1/a} \left(\int_G |f(y)|^r \cdot |g(-y+x)| \eta(dy) \right)^{1/r} \\ & = \|g\|_{L^1}^{1/a} \left(\int_G |f(y)|^r \cdot |g(-y+x)| \eta(dy) \right)^{1/r}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \int_G \left(\int_G |f(y)| \cdot |g(-y+x)| \eta(dy) \right)^r \eta(dx) \\ & \leq \|g\|_{L^1}^{r/a} \cdot \|f\|_{L^p}^r \cdot \|g\|_{L^1} = \|f\|_{L^p}^r \cdot \|g\|_{L^1}^r. \end{aligned}$$

Thus f and g are convolvable, and

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \cdot \|g\|_{L^1}.$$

□

8.5 Fourier series

In this section, we fix a finite-dimensional vector space E over $\mathbb{R}$ and a sub- $\mathbb{Z}$ -module M of E generated by a basis $(e_i)_{i=1}^d$ of E . Let N be the sub- $\mathbb{Z}$ -module of $E^\vee$ generated by the dual basis. Denote by $\mathbb{T}_M$ the quotient group E/M , equipped with the unique Haar measure η such that $\eta(\mathbb{T}_M) = 1$. Moreover, for any $p \in \mathbb{R}_{\geq 1} \cup \{\infty\}$, we denote by $L^p(\mathbb{T}_M)$ the set of all measurable mappings $f : \mathbb{T}_M \rightarrow \mathbb{C}$ such that $\|f\|_{L^p} < +\infty$, where $\|f\|_{L^p}$ is defined as

$$\left(\int_{\mathbb{T}_M} |f(x)|^p \eta(dx) \right)^{1/p}$$

if $p \neq \infty$, and

$$\|f\|_{L^\infty} := \inf_{\substack{A \in B(\mathbb{T}_M) \\ \eta(A)=0}} \sup_{x \in \mathbb{T}_M \setminus A} |f(x)|.$$

We denote by $\pi : E \rightarrow \mathbb{T}_M$ the mapping of projection. For any $\alpha \in N$, we denote by $\varphi_\alpha : \mathbb{T}_M \rightarrow \mathbb{C}$ the mapping that sends, for any $y \in E$, $\pi(y)$ to $\exp(2\pi i \alpha(y))$. Note that φ_α is an element of $L^\infty(\mathbb{T}_M)$. For simplicity, $\varphi_{e_j^\vee}$ is denoted as φ_j . Thus, for any $(b_1, \dots, b_d) \in \mathbb{Z}^d$,

$$\varphi_{b_1 e_1^\vee + \dots + b_d e_d^\vee} = \varphi_1^{b_1} \dots \varphi_d^{b_d}.$$

8.5.1 Definition. Let f be an element of $L^1(\mathbb{T}_M)$. For any $\alpha \in N$, we denote by $c_\alpha(f)$ the following complex number

$$\int_{\mathbb{T}_M} \overline{\varphi_\alpha(x)} f(x) \eta(dx) = \int_{\mathbb{T}_M} \varphi_{-\alpha}(x) f(x) \eta(dx).$$

The family $(c_\alpha(f))_{\alpha \in N}$ is called the *Fourier coefficients* of f .

8.5.2 Proposition. Let f and g be elements of $L^1(\mathbb{T}_M)$. One has

$$\forall \alpha \in N, \quad c_\alpha(f * g) = c_\alpha(f) c_\alpha(g).$$

Proof. By Fubini theorem, one has

$$\begin{aligned} & \int_{\mathbb{T}_M} \exp(-2\pi i \alpha(x)) \int_{\mathbb{T}_M} f(y) g(x-y) \eta(dy) \eta(dx) \\ &= \int_{\mathbb{T}_M} f(y) \int_{\mathbb{T}_M} \exp(-2\pi i \alpha(x)) g(x-y) \eta(dx) \eta(dy) \\ &= \int_{\mathbb{T}_M} f(y) \int_{\mathbb{T}_M} \exp(-2\pi i \alpha(x+y)) g(x) \eta(dx) \eta(dy) \\ &= \left(\int_{\mathbb{T}_M} f(y) \exp(-2\pi i \alpha(y)) \right) \left(\int_{\mathbb{T}_M} g(x) \exp(-2\pi i \alpha(x)) \eta(dx) \right). \end{aligned}$$

□

8.5.3 Definition. We equip $E^\vee$ with the norm $\|\cdot\|$ defined as

$$\|\lambda_1 e_1^\vee + \dots + \lambda_d e_d^\vee\| := \max\{|\lambda_1|, \dots, |\lambda_d|\}.$$

For any $f \in L^1(\mathbb{T}_M)$, we call Fourier series of f the series of functions of the form

$$S_n(f) := \sum_{\alpha \in N, \|\alpha\| \leq n} c_\alpha(f) \varphi_\alpha, \quad n \in \mathbb{N}.$$

For $n \in \mathbb{N}$, we introduce the *Dirichlet kernel* as

$$\delta_n = \sum_{\alpha \in N, \|\alpha\| \leq n} \varphi_\alpha.$$

8.5.4 Proposition. *Let f be an element of $L^1(\mathbb{T}_M)$. One has*

$$S_n(f) = \delta_n \star f.$$

Proof. For any $\alpha \in N$, one has

$$(\varphi_\alpha \star f)(x) = \int_{\mathbb{T}_M} f(y) \varphi_\alpha(x-y) \eta(dy) = \varphi_\alpha(x) \int_{\mathbb{T}_M} f(y) \varphi_\alpha(-y) \eta(dy).$$

□

8.5.5 Remark. One can compute δ_n as follows:

$$\delta_n = \sum_{(b_1, \dots, b_d) \in \{-n, \dots, n\}^d} \varphi_{b_1 e_1^\vee + \dots + b_d e_d^\vee} = \sum_{(b_1, \dots, b_d) \in \{-n, \dots, n\}^d} \prod_{j=1}^d \varphi_j^{b_j} = \prod_{j=1}^d \sum_{k=-n}^n \varphi_j^k.$$

Note that, for any $j \in \{1, \dots, d\}$, one has

$$\sum_{k=-n}^n \varphi_j^k(x) = \begin{cases} 2n+1, & \text{if } x = 0, \\ \frac{\varphi_j(x)^{n+1} - \varphi_j(x)^{-n}}{\varphi_j(x) - 1}, & \text{if } x \neq 0. \end{cases} \quad (8.1)$$

For any $j \in \{1, \dots, d\}$, we let

$$\delta_{n,j} = \sum_{k=-n}^n \varphi_j^k.$$

By the fact that $\varphi_j^{-k} = \overline{\varphi_j^k}$, we obtain that $\delta_{n,j}$ takes value in $\mathbb{R}$.

8.5.6 Definition. For any subset Θ of $\{1, \dots, d\}$, we denote by τ_Θ the mapping from E to E that sends $a_1 e_1 + \dots + a_d e_d$ to

$$(-1)^{\mathbf{1}_\Theta(1)} a_1 e_1 + \dots + (-1)^{\mathbf{1}_\Theta(d)} a_d e_d.$$

In other words, if we consider the span of vectors in E as a linear combination of $e_1, \dots, e_d$, then τ_Θ switches the sign of coordinates indexed by elements of Θ . Note that, for $x = a_1 e_1 + \dots + a_d e_d \in E$,

$$\varphi_j(\tau_\Theta(x)) = \exp(2\pi i (-1)^{\mathbf{1}_\Theta(j)} a_j) = \varphi_j(x)^{(-1)^{\mathbf{1}_\Theta(j)}}.$$

Therefore, by symmetry we get

$$\delta_{n,j}(\tau_\Theta(x)) = \sum_{k=-n}^n \varphi_j(x)^{k(-1)^{\mathbf{1}_\Theta(j)}} = \delta_{n,j}(x).$$

8.5.7 Lemma (Riemann-Lebesgue). *Let a and b be real numbers such that $a < b$, and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping which is integrable with respect to Lebesgue's measure. Then*

$$\lim_{\lambda \rightarrow +\infty} \int_{\mathbb{R}} f(t) \exp(i\lambda t) dt = 0.$$

Proof. We first consider the case where f is of the form $\mathbb{1}_{]a,b]}$. In this case one has

$$\int_{\mathbb{R}} f(t) \exp(i\lambda t) dt = \left[\frac{\exp(i\lambda t)}{i\lambda} \right]_a^b = \frac{\exp(i\lambda b) - \exp(i\lambda a)}{i\lambda}.$$

Therefore,

$$\lim_{\lambda \rightarrow +\infty} \int_{\mathbb{R}} f(t) \exp(i\lambda t) dt = 0.$$

By linearity, we obtain that, if f belongs to the vector subspace S of $\mathbb{R}^{\mathbb{R}}$ generated by functions of the form $\mathbb{1}_{]a,b]}$, then the statement holds.

For the general case, by Lemma 7.2.4, for any $\varepsilon > 0$, there exists $g \in S$ such that

$$\int_{\mathbb{R}} |f(t) - g(t)| dt \leq \varepsilon.$$

Hence, for any $\lambda \in \mathbb{R}$,

$$\left| \int_{\mathbb{R}} f(t) \exp(i\lambda t) dt - \int_{\mathbb{R}} g(t) \exp(i\lambda t) dt \right| \leq \varepsilon.$$

Therefore,

$$\limsup_{\lambda \rightarrow +\infty} \left| \int_{\mathbb{R}} f(t) \exp(i\lambda t) dt \right| \leq \lim_{\lambda \rightarrow +\infty} \left| \int_{\mathbb{R}} g(t) \exp(i\lambda t) dt \right| + \varepsilon = \varepsilon.$$

Since ε is arbitrary, we get the desired result. $\square$

8.5.8 Theorem (Dirichlet). *We denote by P the parallelotope*

$$\{a_1 e_1 + \cdots + a_d e_d \mid (a_1, \dots, a_d) \in]0, \frac{1}{2}[^d\}.$$

- (1) *Let f be an element of $L^1(\mathbb{T}_M)$. By abuse of notation, we still use f to denote the composite function $f \circ \varpi$ (where ϖ is the projection mapping $E \rightarrow \mathbb{T}_M$) and thus consider f as a M -periodic function on E . For any $n \in \mathbb{N}$, and any $x \in E$ (again we consider $S_n(f)$ as an M -periodic function on E in identifying it with $S_n(f) \circ \varpi$)*

$$S_n(f)(x) = \frac{1}{2^d \text{vol}(P)} \int_P \sum_{\Theta \in \{1, \dots, d\}} f(x + \tau_{\Theta}(y)) \delta_n(y) \text{vol}(dy).$$

(2) Suppose that x is an element such that

(a) for any $\Theta \subseteq \{1, \dots, d\}$, the limit

$$f_{\Theta}(x) := \lim_{y \in P, y \rightarrow 0} f(x + \tau_{\Theta}(y))$$

exists,

(b) the function

$$(y \in P) \mapsto \frac{1}{e_1^{\vee}(y) \cdots e_d^{\vee}(y)} \sum_{\Theta \subseteq \{1, \dots, d\}} (f(x + \tau_{\Theta}(y)) - f_{\Theta}(x))$$

is integrable on P .

Then the sequence $(S_n(f)(x))_{n \in \mathbb{N}}$ converges to

$$\frac{1}{2^d} \sum_{\Theta \subseteq \{1, \dots, d\}} f_{\Theta}(x)$$

Proof. (1) Let D be the parallelotope

$$\{a_1 e_1 + \cdots + a_d e_d \mid (a_1, \dots, a_d) \in [-\frac{1}{2}, \frac{1}{2}]^d\}.$$

Then, by the relation $\delta_n(-y) = \delta_n(y)$, we obtain

$$\begin{aligned} S_n(f)(x) &= \frac{1}{\text{vol}(D)} \int_D f(x + y) \delta_n(y) \text{ vol}(dy) \\ &= \frac{1}{2^d \text{vol}(P)} \sum_{\Theta \subseteq \{1, \dots, d\}} \int_P f(x + \tau_{\Theta}(y)) \delta_n(\tau_{\Theta}(y)) \text{ vol}(dy) \\ &= \frac{1}{2^d \text{vol}(P)} \sum_{\Theta \subseteq \{1, \dots, d\}} \int_P f(x + \tau_{\Theta}(y)) \delta_n(y) \text{ vol}(dy). \end{aligned}$$

(2) We apply (1) to the case where f is a constant function of value 1 to obtain

$$1 = \frac{1}{2^d \text{vol}(P)} \sum_{\Theta \subseteq \{1, \dots, d\}} \int_P \delta_n(y) \text{ vol}(dy). \quad (8.2)$$

Therefore,

$$\begin{aligned} &S_n(f)(x) - \frac{1}{2^d} \sum_{\Theta \subseteq \{1, \dots, d\}} f_{\Theta}(x) \\ &= \frac{1}{2^d \text{vol}(P)} \sum_{\Theta \subseteq \{1, \dots, d\}} \int_P (f(x + \tau_{\Theta}(y)) - f_{\Theta}(x)) \delta_n(y) \text{ vol}(dy). \end{aligned}$$

By (8.1), we obtain that

$$\begin{aligned}\delta_n(y_1 e_1 + \cdots + y_d e_d) &= \prod_{j=1}^d \frac{\exp(\pi i(2n+1)y_j) - \exp(-\pi i(2n+1)y_j)}{\exp(\pi i y_j) - \exp(-\pi i y_j)} \\ &= \prod_{j=1}^d \frac{\sin(\pi(2n+1)y_j)}{\sin(\pi y_j)}.\end{aligned}$$

By our assumption,

$$(y \in P) \mapsto \frac{\sum_{\Theta \subseteq \{1, \dots, d\}} (f(x + \tau_\Theta(y)) - f_\Theta(x))}{\sin(\pi e_1^\vee(y)) \cdots \sin(\pi e_d^\vee(y))}$$

is integrable on P . This is because

$$(y \in P) \mapsto \frac{e_j^\vee(y)}{\sin(\pi(e_j^\vee(y)))}$$

is bounded. Therefore, by Fubini's theorem, Lemma 8.5.7 and dominated convergence theorem, we obtain that

$$\lim_{n \rightarrow +\infty} S_n(f)(x) = \frac{1}{2^d} \sum_{\Theta \subseteq \{1, \dots, d\}} f_\Theta(x).$$

□

8.5.9 Remark. The particular case of (8.2) in the case where $d = 1$ shows that

$$\int_0^{1/2} \frac{\sin(\pi(2n+1)t)}{\sin(\pi t)} dt = \frac{1}{2}.$$

8.5.10 Definition. For any $n \in \mathbb{N}$ and $j \in \{1, \dots, d\}$, we define $F_{n,j} : \mathbb{T}_M \rightarrow \mathbb{R}$ as

$$F_{n,j} = \frac{1}{n+1} \sum_{k=0}^n \delta_{k,j}.$$

Note that

$$F_{n,j}(0) = \frac{1}{n+1} \sum_{k=0}^n (2k+1) = n+1,$$

and

$$F_{n,j}(x) = \frac{\varphi_j(x)^{n+2} - \varphi_j(x) + \varphi_j(x)^{-n} - \varphi_j(x)}{(n+1)(\varphi_j(x) - 1)^2}$$

when $x \neq 0$.

If, by abuse of notation, we consider $F_{n,j}$ as a M -periodic function on E , then

$$\begin{aligned} F_{n,j}(y_1 e_1 + \cdots + y_d e_d) &= \frac{\exp(2\pi(n+2)iy_j) + \exp(-2\pi n i y_j) - 2\exp(2\pi i y_j)}{(n+1)(\exp(2\pi i y_j) - 1)^2} \\ &= \frac{\exp(2\pi(n+1)iy_j) + \exp(-2\pi(n+1)iy_j) - 2}{(n+1)(\exp(\pi i y_j) - \exp(-\pi i y_j))^2} \\ &= -\frac{\cos(2\pi(n+1)y_j) - 1}{2(n+1)\sin(\pi y_j)^2} = \frac{\sin(\pi(n+1)y_j)^2}{(n+1)\sin(\pi y_j)^2} \end{aligned}$$

whenever $y_1 e_1 + \cdots + y_d e_d$ does not belong to M .

The n -th Féjer kernel F_n is defined as the product of $F_{n,j}$ when j varies in $\{1, \dots, d\}$. Namely, as a M -periodic function on E , one has

$$F_n(y_1 e_1 + \cdots + y_d e_d) = \prod_{j=1}^d \frac{\sin(\pi(n+1)y_j)^2}{(n+1)\sin(\pi y_j)^2}.$$

Note that (8.2) in the particular case where $d = 1$ shows that

$$\int_0^{1/2} \frac{\sin(\pi(n+1)t)^2}{(n+1)\sin(\pi t)^2} dt = \frac{1}{2}.$$

8.5.11 Theorem (Féjer). *Let $f : \mathbb{T}_M \rightarrow \mathbb{C}$ be a continuous mapping. For any $n \in \mathbb{N}$, let*

$$\tilde{S}_n(f) = \frac{1}{n+1} \sum_{k=0}^n S_k(f).$$

Then the sequence $(\tilde{S}_n(f))_{n \in \mathbb{N}}$ converges uniformly to f .

Proof. By definition, one has

$$\tilde{S}_n(f)(x) - f(x) = \frac{1}{2^d \text{vol}(P)} \int_P \sum_{\Theta \in \{1, \dots, d\}} (f(x + \tau_\Theta(y)) - f(x)) F_n(y) \text{vol}(dy).$$

For any $\delta \in]0, \frac{1}{2}[$ and any $\Psi \subseteq \{1, \dots, d\}$, we let

$$\begin{aligned} P_\Psi(\delta) &= \{a_1 e_1 + \cdots + a_d e_d \mid (a_1, \dots, a_d) \in]0, \delta[{}^{\{1, \dots, d\} \setminus \Psi} \times [\delta, \frac{1}{2}]^\Psi\}, \\ \varepsilon_1(\delta) &:= \sup_{\substack{(a,b) \in E \times E \\ \|a-b\| \leq \delta}} |f(a) - f(b)|, \quad \varepsilon_2(\delta) = \int_\delta^{1/2} \frac{1}{\sin(\pi y)^2} dy. \end{aligned}$$

Without loss of generality, we may assume that vol identifies with the usual Lebesgue measure when we consider the isomorphism from $\mathbb{R}^d$ to E sending $(a_1, \dots, a_d)$

to $a_1e_1 + \dots + a_de_d$. In this case $\frac{1}{2^d \text{vol}(P)} = 1$. Then one has

$$\begin{aligned} |\tilde{S}_n(f)(x) - f(x)| &\leq \int_{P_\emptyset(\delta)} 2^d \varepsilon_1(\delta) F_n(y) \text{vol}(dy) \\ &+ \sum_{\substack{\Psi \subseteq \{1, \dots, d\} \\ \Psi \neq \emptyset}} 2^{d+1} \|f\|_{\text{sup}} \left(\frac{1}{2}\right)^{d-\text{card}(\Psi)} \left(\frac{\varepsilon_2(\delta)}{n+1}\right)^{\text{card}(\Psi)} \\ &\leq 2^{d-1} \varepsilon_1(\delta) + \sum_{\substack{\Psi \subseteq \{1, \dots, d\} \\ \Psi \neq \emptyset}} 2^{d+1} \|f\|_{\text{sup}} \left(\frac{1}{2}\right)^{d-\text{card}(\Psi)} \left(\frac{\varepsilon_2(\delta)}{n+1}\right)^{\text{card}(\Psi)}. \end{aligned}$$

Hence

$$\limsup_{n \rightarrow +\infty} \|\tilde{S}_n(f) - f\|_{\text{sup}} \leq 2^{d-1} \varepsilon_1(\delta).$$

Taking the limit when $\delta \rightarrow 0+$, we get

$$\limsup_{n \rightarrow +\infty} \|\tilde{S}_n(f) - f\|_{\text{sup}} = 0,$$

as required. $\square$

8.5.12 Lemma. *Let X be a locally compact Hausdorff space. We assume that X has a countable topological basis. Let μ be a Radon measure on X . For any $p \in \mathbb{R}_{\geq 1}$, $C_c(X)$ is dense in $\mathcal{L}^p(X, B(X), \mu)$ (the space of Borel measurable real valued functions f on X such that $\|f\|_{L^p} < +\infty$).*

Proof. Let V be the vector space over $\mathbb{R}$ generated by functions of the form $\mathbb{1}_A$, where A is a Borel subset of X such that $\mu(A) < +\infty$.

Let $f : X \rightarrow \mathbb{R}_{\geq 0}$ be an element of $\mathcal{L}^p(X)$. We could write f as the pointwise limit of an increasing sequence $(f_n)_{n \in \mathbb{N}}$ in V . The sequence $((f - f_n)^p)_{n \in \mathbb{N}}$, which is dominated by f^p , then converges to 0 in $\mathcal{L}^1(X, B(X), \mu)$. Hence

$$\lim_{n \rightarrow +\infty} \|f - f_n\|_{L^p} = 0.$$

It remains to check that, for any Borel set A , $\mathbb{1}_A$ can be approximated in $\mathcal{L}^p(X, B(X), \mu)$ by an element of $C_c(X)$. By Proposition 5.4.3, we obtain that, for any $\varepsilon > 0$ there exist a compact subset K of A and an open subset U of X such that $A \subset U$ and $\mu(U \setminus K) < \varepsilon$. By Uryson's lemma, there exists a continuous mapping $h : X \rightarrow [0, 1]$ such that $h(x) = 1$ for any $x \in K$ and that $\text{Supp}(h) \subseteq U$. Thus

$$\int_X |\mathbb{1}_A - h|^p d\mu \leq \mu(U \setminus K) \leq \varepsilon.$$

Hence $C_c(X)$ is dense in $\mathcal{L}^p(X, B(X), \mu)$. $\square$

8.5.13 Theorem. *Let f be an element of $\mathcal{L}^2(\mathbb{T}_M)$. Then the sequence $(S_n(f))_{n \in \mathbb{N}}$ converges in $L^2(\mathbb{T}_M)$ to f . In particular, $(\varphi_\alpha)_{\alpha \in \mathbb{N}}$ forms an orthonormal basis of $L^2(\mathbb{T}_M)$ (again by abuse of notation we use φ_α to denote its equivalence class in $L^2(\mathbb{T}_M)$).*

Proof. We first treat the case where f is a continuous mapping from $\mathbb{T}_M$ to $\mathbb{C}$. By definition, $S_n(f)$ is the orthogonal projection of f to the vector subspaces generated by φ_α with $\|\alpha\| \leq n$. Therefore, one has

$$\|f - S_n(f)\|_{L^2} \leq \|f - \tilde{S}_n(f)\|_{L^2} \leq \|f - \tilde{S}_n(f)\|_{\sup}.$$

Taking the limit when $n \rightarrow +\infty$, we obtain that

$$\lim_{n \rightarrow +\infty} \|S_n(f) - f\|_{L^2} = 0.$$

We now consider the general case. By Lemma 8.5.12, for any $\varepsilon > 0$, there exists $g \in C_c(\mathbb{T}_M)$ such that $\|f - g\|_{L^2} \leq \varepsilon$. Note that an orthogonal projection has operator norm 1. We obtain that

$$\|S_n(f) - S_n(g)\|_{L^2} \leq \varepsilon.$$

Therefore

$$\|S_n(f) - f\|_{L^2} \leq \|S_n(g) - g\|_{L^2} + \|S_n(f) - S_n(g)\|_{L^2} + \|f - g\|_{L^2} \leq \|S_n(g) - g\|_{L^2} + 2\varepsilon.$$

Therefore,

$$\limsup_{n \rightarrow +\infty} \|S_n(f) - f\|_{L^2} \leq 2\varepsilon.$$

Since ε is arbitrary, we obtain

$$\lim_{n \rightarrow +\infty} \|S_n(f) - f\|_{L^2} = 0.$$

□

8.5.14 Corollary (Parseval's equality). *Let f be an element of $L^2(\mathbb{T}_M)$. One has*

$$\|f\|_{L^2}^2 = \sum_{\alpha \in \mathbb{N}} |c_\alpha(f)|^2.$$

Proof. By definition, one has

$$\|S_n(f)\|_{L^2}^2 = \sum_{\alpha \in \mathbb{N}, \|\alpha\| \leq n} |c_\alpha(f)|^2.$$

Taking the limit when $n \rightarrow +\infty$, we obtain the desired result. □

8.6 Fourier series on the real line

In this section, we apply the theory of Fourier series to the particular case where $E = \mathbb{R}$. In this case the subgroup is of the form $M = \mathbb{Z}T$, and the dual group is of the form $N = \mathbb{Z}T^{-1}$, where T denotes a positive real number. The results obtained in the previous section lead to the following statements.

- (1) The functions

$$\varphi_n(t) = \exp(2\pi int/T), \quad n \in \mathbb{Z}$$

forms an orthonormal basis of $L^2(\mathbb{R}/\mathbb{Z}T)$ equipped with the inner product induced by

$$\forall (f, g) \in \mathcal{L}^2(\mathbb{R}/\mathbb{Z}T)^2, \quad \langle f, g \rangle := \frac{1}{T} \int_0^T \overline{f(t)} g(t) dt.$$

- (2) For any $f \in \mathcal{L}^1(\mathbb{R}/\mathbb{Z}T)$ and any $n \in \mathbb{Z}$, the n -th Fourier coefficient of f is given by

$$c_n(f) := \langle \varphi_n, f \rangle = \frac{1}{T} \int_0^T \exp(-2\pi int/T) f(t) dt.$$

The Fourier series of f is defined as

$$S_n(f) := \sum_{k=-n}^n c_k(f) \varphi_k.$$

In the case where f takes value in $\mathbb{R}$, one has

$$\forall n \in \mathbb{Z}, \quad c_n(f) = \overline{c_{-n}(f)}.$$

In particular, $c_0(f) \in \mathbb{R}$, and, for any positive integer n

$$\begin{aligned} c_n(f) \varphi_n(t) + c_{-n}(f) \varphi_{-n}(t) &= 2 \operatorname{Re}(c_n(f) \varphi_n(t)) \\ &= 2 \operatorname{Re}(c_n(f)) \cos(2\pi nt/T) - 2 \operatorname{Im}(c_n(f)) \sin(2\pi nt/T). \end{aligned}$$

One can then rewrite $S_n(f)$ as

$$S_n(f)(t) = a_0(f) + \sum_{k=1}^n (a_k(f) \cos(2\pi k t/T) + b_k(f) \sin(2\pi k t/T)),$$

where

$$a_0(f) = \frac{1}{T} \int_0^T f(t) dt,$$

and

$$a_k(f) = \frac{2}{T} \int_0^T \cos(2\pi k t/T) f(t) dt, \quad b_k(f) = \frac{2}{T} \int_0^T \sin(2\pi k t/T) f(t) dt.$$

(3) For any $f \in \mathcal{L}^2(\mathbb{R}/\mathbb{Z}T)$, the Fourier series of f , which is defined as

$$S_n(f) := \frac{1}{T} \sum_{k=-n}^n \left(\int_0^T \exp(-2\pi ikt/T) f(t) dt \right) \varphi_k, \quad n \in \mathbb{N}$$

converges in $\mathcal{L}^2(\mathbb{R}/\mathbb{Z}T)$ to f .

(4) Let $f : \mathbb{R}/\mathbb{Z}T \rightarrow \mathbb{C}$ be a mapping. We view f as a T -periodic mapping from $\mathbb{R}$ to $\mathbb{C}$. Let $x \in \mathbb{R}$. We assume that

$$f(x+) := \lim_{\varepsilon > 0, \varepsilon \rightarrow 0} f(x + \varepsilon) \quad \text{and} \quad f(x-) := \lim_{\varepsilon > 0, \varepsilon \rightarrow 0} f(x - \varepsilon)$$

exist, and the function

$$(y \in]0, \tfrac{1}{2}[) \mapsto \frac{f(x+y) + f(x-y) - f(x+) - f(x-)}{y}$$

is integrable with respect to Lebesgue measure. Then the sequence $(S_n(f)(x))_{n \in \mathbb{N}}$ converges to

$$\frac{f(x+) + f(x-)}{2}.$$

8.6.1 Proposition. *Let $f : \mathbb{R}/\mathbb{Z}T \rightarrow \mathbb{C}$ be a continuous mapping. Assume that, viewed as a T -periodic function on $\mathbb{R}$, f is of class C^1 . Then*

$$c_n(f') = \frac{2\pi i n c_n(f)}{T}.$$

Proof. By integration by parts, one has

$$\begin{aligned} c_n(f') &= \frac{1}{T} \int_0^T f'(t) \exp(-2\pi i n t/T) dt \\ &= \left[\frac{f(t) \exp(-2\pi i n t/T)}{T} \right]_0^T - \frac{1}{T} \int_0^T f(t) \frac{-2\pi i n}{T} \exp(-2\pi i n t/T) dt \\ &= \frac{1}{T} (f(T) - f(0) + 2\pi i n c_n(f)) \\ &= \frac{2\pi i n c_n(f)}{T}. \end{aligned}$$

□

8.6.2 Proposition (Wirtinger inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a T -periodic function of class C^1 . Let*

$$m_f = \frac{1}{T} \int_0^T f(t) dt.$$

Then the following inequality holds

$$\int_0^T |f(t) - m_f|^2 dt \leq \frac{T^2}{4\pi^2} \int_0^T |f'(t)|^2 dt.$$

Proof. By adding to f a constant function, we may assume that $m_f = 0$. Since f is of class C^1 , one has

$$c_n(f') = \frac{2\pi i n c_n(f)}{T}.$$

In particular, $c_0(f') = c_0(f) = 0$. By Parseval's equality, one has

$$\begin{aligned} \frac{1}{T} \int_0^T |f'(t)|^2 dt &= \sum_{n \in \mathbb{Z} \setminus \{0\}} |c_n(f')|^2 \geq \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{4\pi^2 n^2}{T^2} |c_n(f)|^2 \\ &\geq \frac{4\pi^2}{T^2} \sum_{n \in \mathbb{Z} \setminus \{0\}} |c_n(f)|^2 = \frac{4\pi^2}{T^2} \cdot \frac{1}{T} \int_0^T |f(t)|^2 dt, \end{aligned}$$

as required. $\square$

8.7 Fourier transform

8.7.1 Definition. Let G be an abelian topological group which is locally compact and Hausdorff. A *character* of G is a continuous group homomorphism

$$\chi : G \rightarrow S^1,$$

where

$$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}.$$

The *Pontryagin dual* of G is the group of characters of G :

$$\widehat{G} = \text{Hom}_{\text{cts}}(G, S^1).$$

It is equipped with pointwise multiplication

$$(\chi_1 \chi_2)(x) = \chi_1(x) \chi_2(x),$$

identity element $1_G^\vee(x) = 1$ for all $x \in G$, and inverse $\chi^{-1}(x) = \overline{\chi(x)}$. For any pair (K, V) of subsets of G and S^1 respectively, such that K is compact in G and V is an open neighbourhood of the identity in S^1 , we define a subset of $\widehat{G}$ in the following way

$$W(K, V) = \{\chi \in \widehat{G} \mid \chi(K) \subseteq V\}.$$

One can check that the sets of the form $W(K, V)$ form a neighbourhood basis of the trivial character $1_G^\vee$ and hence determine a topology on $\widehat{G}$ called *compact-open topology*.

8.7.2 Remark. If G is discrete, then compact-open topology on $\widehat{G}$ agrees with the topology of pointwise convergence.

8.7.3 Definition. Let G be an abelian locally compact topological group and let μ be a Haar measure on G . Let $f \in L^1(G)$. We define $\widehat{f}: \widehat{G} \rightarrow \mathbb{C}$ by the formula

$$\widehat{f}(\chi) = \int_G f(y) \overline{\chi}(y) \mu(dy)$$

8.7.4 Definition. We say that G is *Pontryagin self-dual* if $G \simeq \widehat{G}$.

8.7.5 Example. If we consider the topological group $G = \mathbb{R}$ with addition, then

$$\Phi: G \ni t \mapsto (s \mapsto e^{2\pi i s t}) \in \widehat{G}$$

is an isomorphism of topological groups. It is easy to see that it is a group isomorphism. One has to show, that it is a homeomorphism. Since, ϕ is a group homomorphism, it is enough to show homeomorphicity at 0. To show it, we will prove that

- (A) for any $W(K, V)$ there exists $\delta > 0$ such that $\Phi((-\delta, \delta)) \subseteq W(K, V)$ (Φ is continuous),
- (B) for any $\delta > 0$ there exist K, V such that $W(K, V) \subseteq \Phi((-\delta, \delta))$ (ϕ is open).

First we notice, that

$$V_\epsilon := \{e^{2\pi i s} \mid |s| < \epsilon\}$$

forms a basis neighbourhoods of $1 \in S^1$. It is therefore enough to only use V of this form in our proof. In addition, any compact $K \subseteq \mathbb{R}$ is contained in some $[-M, M]$, so it is enough to only consider compact sets of the form $[-M, M]$ in our proof. For (A), choose $W([-M, M], V_\epsilon)$. Then one can take $\delta = \epsilon/M$. To prove (B), choose $\delta > 0$. Notice that, any M , and ϵ such that $\delta < \epsilon/M$ gives $W([-M, M], V_\epsilon)$ that satisfies the desired property. We can then conclude that $\mathbb{R}$ is self-dual. The Fourier transform in this case takes the following form

$$\widehat{f}(s \mapsto e^{2\pi i s t}) = \widehat{f}(t) = \int_{\mathbb{R}} f(s) e^{-2\pi i s t} ds.$$

8.7.6 Remark. As a consequence of $\mathbb{R}$ being self-dual, we obtain that $\mathbb{R}^d$ is self-dual, for any $d \in \mathbb{Z}_{\geq 1}$. We conclude that any finitely dimensional vector space over $\mathbb{R}$ is Pontryagin self-dual.

8.7.7 Example. We fix a finite-dimensional vector space E over $\mathbb{R}$ and a sub- $\mathbb{Z}$ -module M of E generated by a basis $(e_i)_{i=1}^d$ of E . Let N be the sub- $\mathbb{Z}$ -module of $E^\vee$ generated by the dual basis. Denote by $\mathbb{T}_M$ the quotient group E/M , equipped

with the unique Haar measure η such that $\eta(\mathbb{T}_M) = 1$. Let f be an element of $L^1(\mathbb{T}_M)$. Then the Fourier coefficient

$$c_\alpha(f) = \int_{\mathbb{T}_M} \overline{\varphi_\alpha(x)} f(x) \eta(dx) = \int_{\mathbb{T}_M} \varphi_{-\alpha}(x) f(x) \eta(dx).$$

equals $\widehat{f}(\varphi_\alpha)$. In fact, one can check that N is Pontryagin dual of $\mathbb{T}_M$. In general, one can speak of Fourier series whenever one considers Fourier transform on a compact group. From general theory, the Pontryagin dual of such group is always discrete.

Exercises

Write $C([0, 1])$ for the space of continuous functions on $[0, 1]$. For each integer $r \geq 1$, write $C^r([0, 1])$ for the space of functions $f \in C([0, 1])$ such that for every $1 \leq j \leq r$, the j -th derivative $f^{(j)}$ exists on $(0, 1)$ and extends continuously to $[0, 1]$. Write $C_c^\infty(\mathbb{R}^2)$ for the space of smooth functions with compact support on $\mathbb{R}^2$.

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and identify functions on $\mathbb{T}$ with 1-periodic functions on $\mathbb{R}$. For $n \in \mathbb{Z}$, set $\varphi_n(t) = e^{2\pi i n t}$. Then $(\varphi_n)_{n \in \mathbb{Z}}$ forms an orthonormal basis of $L^2(\mathbb{T})$. We identify $L^2(\mathbb{T})$ with $L^2([0, 1])$ via 1-periodic representatives. For an L^2 -integrable function f on $[0, 1]$, define its Fourier coefficients by

$$\widehat{f}(n) = \langle \varphi_n, f \rangle = \int_0^1 f(t) e^{-2\pi i n t} dt, \quad n \in \mathbb{Z}.$$

Then Parseval's identity states that

$$\|f\|_{L^2}^2 = \sum_{n \in \mathbb{Z}} |\widehat{f}(n)|^2.$$

The notation and the corresponding identities extend naturally to higher dimensions.

For $N \geq 1$, define the operator S_N by

$$S_N(f) = \sum_{|n| \leq N} \widehat{f}(n) \varphi_n(t).$$

The Dirichlet Convergence Theorem states that if f is piecewise smooth and 1-periodic, then the partial Fourier series $\{S_N(f)(x)\}_{N \geq 1}$ converges to $\frac{1}{2}(f(x^+) + f(x^-))$ pointwisely.

1. Let $f(x) = x^2$. For $p \in \{1, 2, \infty\}$, find the best constant $c_p \in \mathbb{R}$ minimizing $\|f - c\|_{L^p}$, and compute $d_p := \min_{c \in \mathbb{R}} \|f - c\|_{L^p}$.

2. Let H be a Hilbert space and $A \subseteq H$ be a subset.

- (a) Prove that $A^\perp$ is a closed subspace of H .
- (b) Prove that $\overline{\text{span}(A)} = (A^\perp)^\perp$.
- (c) Deduce that $\text{span}(A)$ is dense in H if and only if $A^\perp = \{0\}$.
- (d) Recall the Weierstrass approximation theorem: every continuous function on a compact interval can be uniformly approximated by polynomials, that is, approximated in the supremum norm, equivalently the L^∞ -norm on continuous functions. Using this theorem, prove that if $f \in L^2([0, 1])$ satisfies

$$\int_0^1 f(x)x^n dx = 0, \quad n \geq 0,$$

then $f = 0$ in $L^2([0, 1])$.

3. Compute the Fourier coefficients of the following functions on $[0, 1]$, and prove the indicated identities.

- (a) Let $f(t) = t$ for $0 \leq t \leq 1$. Prove that

$$\widehat{f}(0) = \frac{1}{2}, \quad \widehat{f}(n) = -\frac{1}{2\pi in} \quad (n \neq 0).$$

Deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Moreover, using the pointwise convergence of the Fourier series at $t = 1/4$, prove that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

- (b) Let $f(t) = t(1 - t)$ for $0 \leq t \leq 1$. Prove that

$$\widehat{f}(0) = \frac{1}{6}, \quad \widehat{f}(n) = -\frac{1}{2\pi^2 n^2} \quad (n \neq 0).$$

Deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

(c) Let $f(t) = e^{\alpha t}$ for $0 \leq t \leq 1$, where $\alpha \in \mathbb{R} \setminus \{0\}$. Prove that

$$\widehat{f}(n) = \frac{e^\alpha - 1}{\alpha - 2\pi i n}, \quad n \in \mathbb{Z}.$$

Deduce that

$$\sum_{n \in \mathbb{Z}} \frac{1}{\alpha^2 + 4\pi^2 n^2} = \frac{e^\alpha + 1}{2\alpha(e^\alpha - 1)} = \frac{1}{2\alpha} \coth\left(\frac{\alpha}{2}\right).$$

(d) Prove that

$$\log(2 \sin \pi t) = - \sum_{n=1}^{\infty} \frac{\cos(2\pi n t)}{n}, \quad 0 < t < 1.$$

Deduce that

$$\int_0^1 \log^2(2 \sin \pi t) dt = \frac{\pi^2}{12}.$$

4. Let f be a continuous function on $[0, 1]$.

(a) If $f \in C^1([0, 1])$, prove that for $n \in \mathbb{Z}$,

$$\widehat{f'}(n) = f(1) - f(0) + 2\pi i n \widehat{f}(n).$$

(b) If $k \geq 1$ and $f \in C^k([0, 1])$, prove that for $n \in \mathbb{Z}$,

$$\widehat{f^{(k)}}(n) = \sum_{\ell=0}^{k-1} (2\pi i n)^\ell (f^{(k-1-\ell)}(1) - f^{(k-1-\ell)}(0)) + (2\pi i n)^k \widehat{f}(n).$$

(c) If $f^{(j)}(0) = f^{(j)}(1)$ for $j = 0, \dots, k-1$, prove that

$$\widehat{f^{(k)}}(n) = (2\pi i n)^k \widehat{f}(n), \quad n \in \mathbb{Z}.$$

5. Let f, g be 1-periodic functions on $\mathbb{R}$ which is L^2 -integrable on $[0, 1]$.

(a) For $a \in \mathbb{R}$, define $T_a(f)(t) = f(t - a)$. Prove that

$$\widehat{T_a(f)}(n) = e^{-2\pi i n a} \widehat{f}(n).$$

(b) For $m \in \mathbb{Z}$, define $A_m(f)(t) = \varphi_m(t)f(t)$. Prove that

$$\widehat{A_m(f)}(n) = \widehat{f}(n - m).$$

(c) For $0 \leq r < 1$, define the Poisson kernel by

$$P_r(t) = \frac{1 - r^2}{1 - 2r \cos(2\pi t) + r^2}.$$

Prove that for $n \in \mathbb{Z}$,

$$\widehat{P_r}(n) = r^{|n|}, \quad \widehat{P_r * f}(n) = r^{|n|} \widehat{f}(n), \quad \int_0^1 P_r(t)^2 dt = \frac{1 + r^2}{1 - r^2}.$$

(d) Let $h(t) = f(t)g(t)$. Prove that for $n \in \mathbb{Z}$,

$$\widehat{h}(n) = \sum_{m \in \mathbb{Z}} \widehat{f}(n - m) \widehat{g}(m) =: (\widehat{f} * \widehat{g})(n).$$

6. For $t > 0$, define two 1-periodic functions on $\mathbb{R}$ by

$$K_t(x) = \sum_{n \in \mathbb{Z}} e^{-\pi t n^2} e^{2\pi i n x}$$

and

$$G_t(x) = \frac{1}{\sqrt{t}} \sum_{m \in \mathbb{Z}} e^{-\pi(x+m)^2/t}.$$

You may use the Gaussian Fourier integral

$$\int_{-\infty}^{+\infty} e^{-\pi u^2/t} e^{-2\pi i n u} du = \sqrt{t} e^{-\pi t n^2}, \quad n \in \mathbb{Z}.$$

(a) Prove that K_t and G_t are continuous functions on $\mathbb{T}$.

(b) Prove that $\widehat{K_t}(n) = \widehat{G_t}(n) = e^{-\pi t n^2}$ for $n \in \mathbb{Z}$.

(c) Deduce that for $x \in \mathbb{R}$,

$$\sum_{n \in \mathbb{Z}} e^{-\pi t n^2} e^{2\pi i n x} = \frac{1}{\sqrt{t}} \sum_{m \in \mathbb{Z}} e^{-\pi(x+m)^2/t}.$$

(d) Define the theta function by

$$\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi t n^2}.$$

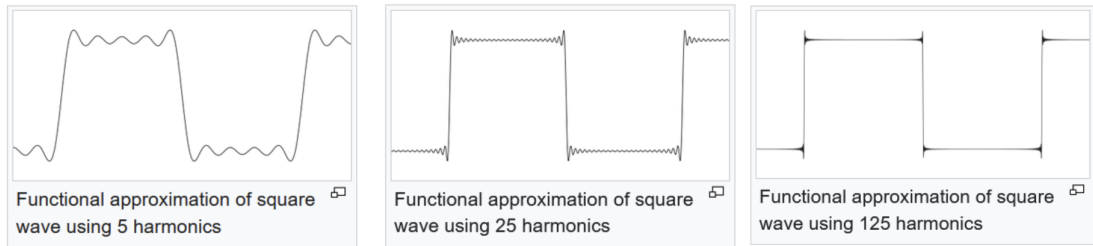
Prove the transformation formula

$$\theta(t) = t^{-1/2} \theta(1/t).$$

(e) Deduce that $\theta(t) \sim t^{-1/2}$ as $t \rightarrow 0^+$.

7. The Gibbs phenomenon is a behavior of the Fourier series of a function with a jump discontinuity and is described as the following:

As more Fourier series constituents or components are taken, the Fourier series shows the first overshoot in the oscillatory behavior around the jump point approaching about 8.95% of the (full) jump and this oscillation does not disappear but gets closer to the point so that the integral of the oscillation approaches zero.



Let h be the function on $[0, 1]$ defined by

$$h(t) = \operatorname{sgn}(1 - 2t), \quad 0 \leq t \leq 1.$$

- (a) Prove that

$$\widehat{h}(0) = 0, \quad \widehat{h}(n) = \frac{1 - (-1)^n}{\pi i n} \quad (n \neq 0).$$

- (b) Deduce that

$$S_N(h)(t) = \frac{4}{\pi} \sum_{\substack{1 \leq n \leq N \\ n \text{ odd}}} \frac{\sin(2\pi n t)}{n}.$$

- (c) Let $N = 2M + 1$ and set $t_M = \frac{1}{4(M+1)}$. Prove that

$$S_{2M+1}(h)(t_M) = \frac{4}{\pi} \sum_{j=0}^M \frac{1}{2j+1} \sin\left(\frac{(2j+1)\pi}{2(M+1)}\right).$$

- (d) Prove that

$$\lim_{M \rightarrow \infty} S_{2M+1}(h)(t_M) = \frac{2}{\pi} \int_0^\pi \frac{\sin u}{u} du.$$

- (e) Numerically, we have

$$\frac{1}{\pi} \int_0^\pi \frac{\sin u}{u} du - \frac{1}{2} \approx 0.08948987 \dots$$

Based on this, provide an explanation of the Gibbs phenomenon.

(f) For $N \geq 0$, let $\tilde{S}_N(h) = \frac{1}{N+1} \sum_{k=0}^N S_k(h)$ be the Féjer mean. Then

$$\tilde{S}_N(h)(t) = (F_N * h)(t) = \int_0^1 F_N(s) h(t-s) ds,$$

where

$$F_N(s) = \frac{\sin(\pi(N+1)s)^2}{(N+1)\sin(\pi s)^2}$$

is the Fejér kernel, which satisfies $\int_0^1 F_N(s) ds = 1$. Based on this, prove that

$$-1 \leq \tilde{S}_N(h)(t) \leq 1$$

for all $N \geq 0$.

8. The Bernoulli polynomials are defined by

$$\frac{ze^{xz}}{e^z - 1} = \sum_{m=0}^{\infty} B_m(x) \frac{z^m}{m!}.$$

Let $B_m = B_m(0)$ be the Bernoulli numbers, then

$$B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0, B_6 = \frac{1}{42}.$$

(a) For $0 \leq m \leq 4$, compute $B_m(x)$.

(b) For $m \geq 1$, prove that

$$B'_m(x) = mB_{m-1}(x), \quad \int_0^1 B_m(x) dx = 0.$$

(c) For $m \geq 2$, prove that $B_m(1) = B_m(0)$.

(d) For $m \geq 0$, compute the Fourier coefficients of $B_m(x)$.

(e) For $k \geq 1$, define

$$\zeta(2k) = \sum_{n=1}^{\infty} \frac{1}{n^{2k}}.$$

Using Parseval's identity for $B_k(x)$, prove that

$$\|B_k\|_{L^2([0,1])}^2 = \frac{2(k!)^2}{(2\pi)^{2k}} \zeta(2k).$$

(f) Prove that for $m, n \geq 1$,

$$\int_0^1 B_m(x) B_n(x) dx = (-1)^{m-1} \frac{m!n!}{(m+n)!} B_{m+n}.$$

(g) Deduce that

$$\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k} B_{2k}}{2(2k)!}.$$

(h) Compute $\zeta(2)$, $\zeta(4)$ and $\zeta(6)$.

9. Let G be an abelian topological group which is locally compact and Hausdorff. A character of G is a continuous group homomorphism

$$\chi : G \rightarrow S^1,$$

where

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

The Pontryagin dual of G is the group of characters of G :

$$\widehat{G} = \text{Hom}_{\text{cont}}(G, S^1).$$

It is equipped with pointwise multiplication

$$(\chi_1 \chi_2)(x) = \chi_1(x) \chi_2(x),$$

identity element $1_G(x) = 1$, and inverse $\chi^{-1}(x) = \overline{\chi(x)}$. We equip $\widehat{G}$ with the compact-open topology inherited from $C(G, S^1)$. Equivalently, since S^1 is metrizable, this is the topology of uniform convergence on compact subsets of G : a basic neighbourhood of $\chi_0 \in \widehat{G}$ is given by

$$U(\chi_0; K, \varepsilon) = \left\{ \chi \in \widehat{G} : \sup_{x \in K} |\chi(x) - \chi_0(x)| < \varepsilon \right\},$$

where $K \subseteq G$ is compact and $\varepsilon > 0$.

(a) Prove that $\widehat{G}$ is an abelian topological group.

(b) Let $G = \mathbb{R}/\mathbb{Z}$. For $n \in \mathbb{Z}$, define

$$\chi_n(t + \mathbb{Z}) = e^{2\pi i n t}.$$

Prove that $\chi_n \in \widehat{G}$ and that the map

$$\mathbb{Z} \rightarrow \widehat{\mathbb{R}/\mathbb{Z}}, \quad n \mapsto \chi_n$$

is an isomorphism of topological groups, where $\mathbb{Z}$ is equipped with the discrete topology. You may use the standard fact that every character of $\mathbb{R}/\mathbb{Z}$ is of the form χ_n .

10. Let $G = \mathbb{Z}/N\mathbb{Z}$ with normalized counting measure and inner product

$$\langle f, g \rangle = \frac{1}{N} \sum_{x \in G} \overline{f(x)} g(x).$$

For $k \in \mathbb{Z}/N\mathbb{Z}$, define

$$\chi_k(x) = e^{2\pi i k x / N}.$$

(a) Prove that $(\chi_k)_{k \in \mathbb{Z}/N\mathbb{Z}}$ is an orthonormal basis of $L^2(G)$.

(b) For $f : G \rightarrow \mathbb{C}$, define

$$\widehat{f}(k) = \frac{1}{N} \sum_{x \in G} f(x) \overline{\chi_k(x)}.$$

Prove the Fourier inversion formula

$$f(x) = \sum_{k \in \mathbb{Z}/N\mathbb{Z}} \widehat{f}(k) \chi_k(x),$$

and Parseval's identity

$$\frac{1}{N} \sum_{x \in G} |f(x)|^2 = \sum_{k \in \mathbb{Z}/N\mathbb{Z}} |\widehat{f}(k)|^2.$$

(c) For $G = \mathbb{Z}/3\mathbb{Z}$, let

$$f(0) = 1, \quad f(1) = 2, \quad f(2) = 0.$$

Set $\omega = e^{2\pi i/3}$ and define

$$\chi_0(x) = 1, \quad \chi_1(x) = \omega^x, \quad \chi_2(x) = \omega^{2x}.$$

Compute $\widehat{f}(0), \widehat{f}(1), \widehat{f}(2)$, express f in the form

$$f = a_0 \chi_0 + a_1 \chi_1 + a_2 \chi_2,$$

and verify Parseval's identity directly.

11. Let $g \in C_c^\infty(\mathbb{R}^2)$ be a smooth function with compact support on $\mathbb{R}^2$, and define

$$\widehat{g}(\xi) = \int_{\mathbb{R}^2} g(x) e^{-2\pi i x \cdot \xi} dx.$$

Define

$$G(x) = \sum_{k \in \mathbb{Z}^2} g(x + k), \quad x \in \mathbb{R}^2.$$

- (a) Prove that G is a smooth function on $\mathbb{T}^2$.
- (b) Prove that for every $m \in \mathbb{Z}^2$, $\widehat{G}(m) = \widehat{g}(m)$.
- (c) Using the Fourier series convergence of smooth functions on $\mathbb{T}^2$, deduce the two-dimensional Poisson summation formula

$$\sum_{k \in \mathbb{Z}^2} g(k) = \sum_{m \in \mathbb{Z}^2} \widehat{g}(m).$$